Aviation Meteorology

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Capt. Pankaj Pahil

First edition · October 2026 · 29 chapters · sketchnote diagrams · practice questions · ghostaviator.com

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About this edition

Aviation Meteorology for the Indian CPL and ATPL examinations.
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Preface

Weather changes what an aircraft can do and what a pilot must prepare for. Begin with the atmosphere, then follow the movement of heat, moisture and air, and finally learn to interpret aviation weather information.

Read the explanation, study the illustrations that support it, and work through the questions. Short reminders help recall; the physical explanation remains the starting point.

The glossary explains common technical terms in plain language. The topic index links back to the relevant chapters.

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Chapter 1: Atmosphere

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1. Composition and Structure of the Atmosphere

📘 Definition — Atmosphere

The atmosphere of the earth is an **envelope of homogeneous mixture of gases**, called **Air**. It surrounds the earth and is attached to it due to gravitation. It moves with the earth at the same speed and direction.

- Total thickness: approximately 480 km
- 80% of the atmosphere is within 16 km
- No exact upper boundary — gradually merges with outer space

Characteristics of the Atmosphere

Property	Detail
Weight	Has weight and hence exerts pressure
Compressibility	Compressible and expandable
Shape	Occupies space, has no definite shape
Mobility	Mobile — allows transfer of heat and moisture
Conductivity	**Poor conductor of heat and electricity**

2. Constituents — Major Gases

Gas	Symbol	% by Volume
Nitrogen	N_2	78.09%
Oxygen	O_2	20.95%
Argon	Ar	0.93%
Carbon dioxide	CO_2	Variable trace gas; specify date and observing dataset for a numerical value

Trace Gases

Neon, Helium, Methane, Krypton, Xenon, Nitrous Oxide, Iodine, H_2, O_3 (0.000007%), Ammonia, Carbon Monoxide, CO_2, SO, Nitrogen Dioxide. The atmosphere also contains water vapour (WV) and solid particles.

⭐ N_2 : O_2 Ratio — Exam Favourite

By Volume: $N_2 : O_2 = 4 : 1$

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By Weight: $N_2 : O_2 = 3 : 1$

⚠ Supplementary Oxygen Required

Pressure and oxygen partial pressure fall with altitude, increasing hypoxia risk (too little oxygen reaching the body). Supplemental-oxygen decisions depend on cabin pressure, exposure and aircraft procedures; 10,000 ft is not a universal meteorological threshold.

Homosphere vs Heterosphere

Layer	Height	Characteristic
Homosphere	Up to ~80 km	Nearly uniform composition; well-mixed
Heterosphere	Above 80 km	Gases separate by molecular weight; composition varies

3. Variable Gases, Greenhouse Gases & Dry/Saturated Air

Variable Gases

The gases like Water Vapour (WV), Carbon Mono Oxide, Sulphur di Oxide, Nitrogen di Oxide, and Methane **vary in amount from place to place**, being concentrated more in industrial areas, cities and water bodies, than in open areas. Though very small in quantity, these gases are very significant for weather and life.

Greenhouse Gases

📘 Greenhouse Effect

Water vapour, CO_2, O_3, Methane and some other gases are **transparent to solar (Short Wave) radiation** but **partially absorb terrestrial (Long Wave) radiation**, and re-radiate it. This keeps the earth warmer than it would have otherwise been.

Greenhouse gases: **H_2O, CO_2, O_3, CH_4 (Methane)**

⚠ Global Warming Warning

Excessive amount of greenhouse gases is causing **Global Warming, which is a serious threat to our life and needs to be limited.**

Dry Air vs Saturated Air

Water in the atmosphere can exist in three states:

- **Solid:** Snow, Ice, Hail
- **Liquid:** Drizzle, Rain, Shower
- **Gas:** Water Vapour (WV)

Concept	Detail
WV concentration	Rapidly **decreases with height**

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Concept	Detail
Warmer air	Can hold **greater amount** of WV
Tropics (23½°N to 23½°S)	Air can hold up to 4% WV by volume
Above 30,000 ft	Very low WV due to very low temperatures
Saturated air	The water-vapour pressure equals the saturation (water-vapour pressure has reached its saturation value at the current temperature) vapour pressure at that temperature: 100% RH. The required amount of vapour changes with temperature.
Unsaturated air	RH < 100%. Unsaturated air may still contain water vapour; “dry air” in composition and gas-law calculations means air without water vapour.

4. Carbon Dioxide (CO_2) & Ozone (O_3)

Carbon Dioxide (CO_2)

- Produced by burning of fuel, wood etc.
- Concentration substantial in **industrial areas**; but carbon dioxide mixes through the whole atmosphere, so its amount is nearly the same everywhere, including over the poles
- Large amount of CO_2 is dissolved in the oceans
- Plants absorb CO_2, use its carbon as food and release O_2 into atmosphere

Ozone (O_3)

📘 Ozone Properties

- Forms in the **upper atmosphere**
- Absorbs **Ultra Violet (UV) radiation** from the Sun → raises temperature
- The ozone distribution depends on chemical formation, destruction and transport by atmospheric circulation.
- Maximum O_3 concentration: 20–25 km
- O_3 found between 10–50 km

✅ Why Ozone Matters for Aviation & Life

Ozone protects us from the ill effects (like skin cancer, sunburn and cataracts) of UV radiation.

Ozone Hole

⚠️ Ozone Depletion

In recent times the O_3 layer has **thinned** and large holes have developed mainly over the Polar and other latitudes due to:

- Atmospheric pollution

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- Excessive use of **halocarbon refrigerants, solvents, chlorofluoro carbons**
- On reaching Stratosphere they release halogen atoms through **photodissociation** → breaks down O_3 into O_2

Solid Particles (Aerosols)

Particles such as salt from evaporating sea water, dust from arid regions, industrial particles:

- Obscure solar radiation and affect visibility and temperature
- Act as **condensation nuclei** → promote condensation (water vapour changes into liquid water and releases latent heat) of WV prematurely
- In larger cities/industrial areas: frequent **fog and smog** (mixture of fog and smoke) due to high concentration of particles

5. Vertical Distribution of Air Mass

Height	Fraction of Atmosphere Contained Below
6 km	1/2 (50%)
10 km	3/4 (75%)
35 km	99%

6. Thermal Structure of the Atmosphere

📘 How the Atmosphere is Heated

Solar radiation heats the earth and the earth in turn heats up the atmosphere by **conduction (heat transfer through direct contact), convection (heat transfer through movement within a fluid), radiation** and release of latent heat (energy absorbed or released during a change of state without a temperature change) by condensation of water vapour. Hence **temperature falls with height**, up to a certain height. **The atmosphere is thus heated from below and not from above.**

Heat Flow from Earth's Surface

Energy transfer	Plain meaning
Sensible heat	Heat transfer that changes temperature, including surface-to-air transfer by conduction and moving air
Latent heat	Energy absorbed or released when water changes state
Radiation	Energy carried by electromagnetic waves; a separate surface energy-budget term

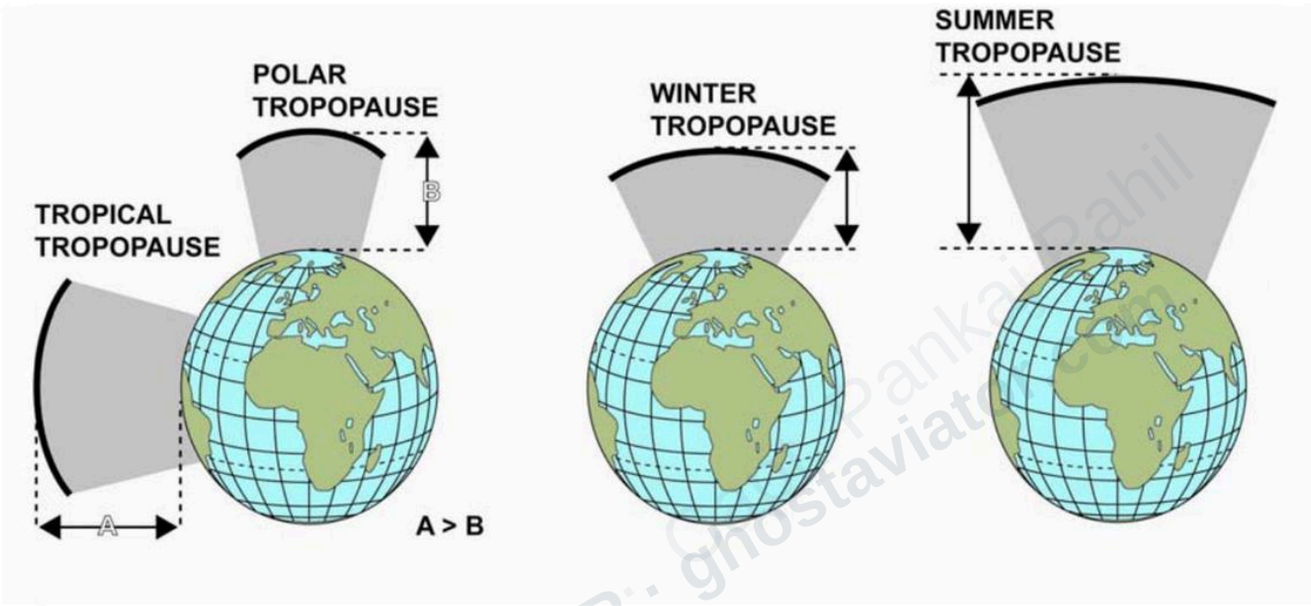
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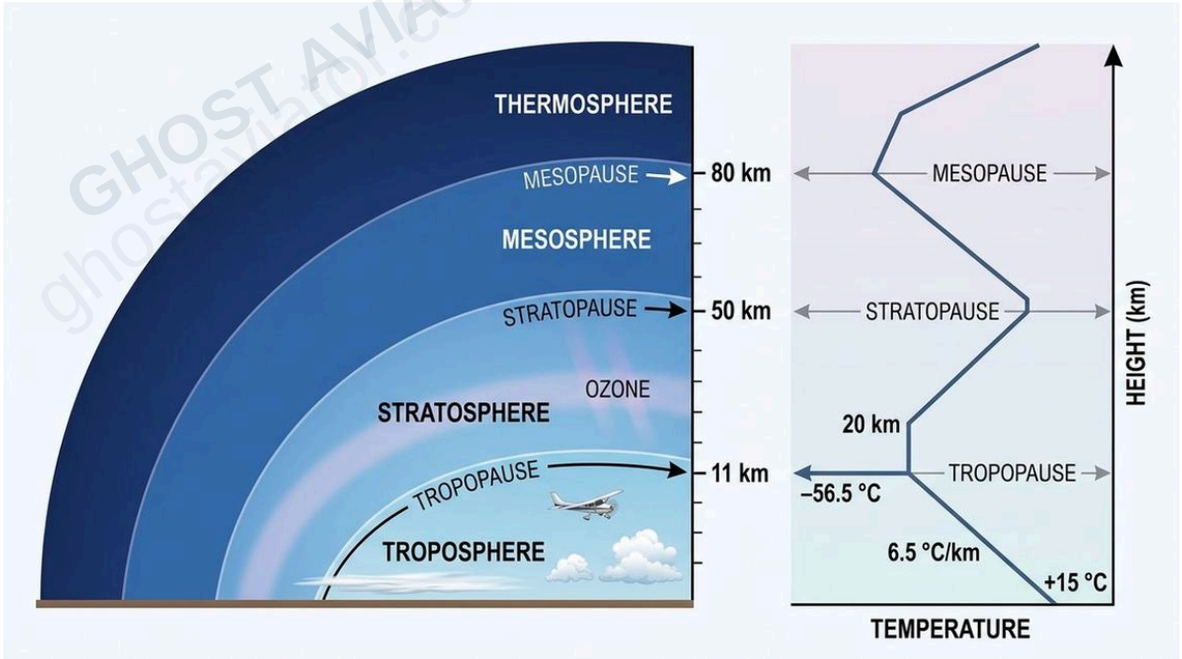
Exam Tip

Sensible heat changes temperature. Latent heat is absorbed or released during changes of state. Their relative contributions depend on the energy budget being discussed; do not apply one surface partition to every atmospheric heat transfer.

7. Atmospheric Layers in Detail



Tropopause height is highest over the tropics, lower in winter and higher in summer



The atmosphere in layers, with the temperature profile beside it

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A. Troposphere

📘 Troposphere

- Lowest portion of the atmosphere
- Extends from surface to 16–18 km at equator and 8–10 km at poles
- Higher height at equator due to **rising hot air convection currents**
- Temperature falls at ~6.5°C/km — called the **Lapse Rate**
- All kinds of weather occur here; turbulent mixing
- Contains 75% of atmospheric mass and 99% of water vapour and aerosols

Sub-Layer	Height Range
Lower Troposphere	Surface to 2.1 km
Mid Troposphere	2.1 km to 7.6 km
Upper Troposphere	7.6 km to Tropopause

B. Tropopause

📘 Tropopause

- Top of the troposphere; marks boundary between Troposphere and Stratosphere
- At tropopause: lapse rate reduces to 1–2°C/km or temperature stops falling
- Controlled by: Surface Temperature, Latitude, Season, Land-Sea distribution, Synoptic Situation
- Temperature at equator: –70° to –75°C
- Temperature at poles: –40° to –45°C
- At lower height in **Winters** than Summers
- **Warmer the surface → Higher the tropopause**

⭐ Tropopause Breaks/Folds

There are two regions where tropopause **abruptly changes height**, called **breaks/folds**:

- At about 40° Lat and 60° Lat
- Break at 40°Lat is more prominent
- **Jet streams occur at these breaks**

Type	Pressure Level	Latitude Range	Height (India)	Temp (India)	Associated Jet
Polar Tropopause	~300 hPa	Lat. 45°–60° polewards	–	–	Polar Front Jet (southern end)
Middle Tropopause	~200 hPa	Between Polar & Tropical	~11.5 km (winters ~23°N)	–45°C	–

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Type	Pressure Level	Latitude Range	Height (India)	Temp (India)	Associated Jet
Tropical Tropopause	~100 hPa	Equator to about Lat. 35°–40°	about 16–18 km	about –75° to –80°C	Subtropical Jet (STJ) near its poleward end

✅ **Importance of Tropopause for Pilots**

- Maximum height clouds can reach
- Jet streams and maximum wind speeds occur **just below it**
- **Clear Air Turbulence (CAT)** occurs just below it

C. Stratosphere

📘 **Stratosphere (Tropopause to ~50 km)**

- Initially (8–10 km above tropopause) **isothermal**
- Temperature then increases slowly and towards the top sharply
- **Inversion produced by absorption of UV radiation from sun by ozone**
- Ozone layer found at height of ~25 km
- Very **dry and stable** region — low humidity, no weather
- Above stormy weather; steady, strong, horizontal winds
- **Nacreous Clouds** (Mother of Pearl Clouds) sometimes seen in higher latitudes in winters

Stratopause: Upper boundary of the Stratosphere.

D. Mesosphere

📘 **Mesosphere (Stratopause to ~80 km)**

- Extends 50–80 km from Stratopause
- Lack of solar radiation absorption + weak vertical mixing → temperatures **fall with height**
- **Noctilucent Clouds** seen occasionally in Polar regions

Mesopause: Top of Mesosphere. Temperatures stop falling. Coldest area of atmosphere. Temperature at 80 km = –100°C (173.15 K). Meteors burn up in this layer.

E. Thermosphere

📘 **Thermosphere / Ionosphere (80 km+)**

- From Mesopause to end of atmosphere
- Temperatures **increase with height**
- Above 60 km also called **Ionosphere**
- Important for **radio wave propagation**
- Serious interference in radio propagation during solar disturbances

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F. Exosphere

📘 Exosphere (outermost layer)

- Outermost layer extending up to 10,000 km from top of Thermosphere
- Molecules, ozone and atoms escape into space
- Air is very thin — few particles; very active with very high temperature but far apart → very little heat present in this layer

8. International Standard Atmosphere (ISA)

📘 What is ISA?

A standard average atmosphere specified for various purposes like design and testing of aircraft, evaluation of aircraft performance, calibration of altimeters. The most widely used is the **ICAO ISA (International Standard Atmosphere)**.

ISA Specifications

Parameter	Value
Air Condition	Dry
Temperature at MSL	15°C (288.15 K)
Pressure at MSL	1013.25 hPa
Density (mass per unit volume) at MSL	1225 g/m³
Acceleration due to gravity	980.665 cm/s²
Lapse Rate (up to 11 km / 36,090 ft)	6.5°C/km (1.98°C/1000 ft ≈ 2°C/1000 ft)
Temperature 11–20 km (up to 65,617 ft)	–56.5°C (constant / isothermal)
Temperature 20–32 km	Rises at 1°C/km with temp –44.5°C at 32 km (104,987 ft)

ISA Deviation Formula

ISA Deviation = Actual Temperature − ISA Temperature

Worked Example 1:
Actual temp at 2 km = 05°C
ISA temp at 2 km = 15 – (6.5 × 2) = 15 – 13 = 2°C
ISA Deviation = 05 – 02 = **+03°C**

Worked Example 2:
Pressure at MSL = 1002.25 hPa
ISA Pressure at MSL = 1013.25 hPa
ISA Deviation = 1002.25 – 1013.25 = **–11 hPa**

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Worked Example 3:
Actual temp at 19 km = –60°C
ISA temp at 19 km = –56.5°C (constant 11–20 km)
ISA Deviation = –60 – (–56.5) = **–3.5°C** → answer –03.5°C

> ⭐ **Key ISA Numbers to Remember**
>
> **15-25-56.5-36090**
>
> MSL Temp = 15°C | MSL Pressure = 1013.25 hPa | Isothermal at –56.5°C | Isothermal starts at 36,090 ft (11 km)

9. Jet Standard Atmosphere (JSA)

> 📘 **Why JSA?**
>
> ISA values are satisfactory for aircraft operating below 30,000 ft, but NOT above. For space flights and very high altitude flying and engine manufacturing, additional specific values are required. JSA was introduced to meet these requirements.

Parameter	ISA	JSA
MSL Temperature	15°C (288.15 K)	15°C (288.15 K)
Lapse Rate	6.5°C/km (1.98°C/1000 ft)	2°C/1000 ft
Tropopause	Yes (at 11 km)	**No Tropopause**

Practice Questions & Answers

Q1Lowest layer of atmosphere is …… (a) Troposphere (b) Tropopause (c) Stratosphere

✅ Correct Answer: (a) Troposphere

The Troposphere is the lowest layer, starting at earth's surface and extending to the tropopause. All weather occurs here.
Distractor Analysis(b) Tropopause is a boundary, not a layer. (c) Stratosphere is above the Troposphere.

Q2Height of Tropopause at equator is …… (a) 8–10 km (b) 16–18 km (c) 14 km

✅ Correct Answer: (b) 16–18 km

At the equator, rising hot air due to convection pushes the tropopause higher — up to 16–18 km.
Distractor Analysis(a) 8–10 km is the tropopause height at the poles. (c) 14 km is intermediate but incorrect.

Q3Height of Tropopause at Poles is …… (a) 12 km (b) 13 km (c) 08 km

✅ Correct Answer: (c) 08 km

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At the poles, the atmosphere is colder; therefore convective activity is minimal and the tropopause is lower, around 8–10 km.
Distractor Analysis(a) and (b) are too high for polar regions.

Q4Higher the surface temperature …… would be the tropopause. (a) Higher (b) Lower (c) Same

✅ Correct Answer: (a) Higher

Warmer surfaces generate more convective activity, pushing the tropopause to a greater height.
Distractor Analysis(b) Lower — incorrect; this is the effect at colder surfaces/poles. (c) Same — temperature definitely affects tropopause height.

Q5Height of tropopause …… (a) Is constant (b) Varies with altitude (c) Varies with Latitude

✅ Correct Answer: (c) Varies with Latitude

Tropopause height depends on latitude (equator ~16–18 km; poles ~8 km) and also on season, surface temperature, synoptic situation.
Distractor Analysis(a) Is constant — definitely not; it varies significantly. (b) Varies with altitude — altitude is not the controlling factor.

Q6. Temperature near the tropopause

The tropical tropopause is generally higher and colder than the polar tropopause. This does not mean that every height above 8 km is colder over the equator: the comparison depends on altitude, atmospheric layer and season.

Q7Atmosphere is heated by …… (a) Solar Radiation (b) Heat from earth surface (c) From above

✅ Correct Answer: (b) Heat from earth surface

The atmosphere is heated from below by the earth's surface (via conduction, convection, radiation and latent heat). The atmosphere is a poor absorber of direct solar radiation.
Distractor Analysis(a) Solar radiation heats the EARTH, not directly the lower atmosphere. (c) From above is incorrect — atmosphere is heated from below.

Q8The temperature at 80 km is …… (a) 173 K (b) 100 K (c) –137°C

✅ Correct Answer: (a) 173 K (= –100°C)

"The temperature at 80 km is about –100°C (173.15 K)." 173 K ≈ –100°C. This is the Mesopause, the coldest point in the atmosphere.
Distractor Analysis(b) 100 K = –173°C, far too cold. (c) –137°C = 136 K, not the correct value.

Q9CO_2 and H_2O are also called …… (a) Green House Gases (b) Rare Earth Gases (c) [other]

✅ Correct Answer: (a) Green House Gases

CO_2 and H_2O (water vapour) along with O_3 and Methane are greenhouse gases — they are transparent to shortwave solar radiation but absorb longwave terrestrial radiation.

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Distractor Analysis(b) Rare Earth Gases (Noble gases like Argon, Neon, Helium) are a different category entirely.

Q10. Tropospheric stability

The troposphere can contain stable, neutral and unstable layers. Compare the environmental lapse rate with parcel cooling rates to judge stability; the name of the atmospheric layer alone does not decide it.

Q11Stratosphere is …… (a) Unstable (b) Neutral (c) Stable

✅ Correct Answer: (c) Stable

The Stratosphere has a temperature inversion (temperature increases with height due to ozone absorbing UV), making it very stable with no convective weather.
Distractor Analysis(a) Unstable — convection and instability occur in parts of the troposphere, not in the stratosphere. (b) Neutral — not the dominant characteristic.

Q12Tropopause is discontinuous at about …… (a) 30°lat (b) 40°lat (c) 80°lat

✅ Correct Answer: (b) 40°lat

The breaks/folds in the tropopause occur at ~40° and ~60° latitude, with the break at 40°lat being more prominent. Jet streams occur at these breaks.
Distractor Analysis(a) 30°lat — not a significant break. (c) 80°lat — too poleward, not where the prominent break is.

Q13Most of atmospheric mass is contained in …… (a) Troposphere (b) Stratosphere (c) Heterosphere

✅ Correct Answer: (a) Troposphere

The Troposphere contains 75% of the atmospheric mass and 99% of water vapour and aerosols.
Distractor Analysis(b) Stratosphere is relatively thin in mass. (c) Heterosphere is the outer part with very low density.

Q14Stratosphere extends from Tropopause to …… (a) 50 km (b) 60 km (c) 40 km

✅ Correct Answer: (a) 50 km

The Stratosphere extends up to ~50 km from the earth's surface (approximately 30–32 km above the tropopause at equator).
Distractor Analysis(b) 60 km is too high — enters Mesosphere region. (c) 40 km is not the correct upper boundary.

Q15The middle atmosphere layer with temperature inversion and stability is …… (a) Troposphere (b) Tropopause (c) Stratosphere

✅ Correct Answer: (c) Stratosphere

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The Stratosphere has a temperature inversion (UV absorption by ozone warms upper layers) making it very stable. It lies between the Tropopause and Stratopause.
Distractor Analysis(a) The troposphere has a normal lapse rate (temperature falls with height), not an inversion; it can hold stable, neutral and unstable layers. (b) Tropopause is a boundary, not a layer.

Q16Mother of Pearl clouds occur in …… (a) Mesosphere (b) Thermosphere (c) Stratosphere

✅ Correct Answer: (c) Stratosphere

Nacreous Clouds (Mother of Pearl Clouds) are sometimes seen in the Stratosphere in higher latitudes in winters.
Distractor Analysis(a) Mesosphere — Noctilucent clouds occur here. (b) Thermosphere — no clouds form here.

Q17The temperature in ISA at 17 km is …… (a) –56.5°C (b) –65.5°C (c) –35.5°C

✅ Correct Answer: (a) –56.5°C

In ISA, from 11 km to 20 km (65,617 ft), temperature is assumed constant at **–56.5°C**. 17 km falls within this isothermal zone.
Distractor Analysis(b) –65.5°C and (c) –35.5°C are both incorrect — no lapse rate change occurs in this zone.

Q18By weight, approximate ratio of O_2 to N_2 in the atmosphere is …… (a) 1:3 (b) 1:4 (c) 1:5

✅ Correct Answer: (a) 1:3 (i.e., O_2 : N_2 = 1:3 by weight)

N_2 : O_2 by weight = 3:1, so O_2 : N_2 by weight = 1:3.
Distractor Analysis(b) 1:4 — that is the VOLUME ratio (O_2:N_2 = 1:4). Don't confuse weight and volume ratios.

Q19By volume, the approximate ratio of O_2 to N_2 in the atmosphere is …… (a) 1:3 (b) 1:4 (c) 1:5

✅ Correct Answer: (b) 1:4 (O_2 : N_2 = 1:4 by volume)

N_2 : O_2 by volume = 4:1, therefore O_2 : N_2 = 1:4. N_2 = 78.09%, O_2 = 20.95% — ratio is almost 4:1.
Distractor Analysis(a) 1:3 is the weight ratio. (c) 1:5 is incorrect.

Q20. Atmospheric carbon dioxide

Carbon dioxide is a variable trace gas. Historical textbooks may quote 0.03%, but this is not a timeless contemporary value. For example, a preliminary Mauna Loa monthly mean is of 427.55 ppm for August 2026, equivalent to 0.042755% dry-air mole fraction. This station value is not the global mean.

Q21In ISA, the mean sea level temperature is …… (a) 15°C (b) 10°C (c) 25°C

✅ Correct Answer: (a) 15°C

ISA MSL temperature = 15°C (288.15 K). This is a fundamental ISA value.
Distractor Analysis(b) 10°C — incorrect. (c) 25°C — too warm, not ISA standard.

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Q22Maximum concentration of ozone is at a height of …… (a) 10–15 km (b) 20–25 km (c) 30–35 km

✅ Correct Answer: (b) 20–25 km

Ozone is found between 10–50 km with maximum concentration at 20–25 km in the lower Stratosphere.
Distractor Analysis(a) 10–15 km — lower end of ozone range, not maximum. (c) 30–35 km — past the maximum.

Q23. Oxygen and increasing altitude

Air pressure falls with altitude. The oxygen proportion remains approximately the same in the well-mixed lower atmosphere, but its partial pressure falls, increasing hypoxia risk (too little oxygen reaching the body). Aircraft altitude alone is not a universal supplemental-oxygen threshold: cabin pressure, exposure and aircraft procedures matter.

Q24CO_2 and H_2O keep the atmosphere …… (a) Warm (b) Cold (c) Have no effect

✅ Correct Answer: (a) Warm

CO_2 and H_2O are greenhouse gases that absorb terrestrial longwave radiation and re-radiate it, keeping the earth warmer than it would otherwise be.
Distractor Analysis(b) Cold — incorrect, they trap heat. (c) Have no effect — incorrect, they are the primary drivers of the greenhouse effect.

Q25Noctilucent clouds occur in …… (a) Thermosphere (b) Mesosphere (c) Stratosphere

✅ Correct Answer: (b) Mesosphere

Noctilucent Clouds are seen on rare occasions in the upper Mesosphere, in Polar regions.
Distractor Analysis(a) Thermosphere — no clouds form there. (c) Stratosphere — Nacreous/Mother of Pearl clouds occur here, not Noctilucent.

Q26Temperature at 2 km is 05°C. What is ISA deviation? Hint: (Actual – ISA) (a) –05°C (b) –02°C (c) 03°C

✅ Correct Answer: (c) 03°C (i.e. +03°C)

ISA temp at 2 km = 15 – (6.5 × 2) = 15 – 13 = 2°C. ISA Deviation = 5 – 2 = **+3°C**.
Distractor Analysis(a) –5°C wrong sign and magnitude. (b) –2°C — wrong direction entirely.

Q27Pressure at MSL is 1002.25 hPa. Find the ISA deviation. Hint: (Actual – ISA) (a) –11 hPa (b) 10 hPa (c) 12 hPa

✅ Correct Answer: (a) –11 hPa

ISA MSL Pressure = 1013.25 hPa. Deviation = 1002.25 – 1013.25 = –11 hPa. Negative means pressure is lower than ISA standard.
Distractor Analysis(b) 10 hPa and (c) 12 hPa — wrong sign and magnitude.

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Q28In actual atmosphere temp. at 19 km is –60°C. Find the ISA deviation. Hint: (Actual – ISA) (a) –4.5°C (b) –05.5°C (c) –03.5°C

✅ Correct Answer: (c) –03.5°C

ISA temp at 19 km = –56.5°C (isothermal zone 11–20 km). Deviation = –60 – (–56.5) = –60 + 56.5 = **–3.5°C**.
Distractor Analysis(a) –4.5°C and (b) –5.5°C are arithmetic errors — make sure signs are handled correctly.

Q29. Nacreous clouds

Nacreous clouds are polar stratospheric clouds. They occur in the lower stratosphere, rather than being a general feature of the upper stratosphere. Their pearly colours are an optical effect.

Q30. The homosphere

Turbulent mixing keeps the proportions of the major gases nearly uniform through much of the lower atmosphere, called the homosphere. Variable constituents, including water vapour and ozone, are exceptions. Gravity creates atmospheric weight and pressure; it does not explain this mixing. Avoid treating an approximate transition height as a perfectly sharp boundary.

Q31. Height containing half the atmospheric mass

In a hydrostatic atmosphere, pressure represents the weight of the overlying air column. About half the atmospheric mass lies below the level where pressure is half its sea-level value. In ISA this is about 18,000 ft; it is an approximate standard-atmosphere result, not exactly 20,000 ft.

Q32In jet standard atmosphere the Lapse Rate is …… (a) 2°C/1000 ft (b) 2°C/km (c) 5°C/km

✅ Answer: (a) 2°C/1000 ft

The Jet Standard Atmosphere assumes a constant lapse rate of 2 °C per 1000 ft and has no tropopause.
❌ (b) 2 °C/km is far too small a lapse rate. (c) 5 °C/km is not a standard rate.

Q33The rate of fall of temperatures with height, called …… (a) Isothermal rate (b) Inversion Rate (c) Lapse Rate

✅ Correct Answer: (c) Lapse Rate

Lapse rate is the rate at which temperature decreases with increasing altitude, approximately 6.5°C/km in the ISA troposphere.
Distractor Analysis(a) Isothermal rate — temperature does NOT change with height (isothermal). (b) Inversion Rate — temperature INCREASES with height (inverse of normal).

Q34In actual atmosphere the lapse rate could …… (a) Assume any value (b) Fall up to 8 km. (c)

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Rise up to 50 km

✅ Correct Answer: (a) Assume any value

In the actual (real) atmosphere — unlike ISA — the lapse rate can assume any value; it can be positive, zero (isothermal), or negative (inversion) depending on atmospheric conditions.
Distractor Analysis(b) and (c) describe fixed behaviours. The actual atmosphere is highly variable.

Q35Tropical Tropopause extends from the equator to Lat. 35°–40°. Over India it is at …… (a) 20–21 km (b) 14–15 km (c) 16–16.5 km

✅ Answer: (c) 16–16.5 km

The tropical tropopause lies near the 100 hPa level, roughly 16–16.5 km, and extends from the equator to about 35°–40° latitude. It is the tropopause over most of India in the summer.
❌ (a) 20–21 km is higher than any tropopause. (b) 14–15 km is lower than the ~100 hPa tropical level.

Q36Lapse rate in the troposphere is produced by …… and in the stratosphere by …… (a) Evaporation; condensation (b) Condensation; solar radiation (c) Terrestrial radiation; solar radiation (d) Solar radiation; convection

✅ Correct Answer: (c) Terrestrial radiation; solar radiation

In the Troposphere: cooling with height is due to terrestrial radiation heating from below. In Stratosphere: temperature inversion is due to solar radiation (UV) being absorbed by ozone — causing warming with height.
Distractor AnalysisOther combinations mix up the mechanisms. Condensation releases latent heat but is not the primary cause of the lapse rate.

Q37. Vertical distribution of water vapour

Water-vapour concentration generally decreases rapidly with height. Much of it is in the lower troposphere; this is different from specifying a sharp upper limit. A 30,000 ft textbook approximation must not make the lower-troposphere description a false distractor.

Q38Negative lapse rate of temperature is …… (a) Isothermal rate (b) Temperature rise with lowering height (c) Temperature rise with height (d) Temperature fall with height

✅ Correct Answer: (c) Temperature rise with height

A "negative lapse rate" means temperature increases with altitude — this is called an **inversion**. This occurs in the Stratosphere and in inversions within the Troposphere.
Distractor Analysis(a) Isothermal = zero lapse rate (no temperature change). (d) Temperature fall with height = positive lapse rate (normal). (b) is phrased incorrectly.

Q39In ICAO ISA the atmosphere is assumed to be isothermal …… (a) in stratosphere (b) 11 to 16 km (c) 11 to 20 km (d) 11 to 32 km

✅ Correct Answer: (c) 11 to 20 km

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In ICAO ISA, temperature is assumed constant at –56.5°C from 11 km (36,090 ft) to 20 km (65,617 ft). This is the isothermal (stratospheric) zone.
Distractor Analysis(b) 11–16 km — too narrow. (d) 11–32 km — too wide; from 20 km temperature starts rising again. (a) "in stratosphere" is vague and incorrect if taken to mean the whole stratosphere.

Q40One of the characteristics of our atmosphere is …… (a) Poor conductor of heat and electricity (b) Equator is warmer than poles above 10 km (c) Lapse rate is positive in stratosphere (d) Density is constant above 8 km

✅ Correct Answer: (a) Poor conductor of heat and electricity

The atmosphere is a poor conductor of heat and electricity — one of its key characteristics listed at the beginning of the chapter.
Distractor Analysis(b) Above 10 km, poles are actually warmer than equator at the tropopause (poles –40 to –45°C vs equator –70 to –75°C). (c) Lapse rate in stratosphere is negative (inversion). (d) Density is NOT constant above 8 km.

Q41. Heating the troposphere

The Sun supplies energy. Surface radiation, conduction, convection and latent-heat release all contribute to tropospheric heating. Convection carries warmer air upward; condensation releases latent heat. The largest contribution depends on the process and region being compared, so there is no universal answer to the original unqualified ranking.

Q42The knowledge of the height of tropopause is important for a pilot because …… (a) weather is mainly confined up to this level (b) clouds rarely reach up to this height due to jetstream (c) stratosphere starts at this height where all solar radiation is absorbed

✅ Correct Answer: (a) weather is mainly confined up to this level

The tropopause is near the upper limit of most weather. Strong convective cloud can extend above it, and turbulence can occur near and above it. Its height helps assess upper-level hazards; it does not guarantee a weather-free cruise.
Distractor Analysis(b) Clouds do grow to just below tropopause — not because of jet streams. (c) Solar radiation is absorbed by ozone in the stratosphere, not at the tropopause itself.

Q43In ISA atmosphere the tropopause occurs at a height of …… (a) 8–10 km (b) 11 km (c) 16–18 km

✅ Correct Answer: (b) 11 km

In ICAO ISA, the tropopause is fixed at **11 km (36,090 ft)**, where the lapse rate stops and isothermal zone begins (–56.5°C).
Distractor Analysis(a) 8–10 km = actual polar tropopause. (c) 16–18 km = actual equatorial tropopause. ISA is a standard — fixed at 11 km.

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Q44. Sensible and latent heat

Sensible heat changes temperature; latent heat is absorbed or released during a change of state. Evaporation absorbs latent heat and condensation releases it. A specified surface energy partition does not describe every heat transfer within the atmosphere.

Chapter 2: Atmospheric Pressure

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1. Static and Dynamic Pressure

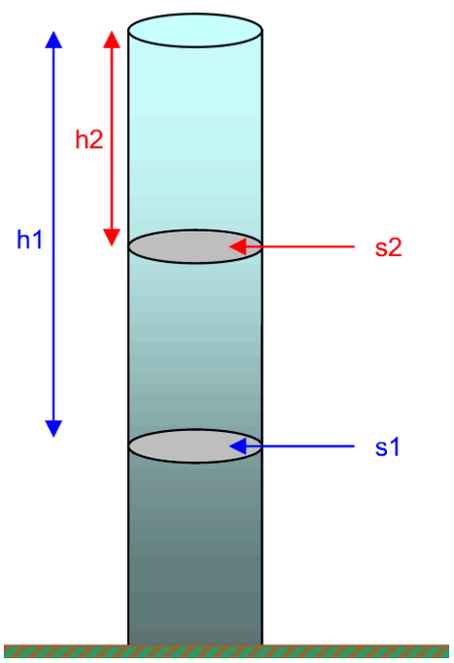
Definition: Pressure is the force per unit area exerted by molecules of air in random motion, acting uniformly in all directions. This is called *Static Pressure* or *Barometric Pressure*.

When air is in motion, an additional pressure is exerted in the direction opposite to the flow. This additional component is called **Dynamic Pressure** or **Wind Pressure**.

🎯 **Exam Tip:** "Static = air at rest. Dynamic = air in motion (opposes flow)." Think: *Static Stops, Dynamic Does move.*

2. Pressure as Weight of Air Above

The atmospheric pressure at any level is the weight of the column of air of unit cross-section extending vertically to the top of the atmosphere. As the amount of air in the column decreases with height, the pressure also decreases with height.



Pressure falls with height: there is less air above the upper level than above the lower level, so the pressure at the upper level is lower.

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Units adopted by ICAO: Hectopascal (hPa)
1 hPa = 1000 dyn/cm²
Unit of force is Newton (N).

Standard Sea Level Pressure:
1013.25 hPa = 760 mm = 29.92 in
This is one standard atmosphere, slightly more than 1 bar.
1 Bar = 100,000 N/m²
1 Bar = 1000 mb (millibars) = 1000 hPa
Heecto = 100; so 1 hPa = 100 Pa

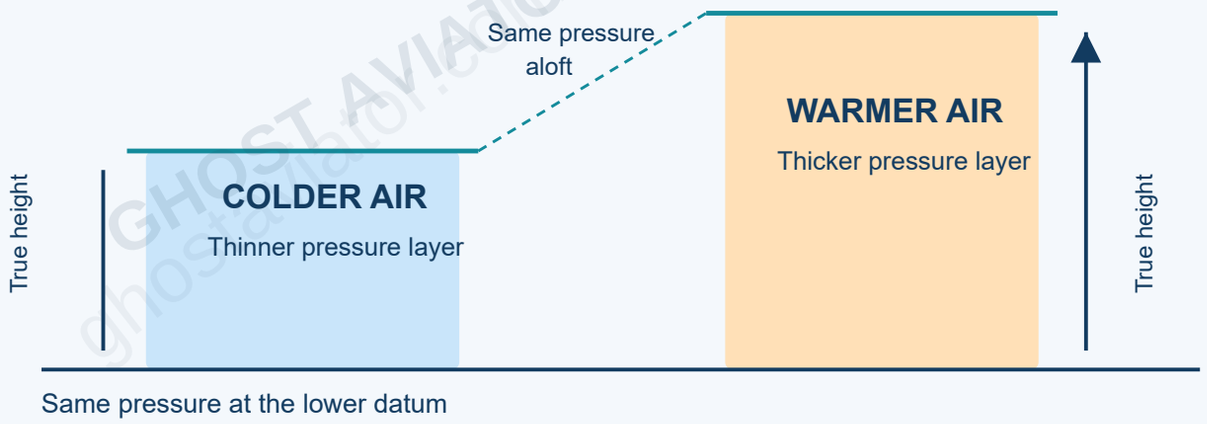
Conversion: hPa → Inches
Multiply hPa value by 0.02953
Example: 1013.25 hPa × 0.02953 = **29.92 inches**

⚠ **Important:** Air pressure changes cause ears to pop up when traveling over significantly varying topography and during rapid descent of an aircraft.

3. Vertical Variation of Pressure

The pressure decreases with height at a decreasing rate:

Equal pressure layers have different thicknesses



Matching pressure readings do not guarantee matching true heights. Schematic; not to scale.

For matching pressures at the bottom and top, a colder air column is thinner. Equal indicated levels can therefore correspond to different true heights. Schematic; no fixed temperature or height difference is implied.

Height Range	Approximate Rate of Decrease per 300 m (about 1,000 ft)
Sea level to 600 m	4%
Up to 1.5 km	3%
Up to 3 km	2.5%
At 6 km	Reduces to half the sea level value

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Height Range	Approximate Rate of Decrease per 300 m (about 1,000 ft)
At 100 km	Negligible — regarded as **vacuum**

Height Difference with 1 hPa Change

Formula:
Height difference = 96 × T / p feet
(where T = temperature in Kelvin, p = pressure in hPa)
Worked Example (a warm day, not ISA):
At 1000 hPa, T = 300 K:
Height = 96 × 300 / 1000 = **28.8 ft**

In ISA conditions, **1 hPa pressure change** corresponds to approximately:

Level	MSL	2,000 ft	20,000 ft	40,000 ft
Height equivalent	27 ft	30 ft	50 ft	100 ft

⚠ **Critical Rule:**
• If air is **warmer than ISA** → height change for 1 hPa will be **MORE** than above values
• If air is **colder than ISA** → height change will be **LESS** than above values

4. Variation of Pressure – Warm vs Cold Air

Cold air is denser than warm air. Pressure falls at a faster pace over a cold column of air than over a warm column.

At any given height, pressure at 850 hPa will be at a **higher height over warm air** than over cold air.

Key Rule: *Pressure at any given height will be higher over warm air than over cold air.*

• **Isobars Dip** going from High to Low pressure
• **Isobars Rise** going from Low to High pressure
• **Isobar (a line joining places with the same pressure at the same level and time)** = line joining places of equal pressure

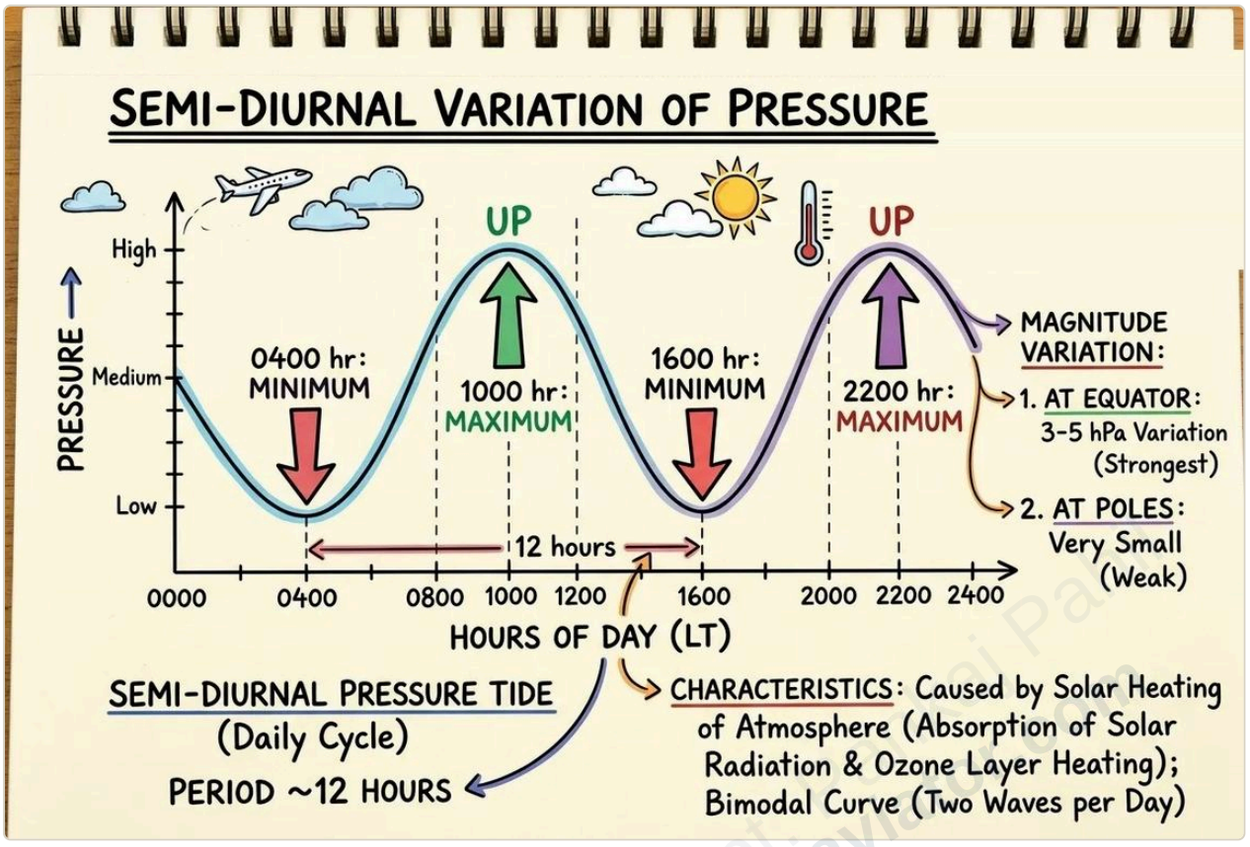
5. Semi-Diurnal Variation of Pressure

Atmospheric pressure follows a **bimodal curve** during a day:
• **Maxima:** 1000 hr and 2200 hr (local time)
• **Minima:** 0400 hr and 1600 hr (local time)

This semi-diurnal variation is due to **solar influence**.
Variations are: very small at the Poles and about 3–5 hPa at the Equator.

This is probably a natural oscillation of the atmosphere with a period of almost 12 hours.

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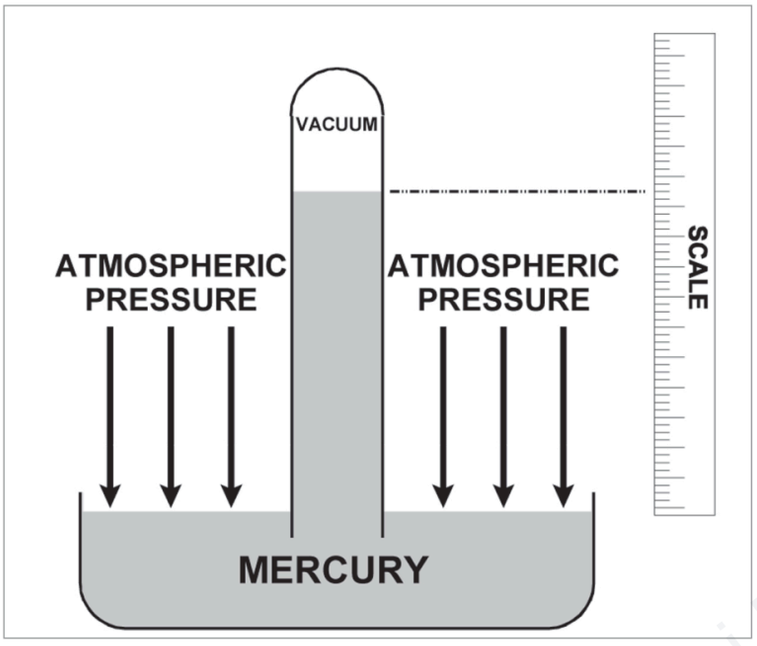


The semi-diurnal pressure wave: two highs (1000 and 2200) and two lows (0400 and 1600) each day, largest near the equator.

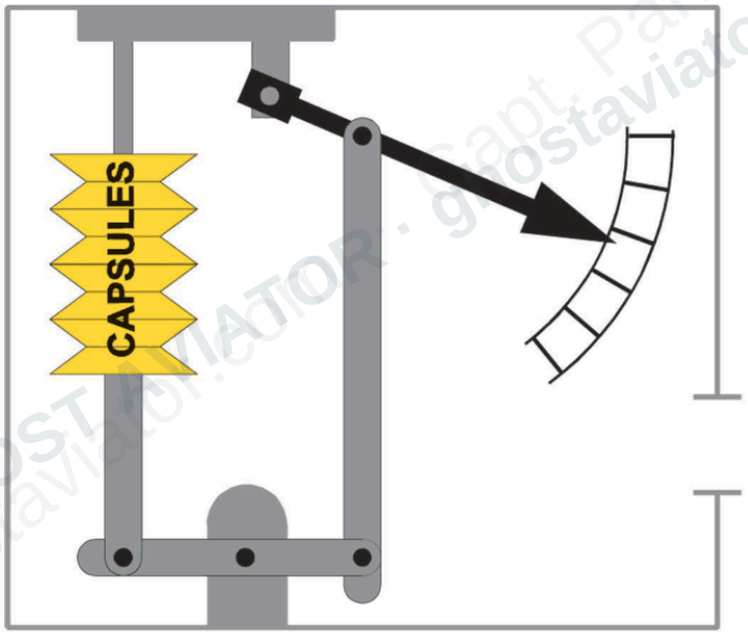
6. Measurement Instruments

Instrument	Type	Description
Mercury Barometer (an instrument that measures air pressure)	Accurate	Atmospheric pressure balanced by height of mercury column
Aneroid Barometer	Practical	Uses metal capsule; not as accurate as mercury; scale graduated in pressure (replaces mercury for ease of handling)
Altimeter	Aneroid type	Aneroid barometer with scale graduated in altitude instead of pressure; can be set to desired pressure value
Barograph	Recording	Continuously records pressure; can be Daily Barograph or Weekly Barograph

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The mercury barometer



The aneroid barometer

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7. Altimeter Correction (Alticor / D Value)

The correction applied to indicated altitude to obtain true altitude is called **Altimeter Correction** (also called **Alticor**).

This is frequently required for **mountain flying** and **bombing operations**.

D Value = True Altitude − Indicated Altitude
Correction to obtain true altitude = True Altitude − Indicated Altitude

D Value decreases when aircraft flies from High to Low pressure.

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Rough Calculations of Alticor

(a) Pressure Correction:
Add 30 ft for every 1 hPa when MSL pressure is **higher than 1013.2 hPa**
Subtract 30 ft per 1 hPa when MSL pressure is **lower than 1013.2 hPa**

(b) Temperature Correction:
Add 1% of indicated altitude for every 3°C temperature is **higher than ISA**
Subtract 1% per 3°C when temperature is **lower than ISA**

(c) Add these algebraically for final correction.

8. Contours, Thickness and Pressure Gradient

Contours

Lines joining places of equal height are called **Contours**. Contour lines may also be regarded as isobars.

In **Contour Charts**, contours are drawn at an interval of 40 gpm (geopotential metre) in 700 hPa and 500 hPa level charts; and at 80 gpm in 300 hPa and 200 hPa level charts.

1 gpm = 9.8 J/kg (gravitational potential energy per unit mass).
Contour lines numbered in geopotential decametres, e.g. 5280 gpm = 528.
A geopotential height of 700 gpm is approximately 700 m. A chart label 700 in geopotential decametres represents 7000 gpm; always check the chart units.
Centres of Low and High are marked as in surface pressure charts.

Thickness

The height interval between the lower and upper level is called **Thickness**. Low thickness = lower mean temperature; High thickness = higher mean temperature. The **isopleths of thickness coincide with the isotherms of mean temperature** of the layer.

Pressure Gradient

Pressure gradient is the horizontal rate of change of pressure perpendicular to the isobars, directed from high to low pressure.
• **Isobar** = line joining places of equal pressure.
• Steep pressure gradient → isobars are close together → **strong winds**.
• Flat/weak gradient → isobars far apart → **light winds**.

Reading Contours (Isohypses) on an Upper-Air Chart

A surface chart shows the *pressure* at one height (mean sea level) with isobars. An upper-air chart works the other way round. It takes one pressure, for example 500 hPa, and shows the *height* at which that pressure is found. This is a **constant pressure chart**, also called a **contour chart**.

The lines on it are **contours**. Another name is **isohypses** (lines joining points of equal height of the pressure surface). Heights are measured from mean sea level.

How to read them:

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- **Low contour height** = the pressure surface is low = low pressure aloft (upper low or trough).
- **High contour height** = the pressure surface is high = high pressure aloft (upper high or ridge).
- **Wind blows parallel to the contours.** Use them like isobars.
- **Contours close together** = steep gradient = strong wind. Contours far apart = light wind.
- Cold air makes pressure fall faster with height. So a mass of cold air shows as low heights on the upper chart, even when the surface chart shows nothing.

Exam example: a contour marked 522 on a 500 hPa chart is in decametres. The 500 hPa level is at 5220 m there.

Do not mix up contours and thickness lines. A contour gives the height of *one* pressure level. Thickness gives the vertical distance *between two* levels.

9. Pressure Tendency and Isallobars

The change of pressure with time is called **Pressure Tendency**. In India, pressure tendency is worked out for the past 24 hr and in higher latitudes for the past 3 hours of the current observation.

Isallobars = lines joining places of equal pressure change. At a glance, isallobars indicate areas of rising or falling pressure tendency.
• **Isallobaric Low** = region of greatest fall enclosed by isallobars.
• **Isallobaric High** = region of highest rise.

Surface Lows tend to move towards **Isallobaric Low**.
Isallobaric High indicates **weakening** of a pressure system.

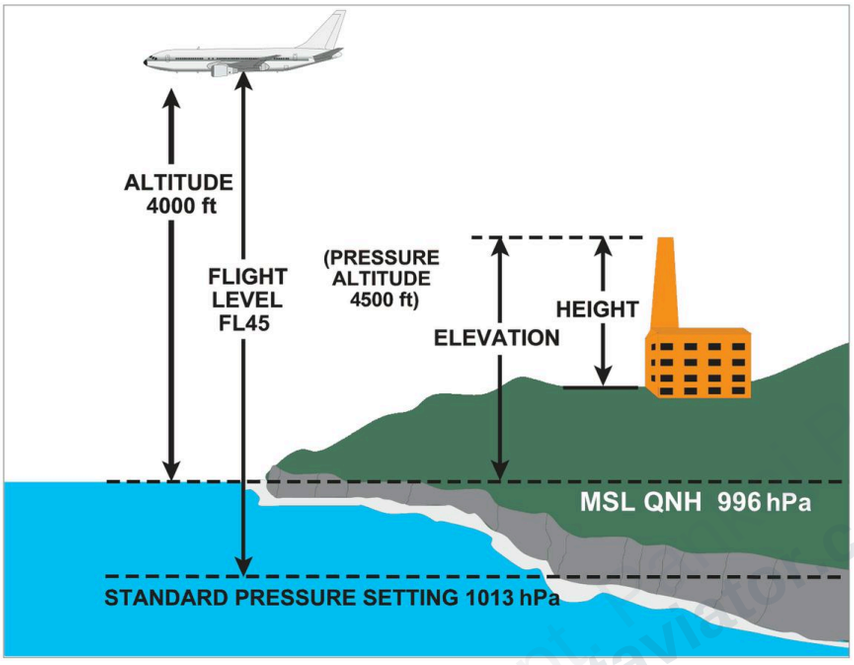
10. Altimetry — Definitions

Term	Definition
Mean Sea Level (MSL)	A reference level taking average of high and low tides.
Altitude	Vertical distance from **MSL**.
Height	Vertical distance from a specific datum (e.g., from the ground).
Elevation	Vertical distance of a point or level on the surface of the earth from MSL.
Transition Altitude (TA)	The altitude at or below which the vertical position of an aircraft is given as an altitude, with QNH set (QNH = aerodrome pressure reduced to mean sea level using the standard atmosphere). In India no transition altitude is below 4000 ft.
Transition Level (TL)	The lowest flight level available for use above the transition altitude. At or above it the altimeter is set to the standard pressure 1013.2 hPa and vertical position is given as a flight level (hundreds of feet).
Transition Layer	The airspace between Transition Altitude (TA) and Transition Level (TL).
Flight Level (FL)	Levels of constant pressure at or above TL, separated by pressure intervals corresponding to MSL pressure. E.g., FL50 = 5,000 ft, FL300 = 30,000 ft, FL200 = 20,000 ft.

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Term	Definition
Pressure Altitude	When the altimeter sub-scale is set to 1013.2 hPa, the altimeter indicates Pressure Altitude. Expressed as FL35, FL190, FL400 (avoid last two zeros).



Altitude, height, elevation and flight level

11. Pressure Settings

Corrections applied to barometer readings to reduce to a common level (MSL) are called **Pressure Setting**. The corrections are: **Index correction** (instrumental error), **Gravity correction** (varies with latitude), and **Temperature correction** (varies with temperature).

Pressure settings: **QFE (atmospheric pressure at the aerodrome reference point), QFF (station pressure reduced to mean sea level using actual temperature)** and **QNH**

Setting	Definition	Use
QFE	Pressure at the specified aerodrome or runway pressure datum. With QFE set, the altimeter reads approximately zero at that datum.	Height above aerodrome
QFF	Barometric pressure of aerodrome reduced to MSL using the *actual temperature* of the place (column of air extending up to MSL).	Plotting synoptic charts and drawing isobars
QNH	Station pressure reduced to mean sea level using the standard atmosphere. With QNH set, the altimeter reads the aerodrome elevation at its datum. Absolute altitude means height above ground and is a different quantity.	Vertical separation of aircraft and terrain
Regional QNH	Forecast value of the lowest pressure expected in an Altimeter Setting Region (ASR). Issued every hour, valid for one hour. This is a United Kingdom arrangement; India does not publish a regional QNH.	Ensuring adequate terrain clearance

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Setting	Definition	Use
QNE	Altitude indicated when sub-scale set to 1013.25 hPa (29.92 in). Used for high altitude airfields.	High altitude airfields; Pressure Altitude reference

12. Relation between QNH, QFE and QNE

Station Elevation = (QNH − QFE) × 30 ft

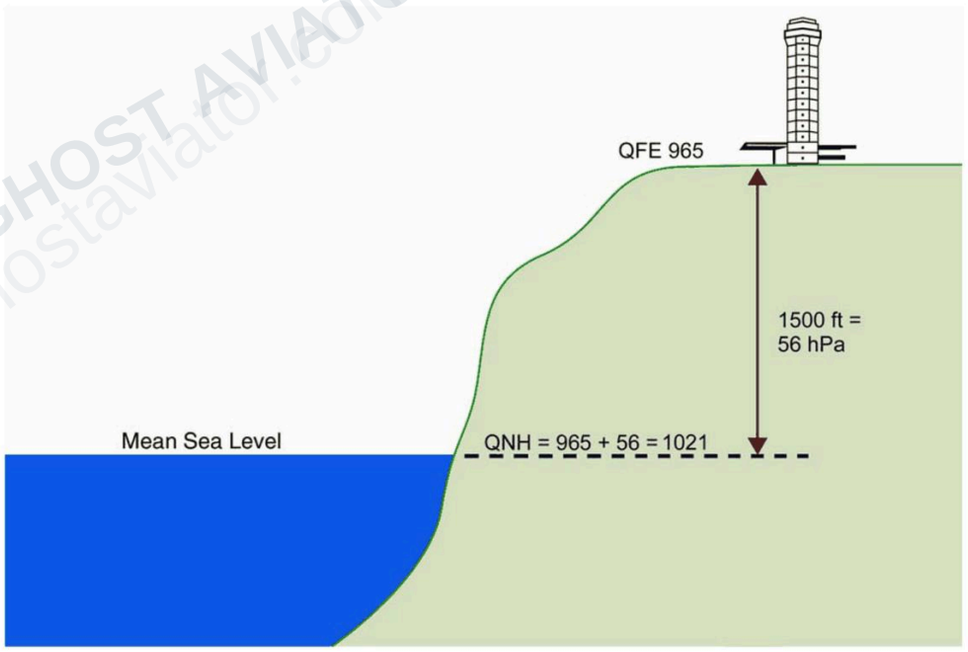
Example(i): an explicitly assumed30 ft perhPa approximation
QFE = 950 hPa, QNH = 1000 hPa, 1 hPa = 30 ft
Elevation = (QNH − QFE) × 30 = (1000 − 950) × 30 = 50 × 30 = **1500 ft**

Example (ii): Altimeter set to QNE = 1013 hPa
Station level would be = (QNE − QFE) × 30 ft = (63 hPa) × 30 = **1890 ft**

Example (iii): QFE 970 hPa, QNH 1020 hPa
Station elevation = (QNH − QFE) × 30 = (50 hPa) × 30 = **1500 ft**

Example (iv): QFE = 1000 hPa, Station elevation = 270 ft, 1 hPa = 27 ft
270 / 27 = 10 hPa → QNH = QFE + 10 = 1000 + 10 = **1010 hPa**

Example (v): Airfield elevation 400 ft, QNH 1010 hPa, lands at elevation 600 ft without resetting, altimeter reads 700 ft. Correct QNH at B?
At B, the indication is 700 ft but the elevation is 600ft: the altimeter over-reads by 100 ft.
Using the stated 27 ft perhPa approximation, reduce the setting by 100 / 27 ≈ 3.7 hPa.
EstimatedQNH at B≈ 1010−3.7=**1006 hPa**.



Finding QNH from QFE

L-H-L Rule for QNH/QFE Relationship

Hotter than ISA: QFF < QNH
At MSL: QFF = QNH

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Pressure set on altimeter < QNH → Indicated Altitude < True Altitude
Pressure set on altimeter > QNH → Indicated Altitude > True Altitude

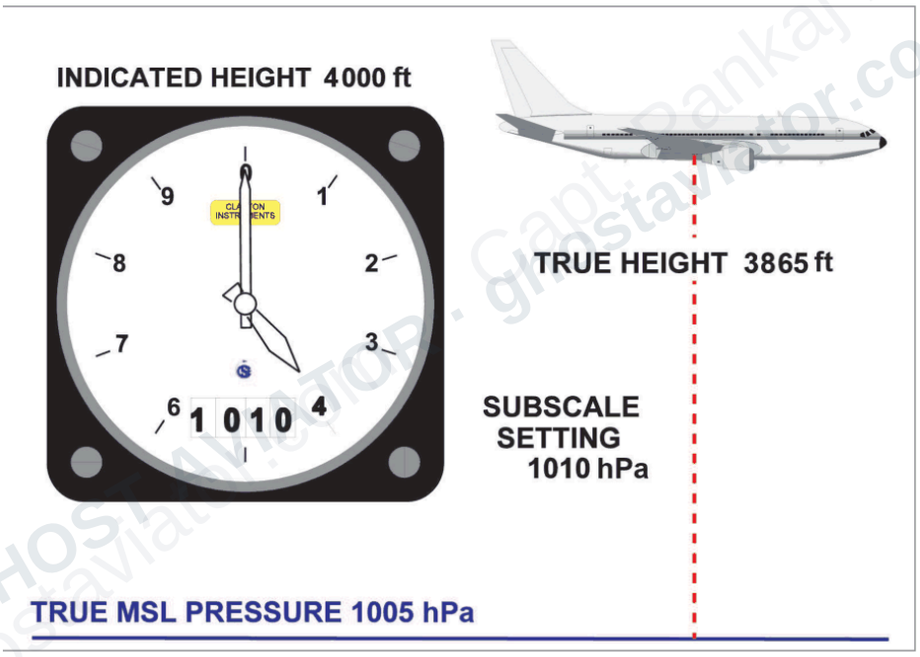
Standard Isobaric Levels

Corresponding to ISA pressure, the Pressure Altitude (Ft) and Flight Levels are:

Level (hPa)	850	700	500	400	300	200	100
Pressure Altitude (ft)	5,000	10,000	18,000	24,000	30,000	about 38,700	53,000
Flight Level	FL50	FL100	FL180	FL240	FL300	FL390	FL530

13. Under-Reading / Over-Reading

When a pressure value is set on the sub-scale, the altimeter indicates height as per ISA specifications.



Altimeter error when the true pressure differs from the setting

H-L-H (Over Read)
High to Low → Over Read
Warm to Cold → Over Read
If QNH is High → True Alt is High
Altimeter reads too high — you are lower than indicated!

L-H-L (Under Read)
Low to High → Under Read
Cold to Warm → Under Read
Altimeter Setting is Low → Indicated Alt will be Low
You are higher than indicated for this pressure-error comparison. Terrain clearance still requires the correct setting and other applicable corrections.

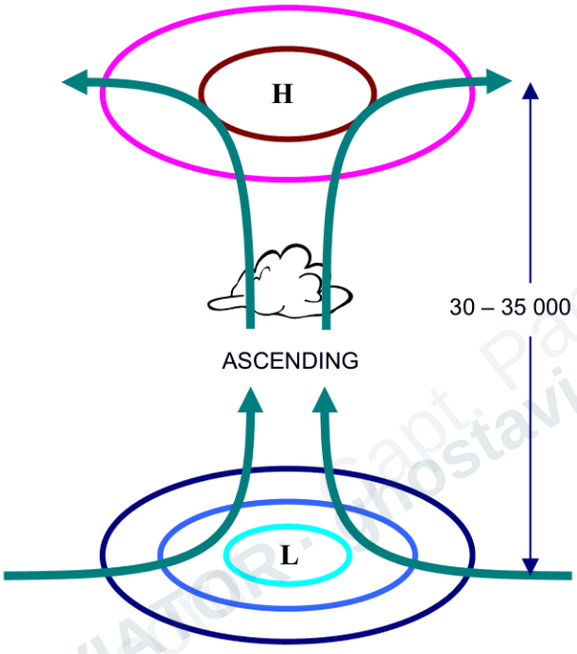
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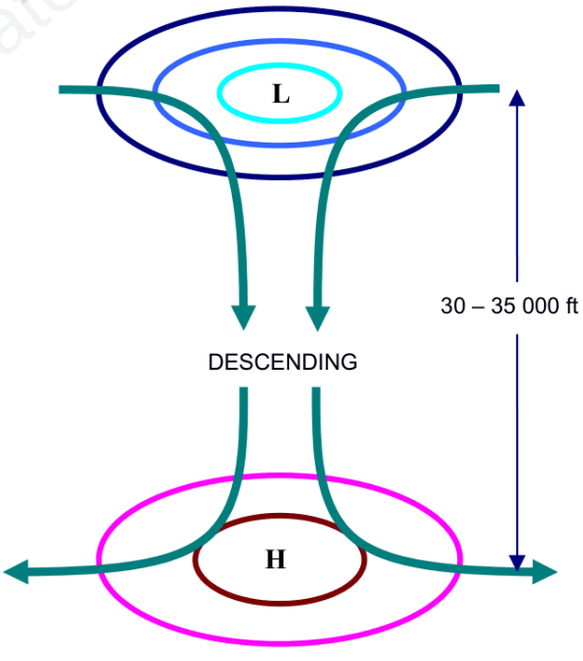
⚠ **Critical:** In an ideal Northern Hemisphere geostrophic pressure pattern, wind from port can accompany travel towards lower pressure. With the pressure setting unchanged, that pressure change makes the altimeter over-read. The association reverses in the Southern Hemisphere. Wind direction alone is not a reliable altimeter correction.

14. Pressure Patterns

To represent the spatial distribution of pressure at a given time, all pressures are reduced to MSL. Pressures at stations are plotted on a chart, and isobars drawn at 2 hPa intervals.

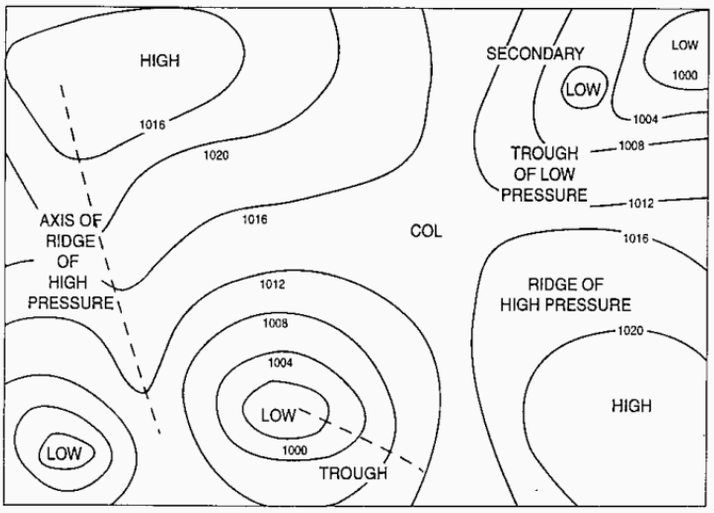


Low pressure: air converges and rises



High pressure: air descends and spreads out

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LOW, CYCLONE OR DEPRESSION, AND TROUGH

Pressure systems on a surface chart

Pattern	Description	Weather	Hemisphere Note
Low (L) / Cyclone	Area enclosed by isobars with lowest pressure at centre. Two or more closed isobars = **Depression**. Severe = **Cyclone**.	Cloud and precipitation can occur with rising moist air; visibility varies	N Hemisphere: winds blow anticlockwise, converging & upward
High (H) / Anticyclone	Region enclosed by isobars with highest pressure at centre.	Fair weather but visibility may be poor	N Hemisphere: winds blow clockwise, veer and diverge outwards
Trough	"V" shaped extension of isobars from a low. Wind direction abruptly changes and backs.	Bad weather, associated with monsoon trough along Indo-Gangetic plains	—
Ridge	Wedge-like extension of isobars from high. Isobars rounded, not "V" shaped.	Fair weather; winds **veer** at a ridge	—
Col	Between two highs and two lows; 2 hPa higher than lows, towards highs.	Mixed — light variable winds	Winds light variable; pressure uniform
Cold Anticyclone	Forms in cold region behind a cyclone, compresses air as it moves away.	Warms up after sometimes	Temporary; it travels with the depressions and does not block them. (A **Blocking Anticyclone** is different: a warm high that extends from the subtropical high, stays for several days and holds up or diverts the depressions.)

🎯 **Quick Reference:**
Low = Cyclone = Convergence = Rising air = Bad Wx

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High = Anticyclone = Divergence (airflow spreading apart) = Subsidence = Good Wx (but poor vis)
Trough = V-shape = Bad Wx | Ridge = Rounded = Good Wx | Col = Uniform = Light winds

Uniform (Flat) Pressure Pattern

In a uniform pressure pattern the pressure is almost level over an area, so the pressure gradient is slack (very weak). A **col** is the common example: a region of almost level pressure between two highs and two lows. It is an area of stagnation (air that hardly moves).

Wind: always very light and variable. On a sunny day with a slack pressure gradient the land heats quickly, which favours a sea breeze on the coast.

Weather depends on the season:

Season	Typical weather
Autumn and winter	Poor visibility and fog. Radiation fog forms inland with a slack pressure gradient.
Summer	Thunderstorms are common.

For the pilot:

- Do not expect a steady wind. It is light and variable.
- Expect fog in the cool season and thunderstorms in the hot season.
- Cols last only a few days. They are then absorbed into other systems.

Practice Q&A — Atmospheric Pressure

Practice questions with answers.

Q1. Winds in a low pressure area
(a) Converge (b) Diverge (c) Go straight

✅ **Answer: (a) Converge**
In a Low, air flows inward (converges) and rises.

❌ (b) Diverge — This happens in a High pressure area. (c) Go straight — only in absence of pressure gradient.

Q2. Low pressure is associated with
(a) Good Weather (b) Bad Weather (c) None

✅ **Answer: (b) Bad Weather**

❌ (a) Good Weather — this is associated with High pressure areas. (c) None — wrong.

Q3. In a high pressure area winds are
(a) Normal (b) Strong (c) Weak

✅ Answer: (c) Weak — In a high (anticyclone) the isobars are widely spaced, so the pressure gradient is slack and the winds are light.

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❌ (a) 'Normal' is not a description of wind strength. (b) Strong winds go with closely packed isobars, as in a deep low.

Q4. Flying from Low to High, an altimeter would read
(a) Over (b) Under (c) constant

✅ Answer: (b) Under — Flying from low towards high pressure with the subscale unchanged, the altimeter under-reads: the aircraft is higher than the altimeter shows.

❌ (a) Over-reading is the error when flying from high to low pressure. (c) The reading does not stay correct: the error grows as the pressure changes.

Q5. Isallobars are lines of equal
(a) Pressure (b) Temperature Tendency (c) Pressure Tendency

✅ **Answer: (c) Pressure Tendency**

❌ (a) Pressure — that's an Isobar. (b) Temperature Tendency — that would be Isalloterms.

Q6. What kind of a barometer is an altimeter?
(a) Aneroid (b) Mercury (c) Alcohol

✅ Answer: (a) Aneroid — An altimeter is an aneroid barometer: a sealed, partly evacuated capsule that expands and contracts with pressure, with its scale marked in height.

❌ (b) A mercury barometer is a fixed, fragile laboratory instrument, not used in aircraft. (c) Alcohol is not used as a barometer liquid in aviation.

Q7. A region between two Lows and two Highs is called
(a) Depression (b) Secondary Low (c) Col

✅ **Answer: (c) Col**

❌ (a) Depression — enclosed area of low pressure. (b) Secondary Low — a smaller low near main low.

Q8. Bad weather and better visibility is associated with
(a) High (b) Low

✅ Answer: (b) Low — A low brings cloud, rain and strong winds, yet the rain washes dust and haze out of the air, so visibility between showers can be good. A high gives fair weather but, with sinking air and light winds, haze and smoke are trapped under the inversion.

❌ (a) High: fair weather and often poor visibility, the opposite pairing.

Q9. The relationship between height and pressure is use in construction of
(a) Altimeter (b) ASI (c) V S I

✅ Answer: (a) Altimeter — The altimeter senses static pressure and converts it to height using the pressure–height relationship of the standard atmosphere.

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❌ (b) The ASI works from the difference between pitot and static pressure, not from height. (c) The VSI works from the rate of change of static pressure.

Q10. Altimeter pressure references

With standard pressure set, an altimeter indicates pressure altitude. With QFE set it reads approximately zero at the aerodrome reference point; with QNH set it reads aerodrome elevation there. Its reference therefore depends on the pressure setting.

Q11. Two aircraft flying at same indicated altitude with altimeters set to 1013.2 hPa. One is flying over cold air mass, other over warm air mass. Which has greater altitude?
(a) Ac flying over warm air mass (b) Ac flying over cold air mass (c) Both

✅ Answer: (a) Aircraft flying over warm air mass — Warm air is less dense, so pressure falls more slowly with height and a given pressure level lies higher. With 1013.2 hPa set, both aircraft are on the same pressure surface, but over warm air that surface is higher, so the aircraft over the warm air mass has the greater true altitude.

❌ (b) Over cold air the pressure levels are squashed downwards, so the same indication means a lower true altitude.

Q12. Rate of fall of pressure with height in cold air mass compared to warm air mass will be
(a) Same (b) More (c) Less

✅ **Answer: (b) More**
Cold air is denser, so pressure falls more rapidly with height over a cold column.

Q13. A near-sea-level pressure approximation

If the question explicitly assumes 30 ft per hPa, a 1000 ft climb gives an approximate pressure decrease of 1000/30 = 33 hPa. This is a stated classroom approximation; the ISA value near sea level is closer to 27 ft per hPa and changes with height and temperature.

Q14. Lines drawn through places of equal pressure are known as
(a) Isobars (b) Isotherms (c) Isoheights (d) Isoclinic

✅ **Answer: (a) Isobars**

❌ (b) Isotherms = equal temperature. (c) Isoheights = equal height (contours). (d) Isoclinic = equal magnetic dip.

Q24. 300 hPa in ISA corresponds to the level
(a) 20,000 ft (b) 30,000 ft (c) 35,000 ft

✅ **Answer: (b) 30,000 ft**
Standard: 300 hPa = 30,000 ft = FL300

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Q36. QNH to QFE using the given approximation

For an aerodrome 160 m above mean sea level, with QNH 1005 hPa and the supplied approximation 8 m per hPa, the reduction is 160/8 = 20 hPa. QFE = 1005 − 20 = 985 hPa. A result of 990 hPa would imply 120 m under this approximation, not 200 m.

Q37. Steep pressure gradient would mean
(a) Contours far apart, weak wind
(b) Contours far apart, strong wind
(c) Isobars far apart, strong wind
(d) Isobars closely packed and strong wind

✅ **Answer: (d) Isobars closely packed and strong wind**

❌ (a) Far apart = weak gradient = light winds. (c) Far apart = weak, not strong.

Q38. What type of inversion occurs when a stable layer lies in a high pressure area?
(a) Negative (b) Radiation (c) Subsidence (d) Airmass

✅ Answer: (c) Subsidence — In a high-pressure area the air sinks, is compressed and warms, forming a warm layer above cooler air below: a subsidence inversion. It traps haze and smoke and is a key cause of poor visibility.

❌ (a) 'Negative' is not a type of inversion. (b) A radiation inversion forms on clear nights by ground cooling, not by sinking air.

Q39. Which of the following would cause true altitude to decrease when the altimeter indicates a constant altitude? (a) Warm/Low (b) Cold/High (c) Cold/Low (d) Warm/High

✅ Answer: (c) Cold/Low — True altitude is less than indicated when the air is colder than standard (the pressure levels are squashed downwards) and when you fly into lower pressure than the subscale setting. Cold and Low together lower the true altitude.

❌ Warm and High both raise the true altitude above the indication; Warm/Low and Cold/High have one effect each in opposite directions.

Q41. An aerodrome is at MSL. Its QNH is 1014.0 hPa. Its QFF will be
(a) 1014.0 hPa (b) 1013.25 (c) Difficult to tell (d) More than QNH

✅ **Answer: (a) 1014.0 hPa**
If the aerodrome is at MSL, then QNH = QFF = 1014.0 hPa (no height correction needed, both reduce to the same level — MSL).

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Chapter 3: Temperature

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1. Definition & Temperature Scales

Temperature is a measure of how hot or cold a body is. It is not the same as heat. It is measured by a thermometer in degrees Celsius (Centigrade) or Fahrenheit.

Celsius Scale: 0°C = melting point of ice; 100°C = boiling point of pure water at normal pressure. Used internationally, in aviation and science.
Fahrenheit Scale: 32°F = melting point; 212°F = boiling point. Used in a few English-speaking countries.
Absolute Zero: The minimum temperature attainable for all substances. Value = −273.15°C = 0K.

Temperature Conversion Formulae:

F = (9C/5) + 32
C = 5/9 × (F − 32)
K = C + 273

Special value — same on both scales:
–40°C = –40°F

Worked Example:
Convert 100°C to Fahrenheit:
F = (9 × 100)/5 + 32 = 180 + 32 = **212°F**

Convert 32°F to Celsius:
C = 5/9 × (32 − 32) = 5/9 × 0 = **0°C**

Convert 15°C to Kelvin:
K = 15 + 273 = **288 K**

🎯 **Exam Tip:** Freezing point of water = 0°C = 32°F = 273 K. Boiling point = 100°C = 212°F = 373 K. The unique crossover: −40°C = −40°F.

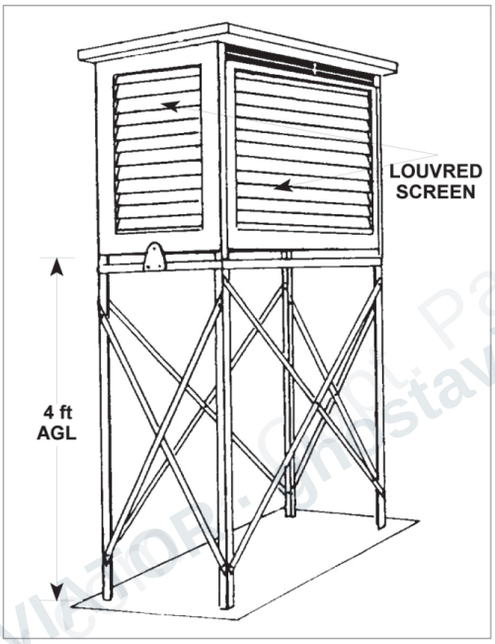
2. Instruments for Measurement

Instrument	Description	Special Notes
Dry Bulb Thermometer	Standard mercury thermometer measuring ambient air temperature.	One part of the psychrometer pair
Wet Bulb Thermometer	Bulb covered with muslin cloth kept moist. Evaporation (liquid water changes into vapour and absorbs latent heat) cools the bulb. Used to measure humidity.	Wet bulb temp always ≤ dry bulb temp

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Instrument	Description	Special Notes
Maximum Thermometer	Like a Doctor's thermometer — records maximum temperature attained. Has a constriction to prevent mercury from falling back.	Reset by shaking
Minimum Thermometer	Uses **Alcohol** instead of mercury (alcohol doesn't freeze). Has a dumbbell-shaped iron index. When temperature falls, alcohol drags index back; index stays at minimum position.	Concave meniscus drags index back to indicate minimum temperature
Thermograph	Gives a continuous record of temperature. Uses bimetallic strips in Upper Air measurements.	Daily or weekly recording



The Stevenson screen

Stevenson's Screen: Surface temperature is recorded at a height of 1.25 m to 2 m above ground in shade, inside a Stevenson's Screen — a louvred white-painted wooden box that protects instruments from direct radiation while allowing free air circulation.

3. Types of Temperature

Type	Definition
Surface Temperature	Temperature recorded at 1.25 m to 2 m above ground in shade (inside Stevenson's Screen).
Ambient Temperature	Temperature of the surroundings.
Virtual Temperature (VT)	In a thermodynamic process, the temperature at which dry air parcel would have the same pressure and density (mass per unit volume) as a moist parcel of air. VT allows use of dry air equation of state for moist air also. VT is always higher than actual temperature.

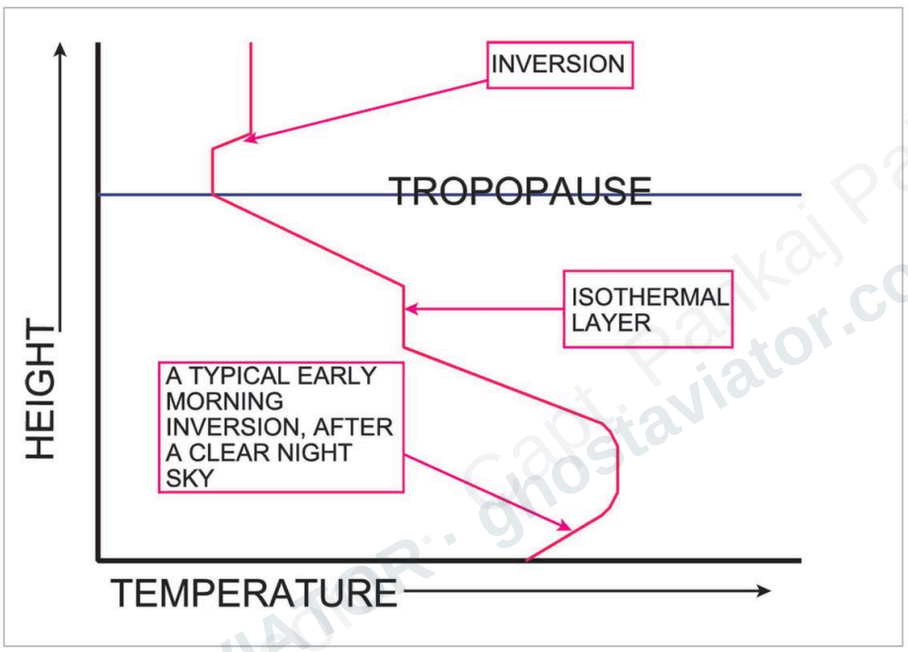
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4. Heat and Temperature

Heat is energy transferred because of a temperature difference. Internal energy is energy stored in the substance.
Temperature is the **average KE** of all molecules and atoms of a substance.

Heat is energy transferred because of a temperature difference; internal energy is energy stored in a substance. Temperature alone does not tell us the total internal energy of a quantity of material. Keep these three ideas separate when explaining atmospheric heating.



Inversions

🎯 **Exam Tip:** Keep temperature, stored internal energy and energy transferred as heat separate. Temperature alone does not state the internal energy of a whole object.

5. Specific Heat and Latent Heat

Specific Heat

Specific Heat is the heat required to raise the temperature of **unit mass** of a substance by **1°C**.

Substance	Specific Heat (relative)
Water	1 (highest — reference value)
Ice	0.5
Soil (Land)	0.2

⚠️ **Aviation Implication:** Because land has a much lower specific heat than water (0.2 vs 1.0), **land heats and cools much faster than the sea**. This drives sea breezes, land breezes, and monsoon

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circulation.

Latent Heat

Latent Heat (energy absorbed or released during a change of state without a temperature change) is defined as the "amount of heat absorbed or released during change of phase from/to solid/liquid/vapour".
• Absorbed during: solid → liquid, liquid → vapour
• Released during: vapour → liquid (condensation (water vapour changes into liquid water and releases latent heat)), liquid → solid (freezing)

Latent heat is "hidden" — it does not change temperature, only phase.

6. Evaporation, Condensation and Latent Heat

When water changes to vapour, a certain quantity of heat is supplied. To change boiling water into vapour requires **more than five times** as much heat as needed to bring ice cold water to the boil.

Once boiling has begun, the temperature remains constant — heat supplied at this stage is latent. It is released as latent heat **when vapour condenses to water**.

The latent heat absorbed by melting of ice or evaporation of water at the earth's surface is subsequently released in the atmosphere by **condensation or freezing**.

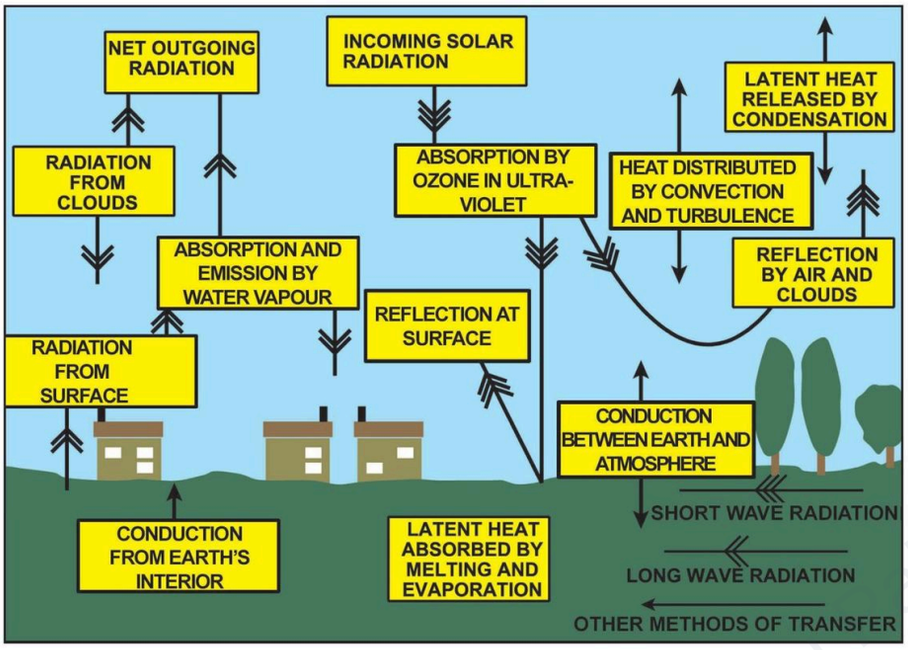
🎯 **Key Point:** Evaporation = heat absorbed from surroundings = cooling effect. Condensation = heat released = warming effect. This is why clouds release heat (Latent Heat Release = LHR) — critical for thunderstorm development!

7. Methods of Heat Transfer

Heat is transferred by **Conduction (heat transfer through direct contact), Convection (heat transfer through movement within a fluid), Radiation**, and other methods. In the atmosphere, **Radiation plays the most significant role**.

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Heat processes in the atmosphere

Method	Definition	Significance in Atmosphere
Conduction	Physical transfer by molecular contact	Important very close to the ground. Heats lowest air layer in contact with warm surface.
Convection	Bodily transfer of fluid to colder part of the fluid. More than 70% of earth covered by water → convection highly important.	**Free Convection** = intense solar heating. **Forced Convection** = topography. Air lifts to higher levels and transfers heat.
Radiation	Every body radiates at its temperature. Medium neither affected nor required. Solar radiation heats earth directly.	**Most important** in the atmosphere. Long and short wave radiation.
Advection (horizontal transport by moving air)	Horizontal motion of the atmosphere transferring heat.	Transfers heat horizontally by winds (e.g., warm/cold advection).
Turbulence	Irregular eddy motion of the atmosphere causes redistribution of heat.	Important in the lowest atmospheric layers.
Latent Heat Release	Latent heat absorbed by evaporation is released by condensation or freezing.	Released in the atmosphere — major heat source for cloud systems and thunderstorms.

In the Troposphere: All above processes transfer heat.
In the Stratosphere: Neither convection nor latent heat has influence. Short wave radiation heats it up due to **ozone absorption**.

8. Insolation

Insolation is the total amount of solar radiation received over a particular area.

Insolation depends on the **obliquity (angle) of sun's rays**:

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- Maximum when sun is **overhead** (vertical rays cover less area → more energy per unit area)
- Least at **sunrise and sunset** (oblique rays spread over large area)
- Maximum insolation in the **Tropics** (between Tropic of Cancer and Tropic of Capricorn)
- Least at the **Poles**

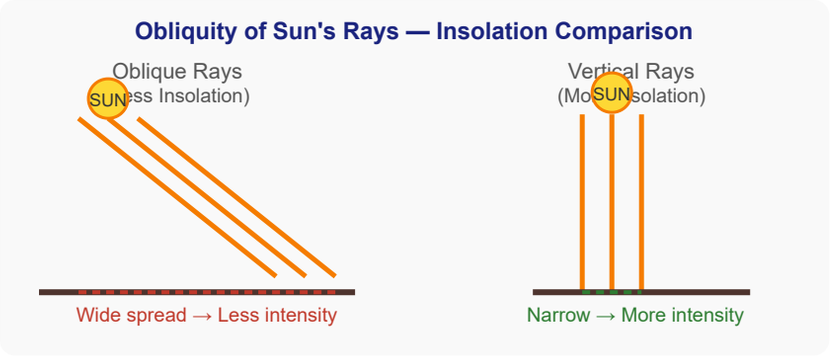


Fig 3.1: Insolation and Obliquity of Sun's Rays

9. Laws of Radiation

Black Body: Objects emit thermal radiation. An ideal **blackbody** absorbs all incident radiation; real objects need not behave as perfect blackbodies.

Law	Statement	Formula / Key Point
Stefan Boltzmann's Law	The total amount of energy radiated by a black body is proportional to the **fourth power of its absolute temperature**.	$E \propto T^4$ Hence intense radiation is emitted by hot bodies like the sun.
Wien's Law	The wavelength of most intense radiation is **inversely proportional** to the absolute temperature.	Hot bodies (like sun) → **Short Waves** Colder bodies (like earth) → **Long Waves**
Planck's Law	Describes distribution of radiated energy with absolute temperature.	Curve is a right-skewed central distribution.

Stefan Boltzmann's Law:
$E \propto T^4$
If temperature doubles: E increases by 2^4 = **16 times**
Example: Sun (T ≈ 6000K) radiates much more intensely than Earth (T ≈ 300K).
Ratio = $(6000/300)^4 = 20^4$ = 160,000 times more energy per unit area from the Sun.

10. Solar Radiation

The temperature of the surface of the sun is about 5800 K (about 5500°C). Solar radiation is therefore mainly **Short Wave** radiation.

The solar spectrum (visible = VIBGYOR — Violet, Indigo, Blue, Green, Yellow, Orange, Red):

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Type	Percentage of Solar Radiation	Nature
IR (Infra-Red)	46%	Beyond red end of spectrum; invisible
Visible Light	45%	VIBGYOR — white light
UV (Ultra-Violet)	9%	Beyond violet end; absorbed by ozone

Complete solar radiation (including visible, UV, and IR) is responsible for all the heat the earth receives from the sun as **short wave radiation**.

11. Terrestrial Radiation & Nocturnal Radiation

Terrestrial Radiation

The earth radiates at its own temperature and loses heat. These are called **Terrestrial Radiation**. They are **Long Wave Infra-Red (IR) radiation** and are **invisible**.

The earth receives heat as **short-wave radiation** from the sun and loses heat as **long-wave radiation**.

Nocturnal Radiation

At night, the short-wave radiation from the sun is absent. Only the earth radiates and loses heat. The radiation emitted by the earth at night is called **Nocturnal Radiation**.

⚠ **Aviation Significance:** Nocturnal radiation causes surface cooling at night → ground fog formation, radiation fog, frost → hazardous for takeoff/landing!

12. Radiation and Heat Budget / Albedo

Radiation and Heat Budget

Since the mean temperature of the earth has remained almost unchanged over a long period, the heat received from the sun as short-wave radiation is returned to space as terrestrial radiation and are equal.

Fate of Incoming Solar Radiation	Percentage
Absorbed by Earth Surface	51%
Absorbed by Water Vapour, Dust, Ozone	16%
Absorbed by Clouds	3%
Total Absorbed	70%
Back-scattered by Air	6%
Reflected back by Clouds	20%
Reflected back by Earth Surface	4%

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Fate of Incoming Solar Radiation	Percentage
Total Reflected/Back-scattered	30%
Grand Total	100%

Albedo

Albedo = Reflected Radiation / Incident Radiation

The 30% of solar radiation reflected back to space by the earth and clouds is the **reflecting power** of earth, called **Albedo**.

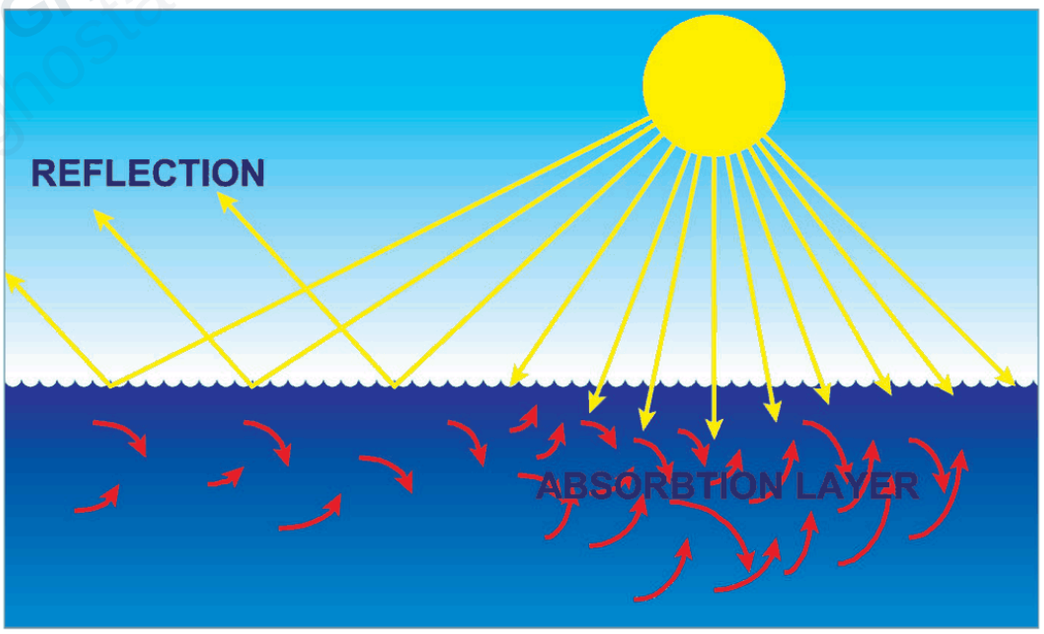
In clear weather, about 5/6 of the solar radiation reaches earth surface → about 1/6 (≈17%) is reflected. Earth surface reflects about 10% in clear conditions.
Snow surface reflects about 80% of incident energy (very high albedo).

🎯 **Exam Tip:**
"Albedo = Reflecting Power of Earth" — 30% of total solar radiation is reflected (not absorbed).
Snow = high albedo (80%) → reflects most sunlight → stays cold.
Ocean = low albedo (~10%) → absorbs most sunlight → stays warmer relative to snow.

13. Diurnal Variation of Surface Temperature

Sea vs Land: The sea surface temperature shows a variation of less than 1°C from day to night. Over land, diurnal variation may average as much as 20°C.

Near the coast, the diurnal variation near the coast may be as large as inland, but with a wind off the sea it will be small. **Sea breezes have a pronounced cooling effect**.



Why the sea has a small diurnal range

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Why is the diurnal variation small over the sea?
1. Higher specific heat of water (5× that of land)
2. Larger mixing depth — surface water mixes with deeper layers
3. Evaporation from sea surface moderates temperature changes
4. Diurnal variation is maximum when wind is **calm**; with strong winds, surface air mixes with air above and heat spreads through the Friction Layer (up to 1 km above).

Timing of Daily Temperature Extremes

Parameter	Timing	Reason
Minimum Temperature	½ to 1 hour after sunrise	Nocturnal cooling continues slightly past sunrise until incoming solar radiation exceeds outgoing terrestrial radiation.
Maximum Temperature	Early afternoon (2–3 hours after noon)	Highest insolation at noon, but 2–3 hours needed for heat to transfer to screen level. Surface is highest at noon; screen level maximum occurs 2–3 hours later.

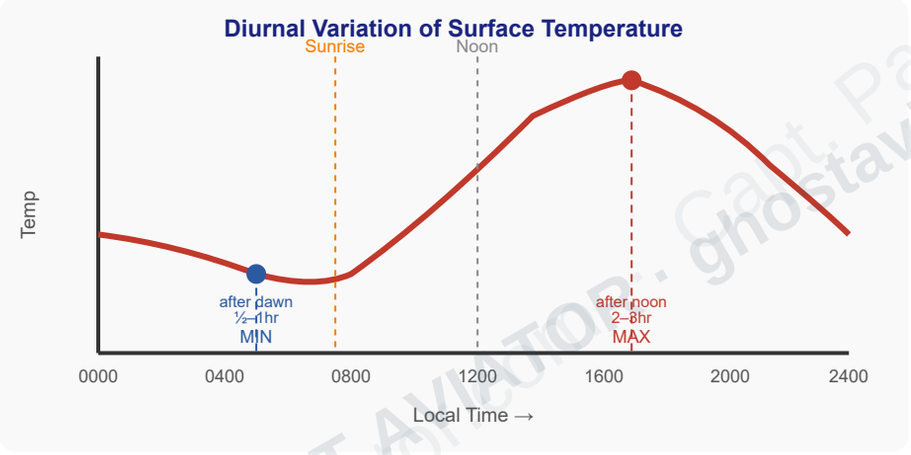


Fig 3.3: Diurnal Variation of Surface Temperature

14. Effect of Clouds on Surface Temperature

A cloud cover:

- Can cut off a fair proportion of solar radiation during the day → reduces daytime heating
- A thick layer of stratus cloud can block radiation and reduces a major part going back to space
- A thick lower cloud also blocks most of the outgoing terrestrial radiation at night

Net effect: Cloudy nights are WARMER than clear nights.
The lower the cloud, the more effectively it reduces the nocturnal cooling.

Day with Clouds:
Reduced insolation → less daytime heating
Diurnal variation is **small**

Night with Clouds:
Clouds act as blanket — trap outgoing IR

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Night temperature stays **HIGHER**
Minimum temperature NOT as low as clear night

🎯 **Exam Tip:** "Cloudy nights are warmer" is a frequently tested concept. The cloud acts like a greenhouse roof — letting sunlight in (partially) and trapping the earth's heat radiation at night. Minimum temperatures on cloudy nights are higher than on clear nights.

Practice Q&A — Temperature

Practice questions with answers.

Q1. Diurnal variation of temperature is greatest when wind is
(a) calm (b) light (c) strong

✅ Answer: (a) calm — The diurnal range is largest with clear skies and calm or very light winds: the ground heats strongly by day and radiates freely at night, and there is no mixing to even out the temperature.

❌ (c) A strong wind mixes the air and reduces the range. (b) Even a light wind begins to blur the extremes; the largest range is in clear, calm conditions.

Q2. Diurnal variation of temperature is maximum over
(a) forest (b) ocean (c) land

✅ **Answer: (c) land**
Land has low specific heat (0.2) — heats and cools rapidly. Ocean has high specific heat (1.0) and large mixing depth.

❌ (b) Ocean — least diurnal variation (less than 1°C). (a) Forest — moderate, less than open land due to canopy and moisture.

Q3. On a clear day the amount of solar radiation received by earth surface is
(a) ¼ (b) 30% (c) 5/6

✅ **Answer: (c) 5/6**
About 5/6 of solar radiation reaches the earth surface on a clear day (the rest is reflected, scattered or absorbed by the atmosphere).

Q4. ALBEDO is
(a) Radiation received by earth (b) Amount of heat (c) Reflecting power of earth

✅ **Answer: (c) Reflecting power of earth**
Albedo = Reflected Radiation / Incident Radiation. It is the reflecting power.

❌ (a) Radiation received = Insolation. (b) Amount of heat = vague, not definition of albedo.

Q5. During Day the ambient temperature is …… than ground

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(a) Lower (b) Higher (c) Same

✅ Answer: (a) Lower — By day the ground heats faster than the air above it, so the air (screen) temperature is lower than the ground surface temperature. At night the ground cools faster and the air is the warmer of the two.

❌ (b) Higher and (c) Same: the ground absorbs the sunshine first and passes heat up to the air, so it is hotter than the air by day.

Q6. Diurnal variation of temperature over ocean is
(a) More than land (b) Above 3°C (c) Less than 1°C

✅ **Answer: (c) Less than 1°C**
Sea surface temperature varies by less than 1°C from day to night due to high specific heat and mixing.

❌ (b) Above 3°C — this is typical of land. (a) More than land — the opposite is true.

Q7. At a coastal station the diurnal variation of temperature depends on
(a) Wind direction (b) Wind speed (c) Radiation

✅ Answer: (a) Wind direction — At a coastal station the daily temperature range depends mainly on the wind direction: with a wind off the sea the range is small, with a wind off the land it can be large.

❌ (b) Wind speed and (c) radiation matter everywhere, but at the coast it is the direction, sea or land, that decides the range.

Q8. Snow surface reflects about ..... % of solar radiation.
(a) 75% (b) 80% (c) 90%

✅ **Answer: (b) 80%**
Snow has a very high albedo — reflects about 80% of incident solar radiation.

❌ (a) 75% and (c) 90% — close, but not the accepted value. Snow = 80% albedo.

Q9. Amount of Solar radiation received per unit area is
(a) Insolation (b) Convection (c) Radiation

✅ **Answer: (a) Insolation**

❌ (b) Convection — heat transfer method. (c) Radiation — general term for electromagnetic waves.

Q10. Solar radiation received by the earth is
(a) Long Wave (b) Albedo (c) Shortwave

✅ **Answer: (c) Shortwave**
Sun's surface temperature is about 5800 K (about 5500°C) → Wien's Law → short wavelength radiation. Earth (≈15°C) radiates long wave.

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Q11. Rise in temperature of a surface is proportional to its specific heat
(a) Directly (b) Inversely

✅ **Answer: (b) Inversely**
Higher specific heat = more heat needed per degree rise = SLOWER temperature rise. Land (sp. heat 0.2) heats faster than water (sp. heat 1.0).

Q12. Specific heat of land is …… than that of water
(a) Lower (b) Same (c) Higher

✅ Answer: (a) Lower — The specific heat of land (about 0.2) is much lower than that of water (about 1.0), so land heats up and cools down much faster than water.

❌ (b) Same and (c) Higher are wrong: if land had the higher specific heat it would warm more slowly than the sea, the opposite of what we see.

Q13. Minimum temperature is reached at
(a) Sunrise (b) Midnight (c) ½–1 hour after dawn

✅ **Answer: (c) ½–1 hour after dawn**
Nocturnal cooling continues slightly after sunrise until incoming solar radiation exceeds outgoing terrestrial radiation.

❌ (a) Sunrise — cooling continues past sunrise. (b) Midnight — temperature continues falling after midnight through the night.

Q14. Dew point

Dew point is the temperature to which air must be cooled at constant pressure and unchanged vapour content to reach saturation. Lifting an air parcel changes pressure and is not the defining constant-pressure process.

Q15. Cloudy nights are
(a) cold (b) normal (c) warm

✅ **Answer: (c) warm**
Clouds trap outgoing terrestrial (long wave) radiation, acting like a blanket and keeping the surface warmer at night.

❌ (a) Cold — this applies to clear nights with strong nocturnal radiation. (b) Normal — not specific.

Q17. Higher the temperature, …… would be the wavelength of emitted radiation
(a) longer (b) shorter

✅ Answer: (b) shorter — By Wien's law the wavelength of the most intense radiation is inversely proportional to the absolute temperature: the hotter the body, the shorter the wavelength. The Sun (about 5800 K) emits short waves; the Earth emits long waves.

❌ (a) Longer is the opposite: it describes cooler bodies.

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Q21. The solar radiation consists of about 46%
(a) UV (b) IR (c) Visible

✅ **Answer: (b) IR (Infra-Red)**
Solar radiation: IR = 46%, Visible = 45%, UV = 9%.

❌ (a) UV = only 9%. (c) Visible = 45%, not 46%.

Q22. The wavelength of most intense radiation is …… proportional to temperature
(a) Directly (b) Inversely

✅ **Answer: (b) Inversely**
Wien's Law: λmax = constant/T. Higher temperature → shorter (smaller) wavelength. Inversely proportional.

Q27. The flow of heat near earth surface is 77% by
(a) Sensible Heat (b) Latent Heat

✅ Answer: (b) Latent Heat — Latent heat (evaporation, condensation, sublimation) carries about 77% of the heat that flows from the Earth's surface to the atmosphere; sensible heat carries about 23%.

❌ (a) Sensible heat is only about 23% of the total.

Q28. −40°C = −40°F
(a) True (b) False

✅ Answer: (a) True — The Celsius and Fahrenheit scales cross at −40: F = (9/5)(−40) + 32 = −40. So −40 °C = −40 °F.

❌ (b) False would be correct only if the scales never met; they meet at −40.

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Q29. Screen air temperature

Air temperature is measured in a ventilated shelter, protected from direct sunlight and surface heating. In the UK Met Office arrangement, the thermometer is 1.25 m above the ground. Specify the observing arrangement rather than calling this a universal height; 4 ft is 1.2192 m, not exactly 1.25 m.

Q30. A minimum thermometer

A traditional alcohol-in-glass minimum thermometer uses alcohol to record the lowest temperature. The word spirit can also refer to an alcohol liquid, so alcohol and spirit are not suitable mutually exclusive options. Electronic instruments may use a different measuring method.

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Q31. Freezing point of water is
(a) 0°F (b) 12°F (c) 22°F (d) 32°F

✅ **Answer: (d) 32°F**
Freezing/melting point of water = 0°C = 32°F = 273 K.

Q35. Convert 68°F to kelvin

Celsius temperature = (68 − 32) × 5/9 = 20°C. Kelvin temperature = 20 + 273.15 = 293.15 K, or 293 K when rounded to the nearest kelvin. Kelvin is written K without a degree sign.

Chapter 4: Air Density

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1. Introduction & The Gas Equation

📘 **Core Concept** Air density (ρ) is the mass of air per unit volume. It is a fundamental parameter that directly affects aircraft performance — engine power output, aerodynamic lift, and propeller/rotor efficiency all depend on air density (mass per unit volume).

The density of air can be derived from the fundamental gas equation:

PV = RT

P	Pressure in pascals (Pa); 1 hPa = 100 Pa
V	Specific volume (volume per unit mass), in m³/kg
R	Gas constant for the particular gas
T	Absolute temperature (Kelvin)

Since for a unit mass of gas, density ρ is the reciprocal of volume V, therefore:

P/ρ = RT or ρ = P/RT

ρ	Air density (kg/m³)
P	Pressure (Pa; multiply hPa by 100)
R	Gas constant (specific to the gas)
T	Absolute temperature (Kelvin)

⚡ **Key Relationship** ρ = P/RT — Density is **directly proportional to pressure** and **inversely proportional to absolute temperature**. Increase pressure → higher density. Increase temperature → lower density.

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2. Density of Dry Air

For dry air, the gas constant R = 287 J/(kg·K).

🔢 **Worked Example — ISA Surface Density**

Given:

- ISA surface pressure = 1013.25 hPa
- ISA surface temperature = 15°C = 288.15 K
- R for dry air = 287 J/(kg·K)

Calculation: ρ = P/(RT) = 101325 / (287 × 288.15) ≈ 1.225 kg/m³ = 1225 g/m³.

Result: ISA surface air density = 1225 g/m³

3. Density of Moist Air

📘 **Theory** Water vapour obeys the fundamental gas equation. The gas constant for water vapour is 8/5 times that for dry air. The total pressure p of moist air = pressure of the dry air (p − e) + pressure of the water vapour (e).

Component	Pressure	Density Expression
Water Vapour	e	5e/8RT
Dry Air (partial)	(p−e)	(p−e)/RT
Moist Air (total)	p	(p − 3e/8) / RT

⚠️ **Critical Principle Moist air is less dense than dry air** under similar conditions of pressure and temperature. This is because water vapour (H_2O, mol. wt. 18) replaces heavier nitrogen (N_2, mol. wt. 28) and oxygen (O_2, mol. wt. 32) molecules.

✈️ **Aviation Note** The effect of humidity on density is small and can be **ignored for aviation purposes** in calculations — unless humidity is very high (tropical conditions), in which case: add 10% to computed take-off distance and anticipate reduced climb rate.

4. Factors Affecting Air Density

📘 **Three Key Factors** Air density is affected by three factors: Altitude, Temperature, and Humidity.

Factor	Effect on Density	Aviation Implication
Altitude ↑	Density ↓ (Less dense)	Reduced engine power, longer T/O run, reduced climb rate
Temperature ↑	Density ↓ (Less dense)	Hot day = less dense air = degraded performance
Humidity ↑	Density ↓ (Less dense, minor effect)	Add 10% to T/O distance in high humidity; anticipate reduced climb

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5. Variations in Surface Density

📘 **Principle** At a given pressure, density is **inversely proportional to absolute temperature**. Warm air is comparatively lighter; cold air is heavier.

Diurnal (Daily) Variation: The lowest densities occur in the afternoon and the highest just after sunrise. This is due to the diurnal variation of temperature.

Seasonal Variation: Seasonal density changes occur due to variations in temperature and pressure.

⚠ **Exam Critical — 1% Rule for Surface Density** A decrease of density of about 1% is produced by:

- A fall of pressure of 10 hPa, OR
- An increase of temperature of 3°C, OR
- An increase in height of 300 feet

6. Variation of Density with Height

📘 **ISA Density Profile** In ISA, density decreases with height at all levels. The decrease at lower levels for 1000 ft is approximately 3% of the value for any given level. This rule gives a good approximation up to 20,000 feet.

If the density of air were to remain uniform with height, the atmosphere would extend up to 8 km. But since density decreases with height, at every 5–6 km it reduces to its previous half value.

Altitude	Density as Fraction of Surface Value	Key Note
0 km (Surface)	1/1 (Full density)	ISA: 1225 g/m³
6 km	1/2	Density halved
11 km	1/4	Tropopause region
17 km	1/8	Stratosphere
240 km	Still offers resistance	Density sufficient for drag
800 km	Atmosphere effectively ends	Upper limit

7. Latitudinal Variation of Density

📘 **Theory** Density of air at sea level is **lowest near the equator** and **greatest at the poles**. This distribution is maintained until about 8.0 km. **Above 8.0 km, a reversal occurs** — density becomes greater near the equator than at poles (higher latitude).

Altitude Band	Highest Density Location	Lowest Density Location
Surface to ~8 km	Poles (cold, dense air)	Equator (warm, less dense)
Above ~8 km	Equator (reversal)	Poles (reversal)

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Aviation Implication — Aircraft Type Matters

- **Piston-engine aircraft** (low cruise altitude): Operational efficiency greater at **high latitudes** (poles) — denser air below 8 km.
- **Jet aircraft** (high cruise altitude): Operational efficiency greater in the **tropics** (equator) — denser air above 8 km reversal.

8. Density Altitude — Key Concept

Definition — Density Altitude The altitude in the **International Standard Atmosphere (ISA)** at which the **prevailing density occurs** in the actual atmosphere. It is the pressure altitude corrected for non-standard temperature.

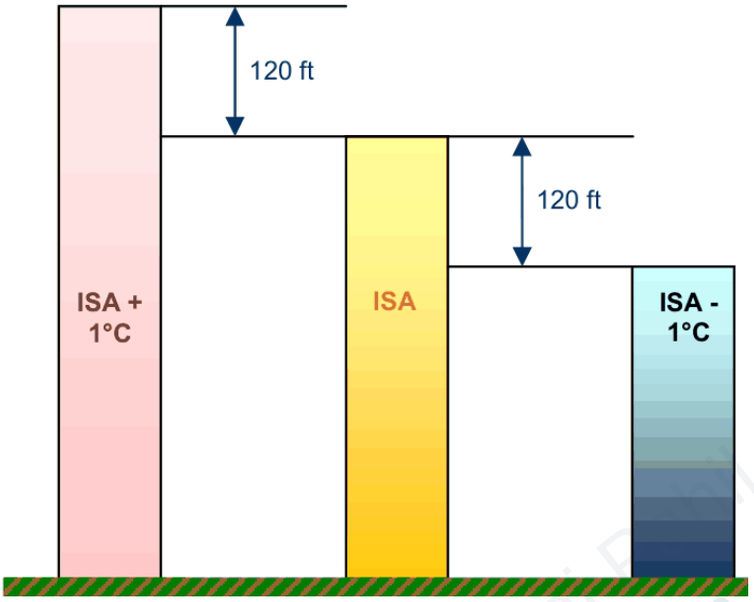


Why a hot, high, humid day lengthens the take-off

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CALCULATING DENSITY ALTITUDE



Each 1°C that the air is warmer than standard raises the density altitude by about 120 ft.

⚠ **Performance Impact** High density altitude = aircraft performs as if it were at a much higher altitude. Hot + high + humid = worst case for aircraft performance. All distances increase; all climb rates decrease.

✅ **Density Altitude Calculation** For every 1°C change in temperature, density altitude differs from pressure altitude by approximately 120 ft.

🔢 Density Altitude Example

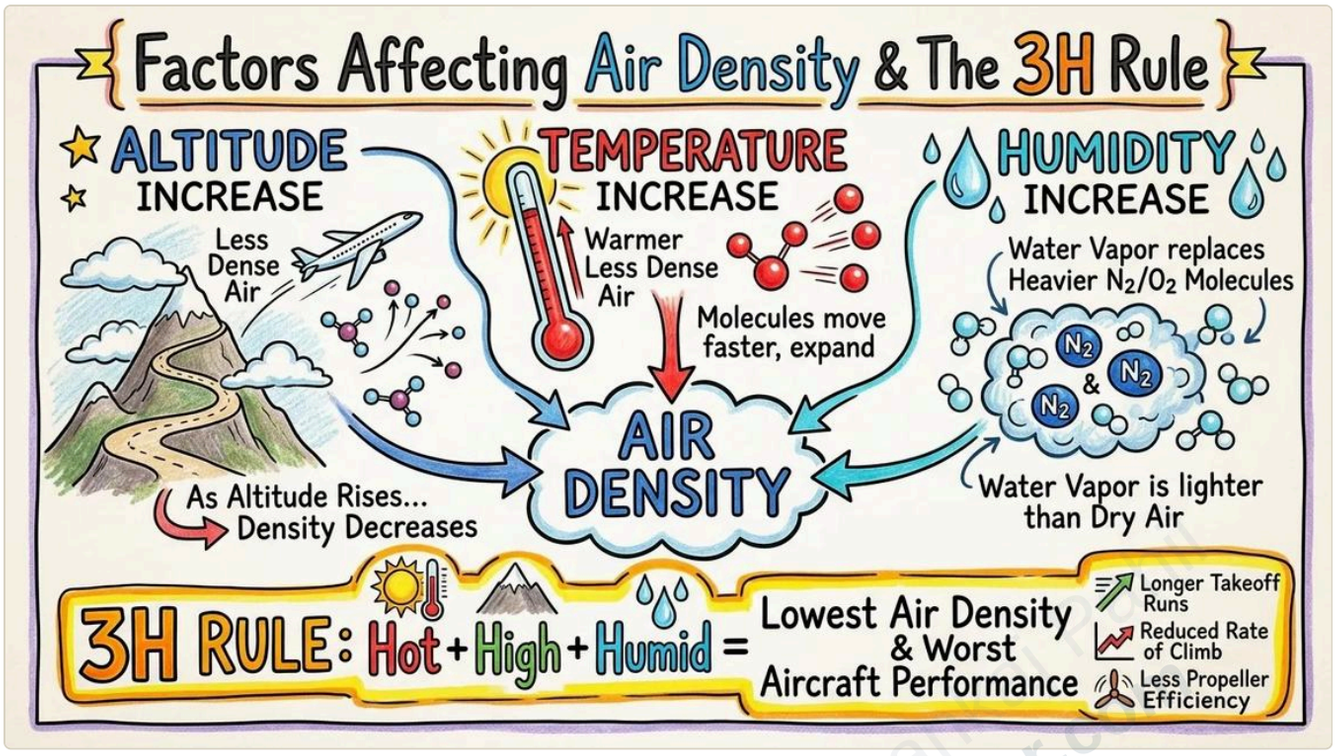
Aerodrome elevation: 2000 ft. QNH is 1013 hPa, so the pressure altitude is also 2000 ft. (With any other QNH, first find the pressure altitude: it is what the altimeter reads with 1013 hPa set.) Temperature is 10°C above ISA standard for that level.

Correction: 10°C × 120 ft/°C = 1200 ft addition

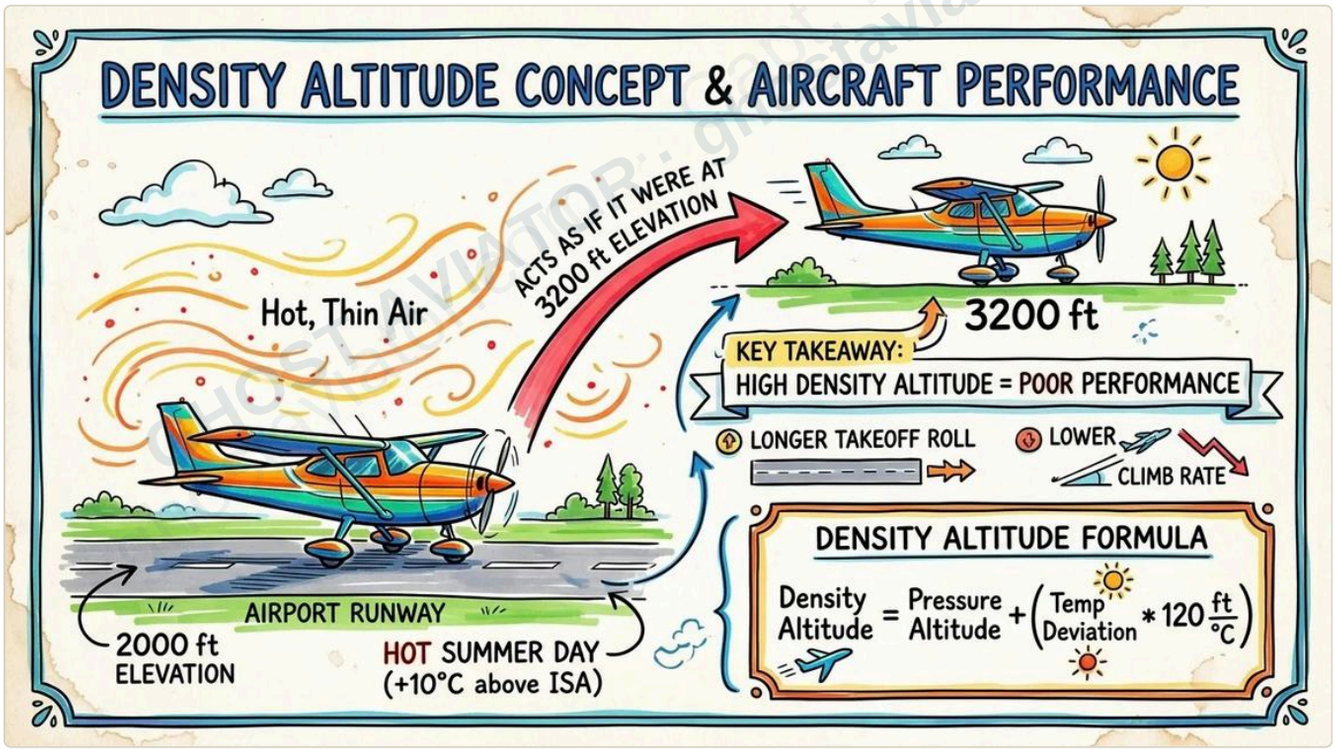
Density Altitude = 2000 + 1200 = 3200 ft

Aircraft will perform as if operating from a 3200-ft elevation aerodrome.

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Hot, high and humid: each lowers air density, and together they give the poorest aircraft performance.



Density altitude: on a day 10°C warmer than standard, an aerodrome with a pressure altitude of 2000 ft behaves like one at about 3200 ft.

10. Practice Q&A — DGCA Examination Style

📘 **Instructions** Cover the answer and attempt each question before revealing. Answers: Q1=a, Q2=b, Q3=a, Q4=b, Q5=b, Q6=b, Q7=d, Q8=b, Q9=a, Q10=c

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Q1. Density is …… at poles than equator
(a) Higher (b) Lower (c) Same

✅ Correct Answer: (a) Higher

Explanation: At sea level, polar air is much colder than equatorial air. Since ρ = P/RT, lower temperature (T) means higher density (ρ) at the same pressure. Cold polar air is denser than warm equatorial air below 8 km altitude.
❌ **Distractors:** (b) Lower — incorrect; cold air is denser, not less dense. (c) Same — incorrect; temperature differences create significant density differences.

Q2. Above 8 km density is …… at poles than at equator
(a) Higher (b) Lower (c) Same

✅ Correct Answer: (b) Lower

Explanation: Above 8 km, a reversal of the latitudinal density gradient occurs. The troposphere is deeper (taller) over the equator, meaning more mass of air extends to greater heights near the equator. Above 8 km, equatorial air becomes denser than polar air, so poles have lower density above this level.
❌ **Distractors:** (a) Higher — only true below 8 km. (c) Same — never the case; latitude always creates density differences.

Q3. The altitude in ISA at which the air density is the same as the observed density is:
(a) Density Altitude (b) ISA Density (c) Real Density

✅ Correct Answer: (a) Density Altitude

Explanation: Density Altitude is defined as the altitude in the International Standard Atmosphere (ISA) at which the prevailing (observed) density occurs. It is the key parameter for aircraft performance calculations.
❌ **Distractors:** (b) ISA Density — not a standard term. (c) Real Density — refers to actual density value, not an altitude concept.

Q4. Density is usually expressed as ……
(a) Kg/sq m (b) g/cu m (c) N/sq m

✅ Correct Answer: (b) g/cu m (grams per cubic metre)

Explanation: Air density is a measure of mass per unit volume, expressed in grams per cubic metre (g/m³). ISA surface density = 1225 g/m³. In SI units it can also be expressed as kg/m³ (1.225 kg/m³), but g/m³ is the unit used in this chapter.
❌ **Distractors:** (a) Kg/sq m — mass per unit area, not mass per unit volume, so it is not density. (c) N/sq m — the pascal, a unit of pressure (force per unit area), not density.

Q5. Higher density altitude means …… density
(a) Higher (b) Lower (c) Same

✅ Correct Answer: (b) Lower

Explanation: A higher density altitude means the air corresponds to a higher standard altitude in ISA — and at higher altitudes, density is lower. High density altitude = low actual air density = poor aircraft performance.

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❌ **Distractors:** (a) Higher — a common trap! "Higher altitude" in ISA means LESS dense air. The higher the density altitude, the worse the performance.

Q6. For given pressure and temperature, moist air has density:
(a) Higher (b) Lower (c) Same

✅ Correct Answer: (b) Lower

Explanation: Moist air is less dense than dry air at the same temperature and pressure. Water vapour (mol. wt. 18) is lighter than the nitrogen (mol. wt. 28) and oxygen (mol. wt. 32) molecules it replaces. The density formula (p − 3e/8)/RT confirms this — the term 3e/8 reduces the effective pressure, reducing density.
❌ **Distractors:** (a) Higher — a very common misconception! "Humid air" feels heavy but is actually less dense. (c) Same — incorrect; water vapour does affect density.

Q7. Air is less denser in:
(a) High Altitudes (b) Warm Air (c) High humidity (d) All these

✅ Correct Answer: (d) All these

Explanation: All three factors — high altitude, high temperature, and high humidity — reduce air density. This is the essence of density altitude: the combined effect of all three parameters on air density and therefore aircraft performance.
❌ **Distractors:** (a), (b), (c) — each is individually correct, but (d) is the complete and correct answer. Partial answers would miss the comprehensive picture.

Q8. Density altitude may be defined as:
(a) The altitude in ISA at which the prevailing pressure occurs.
(b) The altitude in ISA at which the prevailing density occurs.
(c) The altitude in the actual atmosphere at which the prevailing density occurs.

✅ Correct Answer: (b) The altitude in ISA at which the prevailing density occurs.

Explanation: Density Altitude is specifically defined as the altitude in the ISA at which the current (prevailing) density exists. It uses the ISA as a reference standard, not the actual atmosphere.
❌ **Distractors:** (a) — this defines Pressure Altitude, not Density Altitude. (c) — density altitude uses ISA as reference, not the actual atmosphere; using the actual atmosphere's altitude gives geometric altitude.

Q9. Temperature being constant, if pressure increases the density altitude:
(a) increases (b) lowers (c) remains the same

✅ Answer: (b) lowers

At constant temperature a higher pressure means denser air (ρ = P/RT). Denser air corresponds to a lower altitude in the standard atmosphere, so the density altitude lowers.
❌ (a) Increases would need the air to become thinner. (c) Density altitude changes whenever density changes.

Q10. For every 1°C change in temperature, density altitude differs from pressure altitude by:

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(a) 33 ft (b) 100 ft (c) 120 ft (d) 210 ft

✅ Correct Answer: (c) 120 ft

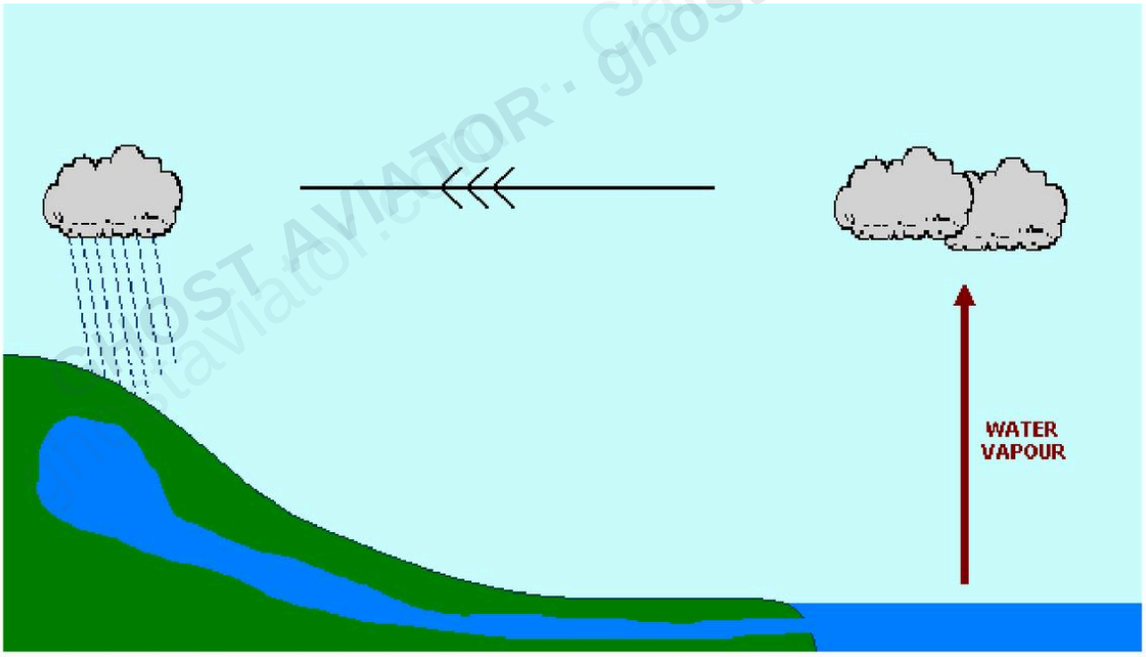
Explanation: For every 1°C deviation of actual temperature from ISA standard temperature, the density altitude differs from pressure altitude by approximately 120 feet. If OAT is above ISA, density altitude is higher than pressure altitude (worse performance). If OAT is below ISA, density altitude is lower than pressure altitude (better performance).
❌ **Distractors:** (a) 33 ft — this is the approximate height equivalent of 1 hPa pressure change. (b) 100 ft — not the correct figure. (d) 210 ft — this is a distractor.

Chapter 5: Humidity

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1. Introduction — Water Vapour & The Water Cycle

📘 **Core Concept** Water vapour is always present in the air to a greater or lesser extent, in the troposphere. This water vapour plays a very important role in all the atmospheric processes.



The water cycle

Water evaporates into the air from oceans, lakes, rivers, and vegetation. It ascends and forms clouds which cause precipitation. This is the **water cycle**.

⚡ **Key Principle** The capacity of dry air to hold water vapour depends largely on **temperature** (higher temperature = higher capacity) and to some extent on **pressure**. Higher the temperature, higher the capacity of air to hold water vapour.

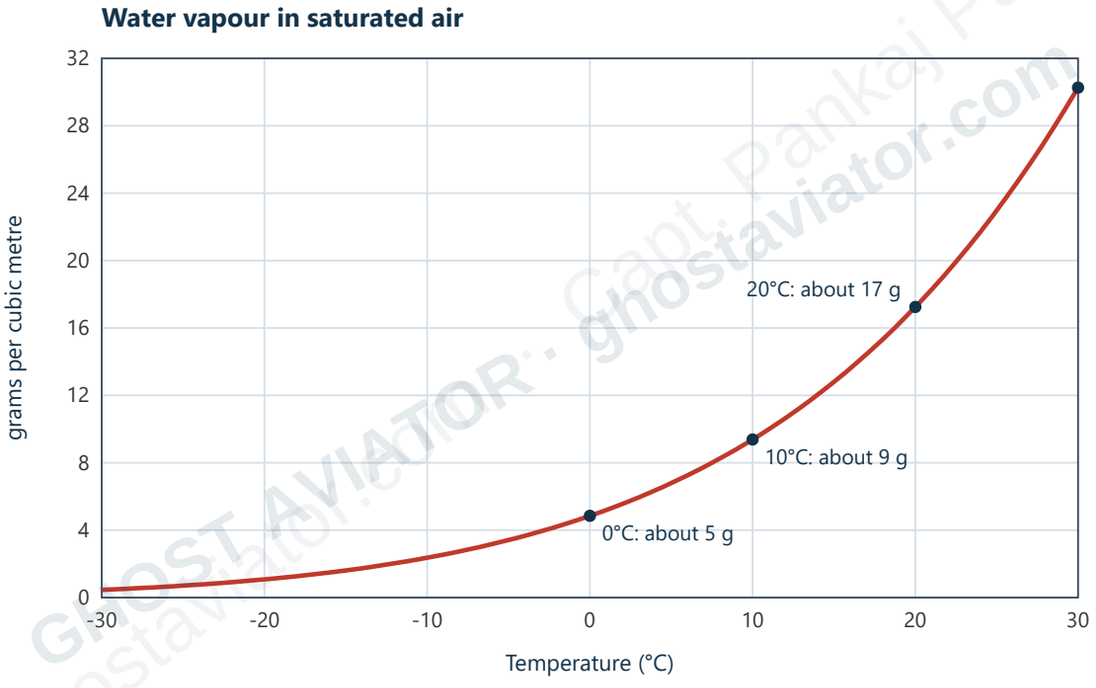
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2. Three Phases of Water

Phase	State	Examples
Gas	Water Vapour	Invisible moisture in the atmosphere
Liquid	Water	Rain, drizzle, shower, super-cooled water droplets
Solid	Ice	Snow, hail, ice crystals, hoar frost

3. Key Humidity Terms & Definitions

📘 **Dry Air** Air that contains no water vapour is called **dry air**. Such an air may exist in the upper troposphere or stratosphere.



Warm air can hold far more water vapour than cold air: about 5 g per cubic metre at 0°C, 17 g at 20°C and 30 g at 30°C.

📘 **Moist Air Moist air** is air containing water vapour. It may be saturated or unsaturated. In gas-law calculations, **dry air** means air without water vapour.

📘 **Saturated Air** Air is like a sponge which can absorb certain amount of water and no more. When the air holds maximum water vapour, it is called **saturated air**.

📘 **Vapour Pressure (VP)** The partial pressure exerted by the water vapour in the air is called vapour pressure. If *p* is the total pressure of air and *e* is the vapour pressure, then **(p − e)** is the pressure of dry air.

📘 **Saturation (water-vapour pressure has reached its saturation value at the current temperature) Vapour Pressure (SVP)** It is the pressure exerted by water vapour when air is saturated.

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📘 **Absolute Humidity (mass of water vapour per unit volume of air, commonly in g/m³)** It is defined as the actual amount of water vapour contained in a given volume of air at a given temperature. It is expressed in g/m³.

📘 **Humidity Mixing Ratio (HMR)** It is the mass of water vapour divided by the mass of dry air. It is expressed as g/kg.

📘 **HMR for Saturated Air** The maximum mass of water vapour that can be contained in a given mass of dry air at a particular temperature and pressure. Also expressed as g/kg of dry air. The saturation humidity mixing ratio (mass of water vapour divided by the mass of dry air, commonly in g/kg) **increases with temperature**.

Term	Definition	Unit	Key Note
Absolute Humidity	Water vapour per unit VOLUME	g/m³	Changes with T and P
HMR	Water vapour per unit MASS of dry air	g/kg	Constant if no moisture added/removed
Vapour Pressure (VP)	Partial pressure of water vapour	hPa	e in gas equations
SVP	VP when air is saturated	hPa	Increases with temperature

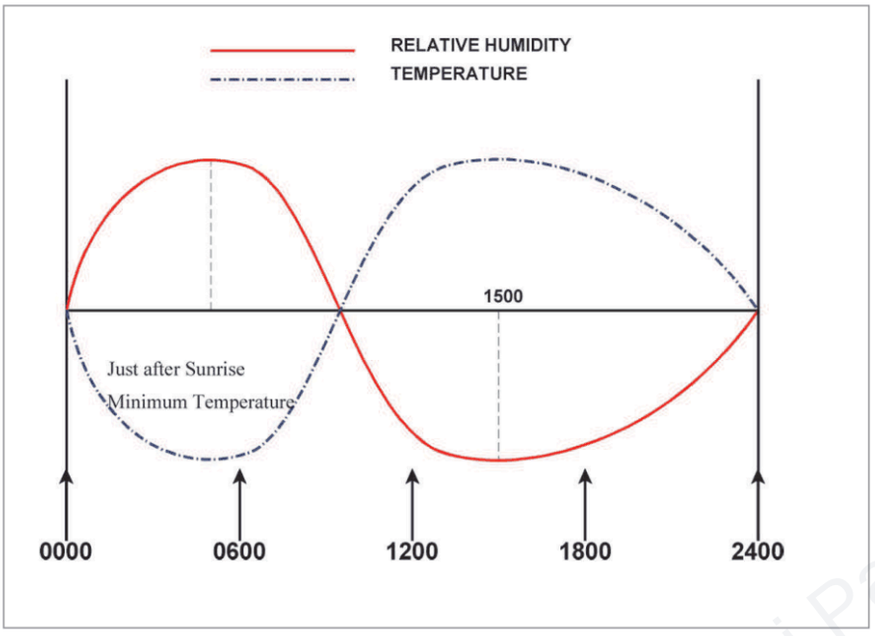
⚠️ **Critical — HMR During Adiabatic Lifting** If there is **no addition or removal of water vapour**, the HMR remains **constant when air is lifted adiabatically**. The mixing ratio is a conservative property during adiabatic processes.

4. Relative Humidity (RH)

📘 **Definition** Relative Humidity is defined as the ratio, in percentage, of the **actual water vapour** in the air to the **maximum it can contain** at the same temperature and pressure.

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Diurnal variation of relative humidity

RH (%) = (HMR × 100) / (HMR for Saturated Air)

OR equivalently:

RH (%) = (VP of Air × 100) / (SVP of Air)

HMR	Humidity Mixing Ratio of actual air
HMR(sat)	HMR for saturated air at same temperature
VP	Actual Vapour Pressure
SVP	Saturation Vapour Pressure at same temperature

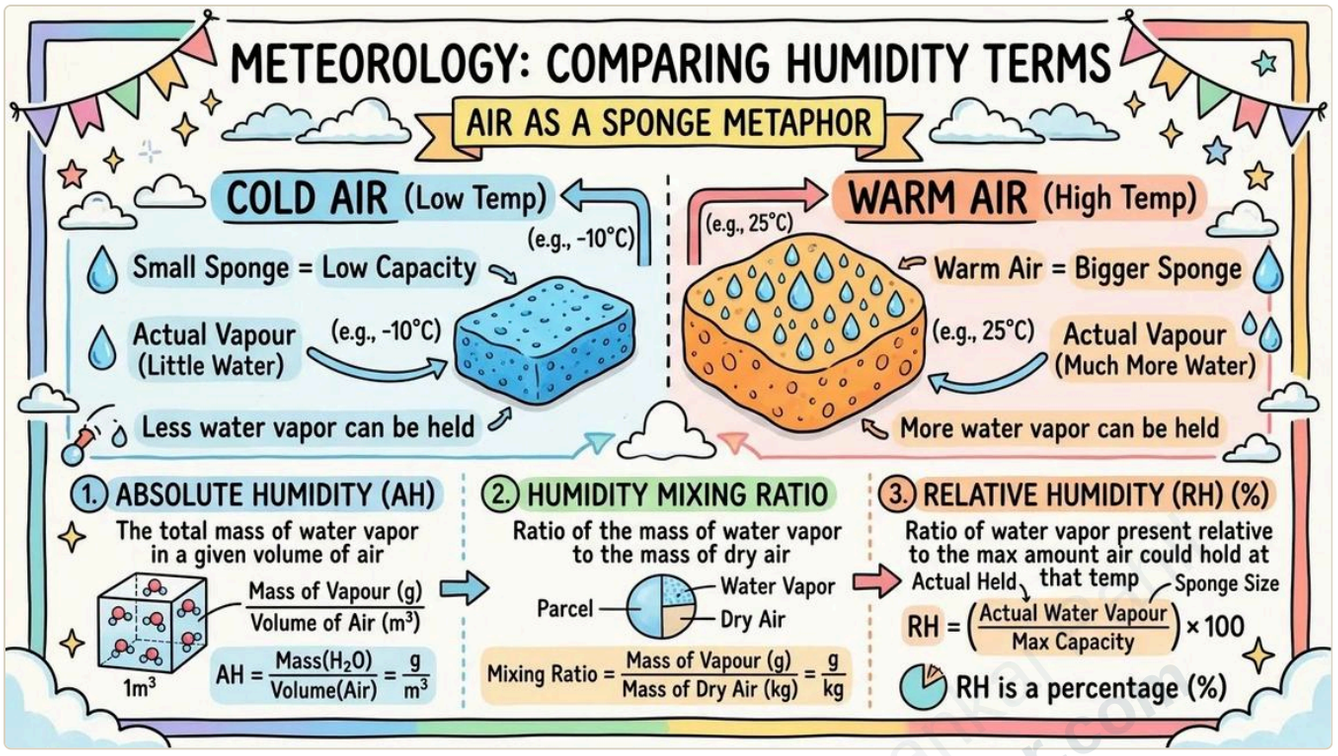
⚡ **RH Measurement** Humidity is measured by the instruments **Psychrometer** and **Hygrometer**, and is recorded by **Hygrograph**.

⚠ **RH vs Dew Point — Key Distinction**

- **RH** is affected by change in *both* water content AND temperature
- **Dew point (temperature to which air must cool to become saturated at unchanged pressure and water-vapour content)** changes when vapour pressure changes; pressure as well as vapour content matters.
- By cooling or warming the air, **RH changes but DP does not change**
- **DP is higher** if air contains more water vapour

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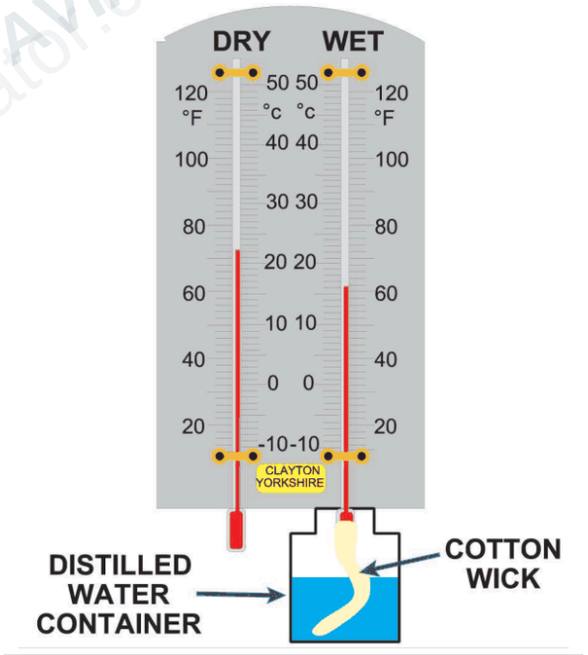
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Three ways of stating humidity. Warm air can hold more water vapour than cold air, like a bigger sponge.

5. Temperature Parameters — Wet Bulb, Dew Point & Frost Point

📘 **Wet Bulb Temperature (Tw)** It is the lowest temperature which air would attain by **evaporating water into it to saturate it**. Desert coolers work on this principle. The drier the air, the more effective would be the cooling.



Wet and dry bulb hygrometer

📘 **Dew Point Temperature (TdTd or Td)** It is the lowest temperature to which air should be cooled at **constant pressure** to **saturate it with respect to water**. Cooling below Dew Point (DP) causes

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condensation (water vapour changes into liquid water and releases latent heat).

📘 **Frost Point** It is the temperature to which air must be cooled to reach **saturation with respect to ice**. Cooling below the frost point causes formation of **hoar frost**.

Temperature	Symbol	Definition	Below This →
Air Temperature	TT	Actual ambient air temperature (dry bulb)	N/A
Wet Bulb Temp	Tw	Lowest T by evaporating water to saturate the air	Not applicable (evaporative limit)
Dew Point	TdTd / Td	T at which air saturates with respect to water at constant pressure	Condensation (dew, fog)
Frost Point	—	T at which air saturates with respect to ice	Hoar frost formation

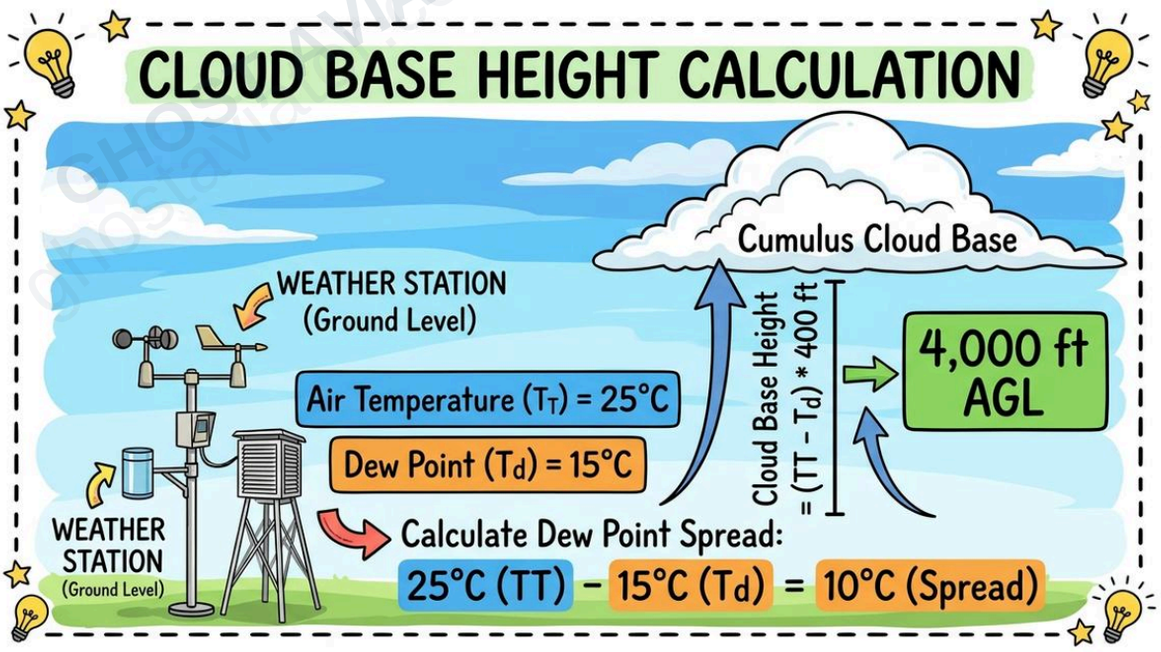
🧠 **Key Relationship — Unsaturated Air**

`TT > Tw > Td (Unsaturated Air)`

`TT = Tw = Td (Saturated Air – Fog, During Rain)`

When RH = 100%, all three temperatures converge.

6. Estimating a Convective Cloud Base



A rule-of-thumb estimate for a well-mixed convective cloud base: a 10°C temperature–dew point spread gives about 4,000 ft above the surface. The result is approximate and does not measure cloud height or apply to every cloud type.

Approximate rule for a well-mixed layer:

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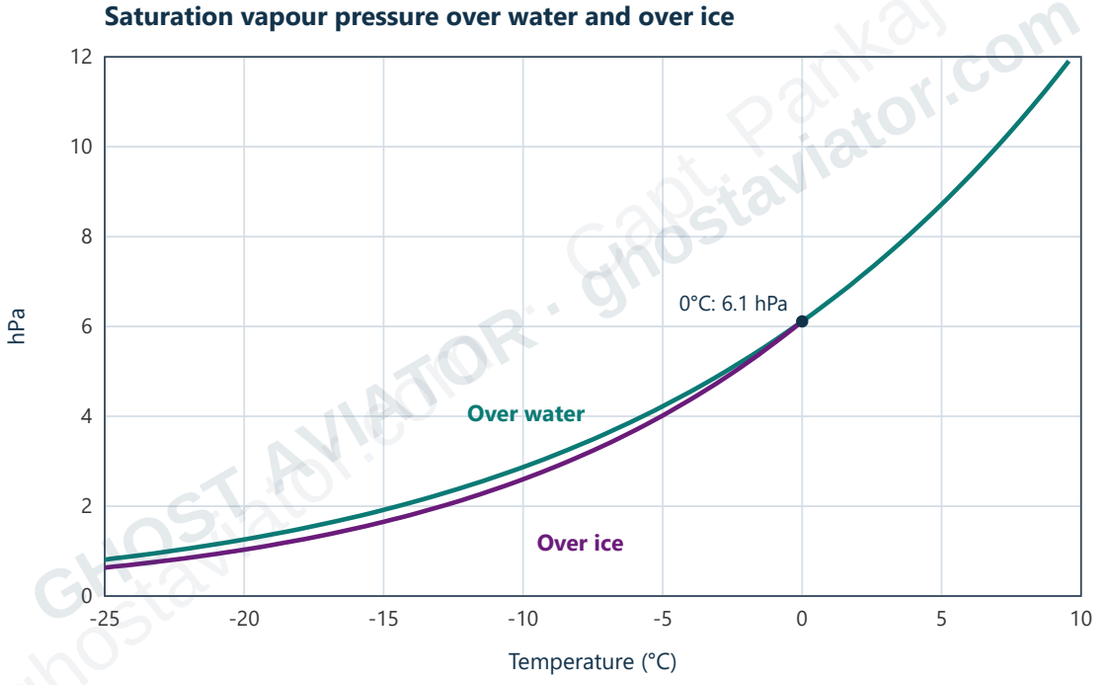
For convective clouds, an approximate base height above ground is 400 × (T − Td)ft, with surface temperature T and dew point Td in °C. The air between ground and cloud base must be well mixed. This estimate is not a measured cloud height and does not apply to every cloud type.

Example:25°C−15°C=10°C; estimated convective cloud base ≈ 10×400=**4000ft AGL**.

A smaller temperature–dew-point spread means the surface air is closer to saturation. A zero spread indicates saturation at the measurement point; it does not by itself establish observed fog or its extent.

7. SVP over Water vs Ice — Sub-Zero Behaviour

📘 **Important Sub-Zero Principle** At sub-zero temperatures, water molecules have more energy and a greater degree of freedom than ice. Consequently, the **saturation vapour pressure over water is more than that over ice particles**.



Below 0°C the saturation vapour pressure over water is higher than over ice. Air that is just saturated for water is already supersaturated for ice.

⚠️ **Bergeron-Findeisen Process — Exam Important** If **water drops and ice particles co-exist** in a cloud:

- SVP over water > SVP over ice at the same temperature
- Water drops evaporate and condense on the ice particles
- Ice particles grow at the expense of water droplets
- This explains **rainfall from clouds which extend above 0°C level** in the atmosphere — mixed-phase clouds produce rain via the Bergeron-Findeisen process
- Both super-cooled water drops and ice crystals can co-exist in this mixed zone

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⚠️ **Super-Cooled Water Droplets — Critical for Aviation Icing** Small water droplets can exist in a **super-cooled** (liquid below 0°C) state:

- In normal clouds: up to −40°C
- In Cumulonimbus (CB) clouds: up to −45°C
- These super-cooled droplets freeze **on contact** with aircraft surfaces → severe icing hazard

8. Temperature Relationships — Saturated vs Unsaturated Air

✅ **Saturated Air (Fog, During Rain)**

`Air Temperature (TT) = Wet Bulb (Tw) = Dew Point (Td)`

When RH = 100%, the air is saturated and all three temperature readings are identical.

⚡ **Unsaturated Air**

`TT > Tw > Td`

In normal (unsaturated) air, air temperature is highest, wet bulb is intermediate, and dew point is lowest.

Condition	TT vs Tw vs Td	RH	DP Spread
Saturated air (fog, heavy rain)	**TT = Tw = Td**	100%	0°C
Unsaturated air (normal)	**TT > Tw > Td**	< 100%	> 0°C
Very dry air (desert)	TT >> Tw > Td	Very low	Large spread

10. Practice Q&A — DGCA Examination Style

📘 **Answer Key** Q1=b, Q2=c, Q3=b, Q4=b, Q5=b, Q6=a, Q7=b, Q8=a, Q9=c, Q10=a

Q1. The ratio in % between the amount of water vapour present in the air to the amount of water vapour that it can hold at the same temperature is:
(a) Humidity (b) Relative humidity (c) Dew point

✅ Correct Answer: (b) Relative Humidity

Explanation: Relative Humidity is precisely defined as the ratio (in %) of the actual water vapour content to the maximum water vapour the air can hold at the same temperature and pressure. It is "relative" because it compares actual moisture to maximum possible moisture.
❌ **Distractors:** (a) Humidity — too vague; humidity is the general term for water vapour content. (c) Dew Point — a temperature, not a ratio.

Q2. The temperature to which air be cooled at constant pressure to become saturated is called:

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(a) Wet bulb temperature (b) Dry bulb temperature (c) Dew point (d) Humidity

✅ Correct Answer: (c) Dew Point

Explanation: The Dew Point Temperature is defined as the lowest temperature to which air must be cooled at constant pressure to become saturated with respect to water. Below this temperature, condensation occurs.
❌ **Distractors:** (a) Wet Bulb — this is the temperature attained by evaporating water INTO the air, not by cooling it. The process is different. (b) Dry bulb — this is just the ordinary air temperature. (d) Humidity — a measure of moisture content, not a temperature.

Q3. Free air temperature, Wet bulb temperature and Dew point temperature are equal when:
(a) Air temperature is 0°C
(b) Relative humidity is 100%
(c) Air temperature is not below 0°C

✅ Correct Answer: (b) Relative humidity is 100%

Explanation: When RH = 100%, the air is fully saturated. There is no difference between actual moisture content and maximum possible moisture content, so TT = Tw = Td. This condition exists in fog, during heavy rain, or in saturated cloud.
❌ **Distractors:** (a) 0°C is not a condition for equality — it could be 0°C with any RH. (c) Temperature above 0°C has no special significance for this relationship.

Q4. Rain and runway performance

Rain can wet a runway and reduce braking effectiveness. Required take-off or landing distance also depends on the aircraft, runway condition, wind, temperature and other factors. A rainy-versus-sunny label alone cannot give a universal distance comparison. Use the applicable aircraft performance data.

Q5. The spread (difference) between Free air temperature and Dew point temperature is ..... when air is saturated:
(a) Large (b) Least (c) Same

✅ Correct Answer: (b) Least (approaches zero)

Explanation: When air is saturated (RH = 100%), TT = Td. The spread between them is zero — the minimum possible. The spread (TT − Td) is also called the "dew point depression." Zero spread means the air is at saturation.
❌ **Distractors:** (a) Large — the spread is largest in very dry air (desert conditions). (c) Same — the spread changes constantly with temperature and moisture content.

Q6. Saturation vapour pressure over water and ice

At the same temperature below 0°C, saturation vapour pressure over liquid water is higher than over ice. This difference helps ice crystals grow at the expense of supercooled droplets in a mixed cloud. State the subzero condition rather than teaching an unrestricted comparison.

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Q7. As the temperature of the air increases, the amount of water vapour required to saturate it:
(a) decreases (b) increases (c) remains same

✅ Correct Answer: (b) increases

Explanation: Warmer air has a greater capacity to hold water vapour. The saturation humidity mixing ratio (HMR-sat) and the saturation vapour pressure (SVP) both increase with temperature. This is why warm tropical air can hold much more moisture than cold polar air.
❌ **Distractors:** (a) decreases — opposite is true. (c) remains same — capacity changes significantly with temperature.

Q8. The actual amount of water vapour contained in a given volume of air at a given temperature is termed as:
(a) Relative Humidity (b) Specific Humidity (c) Absolute Humidity

✅ Answer: (c) Absolute Humidity

Absolute humidity is the actual mass of water vapour in a given volume of air, in g/m³.
❌ (a) Relative humidity is a ratio in per cent, not an amount. (b) Specific humidity is the mass of vapour per unit mass of moist air, in g/kg.

Q9. Mixing ratio during ascent

During adiabatic ascent of an unsaturated parcel, with no mixing or gain or loss of vapour, vapour mixing ratio remains constant. Relative humidity rises as the parcel cools. After saturation, condensation can reduce vapour mixing ratio.

Q10. It is the lowest temperature which air would attain by evaporating water into it to saturate it:
(a) Wet bulb temp (b) Dry bulb temp (c) Dew point

✅ Correct Answer: (a) Wet Bulb Temperature

Explanation: Wet Bulb Temperature (Tw) is precisely defined as the lowest temperature which air would attain by evaporating water into it until it is saturated. The evaporative cooling process lowers the temperature to the wet bulb value. Desert coolers work on this same principle — drier the air, more the cooling effect.
❌ **Distractors:** (b) Dry bulb — this is simply the ordinary air temperature measured by an unmodified thermometer. (c) Dew point — cooling the air without adding moisture until it saturates; different process (temperature-lowering vs. moisture-adding).

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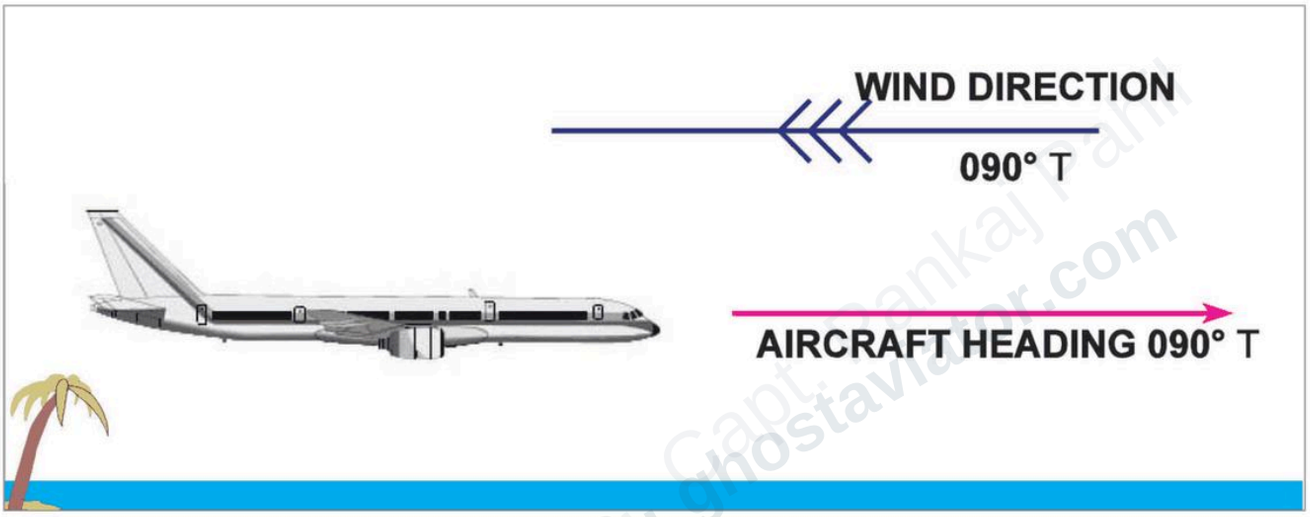
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Chapter 6: Winds

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1. Wind Measurement & Cross Wind

Definition: Wind is the horizontal movement of air from high pressure to low pressure. It is described by its direction (from which it blows) and speed.



Wind direction and aircraft heading

Cross Wind

Runways are oriented along the most prevailing wind directions of a locality based on climatological records. Sometimes — especially during adverse weather and transition seasons — the winds deviate from these directions. A wind 90° to the runway in use is called the **Cross Wind Component**. Critical cross wind components for each type of aircraft are specified. Cross winds tend to swing the aircraft during take-off and landing, especially lighter aircraft.

⚠ **Critical Cross Wind Component:** Great caution is exercised whenever this value is exceeded. Cross winds tend to swing the aircraft during take-off and landing, especially lighter aircraft.

Instruments for Measuring Wind

- **Surface wind speed:** Measured by **Anemometer (an instrument that measures wind speed)**
- **Wind direction:** Measured by **Wind Vane**
- **Upper winds:** Measured by RAWIN and Pilot Balloon equipment
- **Hydrogen-filled balloons:** Tracked by RADAR and Optical Theodolite

Exposure of Wind Instruments

- Anemometer and Wind Vane installed at height of 10 m in area free of obstructions
- Averaged over 10 minutes for all weather observations
- For **take-off and landing** purposes: wind is averaged for 2 minutes

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💡 **Exam Tip:** "10m, 10 minutes for all obs — 2 minutes for T/O and landing." Remember: **2 for T/O/Landing, 10 for general**.

2. Gust, Lull, Squall & Gale

Gust: An irregular and rapid positive fluctuation in wind. Caused by ground friction and uneven heating of the ground (especially on hot afternoons).
Lull: A negative fluctuation in wind (below mean).

Squall

Definition: Sudden increase in wind speed by 16 kt (about 30 km/h), should last for at least 1 minute and speed should increase to 22 kt (about 41 km/h) or more.

- Associated with CB clouds and violent convective activity
- Extends some km horizontally and several thousands of feet vertically
- Both speed and direction in squall may differ widely from prevailing winds
- Violent squalls experienced in Norwesters, Thunderstorms (TS) and Duststorms (DS) from March to June
- Line squalls occur ahead of Cold Fronts and sometimes with Norwesters

⚠ **Squall vs. Gust — Key Difference:** The main difference is **DURATION**. A gust lasts a few seconds; a squall lasts for at least one minute and may be accompanied by marked drop in temperature and precipitation.

⚠ **Strong squalls are more dangerous than strong mean winds.** A parked aircraft may sustain a gale (persistent mean 34 kt or more) but may NOT sustain a squall or sudden gust of say 60 kt, even though the mean wind may not reach gale force.

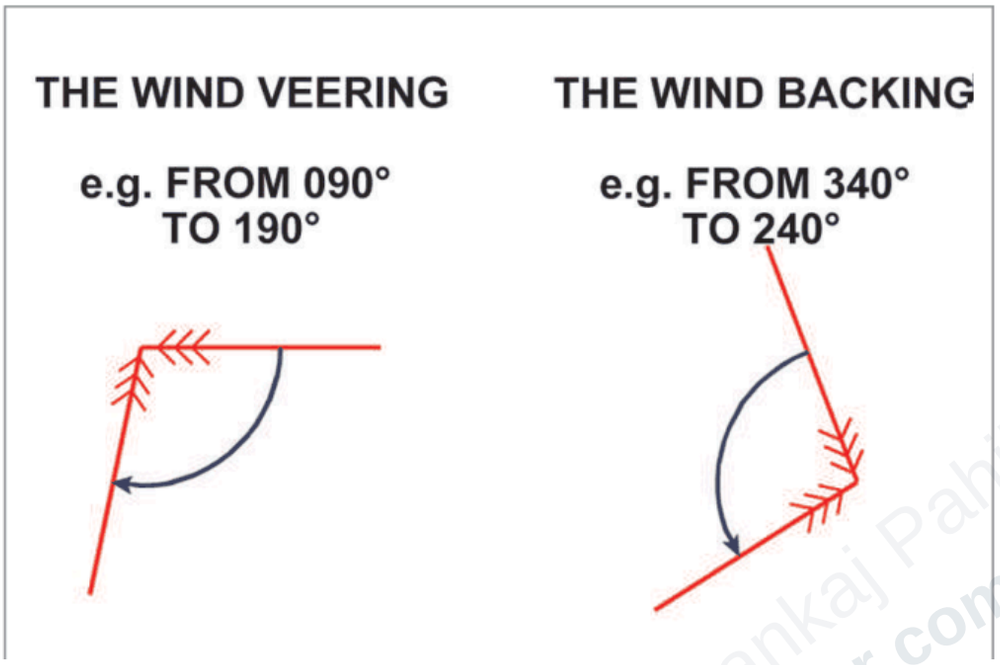
Gale

Definition: Persistent mean wind speed of 34 kt or more. Associated with depressions and cyclonic storms.

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3. Backing & Veering — Buys Ballot's Law



Veering and backing

Backing and Veering

Term	Definition	Example
Backing	Change of wind direction **anti-clockwise**	090° → 060° or 270° → 160°
Veering	Change of wind direction **clockwise**	060° → 090°

In a low pressure area: Wind blows anti-clockwise round the low in the N hemisphere (clockwise in the S hemisphere).
In a high pressure area: Wind blows clockwise round the high in the N hemisphere (anti-clockwise in the S hemisphere).
Note: backing and veering mean a change of wind direction at one place. They do not describe the way wind circles a low or a high.

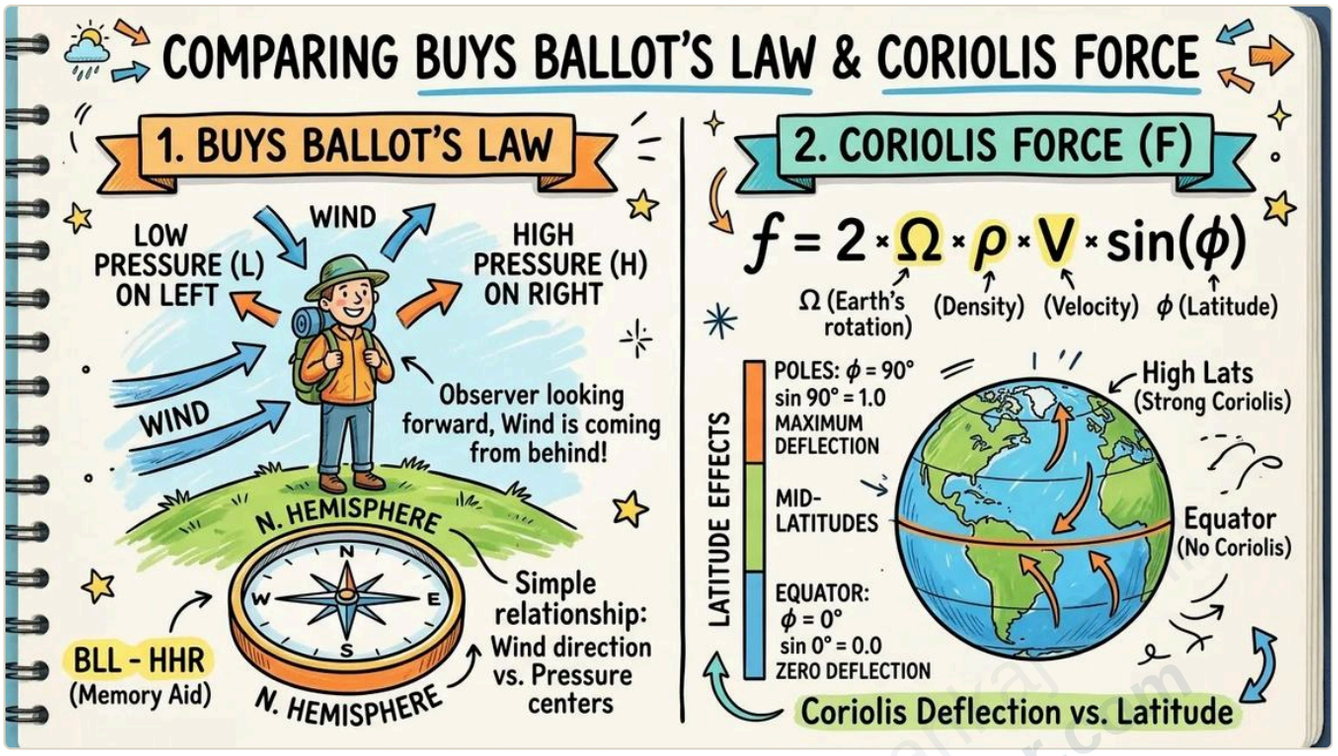
Buys Ballot's Law (Pressure and Wind)

In the N hemisphere: If an observer stands with his back to the wind — Low pressure is to his **LEFT** and High pressure is to his **RIGHT**. (Converse in S hemisphere.)

- N hemisphere: wind blows **anti-clockwise** around a cyclone (Low) and **clockwise** around an anti-cyclone (High)
- S hemisphere: opposite sense

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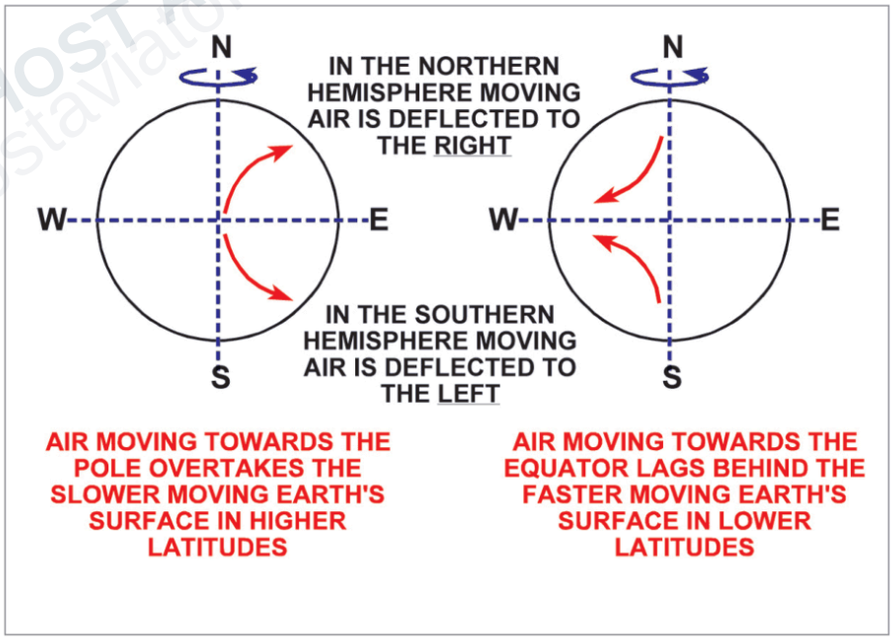
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Buys Ballot's law (northern hemisphere): back to the wind, low pressure is on your left. Coriolis force is zero at the equator and greatest at the poles.

4. Forces Acting on Wind — Coriolis Force

The **pressure gradient force** acts across isobars, from higher to lower pressure. In geostrophic balance, Coriolis balances this force and the wind flows along straight isobars. Due to Earth's rotation (W→E), wind is constantly deflected to its right (N hemisphere) until it attains uniform speed along isobars.



Coriolis deflection

Coriolis Force (f)

Coriolis Force (f) = 2 Ω ρ V sin φ

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Ω = angular velocity of Earth (unaffected by frictional forces)
ρ = density (unit volume of air, magnitude = 2ρV sin φ)
V = wind speed
φ = latitude

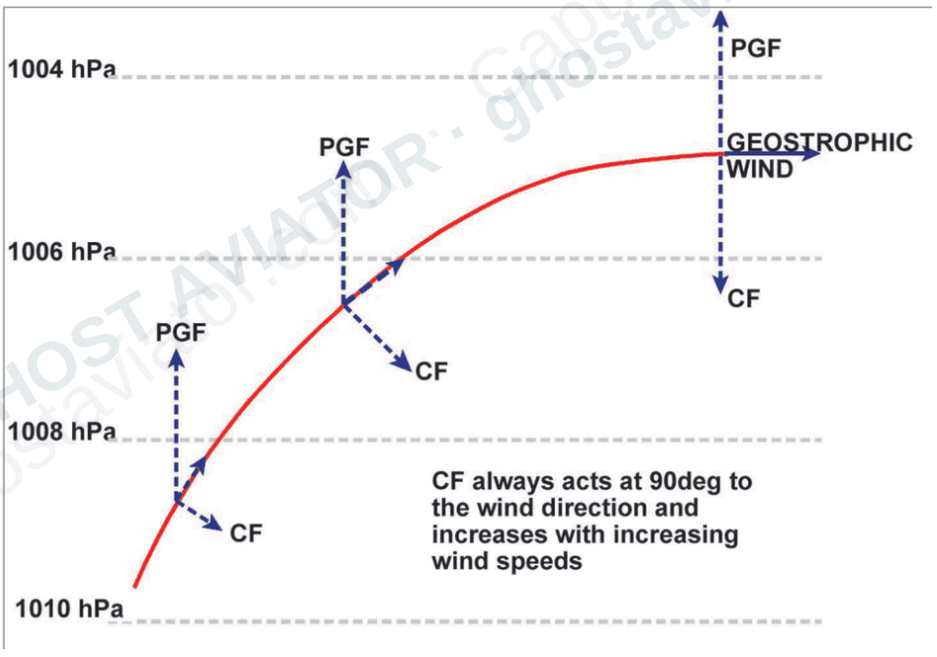
At poles: φ = 90°, sin φ = 1 → Coriolis force MAXIMUM
At equator: φ = 0°, sin φ = 0 → Coriolis force ZERO (minimum)

Coriolis force is directly proportional to wind speed and air density (mass per unit volume). Acts **perpendicular** to the wind direction.

⚠ Key Properties of Coriolis Force:

- Acts perpendicular to the wind direction — does NO work on air
- Zero at equator — geostrophic formula breaks down
- Maximum at poles
- Deflects to the RIGHT in N hemisphere, LEFT in S hemisphere
- Also called the **Geostrophic Force** — it is an apparent force

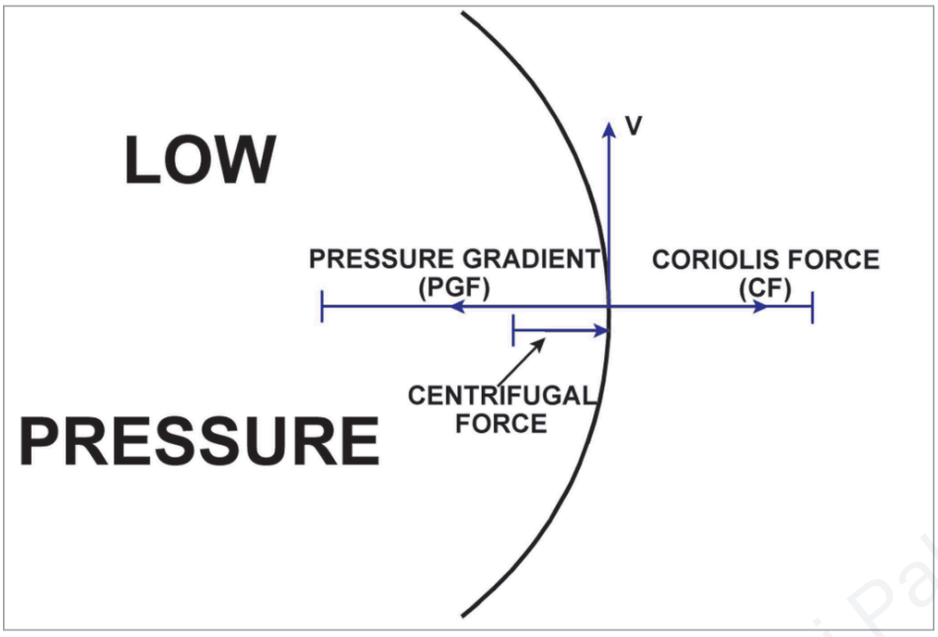
5. Types of Calculated/Balanced Winds



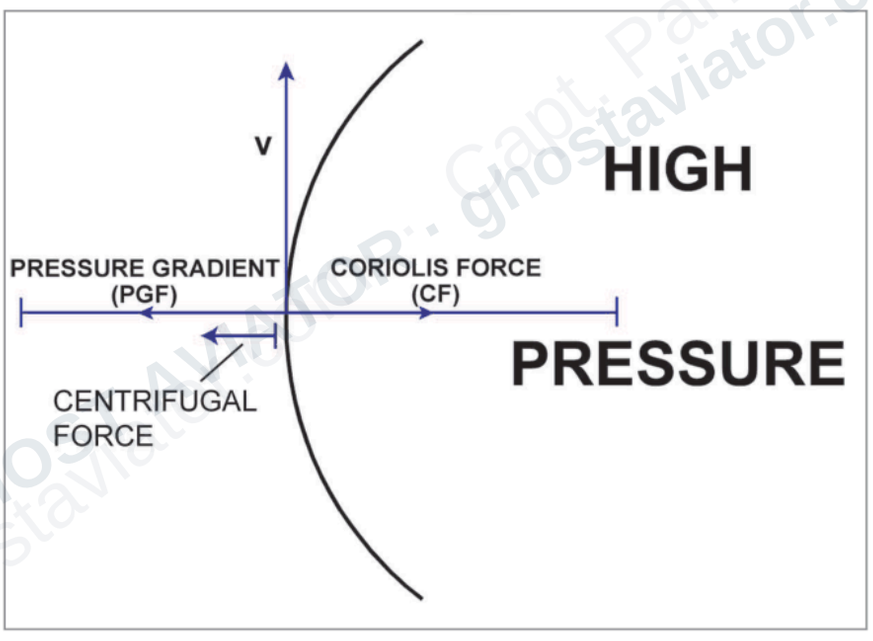
Geostrophic wind

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Gradient wind around a low



Gradient wind around a high

Geostrophic Wind (Vg)

The friction-less wind which blows **parallel to straight isobars** due to balance between Pressure Gradient Force and Coriolis force. It is a *calculated* wind.

$P / \rho = 2 \Omega V \sin \varphi$
$Vg = P / (2 \Omega \rho \sin \varphi)$

Where P/ρ = pressure gradient per unit mass
Closer isobars → stronger wind (inversely proportional to spacing)

⚠ Limitations of Geostrophic Rule:

- Valid only when PGF and Coriolis are in balance and isobars are straight & parallel

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- The geostrophic approximation fails near the **Equator**, where Coriolis is weak. It also requires negligible friction, nearly steady conditions and straight, parallel isobars; it does not automatically fail at the poles.
- Winds like gusts, squalls, local winds, and sea breezes do NOT follow this rule
- Geostrophic wind differs in direction and speed from actual wind (especially middle latitudes)

Cyclostrophic Wind

Wind blowing along **curved isobars** where Coriolis is negligible compared to Pressure Gradient. Centripetal force acts. Balance: **P = Centripetal**

$\rho V^2 / r = P \quad \therefore V = (Pr/\rho)^{1/2}$

A useful approximation for some tightly curved flows, including tornadoes, where Coriolis is small relative to the pressure-gradient and curvature terms. The balance does not require anticyclonic rotation.

Cyclostrophic Wind = markedly curved wind that blows due to balance between **Pressure Gradient and Cyclostrophic force**. Frictional force is disregarded.

Gradient Wind

Blows parallel to curved isobars under balance of Pressure Gradient (P), Coriolis (f), and Centripetal (C) forces. Frictional force disregarded. Middle latitudes: closer approximation to actual wind than geostrophic.

System	Balance	Result
Cyclone (Low)	P = f + C, with C denoting centrifugal force	**Sub-geostrophic** (V < Vg)
Anticyclone (High)	f = P + C, with C denoting centrifugal force	**Super-geostrophic** (V > Vg)

Isallobaric Wind

When pressure changes rapidly, geostrophic and gradient rules do not apply. An additional force called **Isallobaric Force** comes into play, directed from higher isallobar to lower isallobar, deflecting wind towards falling pressure. The resulting wind is the **Isallobaric Wind**.

Inertial Wind

Frictionless flow under balance between **Centrifugal Force** and **Coriolis Force** only. No Pressure Gradient Force.

$V_i = |f| \cdot R$

f = Coriolis parameter (s^{-1}); **R** = radius of curvature (m)
The constant Inertial wind speed is Vi.
The inertial flow is Anticyclonic in BOTH hemispheres.

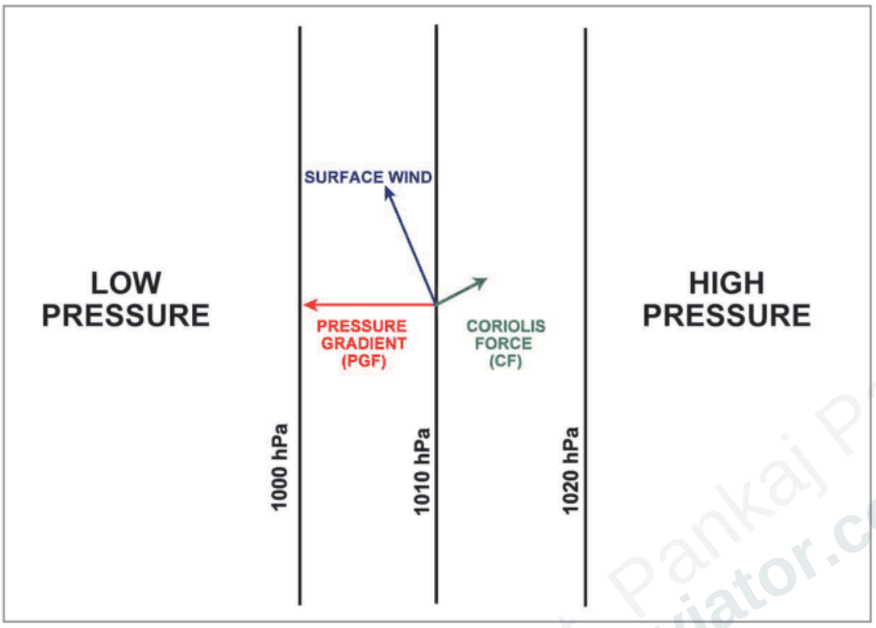
Quick Summary of Wind Types:
Geostrophic = P↔f (straight isobars) | Cyclostrophic = P↔Centripetal (near equator, curved) | Gradient =

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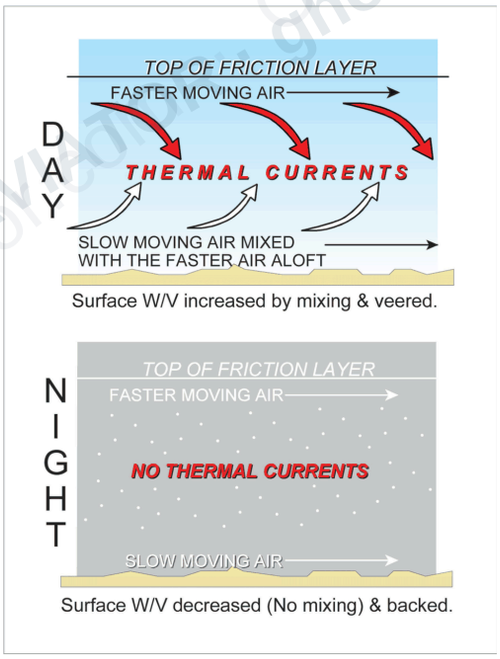
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P↔f↔C (curved, mid-lat) | Isallobaric = rapid pressure change | Inertial = f↔Centrifugal (no PGF)

6. Effect of Surface Friction & Turbulence



Surface wind crosses the isobars



Surface wind by day and night

Friction Layer

- The rough earth causes friction, affecting the atmosphere up to about 1 km
- This layer is called the **Friction Layer**; thickness is variable
- Within friction layer: wind slows down → Coriolis reduces → insufficient to balance PGF → wind is **deflected towards low pressure** → flow becomes **cross isobaric**

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Surface	Inclination to Isobars	Speed (of Vg)
Over sea	About 10° in the cited rough Northern Hemisphere example	About 70% of geostrophic speed in that example
Over Land	~30°	1/3 to 1/2 Vg

Turbulence and Gustiness

- Wind is seldom steady; peak fluctuations = **Gusts**, lowest = **Lulls**
- Width of fluctuations = **degree of gustiness**
- Air flow with such fluctuations = **turbulent**

Types of Turbulence

Type	Cause	Characteristics
Frictional	When stream speed exceeds certain limit → flow unstable → eddies form	Both vertical and horizontal velocities; more over rough terrain/buildings/trees
Thermal	Convection (heat transfer through movement within a fluid) currents due to surface heating; passage of cool air over warmer land/sea	Eddies of large dimensions; stronger gusts; extends to considerable heights when lapse rate is favourable

Factors Affecting Turbulence & Friction Layer

- Flow over buildings, trees, rugged country → vertical and horizontal eddies
- Develop more easily when lapse rate is steep
- Less on cool surfaces and stable atmosphere
- Over land: more by day (steep lapse rate (temperature decrease with increasing height)), least on clear night (inversion)

Diurnal Variation of Surface Wind

Time	Surface Wind	Upper Level Winds	Aviation Implication
Daytime	Strong, gusty; convection stretches friction layer upward; slackens frictional effect	Backs (due to friction aloft)	Gusty conditions, thermal turbulence
Night	Thermals die down; friction layer shrinks to surface; wind weak and **backs**; friction prohibits upper winds from penetrating to surface	Becomes strong above friction layer	**Wind shear (wind speed or direction changing over distance) ~500m** — serious aviation hazard

⚠ **Night Wind Shear Hazard:** The surface wind is, therefore, weak and backs. There may be marked **wind shear around 500 m**, with strong winds above and weak winds below, which is serious as an aviation hazard.

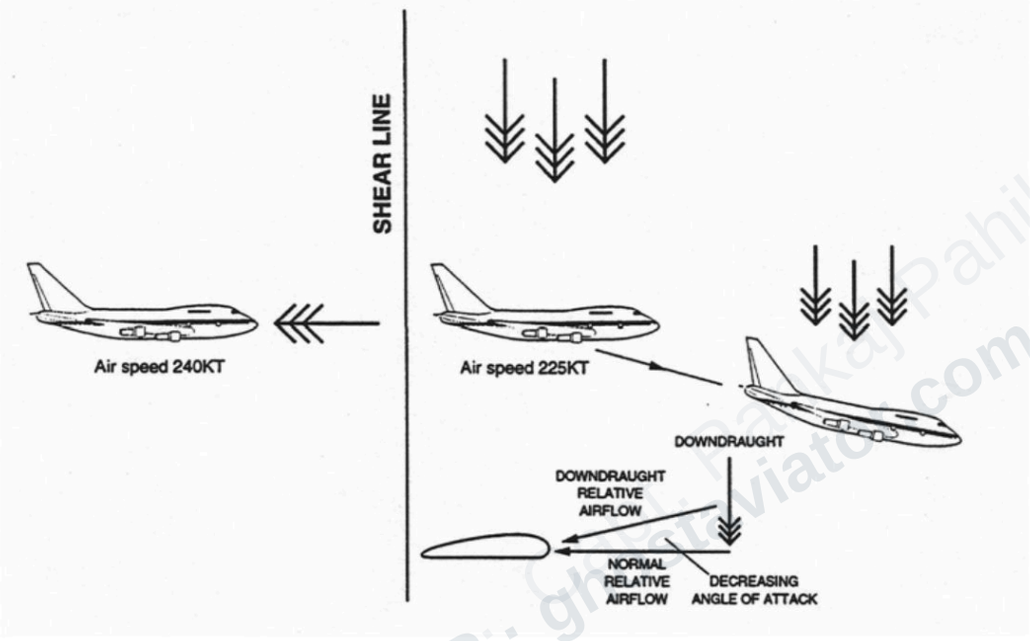
Diurnal variation over sea: Very little (about 1°C); also over land under continuously overcast skies. Surface winds nearly the same 24 hr. Diurnal variation is most apparent in **fine weather, clear night and**

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sunny days.

7. Wind Shear (WS)

Definition: The variation in wind vector along the path of an aircraft, which can displace the aircraft from its intended path.



Downdraught and angle of attack

Type of WS	Definition
Low Level WS	WS along final approach, runway, take-off path, and initial climb out flight path
Vertical WS	The change in horizontal wind vector with height
Horizontal WS	The change in horizontal wind vector with distance
Up-Downdraughts	Change in vertical component of wind vector with distance

Causes of Wind Shear

1. Thunderstorm (TS)

(a) Gust Front (GF): The cold downdraught from TS spreads all around and creates GF as it meets the warm air near the ground. GF may extend as far as 30 km in the direction of storm movement and extend to about 6000 ft from ground. Great turbulence and WS are produced in the area, due to opposing winds and eddies.

(b) Microburst: Highly concentrated downdraught from TS, about 4 km across, lasting for 1-5 minutes. Most hazardous and powerful. Winds can be 90 kt and their directions may also be opposite at the surface.

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Microburst Type	Origin	Description
Wet Microburst	Heavy downpour under TS	Due to evaporation (liquid water changes into vapour and absorbs latent heat) and cooling of cold downdraught
Dry Microburst	High based CB or Anvil (CI) in region of Virga (falling precipitation that evaporates before reaching the surface)	**Virga** = rain shaft evaporating before reaching ground

2. Low Level Inversion

Develops in clear nights; may extend to about 1 km. In friction layer, winds are weak and winds aloft are strong → WS across inversion.

3. Gusty Surface Winds

Lead to strong gusts and lulls → WS.

4. Solar Heating

Intense solar heating causes WS and turbulence due to up and downdraughts.

5. Topography

Strong winds blowing across natural and man-made obstacles (mountain ranges, high rise buildings, flow along valleys, etc.) all lead to WS.

6. Fronts

Abrupt change of temperature and wind at a front. With passage of a front, strong WS may result; cross, head or tail wind. Very strong winds may also be encountered ahead of a Cold Front.

7. Jetstream

Strong vertical and horizontal WS are associated with jet streams in upper air.

Effects of WS on Aircraft

Sudden shear encountered	Initial effect
Decreasing headwind or increasing tailwind	Airspeed and performance decrease.
Increasing headwind or decreasing tailwind	Airspeed and performance increase.

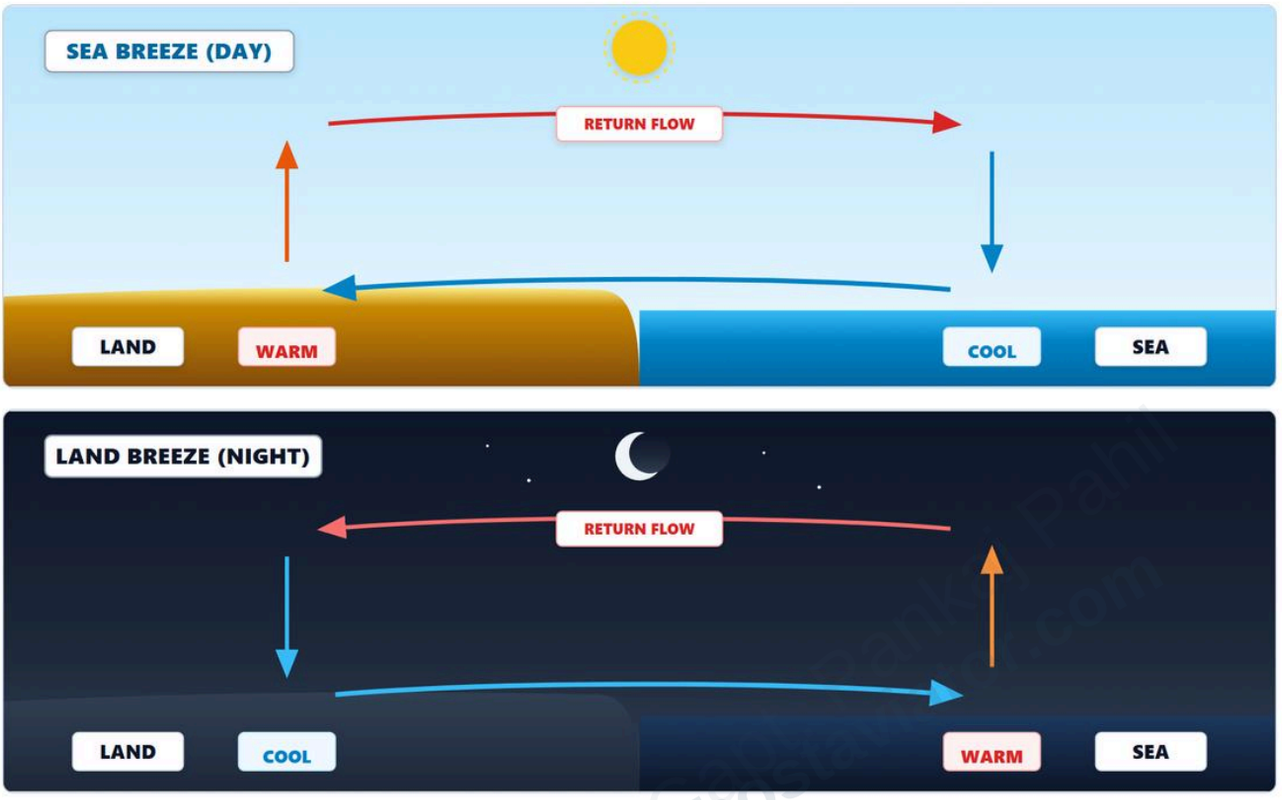
Actual flight-path response depends on the aircraft, vertical wind and crew response. Follow approved aircraft windshear procedures.

⚠ **Critical: Effects of WS on Approach** — These effects are critical during landing and take-offs. Decreasing headwind and increasing tailwind both initially reduce airspeed; the opposite changes initially increase it.

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8. Local Winds Due to Topography



Local coastal circulation: by day, air near the surface moves from the cooler sea towards warmer land; at night the surface flow can reverse. Arrows show an idealised circulation. Background wind and local conditions can alter or prevent it.

Anabatic and Katabatic Winds

Feature	Anabatic (Valley) Wind	Katabatic Wind
Time	Daytime	Night (clear, quiet night)
Direction	Up-slope (mountain slope heated by sun)	Down-slope (mountain slope cools at night)
Character	Warm, lighter; ascends the slope; masked by irregular convection; intensified by sea breeze in funnel valley	Cold, down-slope wind; pools of cold on low-lying ground; causes local frost, mist and fog
Speed	Variable	No more than a few knots normally; **Bora** can exceed 100 kt
Aviation Hazard	Light up-slope wind by day; little hazard on its own	Fog and frost in valleys; fog and TS in the morning over NE India (most marked over snow covered slopes); Bora (100+ kt) dangerous to shipping and low-flying aircraft

⚠ **Bora:** An off-shore wind on northern shores of the Adriatic sea. It is a cold katabatic wind. Sets in suddenly and frequently reaches well over gale force with gusts of over 100 kt. Extremely dangerous to shipping and low-flying aircraft. Similar winds occur on coast of Greenland and Black Sea shores. Over NE India — katabatic winds are common and cause fog and TS in the morning.

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Fohn Wind

If air is forced over a mountain barrier, adiabatic cooling on windward side may lead to cloud and precipitation formation. Within cloud, rising air cools at SALR. Some condensed water falls as precipitation. Air descends on leeward side at the DALR (since cloud has disappeared). The local name in the Alps is the **Fohn**.

- In high mountains, Fohn wind may be 10°C or more warmer than windward side
- The **Chinook** of the Rocky Mountains is an example
- Conditions for Fohn: (a) substantial mountain range, (b) wind blows within 30° of range, (c) high moisture content

💡 **Distinction — Fohn vs. Katabatic:** Fohn is WARM and DRY on leeward. Katabatic is COLD. Both are down-slope winds off high ground.

Ravine Winds

Occur in and near narrow valleys when there is a pressure difference between two sides of the hills. Air is impelled through the valley by the pressure gradient. Such winds may be very strong in the ravine and also after leaving its mouth.

Land Breeze and Sea Breeze

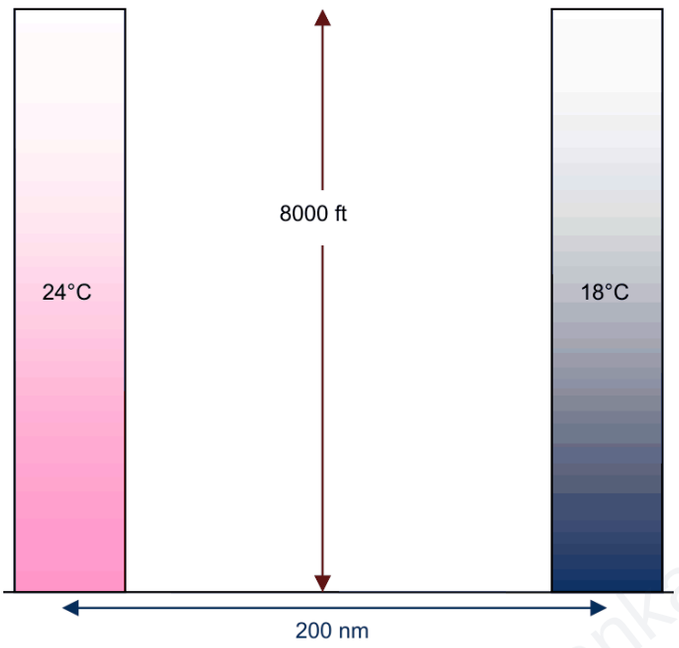
Feature	Sea Breeze	Land Breeze
Time	Day-time	Night
Direction	Sea → Land	Land → Sea
Cause	Land warmer → air rises → pressure aloft greater → upper air drifts to sea → sea level pressure over land drops → sea breeze	Radiative cooling of land → thermals die → friction layer shrinks → land breeze sets in
Extent	Generally 15-25 km on either side of coast; in Pune ~170 km from Mumbai	Shallower than sea breeze; few hundred feet only
Onset	Few hours after sunrise; delayed if off-shore wind	After sunset
Alignment	Starts at right angles to the coast. Later Coriolis turns it (to the right in the N hemisphere, to the left in the S hemisphere) until it blows almost along the coast. With your back to the wind, the land is then on your left in the N hemisphere and on your right in the S hemisphere.	—
Other	In tropics and subtropics: routine and perpendicular to coast; may develop to 3000-5000 ft in tropics → line of small cumuliform clouds	Undetected by powered aircraft; useful to glider pilots

Sea Spray: Under favourable wind conditions, salt particles get sprayed from wave crests; water drops formed can reduce visibility considerably. In rare cases, a thin layer of salt may form on the wind screen of an aircraft flying at low level and render forward visibility to almost zero. Salt spray may also reduce visibility during monsoons and cyclones in coastal areas.

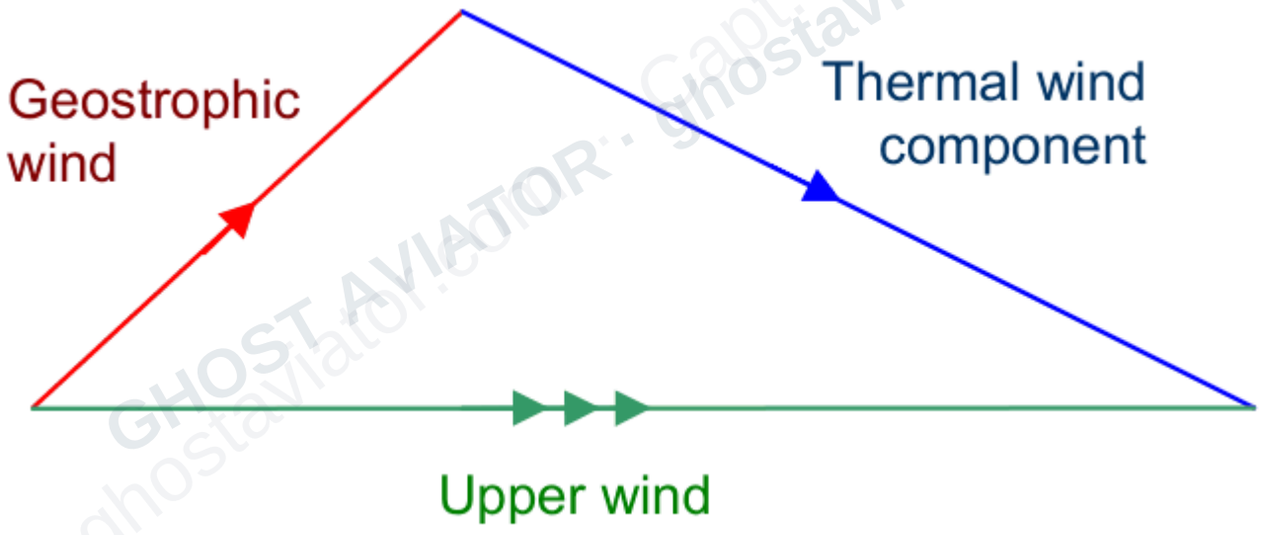
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9. Thermal Wind & Contours



Thermal wind: a warm column and a cold column



The thermal wind component

Thermal Wind

Definition: The thermal wind in a layer is defined as that wind which must be added vectorially to the geostrophic wind at the lower level in order to obtain the geostrophic wind at the upper level. It is a fictitious wind.

$V_t = V_1 - V_0$

$\mathbf{V_t}$ = thermal wind, $\mathbf{V_1}$ = upper level geostrophic wind, $\mathbf{V_0}$ = lower level geostrophic wind

Vector addition: Draw one vector; from head draw the other; join tail of first to head of second = sum.
Vector subtraction: direction of second vector is reversed.

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Example: Calculation of Upper Level Wind

Lower Level Wind	Thermal Wind	Upper Level Wind
270/10kt	270/15kt	270/10kt + 270/15kt = 270/25kt
270/10kt	090/15kt	270/10kt + (−270/15kt) = 090/05kt

- Thermal wind blows parallel to isotherms/thickness lines, keeping **low temperature to the left in N hemisphere**
- Speed is proportional to the temperature gradient
- If south is warm and north cold: temperature gradient acts S→N → thermal wind is **Westerly**
- If such temperature distribution continues, thermal wind will continue to be westerly from level to level → keeps increasing with height
- **Subtropical westerly Jet Stream** is an example of thermal wind

Wind and Contours

Lines joining equal heights on a constant pressure level are called **Contours**. Over warm atmosphere a contour will be at a higher height; over cold at a lower height. Contour lines depict centres of High and Low exactly as isobars do on surface maps. A contour line is horizontal — the pressure is the same at all points on the line. Contour of height say 5520 m = isobar (a line joining places with the same pressure at the same level and time) in the horizontal surface at that height.

- Geostrophic wind blows along contours with **lower value on the left in N hemisphere**, right in S hemisphere
- Contour charts can determine cyclostrophic and gradient winds
- Limitations of geostrophic rule equally apply to contour charts

Variation of Wind with Height

- Temperatures in troposphere decrease from equator to poles → **thermal wind component throughout troposphere blows from Wly (W)**
- **In the upper troposphere the winds are, therefore, mainly Wly**
- Over India: easterlies prevail above 500 hPa during monsoon months (June, July, August, September); strengthen with height, weaker over N India and stronger over S India; tropical easterly jetstream lies near 13N lat at about 15 km
- Outside the tropics: low level westerlies get stronger with height with little change in direction. Low level easterlies get weaker with height and then turn westerly
- Westerly thermal wind: northerly winds would back with height, southerly winds would veer
- In low stratosphere (winters): temperature highest between 40° and 60° lat → westerlies increase with height; equatorwards of temperature maximum → westerlies decrease above tropopause
- Summer hemisphere: temperatures highest in polar regions → easterly thermal wind in stratosphere

10. Vorticity

Vorticity is a measure of rotation or turning. Rotation can be cyclonic or anticyclonic. It plays an important role in formation and development of weather systems such as cyclones, depressions, anticyclones. Change in vorticity causes **divergence (airflow spreading apart) and convergence**.

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Causes of Vorticity

1. **Horizontal Wind Shear:** When a belt of strong winds lies alongside a belt of lighter winds, the faster flow rotates around the slower flow. May be cyclonic or anticyclonic depending on orientation.
2. **Curvature of the Flow:** Flow around a curve has cyclonic vorticity when air deflects to its left (in trough, N hemisphere) and anticyclonic when it deflects to the right (ridge).
3. **Rotation of the Earth:** Earth rotates from W to E (anti-clockwise = cyclonic for N hemisphere). Hence vorticity due to earth is cyclonic in both hemispheres.

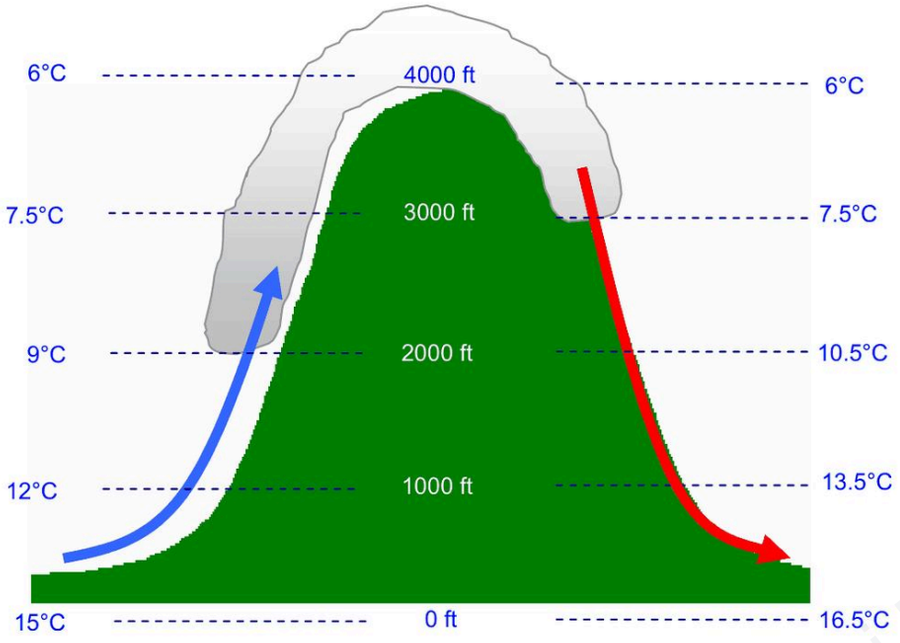
Absolute Vorticity: Sum of all the above vorticities. Absolute vertical vorticity is relative vertical vorticity plus the Coriolis parameter: $\eta = \zeta + f$. Its sign depends on that sum; it is not automatically cyclonic.

11. Names of Winds of the World

Wind Name	Character	Location
Bora	Cold katabatic; originates in mountains of Yugoslavia and NE Italy	Coastal plains of the Adriatic sea
Burans (Russian Burans / Turkish Boran)	Strong NEly wind in Russia and Central Asia; often blows snow ('Purga')	Russia and Central Asia
Chinook	Warm dry Wly wind	Eastern side of Rocky Mountains
Doldrums	Calm winds near Equator; low atmospheric pressure; insignificant wind; cloudy and rainy; also called **ITCZ**	Between ~5°N and 5°S lat
Haboobs (Arabic: blowing furiously)	Any strong wind raising sand into a sand storm	Particularly in Sudan
Harmattan	Dry, dusty northeasterly wind	From the Sahara across West Africa
Khamsin	Oppressive, hot, dry, often laden with sand; Sly wind	Egypt, April–June
Mistral	Cold katabatic — descends from snow-clad Alps down the Rhone River Valley	France and into Gulf of Lyons along the Mediterranean coast
Monsoon	Any markedly seasonal wind	Particularly E and SE Asia
Roaring Forties	Wly winds in both hemispheres between 35° and 60° lat; very stormy nature beyond 40° lat in S hemisphere. Called **Roaring Forties, Furious Fifties, Crying Sixties**	Mid-latitudes S hemisphere
Trades	Steady wind between 10° and 30° from NE in N hemisphere, SE in S hemisphere; 'wind that blows trade' by 18th century navigators	Subtropics both hemispheres

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Foehn effect: air at 15°C cools at the dry rate to the cloud base at 2,000 ft (9°C), then at the saturated rate to 6°C at the 4,000 ft crest. On the lee side it warms at the saturated rate down to 3,000 ft, then at the dry rate, and arrives at 16.5°C.

12. Beaufort Force Scale

BF Scale	Description	Speed at 10m (kt)
0	Calm	<1
1	Light air	1–3
2	Light breeze	4–6
3	Gentle breeze	7–10
4	Moderate breeze	11–16
5	Fresh breeze	17–21
6	Strong breeze	22–27
7	Near Gale	28–33
8	Gale	34–40
9	Strong Gale	41–47
10	Storm	48–55
11	Violent Storm	56–63
12	Hurricane	64 or more

Practice Q&A — Chapter 6: Winds

Q1. In S hemisphere if an observer faces wind, low will be to his ……

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(a) Right (b) Left

✅ Answer: (b) Left — In S hemisphere, Buys Ballot's Law reversed. Face the wind → low is to your LEFT in S hemisphere.

❌ (a) Right: This applies to N hemisphere (back to wind → low on left; face wind → low on right in N hemisphere).

Q2. In N hemisphere due to rotation of earth winds are deflected to ……
(a) Left (b) Right

✅ Answer: (b) Right — Coriolis effect deflects moving objects to the right in the N hemisphere.

❌ (a) Left: This is the S hemisphere effect.

Q3. Local winds and pressure laws

Thermally driven local winds such as a sea breeze are not necessarily geostrophic, so a simple geostrophic pressure-direction rule may not describe their initial flow. Coriolis can turn a developed sea breeze over time; local winds are not universally immune to it.

Q4. Coriolis force acts perpendicular to the …… of wind direction in N hemisphere
(a) Left (b) Right

✅ Answer: (b) Right — Coriolis force acts perpendicular to the wind, to the right in N hemisphere.

❌ (a) Left: Left deflection is S hemisphere.

Q5. Geostrophic wind is due to the balance between the forces ……
(a) Coriolis and Frictional
(b) Pressure gradient and Cyclostrophic
(c) Pressure gradient and Coriolis

✅ Answer: (c) Pressure gradient and Coriolis — P = f is the geostrophic balance.

❌ (a) Coriolis + Frictional: When friction included → actual surface wind, not geostrophic.
❌ (b) PGF + Cyclostrophic: Cyclostrophic balance is P = Centripetal (curved isobars near equator).

Q6. Coriolis force is strongest at ……
(a) Mid latitudes (b) Poles (c) Equator

✅ Answer: (b) Poles — $f = 2\Omega\rho V \sin\varphi$; at poles $\varphi = 90°$, $\sin\varphi = 1$ (maximum).

❌ (a) Mid latitudes: $\sin 45° = 0.707$, intermediate value.
❌ (c) Equator: $\sin 0° = 0$, so Coriolis = zero at equator.

Q7. Geostrophic rule breaks down at ……
(a) Mid latitudes (b) Poles (c) Equator

✅ Answer: (c) Equator — At equator $\sin\varphi = 0$, so Coriolis = 0, Vg formula gives infinity/undefined.

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❌ (a) Mid latitudes: Geostrophic approximation works reasonably well.
❌ (b) Poles: Geostrophic rule gives a good approximation beyond 30°; also breaks at poles due to pressure changes with time but rule is NOT primarily said to break down here.

Q8. Fohn winds are …… on the Leeward side of a mountain.
(a) Dry (b) Cold & Humid

✅ Answer: (a) Dry — Fohn wind is warm AND dry on the leeward side.

❌ (b) Cold & Humid: This is the windward side characteristic. The Katabatic wind is cold.

Q9. The wind sliding down a hill during night is called …… wind.
(a) Fohn (b) Anabatic (c) Katabatic

✅ Answer: (c) Katabatic — Night cooling → slope cools → air sinks → cold down-slope katabatic wind.

❌ (a) Fohn: Fohn is on leeward side of a mountain range, warm and dry — not driven by nocturnal cooling alone.
❌ (b) Anabatic: Anabatic is daytime up-slope wind.

Q10. With the onset of sea breeze there is a …… in temperature and …… in RH.
(a) Fall/Rise (b) Rise/Fall (c) Fall/Fall

✅ Answer: (a) Fall/Rise — Sea breeze brings cooler, moister maritime air inland → temperature falls, RH rises.

❌ (b) Rise/Fall: Temperature doesn't rise with sea breeze onset.
❌ (c) Fall/Fall: RH rises as maritime air is more humid.

Q11. Sea breeze sets in by …… and dies off at ……
(a) Night/Day (b) Day/Night (c) Both Day and Night

✅ Answer: (b) Day/Night — Sea breeze begins during daytime (few hours after sunrise) and dies off after sunset.

❌ (a) Night/Day: That describes the Land Breeze pattern.
❌ (c) Both Day and Night: Sea breeze is specifically a daytime phenomenon.

Q12. If an aircraft in N-hemisphere flies from H to L it will experience
(a) Starboard drift (b) Port drift

✅ Answer: (a) Starboard drift — Buys Ballot: with your back to the wind in the N hemisphere, low pressure is on your LEFT. Flying towards the low, the wind comes from your left and pushes you right → starboard drift.

❌ (b) Port drift: That is flying towards HIGH pressure in the N hemisphere (or towards low in the S hemisphere).

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Q13. In N-Hemisphere if you experience Port drift, altimeter will read
(a) Under (b) Over

✅ Answer: (a) Under — Port drift in the N hemisphere means you are flying towards HIGH pressure. Holding the same altimeter reading you follow a pressure level, and pressure levels are higher over a high, so you climb: true altitude is more than indicated → the altimeter under-reads.

Q14. Lines of constant wind speed drawn on weather charts are called
(a) Isobars (b) Isotachs (c) Isogons

✅ Answer: (b) Isotachs — Lines of equal wind speed = Isotachs.

❌ (a) Isobars: Lines of equal pressure.
❌ (c) Isogons: Lines of equal wind direction.

Q15. Gusts and squalls

Duration distinguishes a brief gust from a squall: a gust is a sudden increase lasting less than a minute, whereas a squall persists for a minute or more and has its own speed criteria. This does not establish that every squall is faster than every gust.

Q16. The thermal wind is:
(a) The wind that blows because of thermals
(b) The warm wind that blows down the hill on the leeward side
(c) The wind which must be added vectorially to the lower level geostrophic wind to obtain the upper level geostrophic wind

✅ Answer: (c) — Thermal wind = vector difference of upper and lower level geostrophic winds. It is a fictitious wind blowing parallel to thickness lines.

❌ (a) Blows because of thermals: That's convective updraft — not thermal wind.
❌ (b) Warm wind on leeward: That is the Fohn wind.

Q17. On a weather map if isobars are closely packed, the surface winds are likely to be
(a) Light and parallel to isobars
(b) Strong and parallel to isobars
(c) Strong and blowing across the isobars

✅ Answer: (c) Strong and blowing across the isobars — close isobars indicate a strong pressure gradient. Surface friction reduces wind speed and Coriolis, so the surface wind crosses isobars towards lower pressure.

Options (a) and (b) describe flow parallel to isobars; surface friction produces a component across them.

Q18. Anabatic wind occurs:
(a) At night (b) Any time of day and night (c) During day

✅ Answer: (c) During day — Anabatic wind is driven by daytime solar heating of mountain slopes.

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❌ (a) At night: Katabatic winds occur at night.
❌ (b) Any time: Anabatic is specifically a daytime phenomenon.

Q19. Anabatic wind is stronger than katabatic
(a) True (b) False

✅ Answer: (b) False — Katabatic wind can be much stronger. Katabatic (e.g., Bora) can exceed 100 kt. Anabatic winds are gentle and often masked by convection.

Q20. Katabatic wind is down slope cold wind due to nocturnal cooling
(a) True (b) False

✅ Answer: (a) True — Katabatic wind is indeed a cold, down-slope wind caused by nocturnal radiative cooling of the mountain slope.

Q21. Katabatic wind occur due to sinking of cold air down the hill slope
(a) True (b) False

✅ Answer: (a) True — Nocturnal cooling → mountain surface becomes cold → adjacent air cools → sinks down slope = Katabatic.

Q22. Anabatic wind occur due to downward movement of air along valley
(a) True (b) False

✅ Answer: (b) False — Anabatic wind moves UPWARD along the valley/slope during daytime. Downward movement is katabatic.

Q23. Sea breeze is stronger than land breeze
(a) True (b) False

✅ Answer: (a) True — Sea breeze is stronger because daytime temperature differential (land vs sea) is greater than night differential. Land breeze is shallower and weaker.

Q24. The wind blows clockwise around a low in N-hemisphere
(a) True (b) False

✅ Answer: (b) False — In N hemisphere, wind blows ANTI-clockwise around a Low. Clockwise around a High.

Q26. The wind blows anticlockwise around a low in N-hemisphere
(a) True (b) False

✅ Answer: (a) True — Confirmed: In N hemisphere, surface winds blow anti-clockwise (cyclonically) around a Low.

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Q27. The wind blows anticlockwise around a low in S- hemisphere
(a) True (b) False

✅ Answer: (b) False — In S hemisphere, wind blows CLOCKWISE around a Low.

Q28. Gradient wind

Gradient wind follows curved isobars when friction is negligible. The pressure-gradient and Coriolis forces provide the net inward acceleration needed for curved motion. In a force-balance description using an outward centrifugal term, these three terms balance. Geostrophic force and cyclostrophic force are not the names of separate physical forces.

Q29. Due to friction, from day to night for an isobaric pattern (in N hemisphere) the surface wind backs and weakens
(a) True (b) False

✅ Answer: (a) True — At night, friction layer shrinks, friction effect at surface increases; surface wind becomes weak and backs.

Q30. The winds which spirals inward counter-clockwise in the N Hemisphere are associated with
(a) Turbulence (b) High pressure area (c) Low pressure area

✅ Answer: (c) Low pressure area — Anti-clockwise inward spiraling in N hemisphere = cyclonic circulation = Low pressure system.

❌ (b) High pressure area: High pressure = clockwise outward spiral in N hemisphere.

Q31. Lower level wind 05010kt, upper level wind 23005kt, what is the thermal wind
(a) 05005kt (b) 23015kt (c) 05015kt

✅ Answer: (b) 23015kt — The thermal wind is the vector difference between the upper and lower winds (upper − lower). Lower 050/10 kt and upper 230/05 kt are in opposite directions, so subtract: 5 kt from 230 plus 10 kt from 230 gives 230/15 kt.

❌ (a) 050/05 and (c) 050/15 have the direction reversed: the thermal wind blows with the colder air on its left in the Northern Hemisphere.

Q32. A change in wind direction from 310° to 020° is
(a) Backing (b) Veering

✅ Answer: (b) Veering — 310° → 020° going clockwise (310→360→020) = clockwise change = VEERING.

❌ (a) Backing: Would be anti-clockwise change (e.g., 310→270→020 the long way around).

Q33. A change from 270° to 250° is

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(a) Backing (b) Veering

✅ Answer: (a) Backing — 270° → 250° is an anti-clockwise change = Backing.

Q34. Sudden change in wind speed from 10 to 30kt and then to 15kt for a few seconds
(a) Gust (b) Squall (c) Gale

✅ Answer: (a) Gust — a sudden increase lasting less than one minute is a gust.

❌ (b) A squall lasts a minute or more at the higher speed. (c) A gale needs a sustained speed above 33 kt.

Q35. Sudden change in wind speed from 10kt to 30kt for 2 - 3 minutes
(a) Squall (b) Gust

✅ Answer: (a) Squall — the increase from 10 to 30 kt persists for 2–3 minutes, meeting the duration criterion for a squall.

Q36. A significant wind shear is generally associated with TS or line squall
(a) False (b) True

✅ Answer: (b) True — Thunderstorms produce gust fronts and microbursts — both are major sources of wind shear.

Q37. Cyclostrophic approximation

In a tightly curved flow, pressure-gradient force can supply the inward acceleration while Coriolis is comparatively small. This cyclostrophic approximation may be useful in tornadoes or parts of intense tropical storms. A fixed height of 2000 ft does not guarantee that friction and other terms can be ignored.

Q38. Rotor clouds have extremely turbulent flying conditions
(a) False (b) True

✅ Answer: (b) True — Rotor clouds form below mountain waves and contain extreme turbulence.

Q39. Surface wind crossing isobars

Friction reduces wind speed and Coriolis, giving surface wind a component towards lower pressure. Rough Northern Hemisphere teaching examples give averages of about 30° over land and 10° over sea. These are approximate examples, not exact universal angles; surface roughness and other conditions matter.

Q40. If the S is warmer than the N level, from surface up to higher levels, then the …… wind will continue to increase with no change in direction in N hemisphere

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(a) Ely (b) Wly (c) Sly (d) Nly

✅ Answer: (b) Wly — If south is warm and north is cold, temperature gradient acts S→N. Thermal wind blows westerly (along thickness lines, low temp to left in N hemi). If this distribution persists, thermal wind continues to be westerly from level to level, increasing with height.

Q42. Gale is:
(a) Persistent strong winds with mean speed 44kt, associated with thunderstorms
(b) Marked increase in wind speed lasting few minutes associated with CB or DS
(c) Persistent strong winds exceeding 33kt, associated with depression

✅ Answer: (c) Persistent strong winds exceeding 33kt (≥34kt), associated with depression — Definition of Gale: persistent mean wind ≥34kt, associated with depressions and cyclonic storms.

❌ (a) 44kt with thunderstorms: A thunderstorm association alone does not define a gale. Assess sustained mean wind speed; assess a squall separately by its sudden increase and duration.
❌ (b) Lasts few minutes with CB: That is the description of squall.

Q43. In N hemisphere thermal wind is parallel to …… with low value to left
(a) Isobars (b) Isotherms (c) Isallobars

✅ Answer: (b) Isotherms — Thermal wind blows parallel to isotherms (or thickness lines), keeping low temperature to the left in N hemisphere.

❌ (a) Isobars: Geostrophic wind is parallel to isobars.
❌ (c) Isallobars: Related to pressure change, relevant to Isallobaric wind.

Q44. The inertial flow is:
(a) cyclonic in both the Hemispheres
(b) anti cyclonic in both the Hemispheres
(c) anti cyclonic around an anticyclone

✅ Answer: (b) Anti cyclonic in both the Hemispheres — Inertial flow has no Pressure Gradient Force; balance of Centrifugal + Coriolis; the resulting flow is anticyclonic in both hemispheres.

Q45. Upper level wind is 24025kt, lower level wind is 06015kt, the thermal wind is?
(a) 16010kt (b) 24040kt (c) 24010kt

✅ Answer: (b) 24040kt — Thermal wind = upper − lower. Upper 240/25 kt; lower 060/15 kt blows from the opposite direction. Subtracting a wind from 060 is the same as adding the same speed from 240: 25 + 15 = 40 kt from 240.

❌ (a) 160/10 and (c) 240/10 do not follow from the vector subtraction.

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Chapter 7: Visibility and Fog

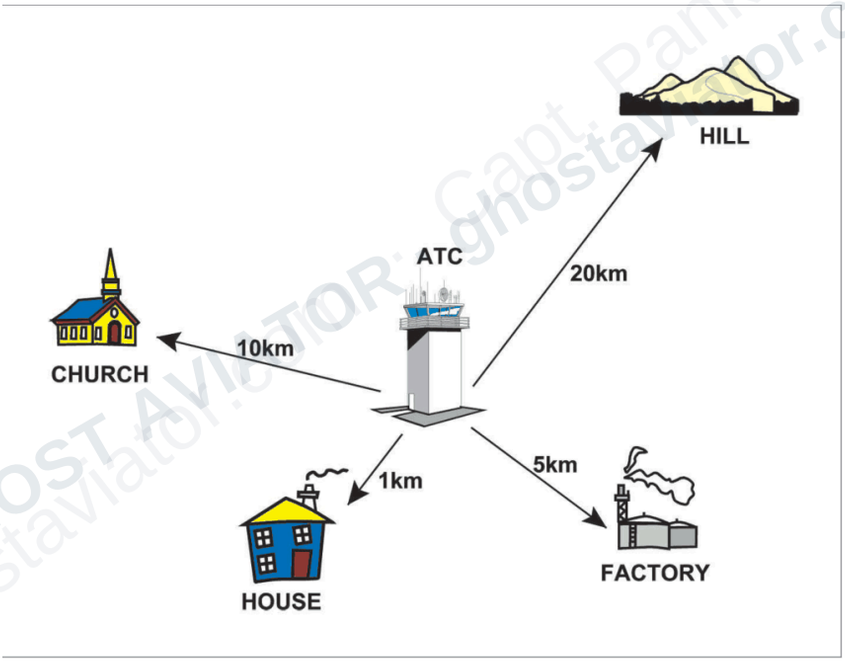
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1. Definition of Visibility

Visibility (Aeronautical Definition): The greatest distance at which a **black object of suitable dimensions**, situated near the ground, can be seen and recognised when observed against the bright background.

Night Definition: The greatest distance at which lights in the vicinity of **1,000 candela** can be seen and identified against an unlit background.

In simple terms: **The distance up to which prominent objects can be seen by naked eye and recognized as such under natural light.**



Visibility from marker objects

Visibility on the ground and from the air are of great importance to a pilot for landing and take-off. The pilot needs to see various land marks, targets, obstructions, beacon lights, other aircraft, runways etc., while in flight and during approach and landing. Poor visibility is a serious aviation hazard. Even at airfields where ILS is installed, a pilot would prefer better visibility conditions for safe landing and take-offs.

2. Visibility Reports & RVR

Visibility Reports

- The visibility given in METAR and SPECI is the **prevailing visibility**: the greatest visibility that is reached over at least half the horizon circle or half the surface of the aerodrome. The lowest visibility is also given, with its direction where possible, when it is below 1500 m, or when it is below half the prevailing visibility and below 5000 m

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- It may be different in different directions due to varying amounts of atmospheric obscurities
- This visibility may also differ from **vertical and slant visibilities** as estimated by an aircraft while in-flight

Method	Daytime	Night-time
Estimation Method	Seeing runway markings and objects at pre-measured distances along the runway, within and around the airfield	From existing lights at known distances in and around the airport — called **Visibility Land Marks**

Runway Visual Range (RVR)

Definition: The range over which the pilot of an aircraft on the centre line of the runway can see the runway markings or the lights delineating the runway or identifying its centre line.

- RVR equipment installed along various runways, close to the take-off and landing areas
- RVR is reported when the visibility is **less than 1500 m**
- RVR 2000 and 1500 are included in METAR as supplementary information
- RVR 2000 and 1500 are **NOT** disseminated internationally

💡 **Exam Tip:** RVR is reported whenever the visibility or the RVR is below 1500 m. The lowest value reported is 50 m and the highest is 2000 m. The instrument is called **Scopograph or Transmissometer**.

3. Causes of Poor Visibility

The surface visibility is reduced by atmospheric obscurities, such as smoke particles, sea spray, dust, industrial particles, vehicular emission, volcanic ash, etc.

Obscurity	Definition	Visibility	Special Notes
Haze	Suspension of very small particles of smoke, dust, water, etc. Gives a milky appearance	≤5000 m	—
Dust Haze	Dust kicked up by strong winds, especially in desert/semi-arid areas; suspended in air; causes Dust Haze. Summer hazard. Thicker during day.	Widely reduced; diversionary hazard	Extends from Rajasthan to Punjab, Haryana, UP, Bihar; vertically may extend 6–8 km
Smoke Haze	From industrial/domestic sources; spreads as haze layer, especially when wind is calm or very light and strong ground inversion	Widely reduced	—
Mist	Suspension of water droplets in atmosphere	≥1000 m but ≤5000 m	RH is almost 100%
Fog	Suspension of water droplets or ice crystals; RH ~100%	<1000 m	Classified as: Thick, Moderate, Light. **Fog is the cloud sitting on ground.**
Smog	When fog and smoke co-exist	Very poor	Severely limits visibility; health hazard

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Obscurity	Definition	Visibility	Special Notes
Dust Storm	—	≤5000 m	If followed by rain visibility improves fast
Sea Spray	Under favourable wind conditions, salt particles sprayed from wave crests; water drops form; thin layer at low level and on wind screen	Considerable reduction	In rare cases reduces forward visibility to almost zero on wind screen; also during monsoons and cyclones
Volcanic Ash (VA)	Solid matter from volcanoes; carried by winds; suspended for long time	Surface and in-flight: almost nil	Severely damages engine and other aircraft parts. Indian Volcano: **Barren Island (12.16N 93.51E)** — first erupted 1787, most recent 2020. **Baratang Island** lies 150 km N of Port Blair.
Precipitation	Rain, showers, drizzle, snowfall	Variable (see table below)	—

Visibility in Precipitation

Precipitation Type	Intensity	Visibility
Rain	Moderate	1–3 km
Rain/Showers	Heavy	<1000 m
Drizzle	Any	<1000 m
Snow	Any	may fall <1000 m

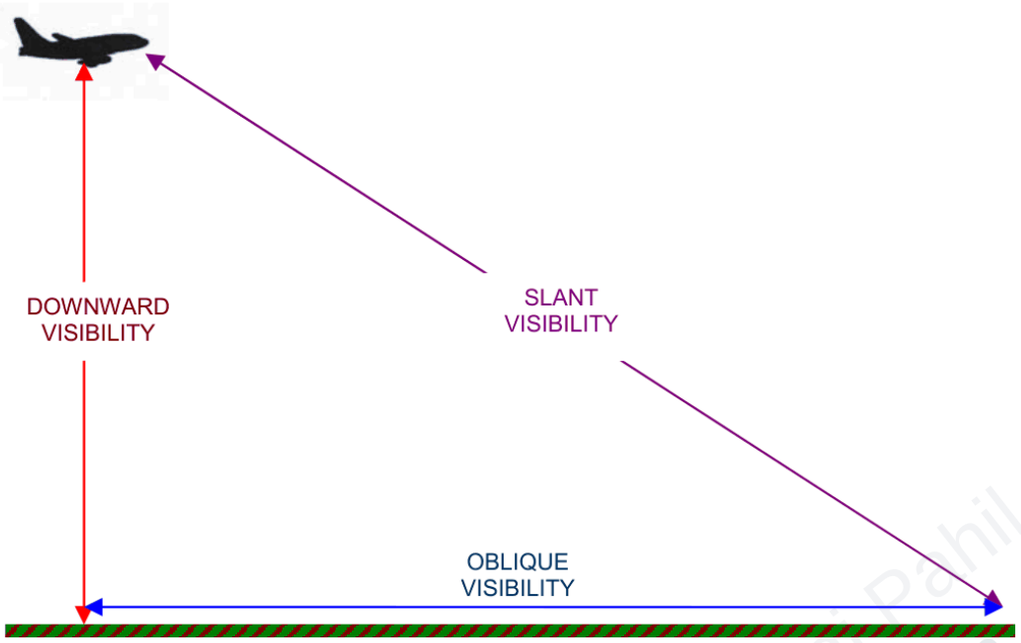
⚠ **Volcanic Ash — Aviation Warning:** VA reduces surface and in-flight visibility to almost nil. It severely damages engines and other aircraft parts. Volcanoes are active in Japan, Philippines, Indonesia, America, Mexico, Iceland, Russia, etc. If VA is encountered, pilots must exit the area immediately. **Indian context:** Barren Island (12.16N 93.51E) — first erupted 1787, most recent 2020. Baratang Island (mud volcano) is 150 km N of Port Blair.

4. Vertical and Slant Visibility

Haze and fog layers are generally confined to a few thousand feet above ground level. In a shallow layer of haze or fog, the **horizontal visibility is poor**. A pilot flying above such a layer may see the airfield clearly through a vertically thinner layer of fog, but on approach for landing, as he has to see through an elongated layer of fog, he may lose sight of the runway. In such a case:

- **Vertical Visibility** may be good
- **Slant Visibility** is poor

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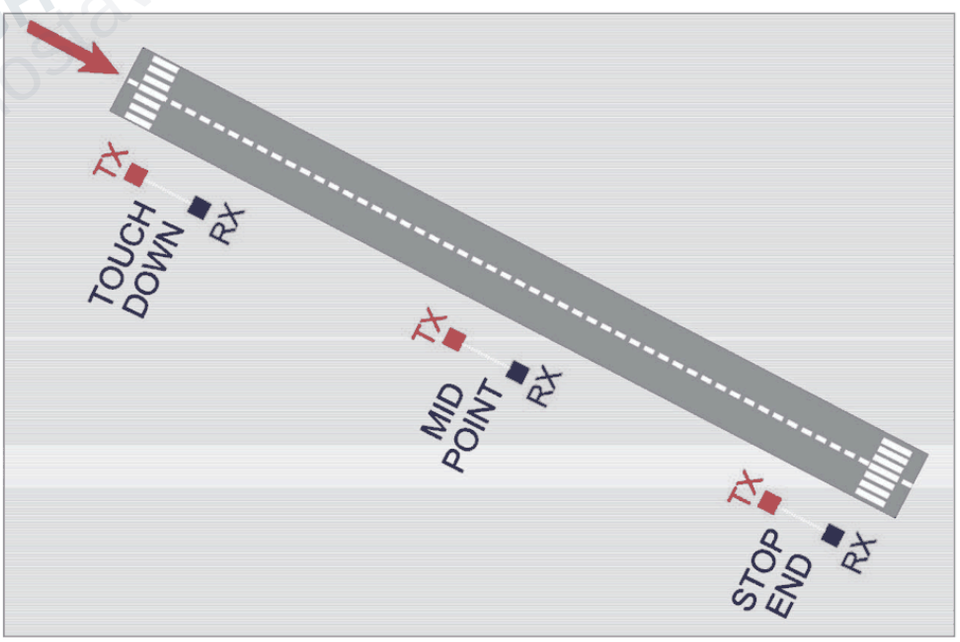


Types of visibility

As one goes higher and higher, the transparency of the atmosphere increases considerably, except in cloud and precipitation. **Visibility is lower looking towards the sun** due to reflection of the moon light (also applies to looking towards the sun during daytime).

5. Instruments for Measuring Visibility

Scopograph or Transmissometer: Used for measuring visibility along the different runways on an aerodrome. RVR equipment installed along various runways close to take-off and landing areas. Consists of a light source and a photoelectric receiver; equipment generates electric current according to the obscurities present between the receiver and the source. Current generated gives a measure of RVR. Display units located in the Met Office and ATC Tower.



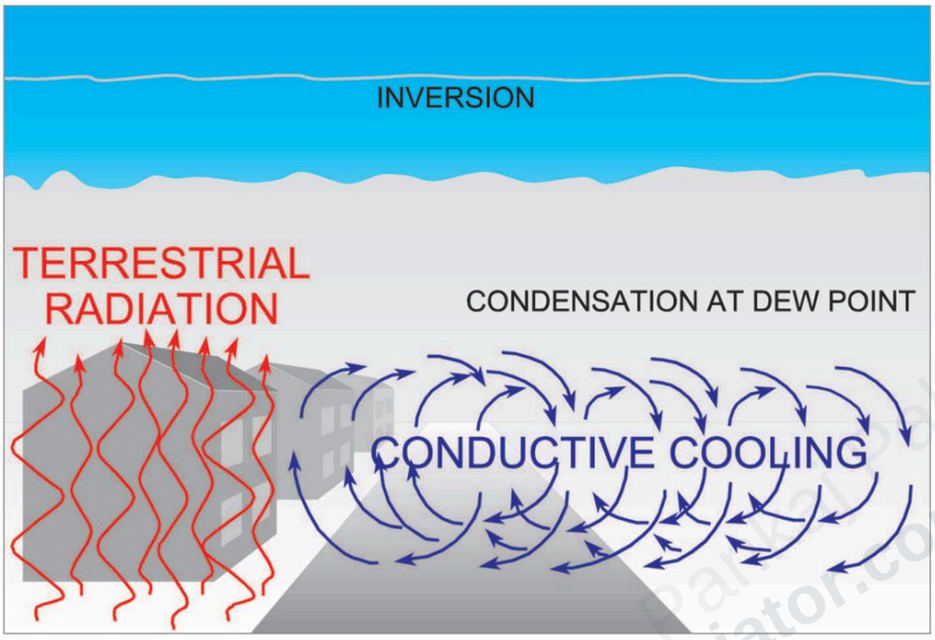
Instrumented RVR

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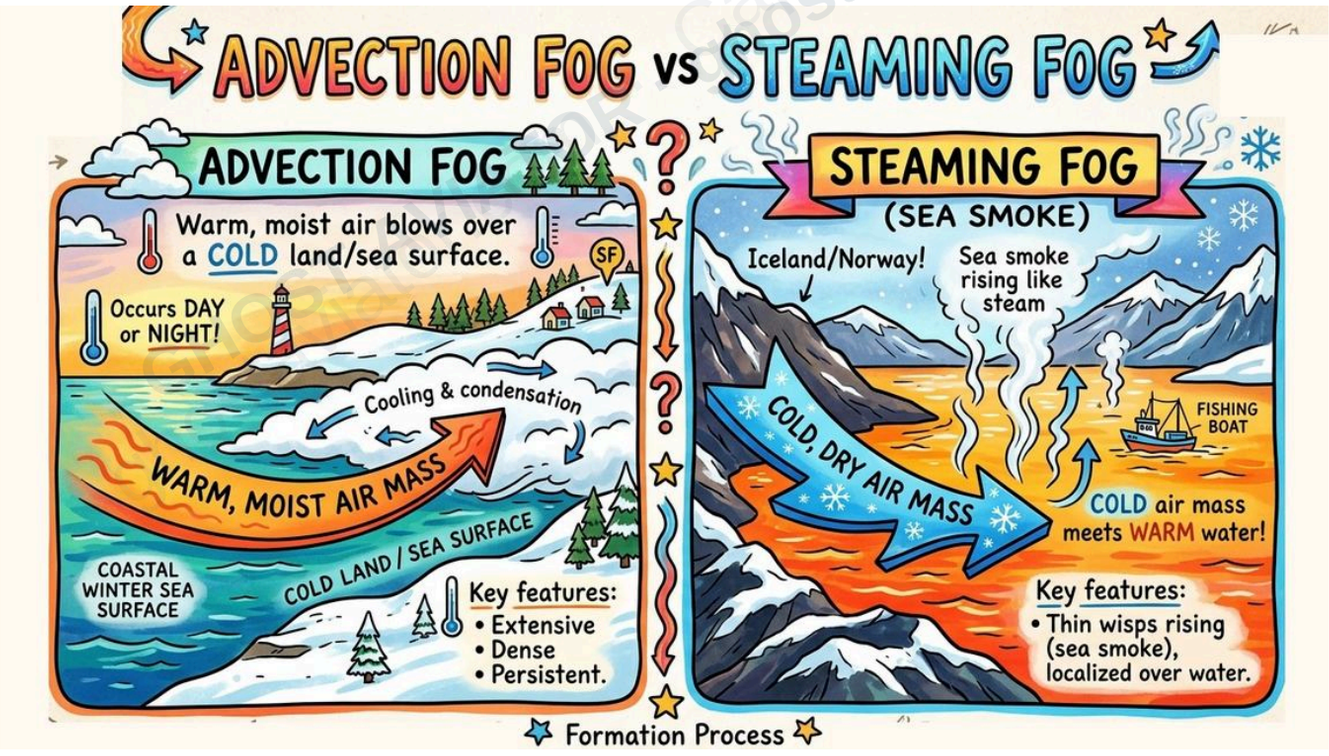
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- At some Indian Met Offices: **Automatic Visual Range Assessor (AVRA)** is used for measuring RVR

6. Types of Fog

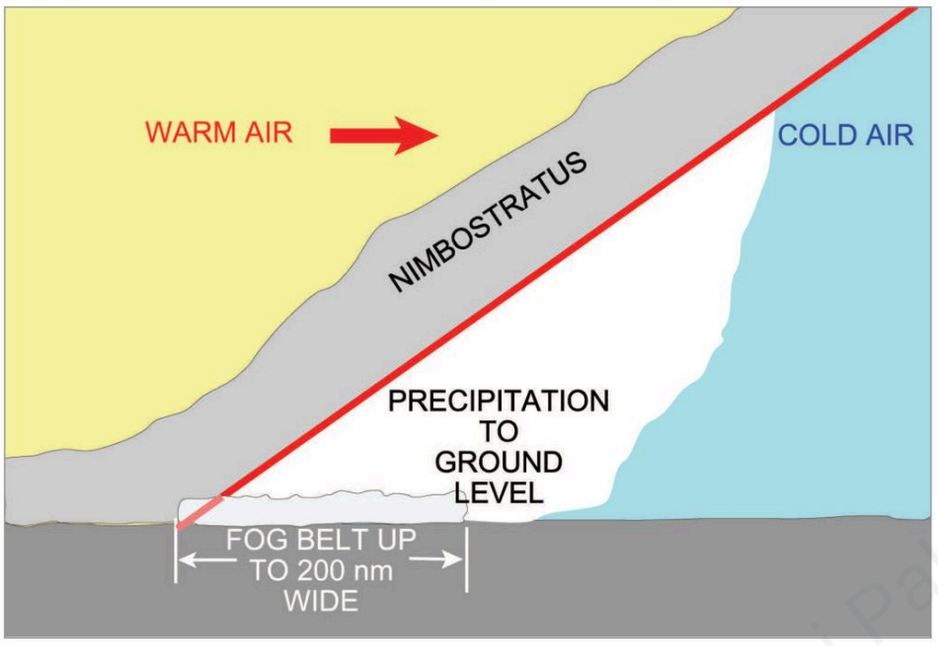


Radiation fog



Two different processes: advection fog forms when moist air is cooled by a colder surface; steam fog forms when evaporation from warmer water moistens colder air above it. The pictured scenes are examples; extent and persistence depend on conditions.

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Frontal fog

6.1 Radiation Fog

Formation: Radiation fog forms due to **nocturnal cooling** of the ground and the adjacent air layers. It is essential that the cooling should lower the temperature below the dew-point. If the air is calm, there is no turbulent mixing. The cooling of the ground is unable to extend upwards at sufficient rate — moisture close to the ground settles down as dew. If the wind is strong, the cooling is spread through too deep a layer. The temperature may not fall enough and fog does not form. In such cases **stratus cloud** may form at the top of the layer of turbulence.

Favourable Conditions for Radiation Fog Formation

Condition	Reason
(a) High Relative Humidity (actual moisture compared with saturation at the current temperature, expressed as a percentage)	Little cooling needed to reach dew-point; condensation (water vapour changes into liquid water and releases latent heat) of water vapour can occur
(b) Clear Sky	Clear skies permit maximum nocturnal cooling of ground and adjacent air layers
(c) Calm or light wind; a few knots can help deepen the cooled moist layer	Enables turbulent mixing and cooling of layers close to ground; brings their temperature to dew-point
(d) Stability	Generally stable atmosphere, so mixing and cooling are confined to a shallow layer

⚠ **Why Calm Wind Can Prevent Radiation Fog:** If the wind is too light (calm), there is no turbulent mixing. The cooling of the ground is unable to extend upwards fast enough — dew may form and fog may remain shallow; calm conditions do not universally prevent fog.

Why Strong Wind Can Prevent Radiation Fog: Turbulence prevents cooling any one layer enough;

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cooling spreads over too thick a layer and temperature doesn't fall to dew point (temperature to which air must cool to become saturated at unchanged pressure and water-vapour content).

Characteristics of Radiation Fog

- Forms in **anticyclones, ridges, and ground inversions** — wind is light, skies are clear, atmosphere stable
- Fog would form even when RH is less than 100% due to hygroscopic condensation nuclei
- Minimum temperature usually occurs just after sunrise
- **Most frequent at about sunrise or it thickens at that time**
- Radiation fog dissipates about 2–3 hours after sunrise due to heating
- If deep, sun's heat takes longer to penetrate ground; fog may persist to midday
- Usually lifts into low stratus clouds before finally dissipating

Radiation Fog in India

⚠ **Radiation Fog — Indian Aviation Hazard (Winter):**

- A **winter hazard** in India
- Usually forms after the passage of a **Western Disturbance (WD)** which caused rain and then sky cleared
- Almost a daily feature in winters in the valley of the **Brahmaputra river** where adequate moisture is present
- Katabatic flow provides the required turbulence and additional cooling
- Most susceptible areas: **NW India, UP, Bihar, Jharkhand, Chhattisgarh, Bengal and Assam** (particularly Southern banks of Brahmaputra river)
- **Vertical extent:** Few hundred feet to 1500 ft, maximum 1 km
- Upper surface of vertical radiation fog is sharp with clear air above
- **Pressure system:** A High or Col with weak pressure gradient is favourable; fog does NOT form when pressure gradient is steep and low level winds are strong

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6.2 Advection Fog

Formation: Forms when **warm moist air from elsewhere is transported over a cold surface** (ground or water) where the surface temperature is less than the dew point of the advected air. Advection (horizontal transport by moving air) fog forms both over land and sea.

- Coastal fogs of Bengal and Orissa in winters are due to a combined effect of advection and radiation
- Advection fog forms any time of day and night
- More persistent than radiation fog and more extensive
- Common over sea in spring when warm air moves over cooler sea water

💡 **Key Distinction:** Radiation Fog = nocturnal cooling of ground (stationary process). Advection Fog = horizontal transport of warm moist air over cold surface (dynamic process). Advection fog can form DAY or NIGHT; Radiation fog mainly at night.

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6.3 Steaming Fog (Sea Smoke)

Formation: Occurs at high latitudes when cold air mass (a broad body of air with similar temperature and moisture) flows over a warm sea surface. The water vapour from the warm sea condenses as it lifts into the cold air. Looks like steam coming out of the sea surface. Hence called **Sea Smoke**.

- Common in Icelandic and Norwegian regions
- Over sea, often referred to as sea smoke

6.4 Frontal Fog

Formation: Occurs on the sloping boundary between two contrasting air masses due to the saturation (water-vapour pressure has reached its saturation value at the current temperature) of the air near the surface in the area of continuous rain ahead of a front. Also forms due to the lowering of cloud to the surface with the passage of a front.

6.5 Upslope Fog

Formation: Forms due to lifting of moist air along a sloping hill or plains — due to adiabatic cooling.

6.6 Inversion Fog

Formation: Occurs in an anticyclonic area when cold air, due to subsidence, comes in contact with a warm water body.

Fog Types — Comparison Table

Fog Type	Cause	Time	Location	Key Characteristic
Radiation	Nocturnal cooling; radiation loss	Night → morning	Land; valleys, depressions	Clear sky + light 3-7kt wind required; dissipates 2-3hr after sunrise
Advection	Warm moist air over cold surface	Day or Night	Land and sea coasts	More persistent; can be thick; Bengal/Orissa coastal fogs in winter
Steaming (Sea Smoke)	Cold air over warm sea	Any time	High latitude seas	Looks like steam; Iceland/Norway
Frontal	Rain ahead of front saturates air; cloud lowers to surface	With frontal passage	Frontal zones	Especially associated with warm fronts and occlusions
Upslope	Moist air lifts along slope; adiabatic cooling	Any	Hill slopes, plains	—
Inversion	Subsiding cold air meets warm water body	Any	Anticyclonic areas; coasts	In anticyclone with temperature inversion (a layer where temperature increases with height)

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7. Fog Dispersal

Thermal Method

When air is heated to temperature above the dew point, fog or clouds will dissipate. During the Second World War, heaters, burners, fires were used to disperse fog but the method involved large expenditures and was discontinued.

Use of Hygroscopic Nuclei

When hygroscopic particles are injected into fog water drops form. This lowers the RH of the air, leading to dissipation of fog. **Seeding** with dry ice or silver iodide or propane gas produces freezing and precipitation.

💡 **Fog Dispersal Methods:**
1. **Thermal Method** — heat air above dew point (expensive, discontinued after WWII)
2. **Hygroscopic Nuclei / Cloud Seeding** — dry ice, silver iodide, propane gas → freezing and precipitation → lowers RH → fog dissipates

Practice Q&A — Chapter 7: Visibility and Fog

Q1. Fog is reported when visibility is reduced to
(a) Less than 1000m (b) 1000m (c) 1000 to 2000

✅ Answer: (a) Less than 1000m — Fog is defined as visibility reduced to LESS than 1000m, with RH nearly 100%.

❌ (b) 1000m exactly: At exactly 1000m it is still Mist (≥1000m ≤5000m).
❌ (c) 1000-2000m: This is the Mist range.

Q2. RVR is reported in visibility falls below
(a) 500m (b) 1000m (c) 1500m (d) 2000m

✅ Answer: (c) 1500m — RVR is reported when visibility is less than 1500m.

❌ Others: Only c is correct per the text — "RVR is reported when visibility is less than 1500m."

Q3. Radiation fog occurs:
(a) Over land (b) Over sea (c) During day

✅ Answer: (a) Over land — Radiation fog requires nocturnal cooling of the GROUND surface. It forms over land, particularly in valleys and depressions.

❌ (b) Over sea: Radiation fog does not typically form over sea as sea temperature varies very little diurnally.
❌ (c) During day: Radiation fog forms at night and persists until morning; daytime heating dissipates it.

Q4. If visibility reduces between 5000m and 1000m and RH is almost 100%, it is
(a) Mist (b) Haze (c) Fog

✅ Answer: (a) Mist — Mist: visibility ≥1000m but ≤5000m, RH almost 100%. Composed of water

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droplets.

❌ (b) Haze: reported at visibility 5000 m or less, but it is a suspension of extremely small dry particles (dust, smoke, sand), so the air is not near saturation.
❌ (c) Fog: Fog is visibility <1000m.

Q5. Radiation Fog forms over N India during:
(a) May to June (b) Dec to Feb (c) Oct to Nov

✅ Answer: (b) Dec to Feb — Radiation fog in India is a winter hazard, typically December to February, especially after passage of Western Disturbances.

❌ (a) May to June: This is pre-monsoon / summer season — too warm for radiation fog.
❌ (c) Oct to Nov: Transition season; not the primary radiation fog season.

Q6. Warm and moist air moving over a cold surface causes:
(a) Thunder clouds (b) Fog and stratus (c) Frontal clouds

✅ Answer: (b) Fog and stratus — Warm moist air moving over cold surface = Advection Fog. The surface chills the air from below, leading to fog or stratus.

❌ (a) Thunder clouds: Thunderclouds require strong instability and rising motion — not created by cold surface.
❌ (c) Frontal clouds: These require frontal lifting.

Q7. The radiation fog activity increases after the passage of a ……
(a) WD (b) Depression (c) Col

✅ Answer: (a) WD (Western Disturbance) — Radiation fog in India usually forms after the passage of a WD which caused rain and then sky cleared, leaving adequate moisture in the air.

❌ (b) Depression: Generally brings cloud cover and rain — not favourable after depression.
❌ (c) Col: Col can be favourable but WD is specifically mentioned in the text as the trigger.

Q8. The radiation fog forms due to
(a) Heating of the earth during day
(b) Radiational cooling of earth at night
(c) Advection of cold air

✅ Answer: (b) Radiational cooling of earth at night — The ground radiates heat at night, cooling the adjacent air layers below the dew point, forming radiation fog.

❌ (a) Daytime heating: Daytime heating promotes convection and dissipates fog.
❌ (c) Advection of cold air: That is Advection Fog mechanism (warm moist air over cold surface).

Q9. Radiation fog is essentially a ……phenomena
(a) Nocturnal (b) Dusk (c) Day

✅ Answer: (a) Nocturnal — Radiation fog forms at night (nocturnal) due to radiative cooling of the earth's surface.

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Q10. Radiation fog forms over ……
(a) Water (b) Land (c) Both

✅ Answer: (b) Land — Radiation fog requires significant diurnal temperature variation, which occurs over land. Sea temperature is nearly constant.

Q11. The …… fog forms due to horizontal movement of warm moist air over cold surface
(a) Radiation (b) Advection (c) Frontal

✅ Answer: (b) Advection — Advection Fog = horizontal transport (advection) of warm moist air over a cold surface.

❌ (a) Radiation: Radiation fog = vertical (radiative) cooling of stationary air.
❌ (c) Frontal: Frontal fog = saturation from rain ahead of front or cloud lowering.

Q12. The …… fog forms due to horizontal movement of warm moist air over cold surface
(a) Radiation (b) Advection (c) Frontal

✅ Answer: (b) Advection — Same as Q11.

Q13. Advection fog forms during:
(a) Night only (b) Day time only (c) Any time of day and night

✅ Answer: (c) Any time of day and night — Advection fog can form at any time because it depends on the horizontal transport of warm moist air, not on day/night cycle.

❌ (a) Night only: That's a characteristic of radiation fog.
❌ (b) Daytime only: No fog type is daytime-only.

Q14. For the formation of Radiation fog
(a) There should be sufficient moisture in atmosphere, cloudy sky, nil wind
(b) There should be sufficient moisture in atmosphere, clear sky, light wind
(c) There should be sufficient moisture in atmosphere, cloudy sky, strong wind

✅ Answer: (b) Sufficient moisture, clear sky, light wind (3-7kt) — The three essential conditions: high RH, clear sky for nocturnal cooling, and light winds for turbulent mixing down to the surface.

❌ (a) Cloudy sky, nil wind: Cloudy sky prevents radiation cooling. Calm conditions may allow shallow fog; some light mixing can deepen it.
❌ (c) Cloudy sky, strong wind: Cloudy sky prevents cooling; strong wind mixes fog away or too thick a layer cooled.

Q15. Instrument used for measuring visibility is called
(a) Viscometer (b) Transmissometer (c) Ceilometer

✅ Answer: (b) Transmissometer — Also called Scopograph. Measures visibility along runways. Uses light source and photoelectric receiver.

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❌ (a) Viscometer: Measures viscosity of fluids — unrelated.
❌ (c) Ceilometer: Measures cloud base height, not visibility.

Q16. Advection fog forms:
(a) Over sea (b) Over land (c) Both over land and sea

✅ Answer: (c) Both over land and sea — Advection fog forms wherever warm moist air is transported over a colder surface, which can be land or sea.

Q17. Fog is …… cloud on ground
(a) Stratus (b) Strato cumulus (c) Nimbostratus

✅ Answer: (a) Stratus — Fog is essentially stratus cloud sitting on the ground. "Fog is the cloud sitting on ground."

❌ (b) Stratocumulus: Has patchy, lumpy structure — not the flat fog layer.
❌ (c) Nimbostratus: Rain-bearing grey layer cloud at significant height.

Q18. Frontal fog is more common with a: (a) Western Disturbance (b) Cyclone (c) Warm Front (d) Cold Front

✅ Answer: (c) Warm Front — Frontal fog forms at a warm front (or an occlusion): the rain falling from the warm air saturates the cold air near the ground and the cloud base lowers to the surface. The fog is mainly ahead of the front.

❌ (a) A Western Disturbance brings fog mostly after its passage, as radiation fog. (b) A cyclone is a wider system. (d) The cold front is a narrow, fast-moving feature and rarely holds fog.

Q19. Pressure patterns and radiation fog

Radiation fog can be favoured by anticyclones, ridges and cols when moist air, cooling and light wind are present. A col is a region of slack pressure gradient between highs and lows; it does not mean that all its air is sinking. Col weather can include fog in cool seasons and thunderstorms in summer.

Q20. Thermal processes / temperature distribution which favours formation of fog is:
(a) Isothermal (b) Adiabatic (c) Inversion (d) SALR

✅ Answer: (c) Inversion — A temperature inversion traps moisture and prevents mixing. Fog forms in the cool moist air trapped beneath an inversion; the inversion lies above the fog layer and stops it mixing away. Radiative cooling creates a surface inversion which is the typical radiation fog environment.

❌ (a) Isothermal: Temperature constant with height — less conducive to fog than inversion.
❌ (b) Adiabatic: Adiabatic process relates to rising/sinking air — upslope fog only.
❌ (d) SALR: Saturated adiabatic lapse rate is about in-cloud processes.

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Chapter 8: Vertical Motion and Clouds

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1. Causes of Vertical Motion

Definition: Vertical motion in the atmosphere — either upward (ascent) or downward (subsidence) — is driven by several mechanisms. Rising air leads to cooling, condensation (water vapour changes into liquid water and releases latent heat) and cloud/precipitation formation; sinking air produces warming and clear skies.

Five Main Causes

Cause	Mechanism	Effect
Terrain (Orographic)	Air forced up windward slope; eddies/instability on leeward. Mountain waves on leeward with strong winds.	Nacreous clouds occur in the lower stratosphere in very cold polar conditions
Convection (heat transfer through movement within a fluid)	Surface heating → air rises as thermals. Gliders use thermals.	CU and CB form; strong up/down draughts
Pressure Systems	Low/Cyclone → convergence → ascent; High/Anticyclone → divergence (airflow spreading apart) → subsidence	Lows = bad wx; Highs = fine wx
Frontal Zones	Warm front: gradual ascent over wide area; Cold front: sharp/abrupt uplift	WF: steady rain; CF: convection, TS, squally wx
Wind Shear (wind speed or direction changing over distance)	Vertical wind shear creates strong eddies. CAT at jet stream amplified by mountain waves.	Turbulence, CAT

2. Convergence and Divergence

Convergence: Net horizontal *inflow* of air into a region. Air accumulates → ascent in lower levels → possible subsidence above tropopause (stratosphere acts as stable barrier). Lows, cyclones, depressions and troughs → convergence → bad weather, but better visibility except in precipitation.

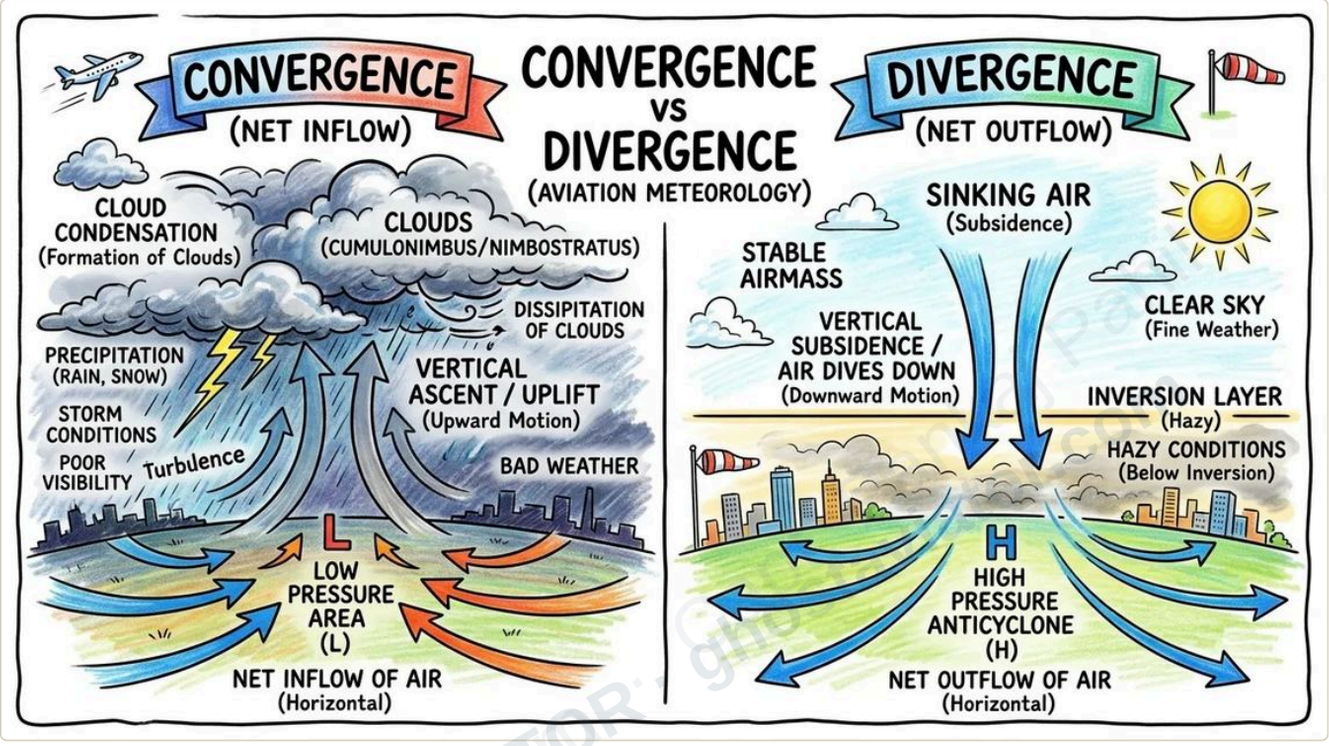
Divergence: Net horizontal *outflow* of air from a region. Example: wind to W is Wly 10 kt and to E is Wly 20 kt → more outflow than inflow = divergence. Anticyclones and ridges → subsidence → fine weather but poor visibility conditions.

Parameter	Convergence	Divergence
Air flow	Inflow	Outflow
Vertical motion (low level)	Ascent (Convection)	Subsidence

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Parameter	Convergence	Divergence
Pressure systems	Lows, Cyclones, Depressions, Troughs	Highs, Anticyclones, Ridges
Weather	Bad weather	Fine weather
Visibility	Better (except in precip)	Poor



Convergence feeds a low: air rises and cloud forms. Divergence drains a high: air sinks and the sky clears, often with haze below an inversion.

3. Clouds — Classification

Cloud: An aggregate of visible water droplets or ice particles. Clouds form by *adiabatic lifting and cooling* until air saturates. The level at which this occurs is the **Lifting Condensation Level (LCL)**. **Deposition** = process by which water vapour directly changes into ice particles.

There are **10 major genera** classified by **Form** (Stratiform, Cumuliform, Cirriform) and **Height** (High, Medium, Low).

Cloud Height by Region

Level	Polar Region	Temperate Region	Tropical Region
High	10,000–25,000 ft (3–8 km)	16,000–45,000 ft (5–13 km)	20,000–60,000 ft (6–18 km)
Medium	6,500–13,000 ft (2–4 km)	6,500–23,000 ft (2–7 km)	6,500–25,000 ft (2–8 km)
Low	up to 6,500 ft (2 km)	up to 6,500 ft (2 km)	up to 6,500 ft (2 km)

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4. High Clouds

High clouds in tropics occur at 6–18 km, consist of **ice crystals**, some may cause precipitation confined to high/medium levels only. Give advance indication of impending weather.

Cirrus (CI)

White and fibrous, delicate filaments or patches, narrow bands. Composed of **ice crystals**. Do **not** cause precipitation or ice accretion.

Cirrostratus (CS)

Thin whitish veil of fibrous or smooth appearance. Covers almost whole sky. Produces **Halo** — a coloured circular ring around sun or moon with red on the inside edge of a 22° halo. When thick may cause any ice fall, remaining confined to high/medium levels. Hardly any ice accretion.

Cirrocumulus (CC)

Thin, white, wave-like very small puffs. Composed of ice crystals. Sky visible in between puffs. Do **not** cause precipitation; hardly any ice accretion.

5. Medium Clouds

In tropics at 2–8 km. Contain water droplets and ice. Can cause snow and rain.

Altocumulus (AC)

White-grey clouds, patches, puffs, rounded masses, rolls, etc. Round masses generally well arranged in long waves, rows or lanes. Sky visible between puffs. **Altocumulus Lenticularis** = lens-shaped clouds at crests of mountain waves on leeward side of high mountains. **No precipitation; may cause ice accretion.**

Altostratus (AS)

Uniformly greyish or whitish cloud sheets/layers of fibrous appearance. Totally or partly cover the sky. If thin → see sun through it like *ground glass*. Optical phenomenon **Corona** occurs in AS. **Thick AS: continuous rain or snow. Ice accretion possible below freezing temperatures.**

Warning: Thick Altostratus causes continuous rain or snow. Ice accretion is possible below freezing temperatures in AS.

6. Low Clouds

In tropics below 2 km. Consist of water droplets or ice crystals.

Stratus (ST)

Grey cloud layer with uniform base. May stretch horizon to horizon. May give **drizzle**, ice prisms or snow grains. ST touching ground = fog. Ragged/patchy ST = **Fracto Stratus**.

Stratocumulus (SC)

Low clouds of large, irregular puffs or rolls, gray or whitish. Regularly arranged. **No serious precipitation.**

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Nimbostratus (NS)

Dark and grey thick cloud layer. Causes **continuous rain or snow**. Thick enough to blot out the sun. Extends both medium and lower levels. **Ice accretion possible below freezing. Ice Pallets** fall from thick NS clouds especially in Warm Fronts. Ice pallets are transparent ice particles, spherical/irregular, about less than 5 mm dia.

Aviation Warning: NS = continuous rain/snow + ice accretion risk below freezing = avoid or be aware during approach/departure.

7. Clouds with Vertical Development

Cloud identification gallery



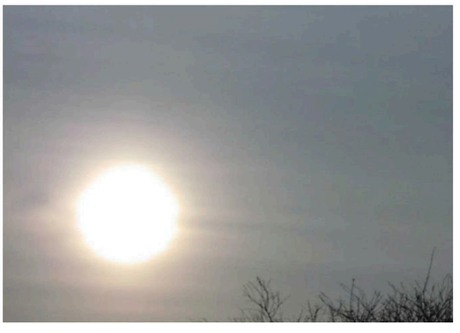
Cirrus



Cirrocumulus



Altocumulus lenticularis (left) and layered cloud with cumulus tops (right)



Altostratus

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Stratus



Stratocumulus



Thick, dark layer cloud of the nimbostratus type



Fair weather cumulus



A large cumulonimbus building; the anvil is not yet clear

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Cumulus (CU)

Detached clouds, dense with sharp outlines. Base dark, nearly horizontal. Develop vertically as rising mounds, domes or towers. Bulging upper part resembles a cauliflower. Sunlit parts brilliant white. Base sometimes ragged = **Fracto Cumulus**. A well-developed CU with great vertical extent = **Towering Cumulus (TCU)**.

Fair Weather Cumulus

Limited vertical extent. Caused by thermal currents due to surface heating. Occur during daytime only, develop in forenoon, reach max in afternoon, dissipate in evening. **Do not cause precipitation.**

Cumulonimbus (CB)

Heavy and dense cloud of large vertical extent, like mountains or huge towers. Upper portion mostly smooth, fibrous and nearly flattened — spreads out like an **anvil**. Base often very dark. Low ragged cumulus fracto or stratus fractus sometimes present below. **Virga (falling precipitation that evaporates before reaching the surface)** = shaft of rain or snow falling from distant cloud which vapourises before reaching the ground.

CB causes ALL types of precipitation: RA, SH, GR, SN.
Hail may be experienced under the anvil of a CB.
CB is the most hazardous cloud for aviation and should be avoided.

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Flying through active CB cloud should be avoided during TS. Severe turbulence, icing, lightning, hail possible 20–30 nm around a CB.

8. Very High Level Clouds

Nacreous Clouds (Mother of Pearl Clouds)

Nacreous clouds are rare polar stratospheric clouds, commonly called mother-of-pearl clouds because of their brilliant iridescent colours. They occur in the lower stratosphere in very cold conditions. They are distinct from noctilucent clouds in the mesosphere.

Noctilucent Clouds

Resemble Cirrus but have bluish, silvery or orange/red colour. Occur in upper Mesosphere between 80–85 km. Visible soon after sunset or before sunrise between 72°N and 45°N, mostly around 55°N, in summers (late May–mid August). Thought to consist of **ice particles** (not meteoric dust, as originally believed).

9. Cloud Base, Amount & Ceiling

Reporting Cloud Base

In METAR, SPECI, TAF and Local Forecast: cloud base reported **from Above Ground Level (AGL)**.
In Area Forecast Charts: cloud base from **Above Mean Sea Level (AMSL)**.

Cloud Amount (Oktas)

Oktas	1–2	3–4	5–7	8/8	Nil	Sky not visible
Term	FEW	SCT (Scattered)	BKN (Broken)	OVC (Overcast)	SKC (Sky clear): old code, not in the current rules. NSC is used when there is no cloud of operational significance and CAVOK does not apply	Sky Obscured
Sky Condition (IMD terms)	Mainly clear	Partly cloudy	Generally cloudy	Cloudy	Clear sky	Sky Obscured

Cloud Ceiling

Cloud Ceiling: Height, above ground or water surface, of the lowest layer of cloud below 6000 m (20,000 ft) covering more than half the sky (5/8 or more).

Cloud Amount in Imaginary Oktas

For expressing cloud amount, the sky is divided into **eight imaginary parts**. Each part is called one **Okta**.

Operational Significance: A cloud with a base below 1500 m (5000 ft) or below the highest minimum sector altitude, whichever is greater, or a cumulonimbus or towering cumulus cloud at any height, is of operational significance.

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10. Cloud Species

Species	Description	Applicable Cloud Types
Fibratus	Nearly straight filaments without tufts or hooks	CI, CS
Lenticularis	Elongated lens-shaped	AC, SC or CC
Castellanus	Cumuliform protuberances connected by common base, arranged in lines	CI, CC, AC or SC
Fractus	Ragged	ST or CU
Congestus	Strong sprouting/great vertical development, bulging resembles cauliflower	CU
Calvus	The bulging tops are starting to lose their sharp cauliflower outline, but no cirrus-like (fibrous) part and no anvil can be seen yet	CB (without distinct anvil)
Capillatus	Distinct anvil. Accompanied by SH, TS, SQ and GR. Produces well defined Virga.	CB

Exam Tip: CB Capillatus = distinct anvil = all precipitation types. CB Calves = developing CB, no distinct anvil yet.

11. Condensation Trails (Contrails)

Condensation trails = visible streaks of condensed water vapour formed in the wake of a moving aircraft.

Type	Description
Wing Tip Trails	Thin, transient, short-lived. Form near wing tips and propeller edges due to aerodynamic reduction of pressure, expansion and consequent adiabatic cooling and condensation.
Exhaust Trails	Form from condensation of moisture from aircraft exhaust at about 9 km and above. Long, persistent, visible. Reveal position and track of aircraft.
Distrails (Dissipation Trails)	Passage of aircraft through a cloud marked by clear lanes. The heat released by aircraft exhaust is sufficient to evaporate the cloud in its wake.

12. Mintra, Drytra and Maxtra Levels

Mintra (Minimum Trail) Level (ML): Below this level, no condensation trails form. The flight level at which temperature is about –45°C gives a good estimate of ML at 100% RH for jet aircraft.

Drytra Level: When temperatures are very low, contrails form even when RH is 0%. It is usually about 2 km above the Mintra level. This is because moisture from the exhaust is sufficient to produce saturation (water-vapour pressure has reached its saturation value at the current temperature).

Maxtra Level: This level lies in the **stratosphere**, above which no contrails form.

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Contrail Layer Summary (bottom to top):
No contrails → Contrails Possible → Contrails Certain (Mintra Level → Drytra Level) → Contrails Possible → No contrails (Maxtra/Sea Level equivalent in stratosphere)

13. Flying in Clouds — Operational Significance

Stratiform Clouds (ST, AS)

Flying is generally smooth. In NS clouds during monsoon with thick AS and NS layers, visibility is reduced due to continuous precipitation. ST clouds reduce visibility and cause difficulty identifying the runway during landings. Ice accretion also occurs in AS and NS clouds above freezing levels.

Cumuliform Clouds (CU, CB)

CC and AC may cause bumpiness. Well developed CU, TCU and CB can cause severe turbulence, downdrafts, gusts, ice accretion, lightning, hail and strong up/down draughts. Flying through CB should be avoided during TS. Hail may be experienced in clear air below the anvil of a CB.

CB Avoidance Rules:
• Avoid flying through active CB
• Avoid flying under CB anvil — hail in clear air possible
• CB causes ALL: RA, SH, GR, SN, TS, severe turbulence, severe icing, lightning
• Operations should be avoided during TS

14. Practice Q&A

Q1. Drizzle occurs from (a) CS (b) ST (c) NS (d) CU

✅ Answer: (b) ST

Explanation: Stratus (ST) gives drizzle, ice prisms or snow grains. It does not give rain.
Distractors: CS (high cloud, no precip); NS (continuous rain, not drizzle specifically); CU (showers, not drizzle).

Q2. Altostratus (AS) is (a) Low cloud of sheet type (b) A medium cloud of sheet type (c) A high cloud of large vertical growth

✅ Answer: (b) A medium cloud of sheet type

AS is a medium-level stratiform cloud. It is a uniformly grey/whitish sheet. In tropics: 2–8 km.
Low cloud = ST/SC/NS. High cloud with vertical growth = CB.

Q3. Showers and convective clouds

Showers can come from sufficiently developed cumulus and cumulonimbus. Cumulonimbus is associated with deep convection and can produce thunder, lightning or hail. Listing CU and CB as separate answers to an unqualified showers question does not provide a unique choice.

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Q4. Heavy icing is possible in (a) CI (b) CS (c) ST (d) CB

✅ Answer: (d) CB

CB contains supercooled water droplets, causing heavy icing especially in middle layers. Also rime and clear ice possible.
CI, CS: mainly ice crystals, negligible icing. ST: possible icing at/below freezing.

Q5. Halos and cloud-entry safety

A halo is produced by light interacting with ice crystals. This identifies an optical mechanism, not permission to enter a cloud or proof that icing is absent. Airframe icing from supercooled droplets and engine icing in suitable ice-crystal conditions are distinct hazards.

Q6. Dark grey cloud causing continuous rain (a) AS (b) NS (c) SC (d) CB

✅ Answer: (b) NS

Nimbostratus is a dark, grey, thick cloud layer causing continuous rain or snow. Thick enough to blot out the sun.
AS: uniformly grey but lighter, may cause rain; SC: low puffs, no serious precip; CB: showers not continuous.

Q7. A uniform layer of cloud resembling fog but not on the ground (a) AS (b) NS (c) ST

✅ Answer: (c) ST

Stratus is a grey cloud layer with uniform base, resembles fog. When ST touches the ground, it IS fog.
AS: medium level; NS: causes heavy rain; ST: low, fog-like.

Q8. The clouds composed of ice crystals having feathery appearance (a) CI (b) CS (c) AS

✅ Answer: (a) CI

Cirrus clouds are white and fibrous — delicate filaments = feathery appearance. Composed entirely of ice crystals.
CS: thin whitish veil; AS: medium level grey sheet.

Q9. NS clouds occur (a) At cold front (b) At warm front

✅ Answer: (b) At warm front

At warm fronts, gradual ascent produces AS and NS, giving widespread continuous rain ahead of the surface front.
Cold fronts produce CU and CB (showers and thunderstorms), not NS.

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Q10. Lenticular clouds indicate presence of (a) Warm Front (b) Cold Front (c) Mountain waves

✅ Answer: (c) Mountain waves

Altocumulus Lenticularis form at crests of mountain waves on the leeward side of high mountains. They are stationary despite strong winds.
Fronts produce layered/cumuliform clouds, not lens-shaped stationary clouds.

Q11. Mintra level

In the contrail terminology used here, Mintra is the lower boundary below which condensation trails do not form for the stated atmospheric and aircraft conditions. Below that level is the key wording; the boundary itself is not a blanket no-contrail altitude for every aircraft or atmosphere.

Q12. Cloud ceiling is the height of the cloud covering... (a) 3–4 oktas (b) 5/8 or more (c) 8/8

✅ Answer: (b) 5/8 or more

Cloud ceiling = height of lowest cloud layer below 6000 m covering 5/8 or more of sky (BKN or OVC).
3–4 = SCT (not ceiling); 8/8 = OVC (that IS a ceiling layer but so is BKN).

Q13. AC cloud with cumuliform protuberances indicates (a) Stability (b) Instability (c) Neutrality

✅ Answer: (b) Instability

Castellaneous AC (ACCas) with tower-like cumuliform protuberances indicates atmospheric instability at medium levels — a precursor to afternoon thunderstorm development.

Q14. Hail may be experienced under the anvil of a CB (a) True (b) False

✅ Answer: (a) True

Hail (GR) can fall in clear air below the anvil of a CB. The anvil spreads out horizontally and can carry hail considerable distances from the storm.

Chapter 9: Stability and Instability

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1. Atmospheric Processes

Air is a poor conductor of heat. A small rising or falling parcel of air in atmospheric processes can be regarded as insulated from its surroundings.

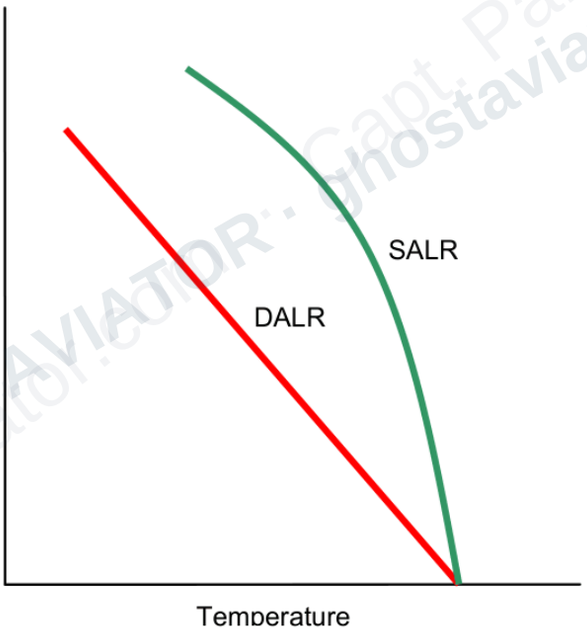
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Process	Definition	Heat Exchange
Isothermal	Heat is allowed to enter or leave the system — temperature remains the same	Yes (with surroundings)
Adiabatic	No heat enters or leaves the system. (Greek: a = not, diabano = pass through)	No (insulated)

Adiabatic Process explained: If a parcel of air ascends adiabatically, it moves into a region of lower pressure and *expands*. The work done in expanding is at the expense of its internal energy → temperature falls. Conversely, descending air is compressed → temperature rises. Adiabatic processes, to a large extent, determine the vertical distribution of temperature in the atmosphere.

2. Lapse Rates

Lapse Rate (LR): Decrease of temperature with height. Taken positive when temperature decreases with height. Average LR in the troposphere is about 6.5°C/km.



DALR and SALR on a temperature-height graph

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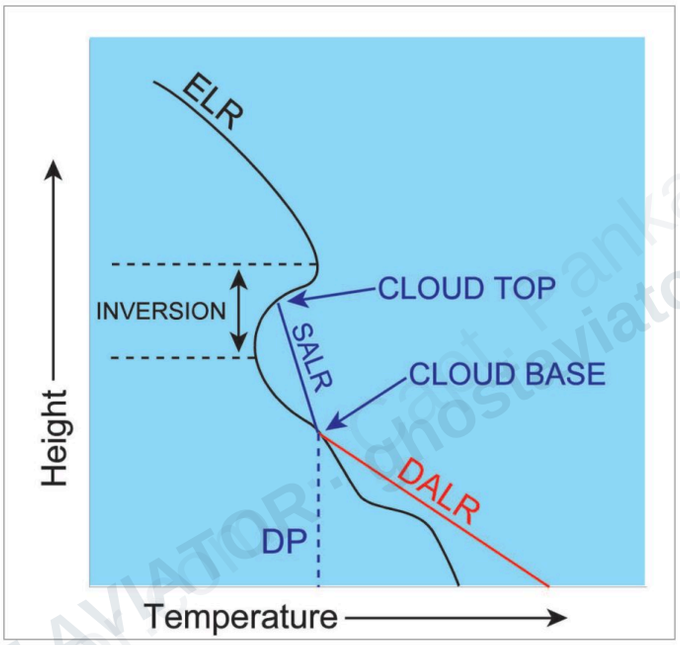
Three Key Lapse Rates

DALR — Dry Adiabatic Lapse Rate (temperature decrease with increasing height) Rate of temperature decrease with height when *unsaturated* air ascends adiabatically. Value = 9.8°C/km = 3°C/1000 ft (In ISA: ELR = 6.5°C/km) **SALR — Saturated Adiabatic Lapse Rate** When saturated air lifts adiabatically → water vapour condenses → latent heat (energy absorbed or released during a change of state without a temperature change) released → offsets adiabatic cooling. Value ≈ 5°C/km (variable — generally increases as the rising air becomes colder, approaching DALR) At higher levels (low temperatures), very little WV → latent heat small → SALR approaches DALR. **ELR — Environmental Lapse Rate** Actual lapse rate existing in the atmosphere. In ISA, ELR = 6.5°C/km. In actual atmosphere: can be any value. It may be super-adiabatic (temperature decreases faster than DALR), zero (isothermal), or negative (an inversion).

Lapse Rate	Value	Condition
DALR	9.8°C/km (3°C/1000 ft)	Unsaturated ascending air
SALR	~5°C/km (variable)	Saturated ascending air
ELR (ISA)	6.5°C/km	Actual atmosphere (ISA standard)

3. Inversion

Inversion: Increase of temperature with height (negative lapse rate). Opposite of normal.



Inversion limiting cloud top

Inversion Layer

An atmospheric layer in which there is an inversion of temperature. Vertical motion through such a layer is inhibited due to stability in the layer. An inversion layer near the Earth's surface occurs during cloudless cool nights. Inversion at higher levels is associated with anticyclones and sometimes with the fronts.

Effects of Inversion

Type	Common Occurrence	Aviation Effect
Low level inversion	Common during winters due to nocturnal cooling. Also associated with ridges, high pressure areas and anticyclones.	Smoke, haze, mist, fog stagnate over ground → **poor visibility**. Often at top of stratified cloud layer.
Higher level inversion	Associated with anticyclones, sometimes fronts	Does not allow surface air to rise and mix with upper winds → smoke/pollution trapped below

Key Point: Inversion indicates **static stability**; turbulence may still occur in wind shear (wind speed or direction changing over distance) or mountain waves in the atmosphere. However, low level inversion

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causes poor visibility — haze, mist, fog, smoke stagnate — especially **radiation fog**. Inversion is often present at the top of a stratified cloud layer.

Valley Inversion

A valley inversion is a low-level inversion that forms on the floor of a valley, mostly at night.

How it develops:

- Air in a valley is more static (still) than air on an open hill. At night it stays in contact with the cold ground for a longer time, so it gets colder.
- The hill slopes also cool at night. The air touching them becomes cold and dense (heavy). It sinks down the slope as a katabatic wind and collects on the valley floor.
- The result is cold air at the bottom and warmer air above. Temperature rises with height. This is an inversion.
- It is most marked with a clear sky and a slack (weak) pressure gradient.

Effects on flying:

- **Fog and mist form first in valleys.** Valley fog is held in by the hills. In calm winter weather it can last for several days.
- Smoke and haze are trapped below the inversion, so visibility is poor.
- A low-level temperature inversion is a known cause of wind shear near aerodromes.

Radiation fog usually clears soon after sunrise as the ground warms. Valley fog can last longer.

4. Stability and Instability

Stable atmosphere analogy: A glass ball placed at centre of a convex glass — displaced and left, it will oscillate and return to original position (like a pendulum). Stratosphere and Thermosphere are **stable** atmospheres.

Unstable atmosphere analogy: A glass ball placed at top of an inverted watch-glass — if displaced will roll down further, will NOT return to original position.

Atmospheric Instability

A layer is **stable** if a parcel of air in it, given a small push upwards, sinks back to its original level. It is **neutral** if it stays at its new level, neither sinking back nor rising further. It is **unstable** if it continues to move up on its own.

To estimate static stability in a layer, changes in KE of a test parcel at various levels are estimated from the Aerological Diagram (**T-phi gram / Tephigram**).

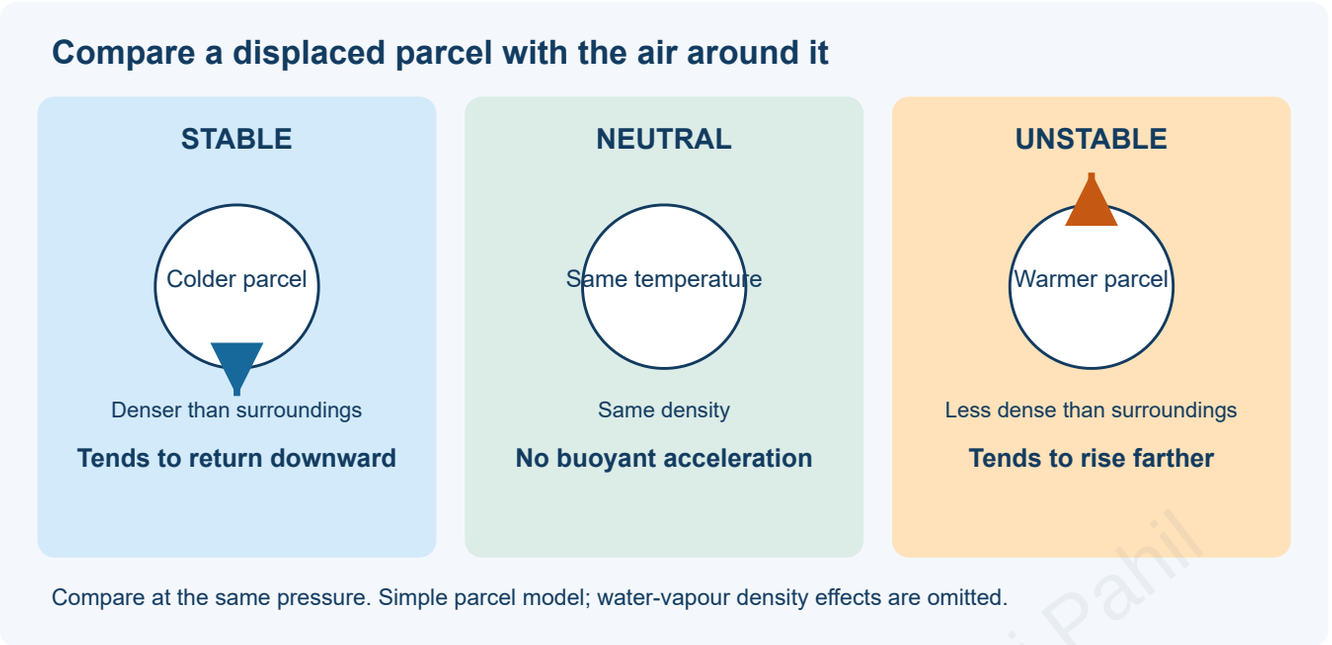
The atmosphere is stable, neutral or unstable according as the KE of the parcel decreases, remains constant or increases — A fall of temperature with height alone does not establish instability: compare the environmental lapse rate with the dry and saturated adiabatic rates.

5. Types of Instability/Stability

Conditional Instability: State of atmosphere in which DALR > ELR > SALR. Atmosphere is unstable for saturated air and stable for unsaturated air.

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At the same pressure, compare a displaced parcel with its surroundings. In this simple model a colder parcel tends to sink back, a warmer parcel tends to rise farther, and an equal-temperature parcel has no buoyant acceleration. Water-vapour density effects are omitted.

Latent Instability: An atmospheric layer in an initially stable parcel of air that eventually becomes unstable due to forced ascent. The parcel possesses latent (hidden) instability. It is **Real Latent** if force required to lift is much less than energy released when it becomes unstable. It is **Pseudo Latent** if energy required to lift > energy released later.

Potential (Convective) Instability: When a layer as a whole if lifted bodily becomes unstable if latent heat released on lifting produces sufficient temperature difference between bottom and top of layer to steepen LR of lifted layer. For potential instability it is essential that **RH is high in lower levels and low in higher levels**.

6. Stability Criteria

STABILITY CRITERIA TABLE

Condition	Stability Status	When (ELR relation)
Dry air is	**Stable**	DALR > ELR
Saturated air is	**Stable**	SALR > ELR
Air (Dry or Saturated) is	**Absolutely Stable**	SALR > ELR (both dry and sat stable)
Air (Dry or Saturated) is	**Absolutely Unstable**	ELR > DALR
Conditional Instability	Unstable for sat; stable for dry	DALR > ELR > SALR

Key Stability Rule:
DALR = 9.8°C/km (fixed)

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SALR varies with temperature and pressure; about 6°C/km is a temperate near-ground example
ELR = actual measured value

If ELR < SALR → Absolutely Stable (AS on both accounts)
If SALR < ELR < DALR → Conditionally Unstable
If ELR > DALR → Absolutely Unstable

Worked Example

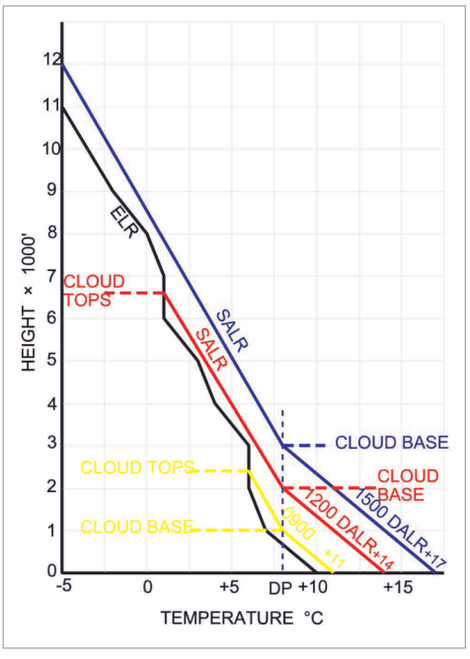
Q: DALR = 9.8°C, ELR = 6.8°C → atmosphere is? For an unsaturated parcel, DALR 9.8°C/km>ELR 6.8°C/km, so it is stable. A saturated parcel requires the SALR: ifSALR is explicitly 5°C/km, this layer is conditionally unstable. Q: SALR = 5.5°C, ELR = 4.5°C → atmosphere is? Compare: SALR (5.5) > ELR (4.5) → both stable → **Absolutely Stable** Q: Surface temp 35°C, DALR → temp at 2 km? T at 2 km = 35 − (9.8 × 2) = 35 − 19.6 = **15.4°C ≈ 15°C**

How to Apply Stability Test (T-phi gram)

When a parcel of dry air is lifted, it follows DALR. Saturated air follows SALR. Comparing the temperature of the lifted parcel (as given by DALR/SALR) with the environmental temperature (ELR) at any level AB (Figure 9.1): if parcel temperature is more than environmental → parcel will shoot up = **Unstable**. If colder → it will sink = **Stable**.

7. Tephigram (T-phi Gram)

Tephigram: A thermodynamic diagram on which upper air temperatures, winds and dew point (temperature to which air must cool to become saturated at unchanged pressure and water-vapour content) temperatures (as obtained from Radiosondes and Rawin) are plotted at various levels. Used for estimating meteorological parameters viz. Instability-Stability in the atmosphere, Mintra Level, LCL, base and top of cloud, Potential and Latent Instability, wet bulb temperature etc.



Finding cloud base and tops on a temperature-height diagram

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Lines on Tephigram (used in India)

Line Type	Represents
Horizontal Lines	Potential Temperature or DALR
Vertical Lines	Temperature
Slanting Lines 45° to DALR	Pressure at different levels
Curved lines slanting left (gradually becoming parallel to DALR at higher levels)	SALR lines
Dashed lines slightly slanting left	Isohygric lines (lines of equal saturation (water-vapour pressure has reached its saturation value at the current temperature) mixing ratio — g/kg)

Separate lines exist for saturation over water and over ice on the Tephigram.

8. Normand Theorem and Normand Point

Normand Theorem: If we draw upwards DALR from dry bulb (TT), SALR from wet bulb (TwTw), and Isohygric from dew point (TdTd) temperatures, the three lines meet at a point called the **Normand Point**. By knowing TT and TdTd, we can calculate TwTw at any level.

LCL (Lifting Condensation Level): The first Normand Point (from the surface) is the **First Normand Point**. This is also called the **Lifting Condensation Level (LCL)** and gives an idea of the **height at which cloud may form**.

Exam Tip: LCL is the level where a lifted parcel first becomes saturated. It estimates a cloud base only when that parcel and lifting process represent the cloud. Found using Tephigram by intersection of DALR (from TT), SALR (from TwTw) and Isohygric (from TdTd).

9. Practice Q&A

Q1. Stability of an unsaturated parcel

If DALR is 9.8°C/km and ELR is 6.8°C/km, an unsaturated parcel cools faster than the environment on ascent and becomes relatively colder: it is stable. To judge a saturated parcel, also compare ELR with SALR. Moisture state must be stated.

Q2. SALR = 5.5°C, ELR = 4.5°C — atmosphere is (a) unstable (b) stable (c) indifferent

✅ Answer: (b) stable

SALR (5.5) > ELR (4.5): even saturated air is stable → Absolutely Stable atmosphere.

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Q3. DALR > ELR > SALR is (a) conditionally unstable (b) latently stable (c) potentially stable

✅ Answer: (a) conditionally unstable

This is the definition of Conditional Instability: DALR > ELR > SALR. Stable for dry air, unstable for saturated air.

Q4. Dry air is unstable when (a) ELR = DALR (b) ELR > DALR (c) ELR < DALR

✅ Answer: (b) ELR > DALR

Dry air is unstable when ELR > DALR (Absolutely Unstable). The environment is cooling faster than the parcel → parcel remains warmer than surroundings → keeps rising.

Q5. The saturated air is said to be unstable if (a) SALR = ELR (b) SALR < ELR (c) SALR > ELR

✅ Answer: (b) SALR < ELR

If SALR < ELR, the environment cools faster than the saturated parcel → parcel remains warmer → unstable.

Q6. If ELR = SALR = DALR the atmosphere is (a) Stable (b) Indifferent (c) Unstable

✅ Answer: (b) Indifferent (Neutral)

When all three lapse rates are equal, the parcel neither accelerates upward nor downward — neutral/indifferent equilibrium.

Q7. DALR means (a) Rate at which temperature of unsaturated parcel of air falls with height when made to ascend adiabatically (b) Rate at which temp falls with height (c) Rate at which ascending parcel of saturated air cools

✅ Answer: (a)

DALR = Dry Adiabatic Lapse Rate = rate of cooling of unsaturated air ascending adiabatically = 9.8°C/km. (b) = ELR definition; (c) = SALR definition.

Q8. Surface temp 30°C. Assuming DALR, temperature at 2 km is (a) 18°C (b) 10°C (c) 42°C

✅ Answer: (b) 10°C

T at 2 km = 30 − (9.8 × 2) = 30 − 19.6 ≈ 10°C

Q9. An Isothermal atmosphere is (a) Stable (b) Unstable (c) Neutral

✅ Answer: (a) Stable

Isothermal = ELR = 0. Since DALR (9.8) > ELR (0), dry air is stable. Since SALR (~5) > ELR (0), saturated air also stable → Absolutely Stable.

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Q10. If ELR is less than SALR, atmosphere is said to be (a) Absolutely unstable (b) Conditionally unstable (c) Absolutely stable

✅ Answer: (c) Absolutely stable

ELR < SALR means environment cools slower than even saturated air → both dry and saturated air parcels are stable → Absolutely Stable.

Q11. DALR is approximately (a) 5°C/km (b) 15°C/km (c) 10°C/km

✅ Answer: (c) 10°C/km (closest to 9.8°C/km)

DALR = 9.8°C/km ≈ 10°C/km. In some textbooks/exams rounded to 10°C/km.

Q12. Why SALR is variable

The saturated adiabatic lapse rate varies with temperature and pressure. A temperate near-ground teaching example uses about 6°C/km near the ground in temperate latitudes, with smaller values in warmer saturated air. Do not treat 5°C/km as a fixed value at every sea-level location.

Q13. SALR tending towards DALR

In colder saturated air, less water vapour condenses and less latent heat is released, so SALR tends towards DALR. This is a gradual dependence on temperature and pressure; −40°C is not a universal switch at which the rates become equal.

Q14. Dry air having a temp of 35°C on surface when forced to rise adiabatically by 1 km would attain a temp of (a) 29°C (b) 25°C (c) 45°C

✅ Answer: (b) 25°C

T = 35 − (9.8 × 1) = 35 − 9.8 = 25.2°C ≈ 25°C

Q15. Inversion in the atmosphere indicates (a) Stability (b) Instability (c) Neutrality

✅ Answer: (a) Stability

Inversion = temperature increases with height = ELR is negative = highly stable condition. Vertical motion is inhibited through an inversion layer.

Q16. Environmental LR can be more than DALR (a) True (b) False

✅ Answer: (a) True

ELR can be more than DALR — this is called super-adiabatic lapse rate, creating Absolutely Unstable atmosphere. Common in very hot surface conditions.

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Q18. The process which to a large extent determines the vertical distribution of temperature in atmosphere is (a) Adiabatic (b) Isothermal (c) Isentropic

✅ Answer: (a) Adiabatic

Adiabatic processes — DALR and SALR — determine vertical temperature distribution in the troposphere. Isothermal processes do not; isentropic is the same as adiabatic in many contexts but adiabatic is the key term here.

Q19. Rise in temperature with height is (a) Lapse (b) Normal (c) Inversion

✅ Answer: (c) Inversion

Lapse = fall with height (normal). Rise with height = Inversion (abnormal, negative lapse rate).

Q20. Inversion is common in (a) Post Monsoon (b) Monsoon (c) Winters

✅ Answer: (c) Winters

Low level inversion is common during winters due to nocturnal cooling of the surface, resulting in surface temperatures lower than air above. This traps fog and haze.

Chapter 10: Optical Phenomena

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1. Introduction

Optical phenomena occur frequently in the atmosphere. They provide useful information about cloud constitution, cloud type, atmospheric turbulence, lapse rate (temperature decrease with increasing height), ice accretion etc. — all important for aviation safety.

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2. Rainbow

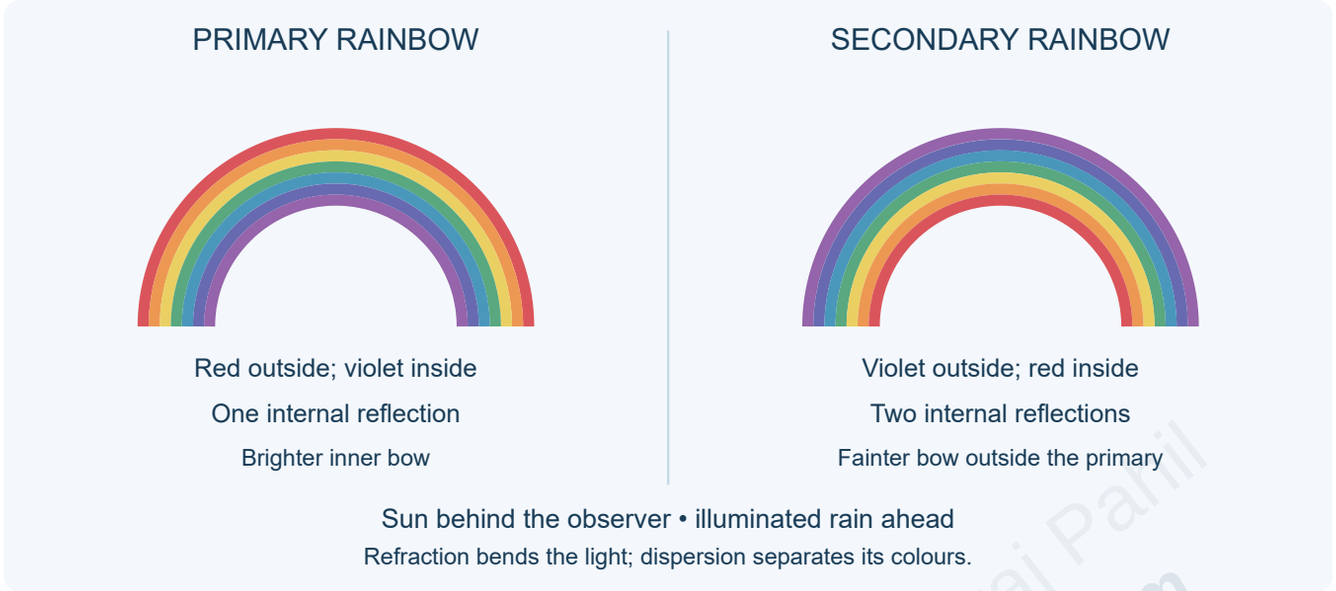


Figure 10.1. Primary and secondary rainbows: compare the colour order and the number of internal reflections.

Rainbow: A group of concentric arcs with colours ranging from violet to red, produced on a background of water drops (raindrops, droplets of drizzle or fog) in the atmosphere by light from the sun or moon. Sometimes two rainbows are observed — Primary and Secondary.

Primary Rainbow

A circle or arc of circles of coloured light in the sky while it is raining. The circle subtends an angle of 42° at the eyes. Seen opposite to the sun or moon. Its centre is on the line joining the luminary and the observer. The rainbow may form a complete ring when seen from a hill, high tower or from an aircraft. Due to **double refraction** on entering the rain drop (once on entering, second time on leaving) with **one internal reflection**. Colours: **VIBGYOR** (red outside, violet inside).

Secondary Rainbow

Concentric with the primary. Radius about 52°. Much less bright than primary with breadth almost double. **Red is on the inside (radius of arc 50°) and violet on the outside (radius of arc 54°)** — colours reversed compared to primary.

Feature	Primary Rainbow	Secondary Rainbow
Radius	42°	52°
Red colour position	Outside	Inside (50°)
Violet position	Inside	Outside (54°)
Brightness	Bright	Much less bright
Internal reflections	One	Two
Breadth	Normal	Almost double

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3. Halo

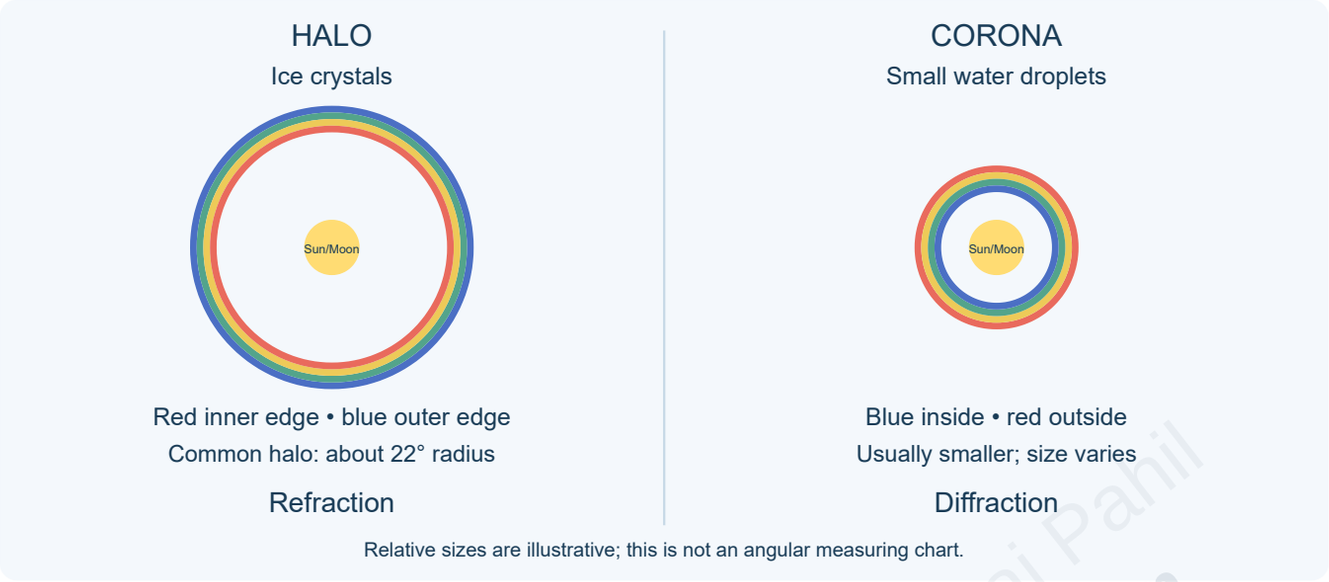


Figure 10.2. A halo and a corona differ in their particles, light process and usual colour order. Their presence alone does not determine an aircraft icing severity.

Halo: A circle of light around the sun or moon. Produced by **refraction** of light through **ice crystals** present in Cirrostratus (CS) clouds.

Small Halo (Most Frequent)

Mostly white luminous ring of 22° radius around the sun or moon at its centre. When well developed, the halo round the sun shows a pure clear halo — the **red on the inside** but other colours usually difficult to recognise. The sky inside the ring is clearly darker than the rest of the sky. A halo indicates ice crystals. It does not establish icing severity or that a cloud is safe to enter.

Large Halo

A circular halo with a radius of 46°. This is the **Large Halo**. It is much less common than the small halo and is always less bright.

Feature	Small Halo	Large Halo
Radius	22°	46°
Colour	Mostly white, red inside	Less bright
Sky inside ring	Conspicuously darker	No special feature
Cloud type	Cirrostratus (CS)	CS
Ice accretion risk	Not established by a halo	Not established by a halo

Aviation significance of a halo: A halo indicates ice crystals. It does not establish icing severity or that the cloud is safe to enter. An optical display is not a substitute for weather information and aircraft operating procedures.

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4. Corona



Figure 10.3. Illustration of a lunar corona: coloured rings close around the Moon. It is not a dated or measured weather observation.

Corona: One or more small coloured rings around the Sun or Moon, formed by diffraction of light by small water droplets or ice particles in thin cloud, mist or fog. The central disk is bluish-white, fading towards reddish-brown at its edge. Diffraction means the spreading and bending of a light wave around small particles.

Aviation significance: A corona identifies an optical effect. It does not establish that droplets are supercooled or determine icing severity. Assess temperature, liquid-water conditions and weather information, and follow aircraft procedures.

Feature	Corona	Halo
Radius	<5° (small)	22° or 46° (large)
Cause	Diffraction (water drops)	Refraction (ice crystals)
Cloud type	AS, thin AC, mist, fog	CS
Icing risk	Not determined by a corona	Not determined by a halo
Red position	Outside	Inside

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5. Mirage

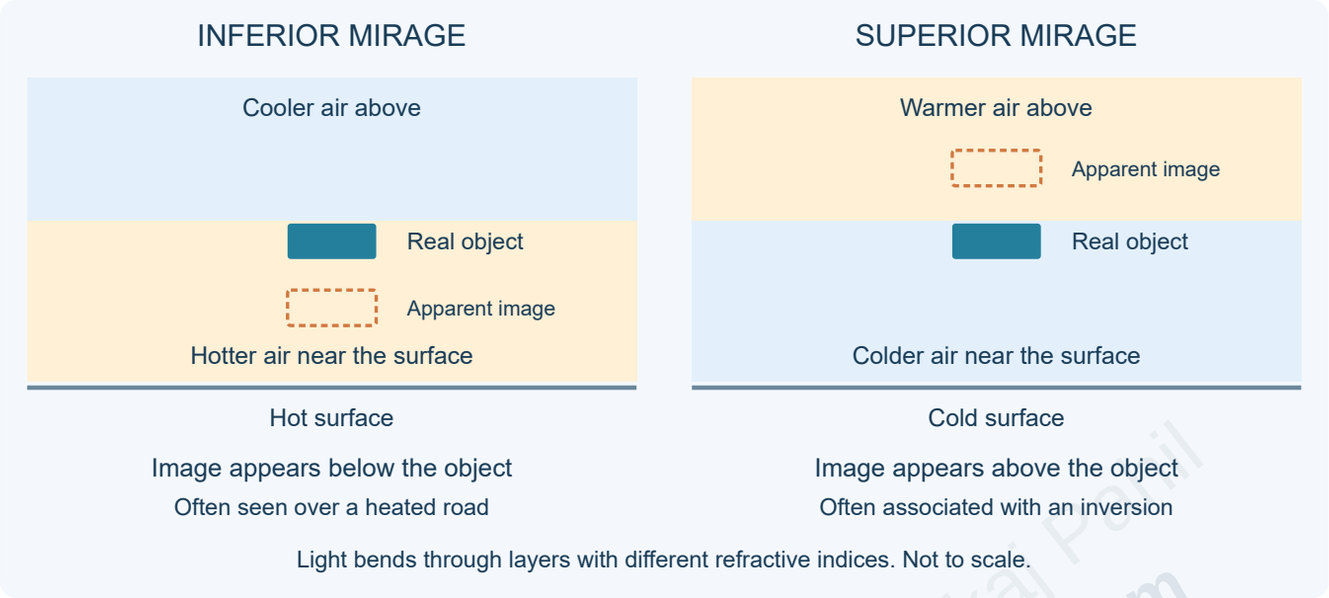


Figure 10.4. Inferior and superior mirages. Dashed shapes show apparent images, not extra objects. The temperature-layer structure refracts light; the diagram deliberately does not prescribe a universal ray path.

Mirage: Light rays are refracted (bent) as they travel from one medium to another of different density (mass per unit volume). In the atmosphere, density reduces with altitude. Light reflected from a distant object travels in a curved path, keeping lower density on the convex side of the curve. Rising sun or moon thus appears slightly higher than they actually are.

Mirages are due to curving of light rays passing through layers of air in which refractive index changes considerably with height due to density differences. Observed when temperature of earth's surface differs markedly from that of air above.

Type	Name	Condition	Appearance
Inferior (Lower) Mirage	Lower Mirage	Lapse rate is steep (strongly heated surface, hot road/desert). The lowest air is hot and less dense; density can increase upward through this shallow layer.	Object appears lower than actual. Pool of water on road. "Water" on hot desert.
Superior (Upper) Mirage	Upper Mirage	Inversion (temperature increases with height). Cold sea surface, cold snow field.	Object appears higher than actual. Levitating ships.

6. Bishop's Ring

Bishop's Ring: A whitish faint ring, centred on the sun or moon, with a slightly bluish tinge on the inside and reddish brown on the outside. Due to **diffraction** of light by **fine dust particles** in the high atmosphere. The dust may also be of volcanic origin. Radius of the ring is about 22°.

Bishop's Ring vs Halo: Bishop's ring is a broad ring: its inner edge is usually 22° to 28° from the sun and the ring is 10° or more wide. The small halo is a thin sharp ring at 22°. Cause and appearance also

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differ. Halo = CS (ice crystals, refraction, bright). Bishop's Ring = fine dust (diffraction, faint, reddish-brown outside).

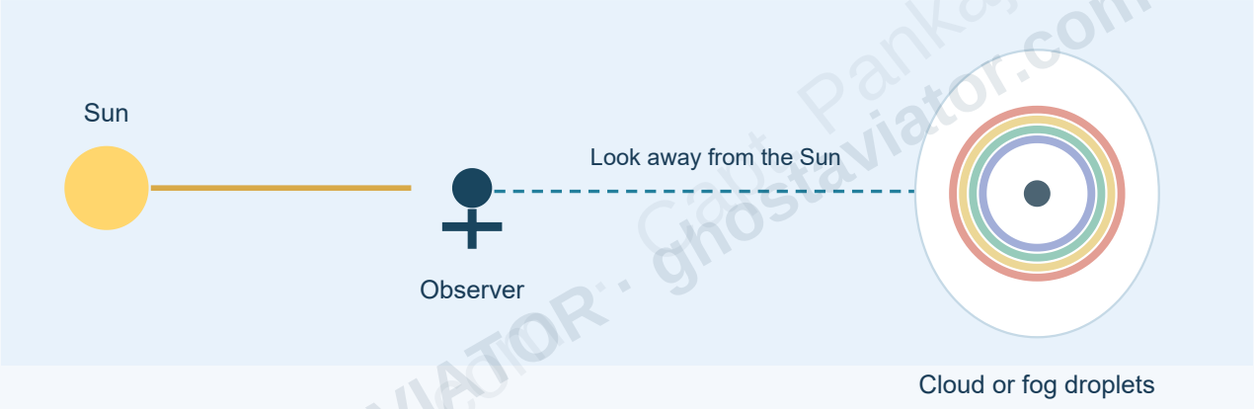
7. Twilight Colours

Various colours in the sky are produced at sunset and at sunrise by **refraction, scattering or selective absorption** of light rays from the sun.

8. Irisation

Sometimes due to **diffraction** of sun light, colours or bands nearly parallel to the margin of the cloud appear, in which pink colour predominates. Occurs in AC lenticularis and other clouds with uniform-sized droplets.

9. Glory



A glory surrounds the antisolar point — opposite the Sun.

An observer's shadow may help locate the centre. Schematic, not to scale.

Figure 10.5. To look for a glory, look towards cloud or fog opposite the Sun. The coloured rings surround the antisolar point, near the observer's shadow when visible.

Glory: Coloured rings centred on the direction opposite the Sun, seen against small water droplets in cloud or fog. Light backscattered by the droplets produces the rings through wave effects. The usual colour order is blue inside and red outside. An airborne observer may see the rings near the aircraft shadow on a cloud below.

10. Crepuscular Rays

Dark bluish streaks which radiate from the sun. They are the **shadows of clouds** at or below the horizon.

11. Aurora

Aurora: (Latin: aurora = dawn). A phenomenon in which visible light is emitted by the high atmosphere, at heights varying from typically about 100–250 km. Aurora are ghostly displays of light in the form of streamers, rays, arches, hands, curtains, draperies, sheets or patches. They appear to shimmer or flit

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across the sky. Aurora are most common in higher latitudes, centering around magnetic poles. They are greenish white or pronounced red or violet.

Feature	Aurora Borealis	Aurora Australis
Name meaning	Northern Lights	Southern Lights
Hemisphere	N hemisphere (generally over NW Greenland)	S hemisphere
Called	Aurora Borealis	Aurora Australis

Cause: Aurora results from bombardment of gases in the rarefied upper atmosphere by electric particles from magnetic storms. These cause excitation of gases with consequent emission of radiation. **They impair radio communication.**

Aurora may be compared with the coloured commercial sign displays in which high voltage electric discharge is passed through tubes containing particular gas under low pressure, from which the colour light characteristic of the gas is then emitted.

12. Atmospheric Electricity and Lightning

In fine weather, the earth is negatively charged and the electric potential is directed towards the earth and has a mean value of 100 volts/m. During a thunderstorm, the direction of current is locally reversed. Atmosphere carries a positive charge towards rain, and air to ground lightning flashes convey predominantly negative charge to ground.

Types of Lightning

Type	Description
Ground Discharge (Thunderbolt)	Lightning between cloud and ground. Usually branches downward from distinct main channel (streak or ribbon lightning). The channel itself is narrow; an illustrative estimate is about 2.5 cm across. A luminous ball observed soon after a ground discharge and disappears with a violent explosion = **ball lightning**. Moves slowly in air/on ground.
Cloud Discharge (Sheet Lightning)	Lightning takes place within the cloud.
Air Discharge	Lightning from cloud to the air — does NOT strike the ground.

13. Saint Elmo's Fire

Saint Elmo's Fire: Sometimes when flying in CU/CB clouds, an aircraft may experience **bluish or greenish luminous discharge** due to strong electrical field in the cloud. The continuous luminous glow is also accompanied by **cracking sound**. Intensity of the glow is **weak to moderate**.

Aviation Warning: Saint Elmo's Fire = strong electrical activity in cloud = in or near CB/CU → precursor to lightning → AVOID the cloud. If experienced, immediately divert.

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14. Practice Q&A

Q1. Aurora Australis occur in the (a) S hemisphere (b) N hemisphere (c) Equator

✅ Answer: (a) S hemisphere

Aurora Australis = Southern Lights = Southern hemisphere. Aurora Borealis = Northern Lights = N hemisphere.

Q2. Aurora Australis are called ..... Lights (a) Northern (b) Southern (c) Temperate

✅ Answer: (b) Southern

Aurora Australis = Southern Lights. Aurora Borealis = Northern Lights.

Q3. Aurora Borealis occur in the (a) S hemisphere (b) N hemisphere (c) Equator

✅ Answer: (b) N hemisphere

Aurora Borealis = Northern Lights, generally over NW Greenland in the N hemisphere.

Q4. Aurora Borealis are called ..... Lights (a) Northern (b) Southern (c) Temperate

✅ Answer: (a) Northern

Q5. Where can a corona form?

A corona can appear when sunlight or moonlight is diffracted by small water droplets or ice particles in a thin cloud, mist or fog. Altocumulus and cirrocumulus are examples. A cloud name alone does not decide whether a corona will be visible.

Q6. Bishop's ring is due to the diffraction of light by fine particles of (a) water (b) dust (c) ice

✅ Answer: (b) dust

Bishop's Ring = diffraction of light by fine dust particles (possibly volcanic) in high atmosphere. Not water drops (that's Corona) or ice crystals (that's Halo).

Q7. The radius of the Bishop's Ring is about (a) 32° (b) 22° (c) 42°

✅ Answer: (b) 22°

Bishop's Ring radius is about 22° — same as Small Halo radius but different cause and appearance.

Q8. Superior Mirage occurs in (a) Lapse (b) Isothermal (c) Inversion

✅ Answer: (c) Inversion

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Superior (Upper) Mirage occurs when there is temperature inversion. Cold sea surface → inversion → light bends downward towards the observer → apparent image is higher than the real object.
Lapse rate steep = inferior mirage. Inversion = superior mirage.

Q9. Inferior Mirage occurs when there is (a) Lapse (b) Isothermal (c) Inversion

✅ Answer: (a) Lapse

Inferior (Lower) Mirage occurs when lapse rate is steep — strongly heated surface, hot road or desert. Surface air is warm and less dense than the cooler air immediately above; light bends upward towards the observer.

Q10. Corona are formed due to ..... of light (a) Refraction (b) Diffraction (c) Scattering

✅ Answer: (b) Diffraction

Corona = diffraction of light by very small water droplets. Halo = refraction by ice crystals. Rainbow = refraction + internal reflection.

Q11. What particles produce a corona?

A corona forms by diffraction of light by small water droplets or ice particles. Thin cloud, mist or fog can provide the particles. This optical effect does not by itself establish temperature or aircraft icing severity.

Q12. Halo is produced by (a) Refraction (b) Diffraction (c) Scattering

✅ Answer: (a) Refraction

Halo = refraction of light through ice crystals in CS cloud. This is a fundamental distinction in DGCA exams.

Q13. Halo is produced when light passes through (a) water particles (b) ice crystals (c) both

✅ Answer: (b) ice crystals

Halo = light refracted by ice crystals in Cirrostratus. Exclusively ice crystals.

Q14. What clouds commonly produce a halo?

Halos form when light interacts with ice crystals, commonly in cirriform clouds such as cirrostratus. A halo indicates ice crystals; it does not prove that the cloud contains no liquid droplets or that it is safe to enter.

Q15. Halo is a luminous ring of ..... radius (a) 22° (b) 32° (c) 42°

✅ Answer: (a) 22°

Small Halo = 22° radius. Large Halo = 46° radius. Primary rainbow = 42°. Secondary rainbow = 52°.

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Q16. Halo round the sun shows a pure clear ..... on the inside (a) Red (b) Yellow (c) Violet

✅ Answer: (a) Red

The red edge of the halo lies closest to the Sun or Moon. Colours grade towards blue on the outer side.

Q17. Halo signifies predominance in the cloud of (a) Supercooled water drops (b) Ice crystals (c) Both

✅ Answer: (b) Ice crystals

A halo indicates ice crystals. The optical display alone does not establish the cloud's liquid-water content or aircraft icing risk.
Halos indicate ice crystals. Coronae arise from diffraction by tiny particles and do not by themselves prove droplet supercooling.

Q18. Can a halo establish icing severity?

No. A halo indicates ice crystals, but an optical display alone does not establish aircraft icing severity or that a cloud is safe to enter. Airframe icing from supercooled droplets and engine icing in suitable ice-crystal conditions are different mechanisms.

Q19. What is the approximate angular radius of the large halo?

Answer: 46°.

The small halo has a radius of about 22° and the large halo about 46°. A nearest numerical option is not a correct answer.

Q20. What produces a halo?

Ice crystals in the atmosphere refract light to produce common circular halos. Cirriform clouds often provide these crystals. The optical observation alone does not establish cloud-entry safety.

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Chapter 11: Precipitation

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1. Types of Precipitation

Precipitation: Liquid water drops or solid water particles falling from clouds to the ground.

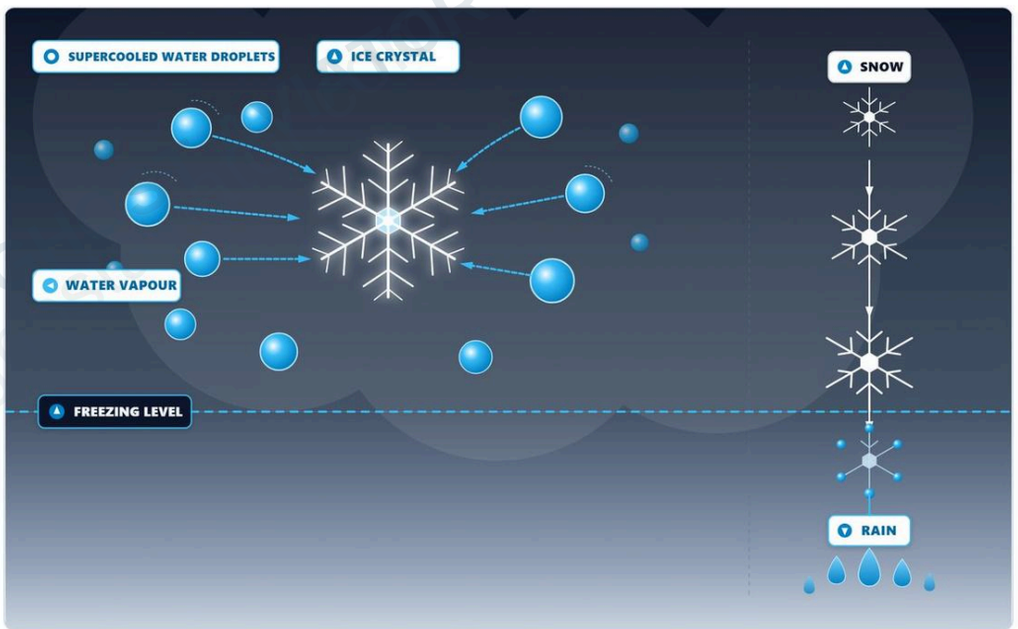
Type	Code	Size / Description
Drizzle	DZ	Very small size, Dia 0.2–0.5 mm

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Type	Code	Size / Description
Rain	RA	Dia 0.5–5 mm
Shower	SH	Not a separate type of precipitation. SH is a descriptor placed before the type (SHRA, SHSN, SHGR, SHGS) to show that it falls as a shower from convective cloud
Hail	GR	Solid balls or pieces of ice (hailstones) with diameters 5–50 mm or more. 1 kg hail has also been reported.
Snow	SN	Assembly of ice crystals, wet or dry snow
Snow grains	SG	Very small, opaque white grains of ice; different from rain mixed with snow.
Ice Pellets	PL	Small transparent ice particles, spherical/irregular, about less than 5 mm dia
Small hail / snow pellets	GS	Used when the largest hailstones are less than 5 mm across (GR is used for 5 mm or more)
Ice Crystals	IC	Also precipitation (diamond dust), but IC is not in the present-weather list of the current Indian requirements

2. Theories of Precipitation Formation

For precipitation to occur, cloud particles must grow large enough to overcome vertical currents inside the clouds. The exact process is not yet fully known. Three theories:



Ice crystals grow at the expense of supercooled droplets

Bergeron Ice Crystal Theory (Cold Cloud Process)

In **Cold clouds** (tops extend well above freezing level), super cooled water and ice particles co-exist. Saturation (water-vapour pressure has reached its saturation value at the current temperature) vapour pressure is more over water drops than over ice crystals → water drops evaporate and sublimate (direct conversion to ice without going through liquid phase) over the ice crystals → ice crystals grow at expense

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of water drops → while falling, bigger ice crystals encounter supercooled water (liquid water below 0°C; droplets can freeze on an aircraft) drops which continually freeze onto them → ice crystals grow to large sizes and fall out of cloud base as snow or rain.

Coalescence Theory (Warm Cloud Process)

The above Bergeron theory is unable to explain rainfall from **Warm clouds** (do not reach freezing level). In tropical areas, showers occur from clouds which do not extend much above the freezing level and sometimes not even up to the freezing level. The coalescence theory: large drops initially form by collision and coalescence of the smaller droplets in the cloud. Coalescence increases markedly due to vertical currents → drops become larger → as large drops fall, pressure falls in their wake → smaller drops sucked in and get attached to falling drops → during fall a large drop collides with the smaller drops on its path and also get attached. Ultimately number of large drops form.

Giant Nucleus Theory

In maritime areas, clouds with lesser vertical development can give rain due to presence of **large number of salt particles** from salt spray. These giant hygroscopic nuclei are carried up to the cloud in vertical currents. They absorb water vapour and form large water drops and initiate the chain reaction of coalescence.

3. Cloud Types and Precipitation

Type of Cloud	Precipitation Type
Stratus (ST)	Drizzle
Altostratus (AS)	Rain and Snow
Nimbostratus (NS)	Continuous precipitation (rain/snow)
Towering Cumulus (TCU) and Cumulonimbus (CB)	Rain, Showers, Hail and Snow
Cumulus (CU) of slight vertical development (fair weather), Altocumulus (AC) and high clouds (CI, CC)	No precipitation
Thick Cirrostratus (CS)	Snow at high to medium levels

Quick Cloud-Precip Table:
ST → Drizzle | AS → Rain + Snow | NS → Continuous | CB/TCU → RA SH GR SN | CS → Snow (high level) | CU (fair wx)/AC/CI/CC → None

4. Snow and Sleet

Snow and Sleet Conditions:
When surface temperature is less than about 4°C AND mean temperature between the base of cloud and the surface level is less than 0°C → precipitation may be in the form of snow or sleet.

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Snowfall is usually in the form of flakes = mixtures of various shapes of minute ice-crystals.
Sleet = rain mixed with snow in UK usage; the term has no internationally agreed definition.

5. Cloud Burst and Flash Floods

Cloud Burst: Very heavy showers or rain over an area in a short period. A sudden rise in the level of rivers or streams causing floods is called **Flash Floods**.

Aviation Warning: Cloud burst associated with intense CB activity. Flash floods may affect low-lying aerodromes and approach paths. Extreme downdrafts and microbursts associated with intense convective systems.

6. Rainy Day

Rainy Day: When the rainfall amount in a day is 2.5 mm or more, it is called a Rainy Day.

7. Diurnal and Seasonal Variation of Rainfall

Tropical Regions

Surface heating favours daytime convection (heat transfer through movement within a fluid) and air-mass thunderstorms over land. Rainfall timing also depends on the weather system and location: frontal and advection (horizontal transport by moving air) thunderstorms can occur by day or night over land or sea. An afternoon maximum is not a universal rule for all tropical rainfall.

Temperate Latitudes

Clouds form due to **convergence caused by depressions or fronts**, frequent in winters. Therefore, rain or snow is maximum in **winters**. India partially follows this pattern — WDs (which are occluded fronts) give high precipitation in winters.

Seasonal Pattern in India

Region/Effect	Maximum Rainfall Time
Daytime convection over land	Often favoured by surface heating; other rainfall systems have different timing
Coastal areas (sea breeze)	Night or early morning
NE India (Katabatic/Anabatic)	Night or early morning
J&K and Western Himalayas	Winter (due to WDs)
Temperate latitudes	Winters (frontal systems)
Post-monsoon/coastal India	After summer

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8. Artificial Rain Making / Cloud Seeding

Cloud seeding: Introducing suitable particles into an existing cloud to try to change precipitation. Hygroscopic particles (particles that attract water) support droplet growth; glaciogenic agents (agents that encourage ice formation) act in suitable supercooled clouds. Results depend on the cloud and weather conditions; rain is not guaranteed within a fixed time.

Seeding must match the cloud: salt particles are used to encourage liquid droplets to grow, while ice-forming agents are used in clouds containing supercooled water. Possible effects downwind remain an area of research; it is not established that a seeded cloud simply takes rainfall away from another area.

Uses of Cloud Seeding

- Fighting fires
- Attempting to enhance precipitation from suitable existing clouds; this cannot be relied on to end a drought.
- Promoting harvests
- Research and operational trials of precipitation enhancement under suitable cloud conditions.
- **Fog dispersal:** Heating or mixing may disperse fog, but can be impractical. Seeding can disperse supercooled fog under suitable conditions; it is not a universal method for clearing every fog.

Cloud seeding: Particles are introduced into a suitable existing cloud to try to alter precipitation. Hygroscopic agents encourage droplet growth; glaciogenic agents encourage ice formation in supercooled cloud. Results depend on the cloud and weather conditions.

9. Classification of Rainfall

Classification (IMD)	Rainfall in 24 hours
Very Light	trace to 2.4 mm
Light	2.5–15.5 mm
Moderate	15.6–64.4 mm
Heavy	64.5–115.5 mm
Very Heavy	115.6–204.4 mm
Extremely Heavy	≥ 204.5 mm

Fog dispersal: Seeding is useful for supercooled fog under suitable conditions. Heating or mechanical mixing can also disperse fog, although practical and cost limits apply.

10. Practice Q&A

Q1. When supercooled water drops and ice crystals coexist, ice crystals can grow at the expense of water drops because:
(a) Saturation vapour pressure over water drops is less than over ice crystals

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(b) Saturation vapour pressure over ice crystals is less than over water drops
(c) Ice crystals convert into water drops

Answer: (b). At the same sub-zero temperature, saturation vapour pressure over ice is lower than over liquid water.

This difference allows water vapour to deposit on ice while liquid droplets evaporate. The original distinct choices have been restored.

Q2. Warm and cold clouds

A warm cloud remains above 0°C throughout. Its droplets can grow by collision and coalescence (joining together). A cold cloud extends into air below 0°C and may contain supercooled droplets and ice crystals. Ice growth can contribute to precipitation; collision and coalescence are not excluded merely because part of a cloud is cold.

Q3. The clouds whose tops do not extend to the freezing level are called (a) Warm Clouds (b) Cold Clouds (c) Moderate Clouds

✅ Answer: (a) Warm Clouds

Warm clouds don't reach freezing level. Rain explained by Coalescence theory — large drops by collision of droplets.

Q4. Coalescence Theory explains occurrence of rainfall from (a) Warm Clouds (b) Cold Clouds (c) Both types of Clouds

✅ Answer: (a) Warm Clouds

Coalescence theory was advanced specifically to explain rainfall from warm clouds (below freezing level) in tropical areas where Bergeron theory fails.

Q5. Ice crystal Theory explains occurrence of rainfall from (a) Warm Clouds (b) Cold Clouds (c) Both types of Clouds

✅ Answer: (b) Cold Clouds

Bergeron Ice Crystal Theory (cold cloud process) applies to clouds with tops above the freezing level where ice and supercooled water co-exist.

Q6. Giant Nucleus Theory explains occurrence of rainfall over (a) Maritime areas (b) Inland areas (c) Hilly areas

✅ Answer: (a) Maritime areas

Giant Nucleus Theory applies to maritime (ocean/coastal) areas where sea salt provides large hygroscopic nuclei that initiate coalescence even in clouds with lesser vertical development.

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Q7. Orographic precipitation

Orographic means caused by air rising over high ground. Moist air forced up a mountain slope cools and may form cloud and precipitation. Stable air can form layered cloud; unstable air can form cumulus or cumulonimbus. Mountain rainfall is not necessarily a very heavy shower.

Q8. Rain shadow area is on the ..... of the mountain range (a) Top (b) Windward side (c) Leeward side

✅ Answer: (c) Leeward side

The leeward (downwind) side of a mountain range receives much less rainfall — called the rain shadow area. Air descends on leeward → warms → dries out.

Q9. Meaning of sleet

In the UK terminology used here, sleet is rain mixed with snow. The term has no internationally agreed meaning, so a definition should specify the reporting convention rather than assume that every country uses the word identically.

Q10. Rainfall in the tropics is more in (a) Winters (b) Summers (c) Post monsoon

✅ Answer: (b) Summers / Afternoons

In tropics, convection (thermal) is most vigorous in summers, particularly in afternoons. Maximum tropical rainfall in afternoon.

Q11. Daytime convection over land

Surface heating over land favours daytime development of air-mass thunderstorms. This does not establish a universal afternoon maximum for all tropical rainfall. Frontal or advection thunderstorms can occur by day or night, over land or sea.

Q12. Rainfall in the temperate latitudes is more in (a) Winters (b) Summers (c) Spring

✅ Answer: (a) Winters

In temperate latitudes, frontal/convergence systems are frequent in winters → maximum rain/snow in winters.

Q13. Over J&K and western Himalayas Rainfall is more in (a) Winters (b) Summers (c) Post monsoon

✅ Answer: (a) Winters

J&K and western Himalayas get maximum rainfall in winters due to Western Disturbances (WDs) — which are occluded frontal systems from the Mediterranean region.

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Q14. Rainfall over coastal areas is more in (a) Evening (b) Night & early morning (c) Afternoon

✅ Answer: (b) Night & early morning

At coastal areas the land breeze blows from land to sea at night and converges offshore with the moist sea air → maximum rain at night/early morning. By day the sea breeze carries the cloud inland.

Q15. Areas to the ..... of Western Ghats of India are rain shadow areas (a) W (b) S (c) E

✅ Answer: (c) E

Western Ghats force SW monsoon air upward on the western (windward) slopes → heavy rainfall. Eastern side = leeward = rain shadow (Deccan Plateau is drier).

Q16. A sudden rise in the level of rivers or streams causing floods is called (a) Cloud Burst (b) Catchments flooding (c) Flash Floods

✅ Answer: (c) Flash Floods

Cloud Burst = very heavy rain in short period. Flash Floods = sudden rise in river/stream levels caused by cloud burst.

Q17. Artificial rain making is also termed as (a) Simulation (b) Cloud seeding (c) Nucleation

✅ Answer: (b) Cloud seeding

Artificial rain-making is called cloud seeding: freezing nuclei such as silver iodide, or dry ice, are introduced into a cloud to start the Bergeron process.

Q18. Fog can also be dispersed for a short period by artificial stimulation (a) True (b) False

✅ Answer: (a) True

Cloud seeding techniques can disperse fog temporarily, either by heating or seeding — to enable aircraft to land and take off. This is used at some airports for CAT III operations improvement.

Q19. Showery precipitation occurs from (a) NS (b) AC (c) CB

✅ Answer: (c) CB

Showers (SH) = intermittent, heavy precipitation from CB. NS gives continuous rain, not showers. AC generally gives no precipitation.

Q20. A day is called Rainy day when rainfall in 24 hr is ..... mm or more (a) 1.5 (b) 2 (c) 2.5

✅ Answer: (c) 2.5

A Rainy Day = rainfall ≥ 2.5 mm in 24 hours. This is the standard meteorological definition used in India.

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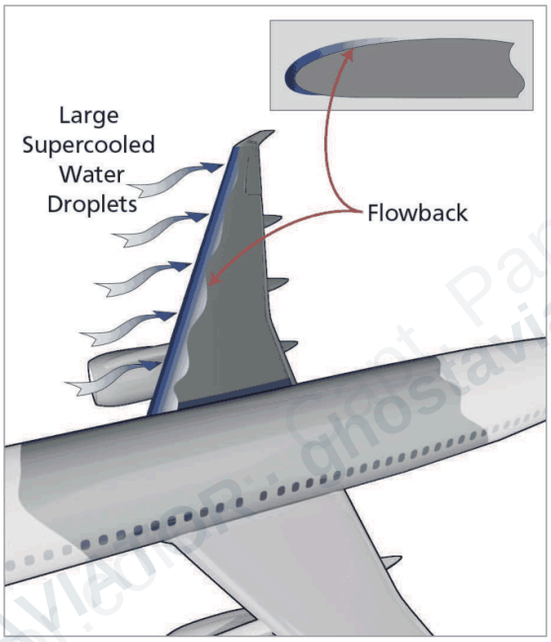
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Chapter 12: Ice Accretion

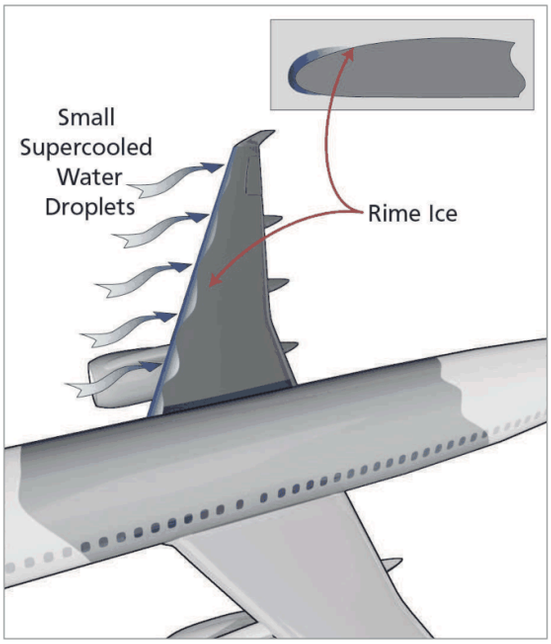
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1. Types of Airframe Ice

Definition — Ice Accretion: The accumulation of ice on airframe surfaces, engine intakes, and instruments due to contact with supercooled water (liquid water below 0°C; droplets can freeze on an aircraft) droplets or water vapour in sub-freezing conditions.



Clear ice formation



Rime ice formation

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Rime Ice

Rime ice forms by the **instantaneous freezing** of small supercooled water droplets on impact. Air is trapped between droplets → *opaque, milky-white, rough* appearance. It is **light and porous** and builds up on leading edges and into the airstream. Less structurally dangerous than glaze but disrupts airflow significantly.

Glaze Ice (Clear Ice / Translucent Rime)

Also called **clear ice** because of its glassy appearance. Forms from **large supercooled water droplets** that spread backwards after impact before freezing. Very little air is trapped → *clear, hard, dense, heavy*. Deposits on leading edges, struts, and propeller blades. Initial deposit may have flat surface; subsequent deposits build up in lumps.

⚠ **Most hazardous type of icing.** Glaze ice:

- Is **sticky** — cannot be easily shaken off; if it breaks it comes away in dangerous lumps.
- Alters aerodynamics and is **heavy** — poses serious aviation hazard.
- May set up **vibrations** due to unequal loading of wings, struts, and propeller blades.
- Comparable to ice cubes in a refrigerator — smooth, hard, transparent.

Hoar Frost

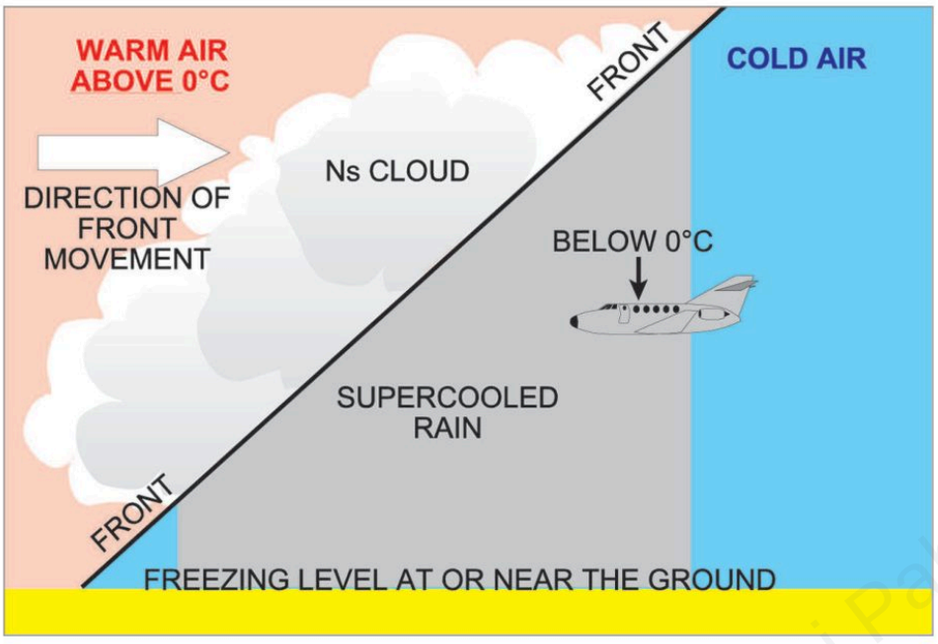
Occurs in **moist, cloudless air** when aircraft surface temperature falls below **0°C** due to **sublimation of water vapour** onto feathery ice crystals. Also forms when a cold-soaked aircraft descends into warmer moist air. It gives a white crystalline coating and is generally less hazardous in flight but can be dangerous on takeoff (disrupts laminar flow).

2. Temperature Range for Airframe Icing

Icing severity cannot be read from a temperature band alone. Assess supercooled liquid water, droplet size, aircraft characteristics and exposure.

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Freezing rain

Factor	Effect on icing
Supercooled liquid water	More available liquid water can increase ice accumulation.
Droplet size	Larger drops may spread and freeze beyond protected areas.
Temperature	Influences freezing and ice type; temperature alone does not determine severity.
Aircraft and exposure	Design, airspeed and time in the conditions affect the result.

Remember: Temperature influences icing, but it does not set the severity by itself. Most structural icing occurs between 0°C and −20°C. Strong cumulonimbus (a deep cloud associated with thunderstorms) updraughts can carry supercooled liquid water to much colder levels, approaching −40°C. High-altitude ice crystals can also threaten turbine engines.

3. Airframe Icing in Relation to Cloud Forms

Ice accretion is dependent primarily on the **size** and **number** of drops in a cloud.

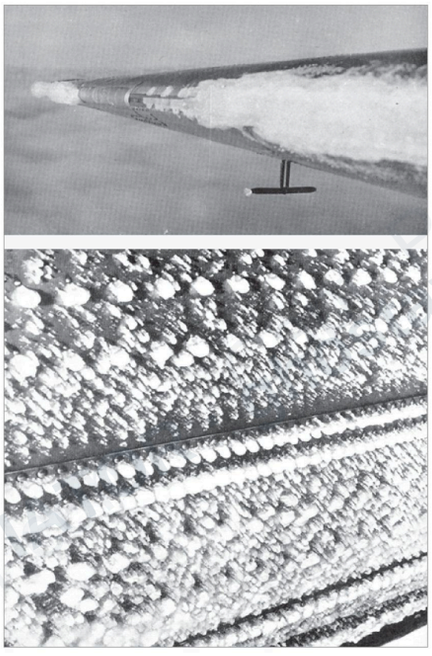
Cloud or precipitation	Teaching point
Cirrus family	Usually dominated by ice crystals, so classical structural icing is less likely. This does not make every high-level encounter harmless.
Layered cloud	Supercooled water can cause structural icing. Severity varies with water content, drop size and aircraft exposure.
Towering cumulus and cumulonimbus	Strong updraughts can sustain supercooled water far above the freezing level, sometimes approaching −40°C. Severe icing is possible.
Freezing rain or drizzle	Large supercooled drops can produce rapid and extensive icing.

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4. Effects of Airframe Icing

⚠ **Airframe icing affects flying characteristics in multiple ways:**

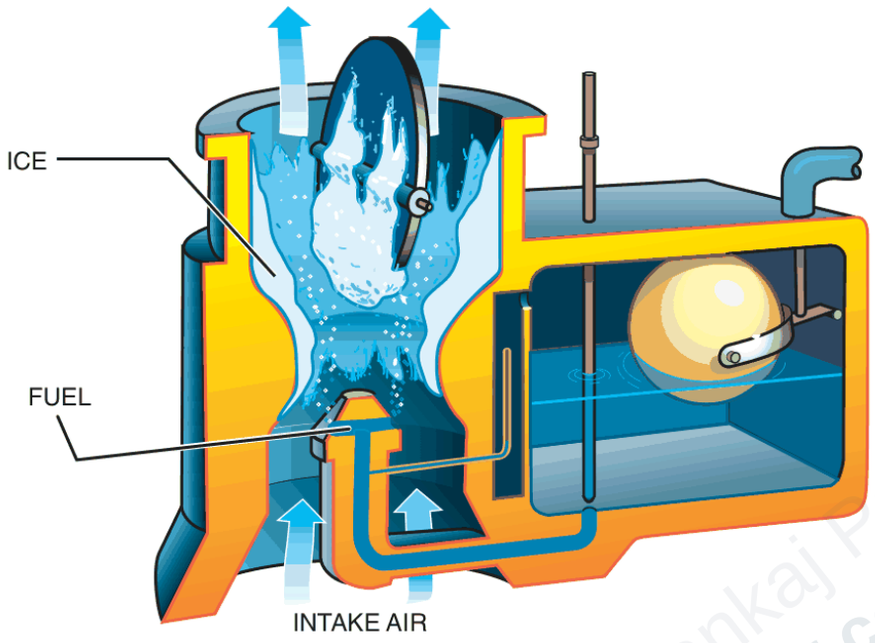
1. Alters **aerodynamics**, increases weight, and **decreases lift**
2. Increases the **stalling speed** appreciably
3. **Increases drag**
4. Unequal loading on propeller blades → **decreases efficiency** and effective power → vibrations
5. Hinges of ailerons, elevator and trimming tabs may get **jammed**
6. **ASI** may give erroneous readings due to icing of pitot tube
7. Icing on aerials may render **communication difficult**



Ice build-up on an aircraft

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5. Engine Icing



Carburettor icing

Impact Icing

Occurs due to the **impact of supercooled water drops on the air intake**. Icing restricts air inflow, which reduces engine power.

Carburetor Icing

Carburettor ice can form as the pressure drop through the venturi and evaporation (liquid water changes into vapour and absorbs latent heat) of fuel cool the incoming air. It is possible in clear air and at warm outside temperatures. Reports extend to about 38°C with relative humidity (actual moisture compared with saturation at the current temperature, expressed as a percentage) as low as 50%; conditions below about 21°C with humidity above 80% are especially favourable. These are guidance values, not safe boundaries. Follow the aircraft handbook for recognition and carburettor-heat procedures.

Remember: Carburettor icing is not limited to cold weather. Pressure reduction and fuel evaporation can cool the intake below freezing even when the outside air is warm.

6. Height of Freezing Level in India

In the North, Central, and Southern parts of India the Freezing Level occurs approximately at the pressure levels in various seasons:

Season	North	Central	South
NE Monsoon	700 hPa	600 hPa	600 hPa
Pre-Monsoon	650 hPa	600 hPa	550 hPa

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Season	North	Central	South
SW Monsoon	550 hPa	500 hPa	500 hPa
Post Monsoon	550 hPa	600 hPa	650 hPa

🧠 **Key pattern:** During SW Monsoon the freezing level is *highest* (lowest pressure = highest altitude) — especially in Central and South India at 500 hPa. During NE Monsoon the freezing level is *lowest* in the North at 700 hPa (lowest altitude, most likely to affect lower-level operations).

8. Practice Q&A

Q1. Hoar frost occurs on airframe in clear air when the temperature of airframe is:
(a) below the frost point (b) frost point (c) just above the frost point

✅ **Answer: (a) below the frost point**
Hoar frost forms by sublimation — water vapour deposits directly as ice when surface temperature is BELOW the frost point (dew point in sub-freezing conditions).

❌ (b) At the frost point — would just be at threshold, deposition is active below it. ❌ (c) Just above — no sublimation occurs above the frost point.

Q2. In clouds at temperatures below 0°C an aircraft may encounter icing of the type:
(a) only Glazed (b) only Rime (c) intermediate between these two

✅ **Answer: (c) intermediate between these two**
In clouds below 0°C, a mix of supercooled water drop sizes and temperatures produces mixed ice — a combination of rime and glaze characteristics.

❌ (a) Only glazed — requires large supercooled drops specifically. ❌ (b) Only rime — requires small drops and very fast freezing.

Q3. Opaque Rime ice is:
(a) Light porous (b) Solid (c) Mixture of porous and solid

✅ **Answer: (a) Light porous**
Rime ice traps air between small droplets during instantaneous freezing → opaque, rough, light, and porous structure.

❌ (b) Solid — describes glaze/clear ice. ❌ (c) Mixture — describes mixed ice.

Q4. Rime is formed by freezing of …… supercooled water droplets on airframe when aircraft is flying through clouds:
(a) Small (b) Large (c) Medium

✅ **Answer: (a) Small**
Small droplets freeze instantly on contact → rime ice. Large droplets spread before freezing → glaze ice.

❌ (b) Large — produces glaze/clear ice. ❌ (c) Medium — produces intermediate/mixed ice.

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Q5. Glazed ice is formed by freezing of …… supercooled water droplets on airframe when aircraft is flying through clouds:
(a) Small (b) Large (c) Medium

✅ **Answer: (b) Large**
Large supercooled drops spread over the surface before freezing → clear/glaze ice.

❌ (a) Small → rime. ❌ (c) Medium → mixed.

Q6. The …… ice poses serious aviation hazard:
(a) Rime (b) Hoar Frost (c) Glazed

✅ **Answer: (c) Glazed**
Glaze/clear ice is the most hazardous — heavy, hard, alters aerodynamics, hard to remove, can break into dangerous lumps.

❌ (a) Rime — less hazardous, lighter, rough but less structurally damaging. ❌ (b) Hoar frost — mostly a takeoff/ground hazard.

Q7. Airframe icing occurs below 0°C. Its probability of occurrence decreases progressively below −20°C as at lower temperatures the proportion of supercooled water drops in a cloud:
(a) Increases (b) Decreases (c) Does not change

✅ **Answer: (b) Decreases**
Below −20°C, supercooled liquid water content drops sharply as more droplets spontaneously freeze into ice crystals → less liquid water → less icing.

❌ (a) Increases — opposite is true. ❌ (c) Does not change — incorrect.

Q8. Ice crystals and aircraft icing

Ice-crystal clouds usually cause little classical structural icing because their particles are already frozen. High ice water content near thunderstorms can still threaten turbine engines. A low structural-icing risk is not a general guarantee of safe flight.

Q9. Icing and cloud type

Cloud type helps assess icing, but it does not give a guaranteed intensity. Nimbostratus can produce moderate to severe icing; altostratus and altocumulus often have lighter icing. Stratocumulus is commonly light to moderate but can give severe icing in mountainous conditions. Assess the actual liquid-water content, temperature and droplets.

Q10. Icing and cloud type

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Q11. Icing in towering cumulus

Towering cumulus can contain abundant supercooled droplets and produce significant icing. Droplet size and liquid-water concentration influence severity. Droplets tend to become smaller at colder levels, but −20°C is not a universal boundary beyond which a convective cloud becomes safe.

Q12. Concept check

Cumulonimbus icing cannot be ruled out by one temperature threshold. The amount and size of supercooled droplets affect the icing risk. Droplets generally become smaller at colder levels, but this is a trend, not permission to enter a thunderstorm.

Q13. Concept check

Icing severity depends on droplet size and liquid-water concentration. A single temperature does not identify a universal peak for every cloud. Cloud type and the vertical distribution of liquid water must also be considered.

Q14. Carburettor icing in warm weather

Fuel evaporation and pressure reduction in the venturi cool the incoming mixture. This allows carburettor ice to form even when the outside air is above freezing, especially with abundant moisture. Published risk ranges describe conditions, not a universal safe boundary. Follow the aircraft-specific guidance for prevention and response.

Q15. …… occurs in a moist cloudless air on an aircraft surface having temp. below 0°C, due to sublimation of water vapour onto feathery ice crystals:
(a) Rime (b) Glazed (c) Hoar Frost

✅ **Answer: (c) Hoar Frost**
Hoar frost = sublimation of water vapour directly onto the cold surface as feathery ice crystals in clear air below 0°C.

❌ (a) Rime — forms in cloud, not cloudless air. ❌ (b) Glaze — requires liquid drops.

Q16. …… occurs in St, Sc, Ac, Cu, Ns at temperature −10 to −40°C and in Cb at temperature −20 to −40°C:
(a) Rime (b) Glazed (c) Hoar Frost

✅ **Answer: (a) Rime**
Rime ice is associated with these cloud types in the temperature range −10°C to −40°C (St, Sc, Ac, Cu, Ns) and −20°C to −40°C (Cb).

❌ (b) Glaze — found closer to 0°C with large drops. ❌ (c) Hoar Frost — requires cloudless air.

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Q17. Mixed icing

Mixed ice combines features of rime and clear ice. Small supercooled droplets freeze rapidly near the impact point, while larger droplets may spread before freezing. A range of droplet sizes can therefore produce both forms together; the ice type is not determined by one temperature limit alone.

Q18. Concept check

Keep two mechanisms separate. Cloud icing begins when supercooled liquid droplets strike the aircraft; their size and freezing behaviour influence the ice type. Hoar frost forms from water vapour depositing directly as crystals on a cold surface in clear, moist air. A cold-soaked aircraft entering a warmer moist layer may develop frost; this does not automatically mean mixed icing.

Q19. Fog droplets and hoar frost

Small supercooled fog droplets can freeze rapidly on a surface to form rime. Hoar frost is a different process: water vapour deposits directly as ice crystals on a cold surface. Identify whether liquid droplets or vapour produced the deposit.

Q20. Icing …… the stalling speed appreciably:
(a) Decreases (b) Increases (c) Does not increase/decrease

✅ **Answer: (b) Increases**
Ice deposits on wings alter the aerofoil shape, reduce lift generation, and increase the minimum speed needed to maintain lift — thus stall speed increases.

❌ (a) Decreases — opposite; icing degrades aerodynamic efficiency. ❌ (c) No effect — incorrect; icing always degrades performance.

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Chapter 13: Thunderstorms

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1. Definition & Basic Facts

Thunderstorm (TS): A weather system produced by strong convection (heat transfer through movement within a fluid) currents in CB clouds. Characterised by lightning flashes, rumbling thunder, blackening of sky, sharp showers, squally winds, and at times hail, snow, sleet, or warm winds. Some TS cause floods, forest fires, tornadoes, waterspouts, and microbursts.

For Aviation: TS is also known as Electrical storm, Lightning storm, or Thundershower. It is reported when thunder is heard. Under suitable conditions, TS may line up = **squall line** or **rain band**.

⚠ **TS is one of the most hazardous meteorological phenomena for aviation.**

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- Diameter of individual CB cells: 1 to 10 km
- In a cluster of CB cells: no narrow cloud-filled lanes between neighbouring cells
- **Severe TS:** accompanied by locally damaging winds, frequent lightning, or large hail

2. Types of Thunderstorms

Type	Characteristics
Heat (Air Mass (a broad body of air with similar temperature and moisture)) TS	Intense heating and convergence due to low pressure. Also occur when cold air passes over a warmer sea. Pop up randomly in warm humid air. Relatively weak, mostly develop in afternoons on plains; nights in valleys and hills.
Frontal TS	Occur at Cold Front (occasionally warm front). Triggered by vigorous uplift of moist air along/ahead of front. More violent than air mass TS. May develop as Squall Line along/ahead of cold front → causes severe weather.
Steady State TS	Associated with Fronts, Converging Winds, and Troughs aloft. Intensified by afternoon heating. Precipitation falls outside updraughts → become stronger in mature stage. Last several hours.
Mesoscale Convective Complex (MCC)	Nearly circular cluster of many interacting TS covering a very large area (cold cloud shield of about 100,000 sq km or more), far larger than a single air mass TS. Thunderstorms keep developing within an MCC. Life: 6 to 24 hr. Produce severe weather, Tornadoes, Flash Floods (sudden rise in rivers/streams). Form near fronts and in tropics in ITCZ.

🧠 **Meso-Scale Weather examples:** TS, DS, Land and Sea breeze, Katabatic wind, etc. = local scale phenomena.

3. Classification by Cell Structure

Type	Structure	Key Features
Single Cell	One main updraught	Rarely severe. Local instability/heating in summers (Heat/Air mass TS). Also in cold front in winters.
Multi Cell Cluster	Cluster of storms due to convective updraughts in/near mountain ranges and strong cold fronts or troughs	Each cell lasts 20 min; the cluster may last hours.
Multi Cell Lines (Squall Lines)	Hundreds of km long; move swiftly along/ahead of cold fronts	Heavy rain, hail, lightning, strong winds, even isolated tornadoes. Unusually powerful = **derecho**. Also near tropical cyclone outer rain band.
Supercell	A long-lived thunderstorm with a persistent rotating updraught called a mesocyclone. Strong vertical wind shear (wind speed or direction changing over distance) supports its organisation.	Can produce damaging wind, large hail and tornadoes; its size and severity vary.

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4. Favourable Conditions & Triggering Mechanisms

Three essential conditions for TS formation:

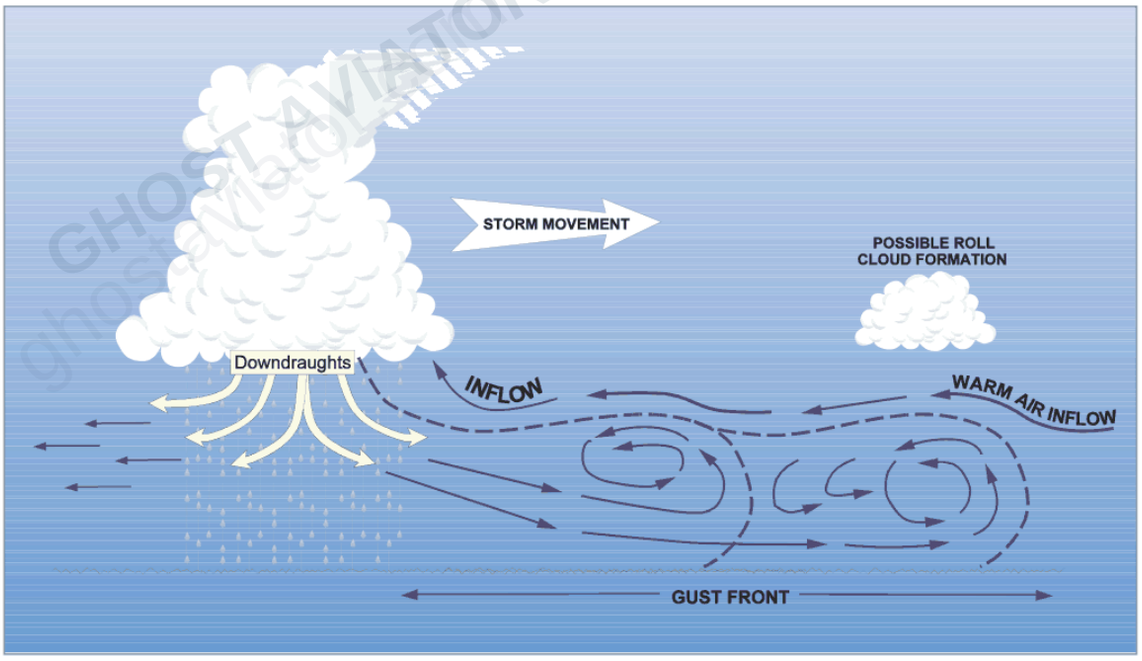
1. **Steep Lapse Rate:** Steeper than SALR throughout a layer at least 5–6 km in depth, permitting development of clouds above 0°C level.
2. **High Humidity:** Adequate supply of moisture from below; high humidity aloft. If humidity aloft is low → growth of cloud is arrested due to evaporation (liquid water changes into vapour and absorbs latent heat) of rising parcel.
3. **Trigger Action:** A lifting mechanism which can produce saturation (water-vapour pressure has reached its saturation value at the current temperature) in the region of the steep lapse rate (temperature decrease with increasing height).

Triggering Mechanisms:

- (a) Insolation / Local Convection
- (b) Frontal Lifting
- (c) Convergence
- (d) Orographic Lifting
- (e) Radiational or Katabatic Cooling

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5. Life Cycle of a TS



Gust front and roll cloud

Stage 1: Cumulus Stage

Trigger lifts air → expands and cools → condensation (water vapour changes into liquid water and releases latent heat) → latent heat (energy absorbed or released during a change of state without a temperature change) released → air warmer than surroundings → low pressure develops → **inflow (entrainment)** takes place through sides at all levels and base of cloud.

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Entrainment: The mixing of environmental air into a cloud. Parcel rises higher and higher as long as it is warmer than surroundings → CU building up into a CB cloud.

Updraught speeds of 30 m/sec or 60 kt are not uncommon.

Rain/snow/hail particles grow inside cloud → fall as precipitation when too large/heavy to be sustained by updraughts.

Aviation hazards from this stage: turbulence and icing.

Stage 2: Mature Stage

Warmed air continues to rise up to tropopause (which is the cap) → forced to spread out as cirrus clouds giving the anvil shape = **False Cirrus** (also called False Cirrus).

Water droplets coalesce into larger → ice particles. Falling precipitation drags adjacent air → **downdraught**. Entrained drier air cools downdraught further and accelerates it. **Both up and downdraughts coexist.**

At ground level: sudden rush of **very strong cold winds** = **squall**.
Downdraught close to ground spreads horizontally → leading edge = **First Gust / Gust Front** (arc-shaped surface).

May be followed by heavy rain/showers. Gust front undercuts warm air at surfaces → generates new TS cells.

Duration in the simple single-cell teaching model: about 15–20 min; organised storms can persist much longer. Lightning and considerable **turbulence** within and around cloud.

Mature stage = production of precipitation + coexistence of up and downdraughts side by side.

⚠ **Mature stage presents serious aviation weather hazards.** This is the most violent stage of a TS.

Stage 3: Dissipating Stage

Updraught ceases → storm dominated by downdraughts. Precipitation may still occur but decreases as moisture is depleted. Downdraughts spread across the lower portion of CB → cloud gradually dissipates.

If little vertical WS: storm rapidly enters dissipating stage.
If sufficient vertical WS: storm may become a super cell — mature stage can sustain itself for several hours; tornadoes may develop.

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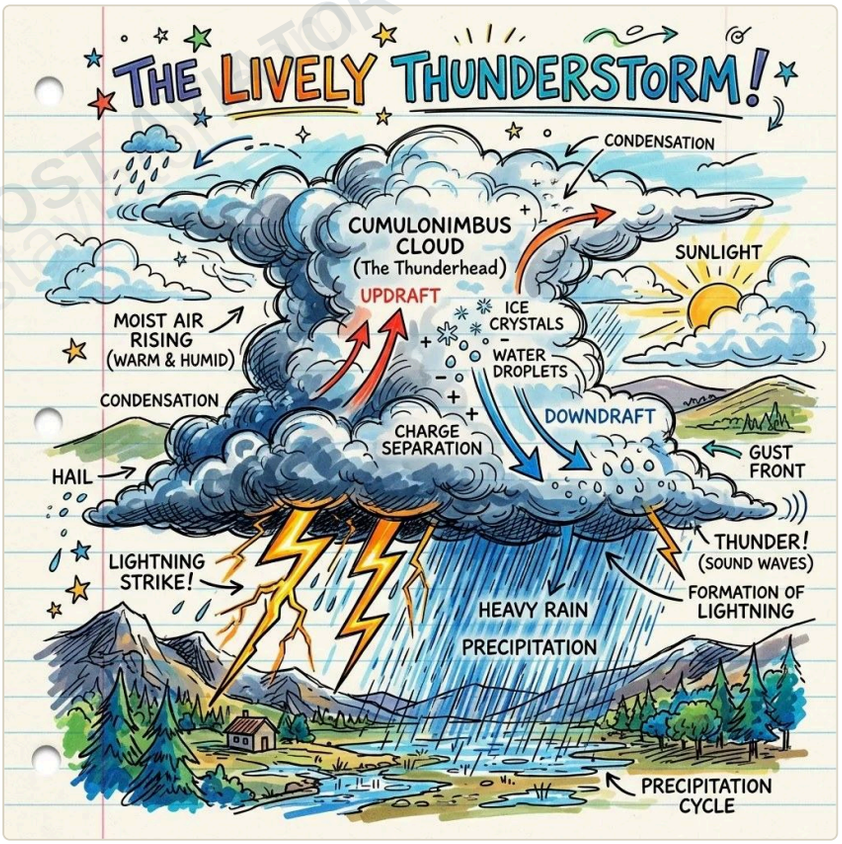
6. Associated Clouds & Features

Feature	Description
Anvil / False Cirrus	Cirrus cloud at tropopause level; air forced to spread out horizontally. Indicates mature/dissipating stage.
Roll Cloud	Elongated cylindrical dark cloud appearing to rotate slowly about horizontal axis. Occurs *behind* the First Gust, *detached* from CB. Seldom associated with severe weather.
Shelf Cloud	Wedge-shaped elongated cloud with flat base. At edge of Gust Front, *attached* to CB. Due to uplift of warm air along Gust Front. Associated with **Severe weather**.
Gust Front / First Gust	Arc-shaped leading edge of cold downdraught spreading along the ground. Causes sudden pressure rise. Undercuts warm air → new TS cells generated.

🧠 **Roll vs Shelf Cloud:** Roll = Detached + Behind gust = less severe. Shelf = Attached to CB + At gust front edge = SEVERE weather. Remember: "Shelf = Serious."

Structure of Severe TS

Severe TS are **tilted** due to vertical WS. Because of tilt, precipitation and downdraughts fall *away* from the updrafts → updraughts cannot be cut off → cell grows to great heights and lasts longer. The **700–500 hPa** level winds influence movement of TS. Upper level Jet Stream produces wind shear, tilts updrafts → favours great vertical development.



Inside a thunderstorm: updraft and downdraft side by side, charge separation, lightning, hail, heavy rain and the gust front.

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7. Intensity, Diurnal & Seasonal Variation

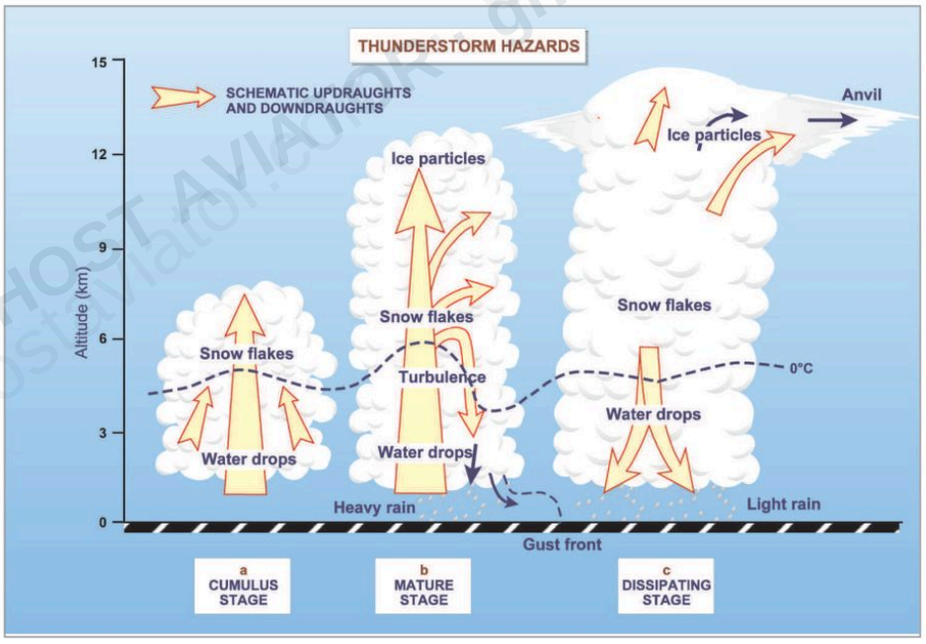
Intensity	Thunder & Lightning	Precipitation	Max Wind
Light	Faint peals at long intervals	Light showers	–
Moderate	Loud peals, frequent flashes	Moderate or heavy showers, light hail	15–40 kt
Severe	Almost continuous	Heavy rain/shower, may be hail	>40 kt

Diurnal Variation:

- Over plains: mostly **afternoon**, die out in evening
- Valley and foot hills: generally **night and early morning**
- Over sea: more frequent at **night**
- Middle latitudes: over land most frequent in **summers**; frontal TS more frequent in **winters** (frequent Cold Fronts)

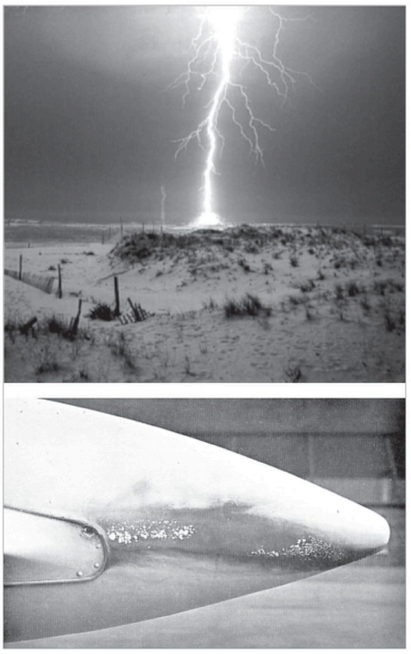
8. Aviation Hazards

⚠ **TS pose a variety of hazards to aircraft on ground and in flight:**



Thunderstorm hazards

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Lightning

(a) Turbulence

Most serious hazard of a TS. Sources: gusts, eddies, up and downdraughts inside CB; large amount of air entrained from outside → strong upcurrents in clear air around CB.

- Moderate to severe turbulence: **inside CB** (at boundary between up and downdraughts)
- Clear air turbulence: 10–20 km around CB in clear air
- Downwind: 25–30 km downwind
- Above CB: **thousands of feet above CB top** (especially when winds are very strong)
- Below cloud base: down to surface — makes landing/takeoff dangerous

Turbulence increases gust load which may exceed stress limit of aircraft. Attempts to regain control → structural failure may occur.

Turbulence Level	Effect Inside Aircraft
Light	Feel slight strain on seat belt; unsecured objects may be displaced slightly; food service can continue and there is little or no difficulty in walking
Moderate	Feel definite strain on seat belt; unsecured objects dislodged; food service and walking are difficult
Severe	Forced violently against seat belt; objects tossed around; walking and food serving are impossible

(b) Wind Shear (WS)

Strong WS often associated with TS. Outflow of TS can cause extreme changes in wind speed and direction near surface during critical phases of flight.

- Cold strong downdraught from CB strikes the surface → rises up to 2000 ft
- WS zone forms between outflow and warm air above (blowing opposite direction toward CB)
- Aircraft initially encounters head wind → then strong horizontal spreading → then tail wind

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- Landing angle may become dangerously steep; accidents during take-off, landing, climb/descent, final approach
- WS hazardous during all critical phases; **strong vertical WS is necessary for severe TS**

(c) Squall

Initial downdraught from cloud spreads all around horizontally on the ground as squall. Speed normally up to 40 kt but can be as high as 100 kt. The strongest speeds are ahead of the direction of movement.

(d) Draughts

Strong up and downdraughts within cloud; greater vertical extent.

- Can suck gliders in; updraughts impose sudden **negative g force**
- Stronger from middle of cloud to the surface; can force even powerful jet aircraft to ground if below cloud base
- Horizontal extent: can be more than 1 km in a TS
- 3–5,000 ft horizontal spreading

(e) Gusts

Near boundary between up and downdraughts, vigorous eddies form. Severe turbulence = combined effect of gusts and draughts. Impose severe load factor. Structural failure may occur if pilot attempts to correct attitude changes too quickly or harshly.

(f) Icing

Always possible in temperature range 0° to −30°C. Affects aerodynamics, jams controls, blocks pitot tube, engine, and carburetor.

- **Glazed clear ice** is worst within first 10,000 ft from the freezing level (large supercooled drops present)
- Towards top: more ice crystals → rime ice
- In piston engine aircraft: carburetor icing may cause loss of power
- Along frontal line: icing may become a serious problem
- Use of **de-icing devices and carburetor heat is essential**

(g) Hail

Hailstone starts as an ice crystal → grows by collision with supercooled water (liquid water below 0°C; droplets can freeze on an aircraft) drops while being tossed up and down in cloud.

- Onion-like structure: alternate layers of rime and clear ice
- Clear ice forms nearer the freezing level; rime ice above
- Worst hail: from freezing level to 25,000 ft
- Hail encounters are rare but if encountered, only in small region of cloud
- Small hail: superficial damage | Large hail: serious damage

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- Damaging hail may occur up to 45,000 ft in CB and under the Anvil

(h) Heavy Showers

May reduce visibility to very low value. Creates thick film of water on runway → hydroplaning and skidding of aircraft. Ingestion of heavy showers in engine → reduces power. Noise disturbs aircrew. Fractostratus clouds below 1000 ft generally accompany heavy showers. Avoid landing and take-off during heavy rain/showers.

(i) Lightning

Lightning: Lightning can damage aircraft skin, electrical equipment and communication or navigation systems. Bonding and protection reduce the risk; they do not make a strike harmless. Volts per metre describe electric field strength, not a voltage difference.

(j–m) Other Hazards

- **(k) Noise:** Hail striking aircraft and windscreen can be very frightening; more so in turbulence
- **(l) Darkness & Disorientation:** Absolute darkness in thick CB → constant eye on Artificial Horizon is essential
- **(m) Instrumental Error:** Below base of growing/mature CB, reduction in pressure → altimeter tends to **over-read** (indicates more height than actual); adequate ground clearance essential. Pitot tube may be blocked by heavy water ingestion → **ASI reading would be misleading**

9. Radar Detection of Thunderstorms

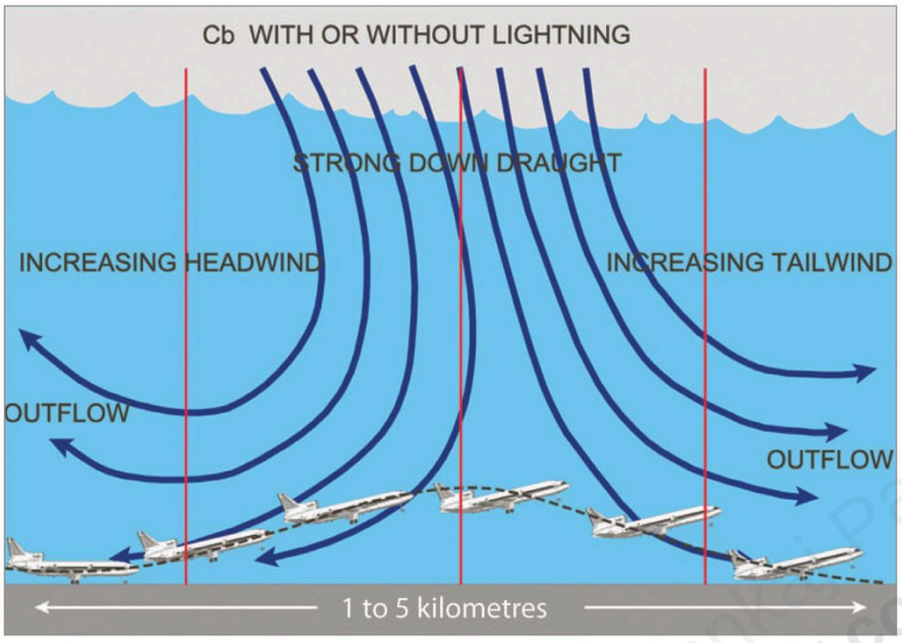
Storm Detection Radars: wavelength 3 to 20 cm. For detecting precipitation: 3 to 20 cm wavelength. Wavelength of 10 cm is widely used. Can be ground-based or airborne.

Band	Wavelength (cm)	Common Use
K Band	0.86	–
X Band	3.0	Airborne weather radar
C Band	about 5	Ground-based weather radar
S Band	10.0	TS detection (widely used)
L Band	20.0	Long-range detection

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10. Air Pocket, Downburst & Microburst



Microburst

Air Pocket

An updraught or downdraught caused by localized regions of warm or cool air and High or Low pressure areas. A pocket of warm air rises (less dense); a low pressure region creates updraft; a high pressure region creates downdraught. Extreme cases = **downburst (a strong descending current that spreads outward near the ground)** and **microburst**.

Downburst

A strong local downdraught from a CB that spreads out on reaching the ground and gives damaging winds. By the size of the outflow it is called a **microburst** (less than 4 km across) or a **macroburst** (4 km or more).

Microburst vs Macroburst

Parameter	Microburst	Macroburst
Diameter	less than 4 km	4 km or more
Life	1–5 min (may prolong 15–20 min)	Longer
Max Wind Speed	May exceed 75 km/h close to ground	Similar or higher
Avg Horiz Wind	90 km/h (may exceed 200 km/h)	–

Cause of Microburst: Dryness of atmosphere → falling rain evaporates rapidly (Virga (falling precipitation that evaporates before reaching the surface)) → evaporation reduces temperature of the air column → column colder than environment → descends and accelerates towards ground as microburst.

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Systems which produce Microburst:

1. CB cloud: about 5% of all TS produce microburst
2. Heavy rain: high probability of microburst
3. Virga: medium probability of microburst

Aviation Hazards from Microburst

Microburst affects aircraft in three stages:

1. **Contact Stage:** Cold downdraught accelerates towards the ground
2. **Outburst Stage:** Downdraught curls upwards after hitting the ground
3. **Cushion Stage:** Wind above curls and accelerates, posing danger to aircraft

An aircraft on approach encounters: first a **head wind** → then strong **horizontal spreading** → then a **tail wind**. The head and tail wind components are due to horizontal spreading of the microburst.

⚠ **Time for recognition and avoidance: as little as 5–15 seconds.**
Best avoided by NOT flying below a CB cloud, through a TS/DS, in heavy rain, and below Virga.
No take-off and landing is advisable during TS/DS.

Effects on aircraft:

- **Head Wind / Tail Wind Response:** Increasing head wind → lifts aircraft above AoA for landing → may overshoot. Increasing tail wind → undershoot.
- **Vertical WS Response:** Alternating up/downdraughts → momentary stick shaking or airframe shuddering; additional strain on pilot.
- **Cross Wind Shear:** Roll or yaw.
- **Turbulence:** Intense; may mask changing airspeed; delay recognition of severe downburst.
- **Rain Effect:** Heavy rain → reduce visibility; increase cockpit noise → avoid.

11. Dust Storms (DS) / Sand Storms (SS)

In the Pre-Monsoon season, surface temperatures very high (35°C or more) over NW India. Atmosphere highly unstable over desert/semi-arid areas. Small moisture incursion → CB cloud formation possible. Convective clouds do not build up high (low humidity aloft) — but if tops extend above freezing level → Duststorms/Sandstorms. Raise dust/sand from ground up to over 10,000 ft. Reduce visibility to 50 m or less. In N India: called **Andhi** (blinding storms).

Category	Wind Speed	Visibility
Light DS	Up to 21 kt	500 to <1000 m
Moderate DS	22 to 40 kt	200 to <500 m
Severe DS	>40 kt	<200 m

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🧠 Aviation hazards of DS/SS are almost same as TS. After DS passes, visibility remains poor for a long time. Moderate to severe vertical WS may be experienced during take-off/landing as DS/SS approaches. Radar echoes of DS are **less intense** than TS echoes.

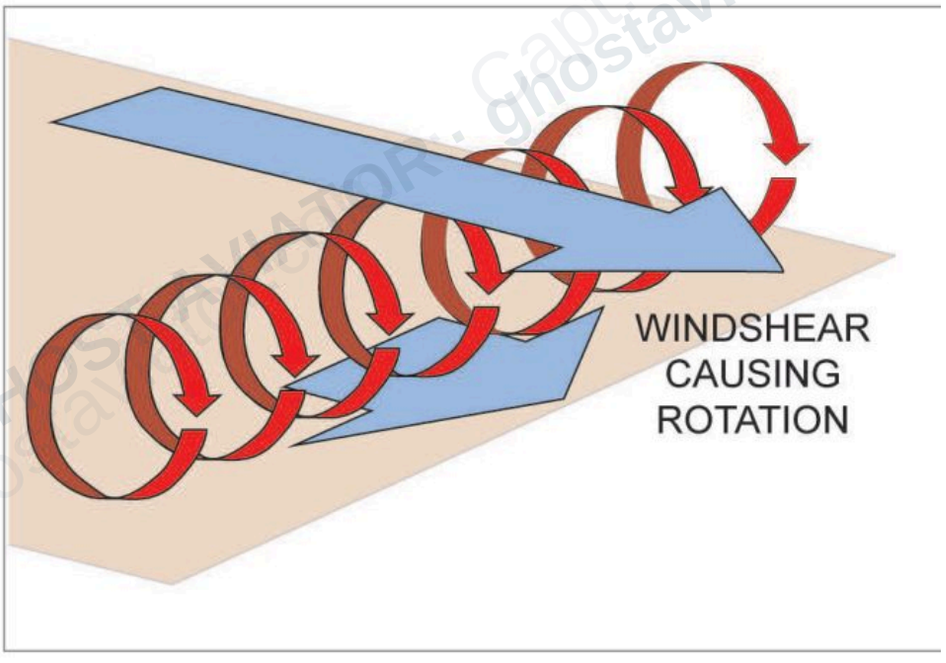
12. Norwesters

During Pre-Monsoon period (**March to May**): West Bengal, Chhatisgarh, Bihar, Jharkhand (Chotanagpur), North-East states, and Bangladesh are affected by violent TS called **Norwesters** because they approach from the **North-West**.

Begin March, increase in frequency till monsoon establishes over NE India. Also known as **Kalbaisakhi** (demon-like destructive character). Often accompanied by strong squalls; sometimes hail; on rare occasions a tornado may accompany them.

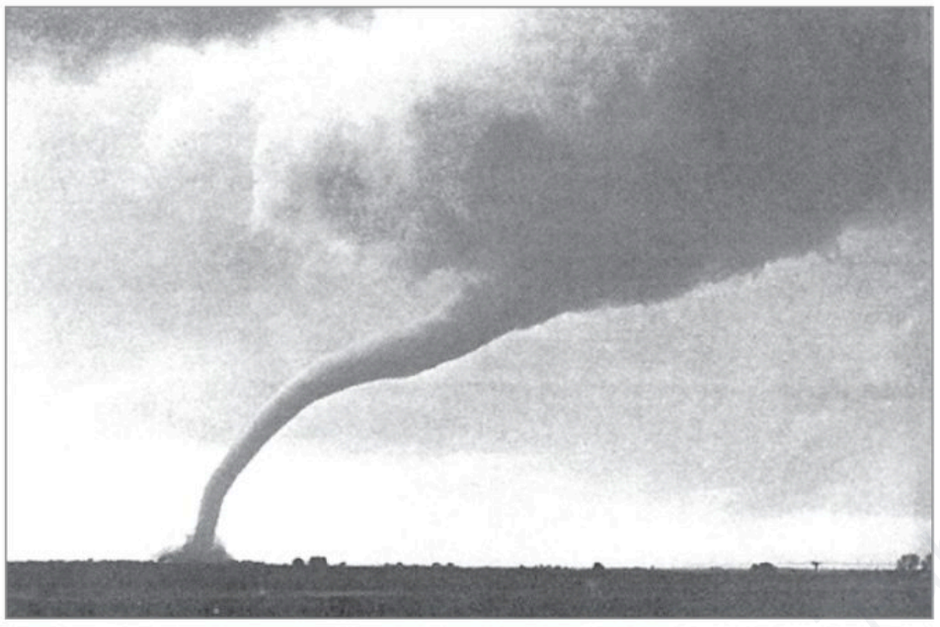
Flying through Norwesters can be extremely dangerous. They may regenerate TS in a line (**Line Squalls**). Circumnavigating or penetrating these may be disastrous.

13. Tornado, Water Spout & Dust Devil



Wind shear and rotation

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A tornado

Tornado

A rotating funnel in which air is sucked up from below the base of a CB. TDO = Tornado.

- Strong rotational wind produced when existing circulation below CB is drawn into its base through convergence; also by low level WS below CB
- Speed of rotating winds: in excess of 150 kt
- Pressure in the core may be **20 to 200 hPa** lower than the surroundings
- Fall in temperature makes tornado visible (condensation)
- Diameter: usually about 100–150 m, but can be from a few metres to more than 1.5 km
- Life: few minutes to more than an hour
- Tornadoes uproot trees, explode buildings, cause devastation in narrow path
- Gives **hook-shaped radar echo**
- Rare in India; on 17 March 1978 a tornado struck North Delhi → severe damage

Water Spout

Tornado over the sea = Water Spout (WTSPT). Funnel sucks up sea water; sometimes rain of fish is observed over the adjoining coastal area.

Dust Devil

Hot summer months, local surface winds converge towards a hot spot. Start rotating about a vertical axis; a whirl of dust raising wind rises from the ground. Vertically may reach 2 km. Diameter: about 10 m. Short-lived phenomenon. Capable of lifting light loose articles like paper.

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15. Practice Q&A

Q1. Thunderstorm ingredients

Thunderstorm development needs sufficient moisture, deep instability and a lifting trigger. A lapse rate steeper than the saturated adiabatic lapse rate through a deep layer extending above the freezing level favours development. Convection, terrain, convergence and fronts can provide lift. Strong wind alone does not supply all three ingredients.

Q2. Hail is most likely to fall from a cloud:
(a) Having layers of ice crystals (b) Composed of ice crystals (c) Having strong vertical development

✅ **Answer: (c)** — Strong vertical development (CB) provides the repeated up/down cycling needed for hailstone growth by collision with supercooled water drops.

❌ (a)(b) Ice crystal clouds (Ci, Cs) do not sustain hail growth — no liquid water present.

Q3. Norwesters are:
(a) The western disturbances which affect NW India (b) Severe thunderstorms which occur over NE India during hot weather period (c) Severe thunderstorms which occur over Peninsula during hot weather period

✅ **Answer: (b)** — Norwesters are violent TS affecting West Bengal, Bihar, NE states during Pre-Monsoon (March–May), approaching from the NW direction.

❌ (a) Western disturbances = extra-tropical cyclones affecting NW India in winter — completely different. ❌ (c) Norwesters affect NE India, not the Peninsula.

Q4. Duststorm occurs over NW India during:
(a) Post-monsoon (b) Winter (c) Pre-Monsoon

✅ **Answer: (c) Pre-Monsoon** — Very high surface temperatures (35°C+) over NW India create intense instability; small moisture incursion → CB → Duststorms.

❌ (a)(b) Post-monsoon/Winter: lower temperatures, less instability over NW India.

Q5. A 'mature' thunderstorm has strong:
(a) updraft only (b) downdraft only (c) updrafts and downdraughts

✅ **Answer: (c)** — The defining characteristic of the mature stage is the COEXISTENCE of both updraughts and downdraughts side by side.

❌ (a) Updraft only = Cumulus stage. ❌ (b) Downdraft only = Dissipating stage.

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Q6. Aircraft icing is most favoured in the cloud at temperatures ranging between:
(a) −20°C and −40°C (b) 0°C and −20°C (c) below −40°C

✅ **Answer: (b) 0°C and −20°C** — Maximum supercooled liquid water content exists in this range, particularly in CB and TCu clouds where severe icing is most likely.

❌ (a) −20°C to −40°C: droplets increasingly freeze → icing reduces. ❌ (c) Below −40°C: virtually all ice crystals, negligible icing.

Q7. A short duration, showery precipitation is associated with:
(a) ST (b) AS (c) CB

✅ **Answer: (c) CB** — CB clouds produce intense, short-duration convective showers and thunderstorms due to strong updrafts and rapid precipitation development.

❌ (a) St — stratus gives drizzle, continuous light rain. ❌ (b) As — continuous moderate rain/snow from layer clouds.

Q8. Hail is:
(a) Solid precipitation commonly occurring over the mountainous regions in winters (b) Frozen or partly frozen rain falling from sheet type of clouds (c) Solid precipitation falling from a deep convective cloud

✅ **Answer: (c)** — Hail is solid precipitation falling from a deep convective cloud (CB) where strong updrafts support the growth of hailstones by repeated cycling through supercooled water drop regions.

❌ (a) Describes ice pellets more than hail. ❌ (b) Frozen rain from sheet clouds = ice pellets or sleet, not hail.

Q9. The most hazardous cloud for aviation is:
(a) CB (b) CU (c) NS

✅ **Answer: (a) CB** — Cumulonimbus combines ALL aviation hazards: turbulence, icing, hail, lightning, wind shear, microburst, and instrument errors.

❌ (b) CU — less developed, fewer hazards. ❌ (c) Ns — gives continuous rain but lacks the violent convection of CB.

Q10. Thunderstorm lifetime

An ordinary single-cell thunderstorm typically lasts about 30 minutes. Multicell systems can persist for hours because new cells replace older ones; supercells can also last for hours. Distinguish the life of one cell from the life of an organised storm system.

Q11. Duration of the mature stage

In a simple single-cell model, the mature stage contains both rising and descending currents and precipitation. The cited teaching model gives about 15–20 minutes for this stage. Real storms vary, and an organised supercell can remain mature for several hours. Do not confuse one stage with the lifetime

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of the whole storm.

Q12. Norwesters occur during:
(a) Jan–Feb (b) Mar–May (c) Jun–Sep

✅ **Answer: (b) Mar–May** — Norwesters are a Pre-Monsoon phenomenon, beginning in March and increasing in frequency till the monsoon establishes over NE India.

❌ (a) Jan–Feb: winter, too cold for this convective activity in NE India. ❌ (c) Jun–Sep: monsoon season — different weather pattern.

Q13. Norwesters occur during:
(a) Winter (b) Hot weather (c) Monsoon

✅ **Answer: (b) Hot weather** — Pre-Monsoon = Hot weather season in India (March–May). Intense surface heating drives convective instability.

❌ (a) Winter: western disturbances dominate, not Norwesters. ❌ (c) Monsoon: monsoon TS are different from Norwesters.

Q14. Norwesters affect:
(a) N India (b) Bengal, Bihar, Orissa and Assam (c) Central India

✅ **Answer: (b) Bengal, Bihar, Orissa and Assam** — Norwesters primarily affect West Bengal, Bihar, Jharkhand, Chhatisgarh, NE states, and Bangladesh.

❌ (a) N India: Western disturbances, not Norwesters. ❌ (c) Central India: not the primary area for Norwesters.

Q15. The trigger action may take place due to:
(a) Clear night sky, no wind (b) Orographic lifting (c) high pressure

✅ **Answer: (b) Orographic lifting** — Orographic lifting forces moist air upward along mountain slopes → trigger action for TS. Other triggers: insolation, frontal lifting, convergence, katabatic cooling.

❌ (a) Clear night sky → radiational cooling → stable surface layer → suppresses convection. ❌ (c) High pressure → subsidence → suppresses convection.

Q16. Local timing of Norwesters

Norwester timing depends on where the storm starts and where it is observed. A study of Gangetic West Bengal describes frequent arrival from the west or northwest during 1200–1500 UTC, which is 1730–2030 IST. An afternoon-only answer therefore needs a clearly specified location and event.

Q17. Norwester development regions

Norwesters affecting Gangetic West Bengal can develop over the plateau region to its west and northwest. The cited study includes the Bihar plateau, southeast Madhya Pradesh and west Odisha, with storms travelling east or southeast. One named plateau should not be treated as the exclusive origin of

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every Norwester.

Q18. Andhi (blinding storms) occur generally over:
(a) S India (b) N India (c) NE India

✅ **Answer: (b) N India** — Andhi (Duststorms) are a North India phenomenon, occurring in the Pre-Monsoon season over the semi-arid/desert regions of NW India (Rajasthan, Punjab, Haryana, UP).

❌ (a) S India — different climate; less dust. ❌ (c) NE India — Norwesters, not Andhi.

Q19. Wind speed in Light DS is:
(a) 25 kt (b) 30 kt (c) up to 21 kt

✅ **Answer: (c) up to 21 kt** — Light DS: wind up to 21 kt, visibility 500 to <1000 m.

❌ (a) 25 kt — falls in Moderate DS range. ❌ (b) 30 kt — also Moderate DS.

Q20. The diametre of Microburst is:
(a) less than 4 km (b) less than 2 km (c) less than 6 km

✅ **Answer: (a) less than 4 km** — Microburst dia <4 km; Macroburst dia ≥4 km.

❌ (b)(c) — not the correct threshold values.

Q21. The diametre of and Macroburst:
(a) <4 km (b) 4 km or more (c) >8 km

✅ **Answer: (b) 4 km or more** — Macroburst diameter is 4 km or more (contrast with Microburst <4 km).

❌ (a) <4 km = Microburst. ❌ (c) >8 km — not the threshold used.

Q22. For detecting precipitation a Radar wavelength in the range …… is suitable:
(a) 30 to 200 mm (b) 400–500 mm (c) 600–700 mm

✅ Answer: (a) 30 to 200 mm — Weather radars used for precipitation work in the 3–20 cm band (30–200 mm), which is the range of X, C and S band radars. Longer wavelengths (400 mm and more) are not used for rain detection.

❌ (b) and (c) are far too long to be reflected by raindrops.

Q24. The wavelength of TS detection X band radar is:
(a) 10 mm (b) 20 mm (c) 30 mm

✅ Answer: (c) 30 mm — An X-band radar works at a wavelength of about 3 cm (30 mm) and is used for storm detection.

❌ (a) 10 mm and (b) 20 mm are shorter wavelengths (K band).

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Q25. The wavelength of storm detection S band radar is:
(a) 50 mm (b) 100 mm (c) 200 mm

✅ **Answer: (b) 100 mm (10 cm)** — S Band = 10.0 cm = 100 mm. The widely used wavelength for TS detection.

❌ (a) 50 mm = 5 cm = C Band. ❌ (c) 200 mm = 20 cm = L Band.

Q26. Over plains TS mostly occur during the:
(a) afternoon (b) night (c) early morning

✅ **Answer: (a) afternoon** — Over plains, maximum surface heating occurs in afternoon → convective instability → TS form in afternoon, die out in evening.

❌ (b) Night: sea TS. ❌ (c) Early morning: valley/foot hills TS and sea TS.

Q27. Over valley and foot hills TS generally occur during:
(a) afternoon (b) night & early morning (c) early morning

✅ **Answer: (b) night & early morning** — In valleys and foot hills, katabatic winds and radiational cooling at night + orographic lifting create TS conditions during night and early morning.

❌ (a) Afternoon: plains TS. ❌ (c) Early morning only: too restrictive.

Q28. Over the sea TS are more frequent:
(a) afternoon (b) night (c) early morning

✅ **Answer: (b) night** — Over the sea, low-level convergence and moisture availability are greatest at night → TS more frequent at night.

❌ (a) Afternoon: land TS. ❌ (c) Early morning: can occur but not the most frequent time for sea TS.

Q29. Organised thunderstorm systems

A mesoscale convective system contains an organised group of thunderstorms. New cells can develop as older cells weaken, allowing the system to persist for hours. Its organisation and lifetime differ from those of a single ordinary thunderstorm.

Q30. Loud peals of thunder, frequent flashes of lightning, moderate or heavy showers accompanied by light hail, and a wind speed of 15–40 kt are characteristic of a TS of intensity: (a) Light TS (b) Moderate TS (c) Severe TS

✅ Answer: (b) Moderate TS — Moderate TS: loud peals and frequent lightning, moderate or heavy showers, light hail, and maximum wind speed of 15–40 kt.

❌ (a) A light TS has distant thunder and light showers. (c) A severe TS has almost continuous thunder and lightning, large hail and winds above 40 kt.

Q31. For a severe TS one of the requirements is strong Wind Shear:

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(a) Horizontal (b) Vertical (c) Slant

✅ **Answer: (b) Vertical** — Strong vertical wind shear tilts the TS cell → precipitation falls away from updrafts → updrafts not cut off → storm grows to great heights and persists → severe TS.

❌ (a) Horizontal WS: affects aircraft at different altitudes but not the storm structure in the same way.
❌ (c) Slant: not a standard classification for this purpose.

Q32. Wind shear and organised thunderstorms

Strong vertical wind shear is one ingredient favouring an organised severe thunderstorm. Wind shear is a change in wind speed or direction with height. The storm structure depends on that wind profile; a fixed northward or southward tilt is not a universal property of severe cells.

Q33. When flying through an active TS, lighting strikes are most likely:
(a) Above 5000 ft and under the anvil (b) In the clear air below the cloud in rain (c) In the temperature band between +10° and −10°C

✅ Answer: (c) In the temperature band between +10° and −10°C — Lightning is most likely within about 5000 ft of the freezing level, in the temperature band from +10 °C to −10 °C, where the electric charges separate most strongly.

❌ (a) and (b) are not the most likely zone for a strike on an aircraft.

Q34. Thunderstorm hazards

A mature thunderstorm can contain strong rising and descending currents, severe turbulence, gust fronts and downbursts. Downbursts can create dangerous wind shear. Lightning and icing are additional hazards. An anvil is a cloud feature, not a substitute for assessing these hazards.

Q35. Hail grows as it freezes as it:
(a) freezes as it falls (b) up and down forces in CU cloud (c) collision with ice crystals (d) collision with supercooled water drops

✅ **Answer: (d) collision with supercooled water drops** — Hailstone grows by collision with supercooled water drops while being tossed up and down by strong updrafts in CB clouds.

❌ (a) Freezing while falling — no updraft cycling. ❌ (b) CU: not deep enough. ❌ (c) Collision with ice crystals — doesn't grow hailstones.

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Chapter 14: Air Masses, Fronts and Western Disturbances

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1. Air Masses — Definition & Classification

Air Mass: A large body of air covering an area of thousands of square kilometres in which the horizontal and vertical distribution of temperature (density (mass per unit volume)) and moisture are nearly uniform.

Air masses move with the wind and carry their properties. Weather associated with air masses depends on their properties and interplay with other air masses. Characteristics differ with origin and travel — during travel they get modified.

Source Region: An air mass (a broad body of air with similar temperature and moisture) forms when a large body of air stagnates over a region having fairly uniform temperature and moisture content for a long period. High pressure areas with light winds are the most suitable source regions. Main source regions = **poles** and **sub-tropical highs**.

Classification: Temperature (density) and humidity are the two main characteristics of air masses.

- Temperature characteristics determined by latitude zone of the source region
- Moisture characteristics determined by whether source region is continental or maritime

→ Polar/Tropical (temperature) × Continental/Maritime (moisture) = four basic types, plus Arctic and Equatorial.

Air Mass	Symbol	Place of Origin	Properties
Arctic	A	Polar regions	Extreme low temp, low humidity in winters; humidity increases in summers
Polar Continental	Pc	Sub-polar continental areas	Low temperatures which increase with movement towards lower latitude; **low humidity**
Polar Maritime	Pm	Sub-polar and Arctic oceanic areas	Low temperatures which increase with movement towards lower latitude; **high humidity**
Tropical Continental	Tc	Subtropical High Pressure land area	High temperatures, **low humidity**
Tropical Maritime	Tm	Southern border of oceanic sub-tropical highs	Moderately high temperatures, **high humidity**
Equatorial Maritime	Em	Equatorial and tropical seas	High temperatures, **high humidity** (SW Monsoon is Em air mass)

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2. Air Masses over Indian Sub-Continent

Tropical Maritime (Tm)

Originates in subtropical highs of the north Pacific Ocean; arrives over India during monsoon season after travel over SE Asian countries. Characterised by:

- High temperature, high relative humidity (actual moisture compared with saturation at the current temperature, expressed as a percentage), high dew point (temperature to which air must cool to become saturated at unchanged pressure and water-vapour content)
- Excellent visibility
- Fairly high diurnal range of temperature
- Cumuliform clouds causing showers or thunderstorms in some places

Tropical Continental (Tc)

Most common air mass over India, particularly in **winter**. Originates in the Siberian high; moves over India as a cold dry current. Associated with fair weather and **poor visibility** conditions.

Equatorial Maritime (Em)

Originates over the Indian Ocean; long travel over water. Prevails over India to the south of 25°N during monsoon season. High humidity → sultry weather. Visibility is good except in showers. Diurnal range of temperature is small.

Seasonal Air Mass Distribution over India

Season	North/NW India	South/Central India	NE India / Himalayan ranges
Winter	Tc (dominant); incursion of Pc after WD passage → Cold Waves	Occasional Tm from south of equator (= Em / equatorial maritime air)	WDs draw Tm from Arabian Sea and Bay of Bengal
Summer / Monsoon	Tc air	Em air (Monsoon Trough established over Gangetic plains by June)	Modified Tm air from north Pacific (different from Em)

3. Fronts — Definition & Terminology

Front: A narrow zone of transition between two air masses of contrasting densities. The contrasting densities are due to different temperatures and humidity of the air masses. At a front there is change in temperature, wind, clouding, and precipitation.

Frontal Surface: The sloping surface along which the two air masses exist and meet; the place where this frontal surface meets the ground is the front.

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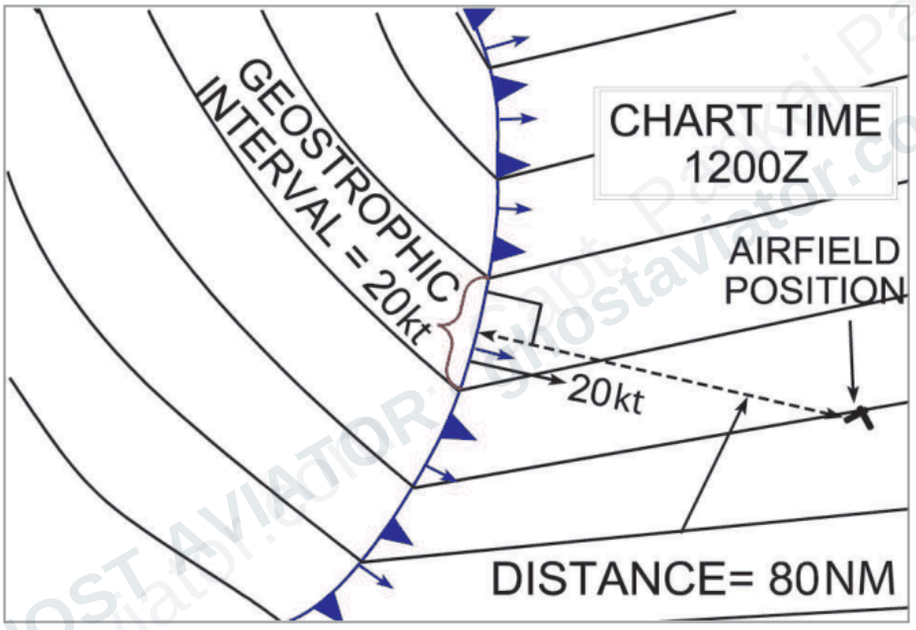
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Frontogenesis: Formation or strengthening of a front. Occurs only when there is a significant density contrast between two air masses.

Frontolysis: Weakening of front with time and dissipation of frontal weather. If density contrast is insignificant, the front passes off unnoticed (except for change in winds).

4. Cold Front

A demarcation zone between **cold air overtaking warm air**. Cold air provides a wedge on which warm air glides. Most cold fronts are associated with **CU and CB clouds**.
Average slope of cold front: 1:50 (becomes shallower at higher levels).



Cold front speed

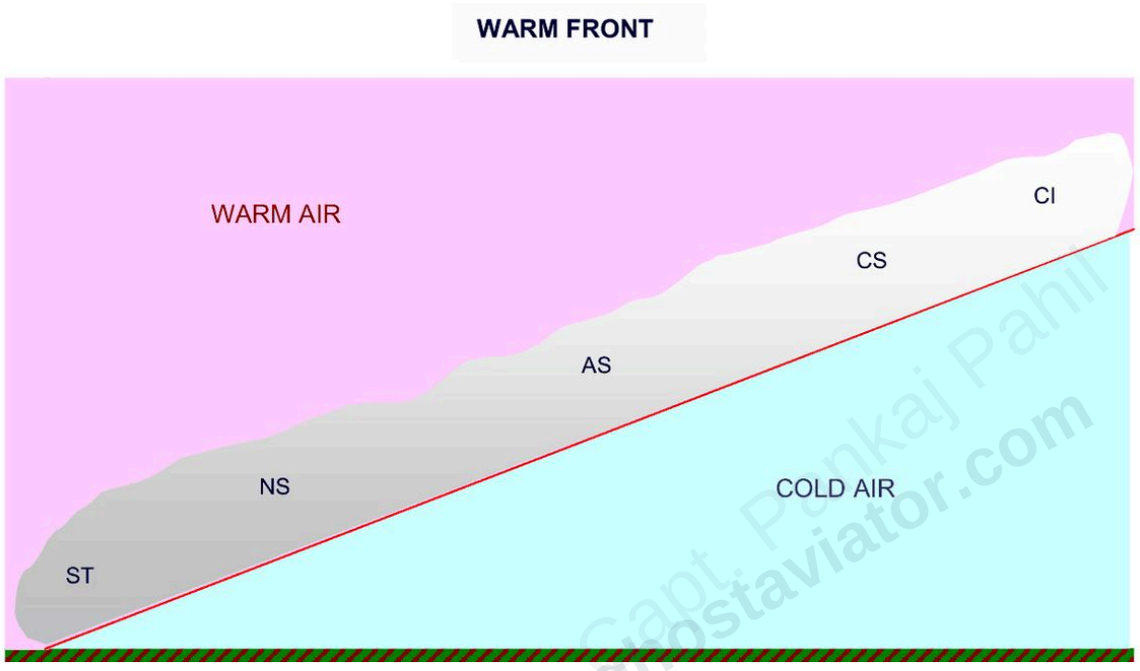
Feature	Kata Type Cold Front	Ana Type Cold Front
Slope	1:120 (shallow)	1:50 (steep)
Speed	About 20 kt	Exceeds 30 kt
Warm Air Motion	General **sinking** (kata = katabatic) at high levels	**Rising** (ana = anabatic) at high levels — instability
Clouds	ST, NS, AS and CU, CB (widespread stratiform)	CU, CB (convective; CB clouds dominate)
Precipitation	Widespread; showery in advance & just behind surface front (due to CB); NS and AS behind	Violent TS, hail, short-duration showers; confined to 100–150 km
Special Feature	Sometimes moves very fast → **line squall** at 100–300 km ahead of front	Behind the front: CU, CB clouds and isolated showers. Sky clears rapidly.

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5. Warm Front

A demarcation zone between **warm air overtaking cold air**. Slope: 1:150 to 1:180. Speed: about 10–15 kt. If warm air is stable and sinks at higher levels = Kata Type warm front. If warm air is unstable = Ana Type warm front.



Warm front cross-section

Feature	Kata Type Warm Front	Ana Type Warm Front
Approach (1000 km ahead)	Ci, Cs	Ci, Cs
400–500 km from front	NS, AS clouds appear	CU and CB embedded in stratiform
Just ahead of front	Drizzle, low stratus, sometimes **frontal fog**	Drizzle from ST clouds; brief spells of heavy precipitation
After passage	Fog dissipates; sky partially clear; warmer and humid	Fog dissipates; skies practically clear

🧠 **Warm Front sequence (Kata):** 1000 km: Ci,Cs → 400-500 km: NS,AS → Just ahead: Frontal Fog + Drizzle → After: Warmer & Humid.
Cloud sequence ahead of warm front: **Ci → Cs → As → Ns → St → Front**

6. Occluded Front

A cold front normally moves at **twice the speed** of a warm front. An occluded front forms when a cold front catches up with a warm front.

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Feature	Cold Occlusion	Warm Occlusion
Condition	Air mass of advancing cold front is **COLDER** than the cool air mass of the warm front	Air mass behind advancing cold front is **LESS COLD (COOL)** than cold air mass of warm front ahead
Process	Advancing cold front **undercuts** and lifts both warm and cool air masses of the warm front	Advancing cold front **overrides** the warm front ahead
Initial weather	Warm front type; during passage = showery (cold front) weather	Similar to warm front throughout
Season	Common in **summers**	Occurs in **winters**, less common

7. Stationary Front & World Fronts

Stationary Front: Exhibits almost no movement. Two contrasting air masses on either side. Winds on either side blow parallel to the front but in **opposite directions**. The Polar Front is an example of a stationary front.

Front	Location	Description
Polar Front	~60°N/S	Transition zone between polar easterlies and mid-latitude westerlies. Not continuous; active only in segments.
Arctic Front	North of polar front	Boundary between arctic and polar air masses. Often displaced southwards into temperate latitudes in winters.
Mediterranean Front	Over Mediterranean Sea	Boundary between Pc air from Europe and Tc air from North Africa. Extends roughly W to E over Mediterranean in winter.

8. Fronts in India

India is not affected by fronts, **except in extreme NW India**. The Western Disturbances which affect NW India are of frontal origin. However, when they reach India they are already in the **occluded stage**.

9. Western Disturbances (WD)

Western Disturbances: Lows or troughs which move from W to E, north of 20°N, and cause weather. When the low has two or more closed isobars at 2 hPa interval, it is termed a **Western Depression**. Troughs above 500 hPa level = **Troughs in Westerlies**.

Origin: Mediterranean Sea, Caspian Sea, and Black Sea; move eastwards across north India.
WDs belong to the **extra-tropical cyclone family**. Over India they are in the occluded state, causing warm/cold front type cloudiness and weather. Intensify over NW India due to orography and moisture feed from Arabian Sea.

Induced Low and Induced Cyclonic Circulation: Under the influence of a WD, sometimes a low develops to the south of the system (*induced low*), and sometimes a cyclonic circulation develops to the

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south (*induced cyclonic circulation*).

Parameter	Value / Detail
Frequency in winters	5–7 per month
Monsoon months	NIL (decreases to nil)
Reappear	From October onwards
Rate of movement	About 250–300 km per day
Track	Pakistan/Rajasthan → NE towards W Himalayas; some glide along foot hills towards E; few move towards UP, MP, Bihar, then NE India
Weather affected area (~50% cases)	J&K, HP, Punjab, Haryana, Uttarakhand, Sikkim, Bhutan, Arunachal
Max clouding and precipitation zone	Always along the **foot hills**
After WD passage	Widespread fog and cold waves are common; fog lifts by forenoon (may persist for a few days)

⚠ **Aviation Hazards from WD:**

- During approach/passage: low clouds, poor visibility, rain, TS, hail, icing
- After passage: widespread fog
- Pre-monsoon WDs: widespread dust haze and dust storms over NW India; TS over northern plains

Trough in Westerlies: A series of upper-level troughs; westerlies keep moving W to E. Large amplitude troughs → upper air divergence (airflow spreading apart) ahead → enhances weather activity considerably. Severe TS/DS, Norwesters, and heavy rains are associated with these troughs.

11. Practice Q&A

Q1. The airmass which originates at sea in high latitudes is called:
(a) Polar maritime (b) Tropical continental (c) Tropical maritime

✅ **Answer: (a) Polar maritime (Pm)** — High latitudes + sea origin = Polar Maritime. Cold temperatures + high humidity (maritime source).

❌ (b) Tropical continental: low-latitude land origin → hot & dry. ❌ (c) Tropical maritime: low-latitude sea → warm & moist.

Q2. The airmass which originates over equatorial region is:
(a) Warm and dry (b) Warm and Moist (c) Cold and dry

✅ **Answer: (b) Warm and Moist** — Equatorial Maritime (Em) originates over equatorial and tropical seas → high temperature + high humidity = warm and moist.

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❌ (a) Warm & dry: Tropical Continental (Tc) — land origin. ❌ (c) Cold & dry: Polar Continental (Pc) — high latitude land.

Q3. If the advancing cold front is colder than the cool air mass of the warm front, the advancing cold front undercuts and lifts both the warm and cool air masses of the warm front. This is:
(a) Warm Occlusion (b) Cold Occlusion (c) —

✅ **Answer: (b) Cold Occlusion** — Cold Occlusion: advancing cold air is COLDER than the cool air ahead → undercuts both warm and cool air masses → lifts them. Common in summers.

❌ (a) Warm Occlusion: advancing cold air is LESS COLD than the cold air ahead → overrides the warm front.

Q4. The airmass which originated over land area located in polar region is:
(a) Warm and dry (b) Warm and Moist (c) Cold and dry

✅ **Answer: (c) Cold and dry** — Polar Continental (Pc): polar/sub-polar land origin → cold + low humidity = cold and dry.

❌ (a) Warm & dry: Tropical Continental. ❌ (b) Warm & moist: Tropical Maritime or Equatorial Maritime.

Q5. If a warm airmass overtakes a cold air mass, it is called:
(a) Cold Front (b) Warm Front (c) Occluded Front

✅ **Answer: (b) Warm Front** — Warm Front: warm air overtaking cold air. The warm air glides up over the cold air (slope 1:150 to 1:180).

❌ (a) Cold Front: cold air overtaking warm air. ❌ (c) Occluded Front: cold front catches up with warm front.

Q6. At warm front:
(a) Warm air overtakes the cold air (b) Cold air undercuts the warm air

✅ **Answer: (a) Warm air overtakes the cold air** — At a warm front, warm air actively overtakes and glides up over the retreating cold air mass.

❌ (b) Cold air undercuts warm air = Cold Front definition.

Q7. Concept check

A cold front lifts warmer air. If that air is unstable, cumuliform cloud, showers and thunderstorms may develop. The presence of a cold front alone does not establish that every layer is unstable or that every cold front produces cumulonimbus.

Q8. Sometimes line squall occurs 100-300 km ahead of a:
(a) Warm front (b) Cold front

✅ **Answer: (b) Cold front** — Kata type cold front sometimes moves very fast → causes a line squall 100–300 km ahead of the front.

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❌ (a) Warm front: no line squall; weather is more gradual and stratiform ahead of warm front.

Q9. Precipitation occurs over a belt of 30–50 km on both sides of front:
(a) Cold front (b) Warm front

✅ **Answer: (a) Cold front** — Cold front precipitation is intense but narrow — confined to a belt around the frontal surface (30–50 km on both sides). Ana type cold front precipitation: 100–150 km.

❌ (b) Warm front: precipitation extends over a much wider belt (hundreds of km ahead of front) — gradual and widespread.

Q10. Cold front moves at ... the speed of a warm front moves:
(a) Same (b) Double (c) Half

✅ **Answer: (b) Double** — A cold front normally moves at twice the speed of a warm front. This is why cold fronts eventually catch up with warm fronts, forming occluded fronts.

❌ (a) Same speed — they would never occlude. ❌ (c) Half — if true, cold fronts would always fall behind.

Q11. Line Squalls occur ... of cold front:
(a) Ahead (b) Behind (c) At the

✅ **Answer: (a) Ahead** — Line squalls occur 100–300 km AHEAD of the cold front (Kata type). They are generated by the advancing cold air mass pushing ahead of the frontal surface.

❌ (b) Behind: behind a cold front, skies typically clear. ❌ (c) At the front: the main precipitation zone is at the front itself.

Q12. Fronts are associated with: (a) Tropical cyclones (b) Monsoon depressions (c) Extra-tropical cyclones

✅ Answer: (c) Extra-tropical cyclones — Fronts form between air masses of different temperature and humidity, which is the situation in extra-tropical (mid-latitude) cyclones. Tropical cyclones and monsoon depressions have no fronts.

❌ (a) and (b) are non-frontal systems with a uniform warm, moist air mass.

Q13. CB, Roll-type clouds, SC, AC with embedded CB are associated with:
(a) Cold Front (b) Warm front

✅ **Answer: (a) Cold Front** — Cold fronts produce CU, CB clouds. Roll clouds occur behind the Gust Front (associated with the cold front). Embedded CB in Ac are associated with Ana type fronts.

❌ (b) Warm front: primarily Ci, Cs, As, Ns, St — stratiform clouds. Embedded CB only in Ana warm front.

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Q14. Polar front

The polar front is a transition between colder polar air and warmer air at middle latitudes. In the broad circulation pattern it lies between the polar easterlies and the mid-latitude westerlies. A tropopause is a different boundary: it separates the troposphere from the stratosphere.

Q15. The air mass which originates from sea area located in lower lat is:
(a) Warm and Dry (b) Warm and Moist (c) Cold and Moist

✅ Answer: (b) Warm and Moist — An air mass that originates over the sea in low latitudes is maritime tropical: warm from the low latitude and moist from the sea.

❌ (a) Warm and Dry is a continental tropical air mass. (c) Cold and Moist is a polar maritime air mass.

Q16. WDs approach India as:
(a) Cols (b) Occluded Fronts (c) Highs

✅ **Answer: (b) Occluded Fronts** — Western Disturbances are of frontal origin but when they reach India they are already in the occluded stage (due to the long travel distance).

❌ (a) Cols: saddle-shaped pressure patterns between highs and lows. ❌ (c) Highs: WDs are lows/depressions, not highs.

Q17. Maximum WDs occur in:
(a) Summers (b) Post Monsoon (c) Winters

✅ Answer: (c) Winters — Western Disturbances are a winter feature. They arrive at 5–7 a month in winter and are almost absent in the monsoon months.

❌ (a) Summer and (b) post-monsoon: far fewer WDs than in winter.

Q18. Ahead of a warm front the surface wind:
(a) Backs & weakens (b) Veers & strengthens (c) Backs & strengthens

✅ **Answer: (c) Backs & strengthens** — Ahead of a warm front (in the Northern Hemisphere), the surface wind backs (shifts anticlockwise) and strengthens as the front approaches and pressure falls.

❌ (a) Backs & weakens: not correct — wind strengthens. ❌ (b) Veers & strengthens: veering (clockwise) occurs after warm front passage.

Q19. On approach of a Warm Front temperature:
(a) Fall (b) Rise (c) Remain same

✅ **Answer: (c) Remain same** — Ahead of the warm front, temperature gradually changes little; the dramatic temperature rise occurs AFTER the warm front passes (warm air replaces cold air).

❌ (a) Fall: temperatures don't fall as warm air approaches. ❌ (b) Rise: the significant temperature rise is after the warm front passage.

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Q20. Clouds near a warm front

In the classic warm-front pattern, high cirrus and cirrostratus are followed by altostratus and nimbostratus as the cloud thickens and lowers. Low stratus may occur near the front. This is a typical pattern, not a requirement that every front displays every cloud type.

Q21. During the passage of a Cold Front winds:
(a) Suddenly become squally (b) Back and weaken (c) Veer and are of moderate strength

✅ **Answer: (a) Suddenly become squally** — At cold front passage, the wind suddenly becomes squally (squall = sudden violent wind from the cold downdraught spreading outward). This is the Gust Front / First Gust effect.

❌ (b) Back and weaken: wind behaviour ahead of warm front. ❌ (c) Veer moderate: after warm front passage.

Q22. Visibility is poor in a Warm Front:
(a) Ahead (b) Ahead & During (c) After & During

✅ Answer: (b) Ahead & During — Visibility is poor ahead of a warm front, in the drizzle, low stratus and frontal fog, and it stays poor while the front passes. After passage the frontal fog clears, though low cloud, drizzle and reduced visibility can persist in the warm sector.

❌ (a) Ahead alone leaves out the passage itself. (c) After passage visibility improves.

Q23. Concept check

Visibility behind a cold front is often good outside showers. Fog requires its own formation conditions, so it cannot be predicted merely from the words “after a cold front”. Assess moisture, cooling and wind conditions separately.

Q24. WD is a ……. front:
(a) Cold (b) Warm (c) Occluded

✅ **Answer: (c) Occluded** — When WDs reach India, they are in the occluded stage. They originate as frontal systems (warm/cold fronts) but by the time they reach India, the cold front has caught up with the warm front.

❌ (a)(b) — WDs have passed the warm/cold front stage by the time they reach India.

Q25. FZRA and FZFG occur ……. of a warm front:
(a) Ahead (b) During (c) After

✅ **Answer: (a) Ahead** — FZRA (Freezing Rain) and FZFG (Freezing Fog) occur ahead of a warm front. Warm rain falls through a sub-zero cold air mass ahead of the approaching warm front → freezes on contact = freezing rain. Freezing fog also forms in the cold air ahead of the front.

❌ (b)(c) — After warm front passage, temperatures rise above freezing in many cases.

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Q26. Concept check

Precipitation need not stop when a warm front passes. The warm sector can contain low cloud, light rain or drizzle. Behind a cold front, showers may continue. Frontal passage alone does not guarantee a dry period.

Q27. Wind ahead of a warm front

In the conventional Northern Hemisphere depression pattern, surface wind ahead of a warm front may increase in speed and back (change anticlockwise). At passage it commonly veers (changes clockwise). These are typical changes, not universal rules for every frontal system.

Q28. A WD with two or more closed isobars, at 2 hPa interval, is termed as:
(a) Troughs in Westerlies (b) Western Depression (c) Western Cyclone

✅ **Answer: (b) Western Depression** — When the low has two or more closed isobars at 2 hPa interval, it is termed a **Western Depression**.

❌ (a) Troughs in Westerlies: troughs above 500 hPa level without closed isobars. ❌ (c) Western Cyclone: not a standard classification used in the text.

Q29. A WD originate over:
(a) Baluchistan (b) Caspian sea (c) Mediterranean, Caspian and Black Seas

✅ **Answer: (c) Mediterranean, Caspian and Black Seas** — WDs originate over the Mediterranean Sea, Caspian Sea, and Black Sea; they move eastwards across north India.

❌ (a) Baluchistan: not a source — WDs pass over this region. ❌ (b) Caspian sea only: incomplete; all three seas are source regions.

Q30. Induced lows develop to the ... of a WD:
(a) N (b) S (c) NE

✅ **Answer: (b) S (South)** — Under the influence of a WD, a low sometimes develops to the SOUTH of the system (induced low), and a cyclonic circulation sometimes develops to the south (induced cyclonic circulation).

❌ (a) North: the WD itself is north of induced systems. ❌ (c) NE: not the stated direction.

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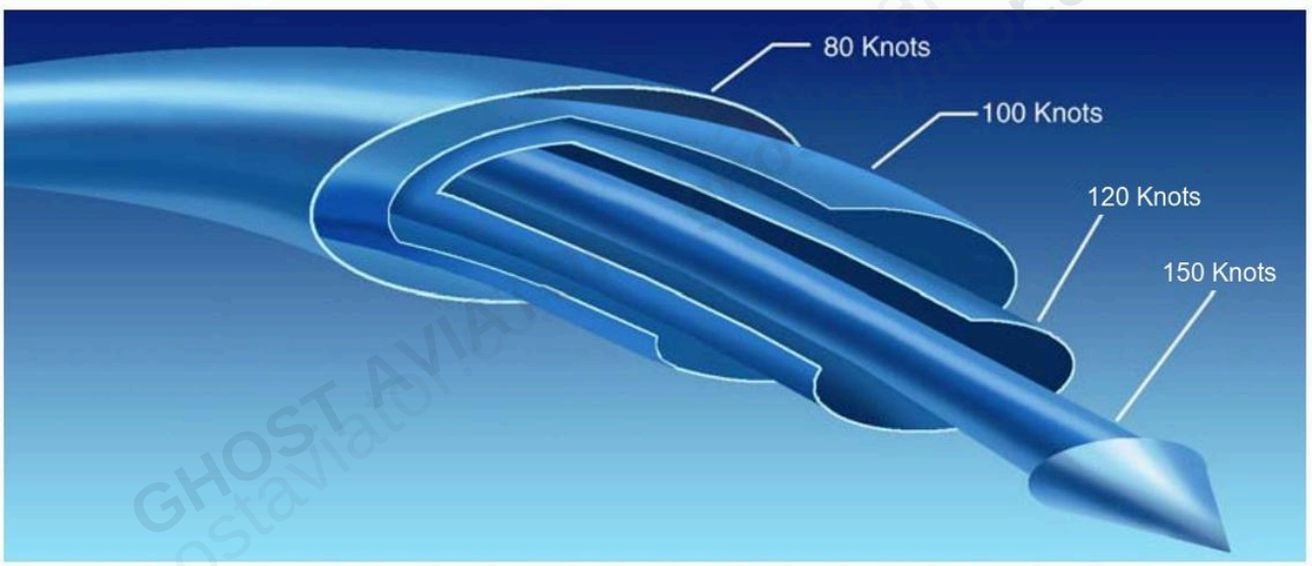
Chapter 15: Jet Streams

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1. Definition & Basic Properties

Jet Stream: A strong narrow current of winds along a semi-horizontal axis in the upper troposphere. Jet streams are present in both hemispheres all year round. The path of a jet stream is zigzag wavy like a serpentine.

- Length: thousands of km
- Width: hundreds of km
- Depth: some km (often less than 5 km)
- **Vertical Wind Shear: about 5 m/s per km**
- **Arbitrary lower limit of jet core velocity: 30 m/sec (60 kt)**



Isotachs across a jet stream

Jet Streaks (Wind Maxima): Along the axis of a jet stream there are centres of high speed winds called **Jet Streaks** — wind maxima. Weaker winds lie in between. Jet streaks retain their strength but move slower (by up to 25 kt) than the jet stream itself. In a wavy jet they are located over or near the **ridge**.

Wind speed along the axis is **NOT always uniform**. At certain places the wind speed reduces below 60 kt → constituting breaks in the jet stream. Sometimes jet streams split into two or more parts, combine, or flow in different directions.

🧠 **Path of maximum speed = the CORE (axis) of the jet stream.**
Jet Streaks = centres of high wind speed along the axis, located over/near ridges in wavy jets.

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2. Types of Jet Streams

There are **four types** of jet streams in the troposphere:

Type	Symbol	Direction	Height / Pressure	Key Features
Arctic Jet Stream	AJ	Westerly	7–8 km, near polar tropopause	Over Arctic region, close to polar tropopause
Polar Front Jet	PFJ	Westerly (mainly); NWly or SWly on occasions	~250 hPa, 9–12 km (30,000–39,000 ft)	Also called mid-latitude jet. N America, Europe, Asia; SH = circles Antarctica. Migrates: winter → 30°N; summer → 60°N. Normal speed: 80–100 kt in winters.
Sub-tropical Jet	STJ	**Westerly**	10–16 km, near mid-latitude tropopause break	Above subtropical high pressure belt. Winter position: S of 22°N; Summer: N of 35°N (weakens). Affects India: Oct–May. Speed: 100–215 kt (up to 398 km/h).
Tropical Jet Stream	TJ	**Easterly**	~15–16 km, ~150 hPa	Also called Easterly Jet Stream. Prominent during monsoons, weak in winters. Speed: 60–80 kt avg; max 150 kt. Axis at 13°N (Chennai lat.) over India in Jul–Aug. **Only over Asia and Africa.** NOT over Atlantic and Pacific.
Stratospheric Jet	SJ	Westerly (winter) / Easterly (summer)	Above 20 km	In Arctic and Antarctic regions. Westerlies in winter; easterlies in summer.

3. Sub-tropical Jet Stream (STJ) over India

STJ seasonal movement over India:

- Appears over N India in **October** after withdrawal of monsoon
- Progressively shifts southwards as winter progresses
- Mean position: roughly at 27°N at height of 12 km
- Reaches southernmost position at 22°N in **February**
- Prevails over Indian subcontinent for about 8 months (Oct to May)
- After May: shifts northwards and moves out of Indian subcontinent

STJ wind speeds over India:

- Mean wind speed: 100 kt
- Maximum: 200 kt
- Lowest (Oct and May): 60–70 kt
- Highest (January): 100–120 kt; on occasions core reaches 200 kt
- Higher speeds more common in January and February than December

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- Level of maximum wind: a **few km below the tropopause**

STJ structure:

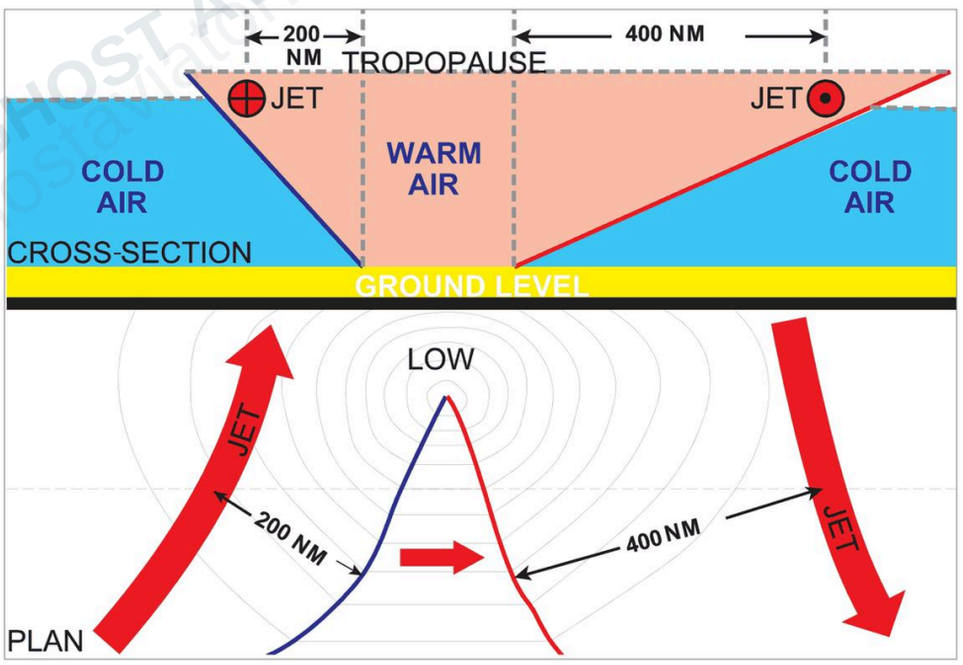
- STJ splits due to Himalayas → one branch south, one north of Himalayas → recombine over China
- STJ has a **layered structure**: often two layers of maximum wind **to the S of the jet core**
- Jet stream strengthens by 10–15 kt downstream, along axis from **Jodhpur to Guwahati**
- STJ located 3–4 km below the tropopause in region where there is a break in tropical and middle tropopause at 300–200 hPa

🧠 **STJ positions at a glance:**
Oct (arrival, 27°N, 12 km) → Feb (southernmost, 22°N) → May (exit northwards) → ABSENT Jun–Sep (monsoon season, TJ takes over)

4. Temperature Distribution & Wind Shear

STJ is caused by a concentration of horizontal temperature gradient below the jet and reversal above it.

- Surface to 300 hPa: horizontal temperature gradient from S to N (particularly at 500 and 300 hPa) → thermal wind builds up → jet maxima reached at 200 hPa
- At 200 hPa: thermal gradient is slack and thereafter reverses
- **Lapse rates more to the S of the jetstream than to the N and above and below.**
- Near the STJ: temperature gradient is **very small or negligible**



Jet streams cross-section

Vertical shear: Wind speed changes rapidly above and below the jet core. For aviation hazards, distinguish this wind-speed profile from the location of strongest clear-air turbulence (CAT): a common

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CAT region is near or just below the core on the cold-air side, with a secondary region above.

Horizontal Shear: In STJ, horizontal shear is more to the **north** than to the south of the jet core.

- At core level: winds weaken by about 35 kt/100 NM to the south
- North side of front is steeper: about 100 kt/100 NM
- Shear is more below the core and to the north (left looking downstream)

5. STJ and Weather

For an ideal straight jet streak in the Northern Hemisphere, upper divergence is favoured in the right entrance and left exit regions; convergence is favoured in the left entrance and right exit regions. Left and right are viewed looking downstream. Curvature and the surrounding weather pattern can change this simple model. Upper divergence (airflow spreading apart) can support rising motion and development of a low-level system.

6. Tropical Jet Stream (TJ)

Also called: Easterly Jet Stream.

- Strong Easterlies blow to the S of 25°N at 100 hPa during monsoon
- **Found only over Asia and Africa; NOT over Atlantic and Pacific**
- During monsoon: 10–20 kt Ely winds at 300 hPa, S of 27°N and Wly to N
- Prominent during monsoons; becomes very weak in winters
- Average core speed: 60–80 kt; maximum: 150 kt
- TJ seen over India: **June to August**; disappears by **September**
- Axis: 13°N (near Chennai/Madras latitude), at 150 hPa, 15–16 km height
- Level of maximum wind: **just below the tropopause**
- Wind shears in TJ much **less than STJ**

Cause of TJ: In summer the Tibetan Plateau and the Asian landmass become very hot, while the Indian Ocean to the south stays cooler. So temperature falls towards the south through the troposphere. This reversed temperature gradient turns the low-level westerlies into easterlies above 500 hPa. Easterlies strengthen with height up to 16 km.

🧠 **TJ vs STJ:**
STJ = Westerly = Oct–May = 22–27°N = 12 km = 100–200 kt (strong)
TJ = Easterly = Jun–Aug = 13°N = 15–16 km = 60–150 kt (monsoon jet)
TJ = only Asia + Africa. STJ = global.

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7. Low Level Jets (LLJ)

Very strong winds appear in the lower levels, between 1 km and 3 km at many places in the sub-tropics. These are called **Low Level Jets (LLJ)**.

A prominent LLJ is noticed over the Indian Peninsula during SW monsoon when low level winds become very strong. This LLJ is probably a continuation of the jet stream off the **Somalia Coast in Africa**. The LLJ **enhances the monsoon activity** over the Indian Peninsula.

8. Cause, Clouds & CAT associated with Jetstream

Cause

Jetstream is caused by the **large horizontal temperature gradient**. The larger the temperature difference between two air masses, the stronger is the jet stream.

Clouds Associated with Jetstream

- Extensive sheets of high clouds on the **warm side**, with sharply defined edges parallel to the jet axis
- **Ci clouds** more common on the **warm (equator) side** of the jet, with a sharp edge along the jet axis
- CC clouds may also be seen — tops just below the tropopause
- Warm side cloud tops: a few thousand feet below the tropopause
- Roll type of Ci or CS may be seen along the length
- Where moisture is less: thin layer of haze on the warm side; clear on the cold side

Clear Air Turbulence (CAT)

CAT occurs around boundaries of Jetstream due to large vertical and horizontal wind shear (wind speed or direction changing over distance). Also more in a developing and rapidly moving jet.

⚠ **Strongest CAT occurs near or just below the axis on the COLDER (low pressure) side, with a secondary maxima above the Jetstream.**

Looking downstream: CAT is to the **LEFT in the Northern Hemisphere**; to the right in Southern Hemisphere.

CAT in a jetstream is most severe in:

1. Strong Winds
2. Curved Jets — the sharper the curve, the stronger the turbulence
3. On the leeward side and above mountain ranges

CAT near a jet: Strong shear on the cold side near and below the core often favours turbulence. No side of a jet guarantees smooth air. Use forecasts and aircraft reports, and follow aircraft-approved turbulence

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procedures when considering a route or altitude change.

10. Practice Q&A

Q1. The arbitrary lower limit of jet core velocity has been assigned by WMO as:
(a) 60 kt (b) 60 m/s (c) 70 m/s

✅ Answer: (a) 60 kt — The WMO lower limit of jet core speed is 60 kt, which is 30 m/s. Anything slower is not classed as a jet stream.

❌ (b) 60 m/s would be about 117 kt, far above the minimum. (c) 70 m/s is higher still.

Q2. Jet stream has:
(a) one maxima (b) one or more maxima (c) only two maxima

✅ Answer: (b) one or more maxima — The wind speed along a jet axis is not uniform: there are one or more maxima, called jet streaks.

❌ (a) and (c) set a fixed number of maxima, which the jet does not have.

Q3. The vertical wind shear in a jet stream is:
(a) 5 m/sec/km (b) 6 m/sec/km (c) 4 m/sec/km

✅ **Answer: (a) 5 m/sec/km** — "The vertical WS is about 5 m/s per km." This is the key defining vertical wind shear value for jet streams.

❌ (b) 6 m/sec/km — not the stated value. ❌ (c) 4 m/sec/km — not the stated value.

Q5. In a jet stream, the path of maximum speed is known as: (a) Core (b) Axis (c) Jet streak

✅ Answer: (b) Axis — The line along which the speed is greatest is the jet axis. The tube of fast air surrounding it is the core.

❌ (a) The core is the tubular volume around the axis. (c) A jet streak is a local maximum of speed along the axis.

Q6. Speed along a jet stream

Wind speed need not be uniform along the jet axis. Faster sections, called jet streaks, occur between weaker sections. A jet stream therefore does not promise a constant headwind or tailwind along an entire route.

Q7. Along the axis of a jet stream there are centres of high speed winds, these are called:
(a) Ridge (b) Jet streaks (c) Between Trough and Ridge

✅ **Answer: (b) Jet streaks** — Centres of high speed wind along the jet axis = Jet Streaks (wind maxima). In a wavy jet, these are located over or near the ridge.

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❌ (a) Ridge: the location of jet streaks in a wavy jet, not the name of the centres. ❌ (c) Not the correct terminology.

Q8. Sub-tropical Jet Stream (STJ) is:
(a) Westerly (b) Easterly (c) Southerly

✅ Answer: (a) Westerly — The sub-tropical jet stream is a westerly jet. The tropical jet stream is the easterly one.

❌ (b) Easterly describes the Tropical Jet Stream (TJ), not the STJ. (c) Southerly is not a jet stream direction.

Q11. The southern most position of STJ is in February:
(a) 22°N (b) 13°N (c) 17°N

✅ **Answer: (a) 22°N** — "It reaches its southern most position of 22°N in February."

❌ (b) 13°N: this is the latitude of the Tropical Jet Stream axis, not the STJ. ❌ (c) 17°N: not the stated value.

Q12. The STJ affects India from:
(a) Oct to May (b) Sep to Oct (c) Jun to Sep

✅ **Answer: (a) Oct to May** — STJ prevails over the Indian subcontinent for about 8 months from October to May. After May it shifts northwards and exits the Indian subcontinent.

❌ (b) Sep–Oct: too short; STJ arrives in October and stays till May. ❌ (c) Jun–Sep: this is the monsoon period when TJ (Easterly jet) prevails — STJ is absent then.

Q13. STJ has a layered structure. There are often two layers of max wind to the … of jet core:
(a) S (b) N (c) SW

✅ **Answer: (a) S** — "There are often two layers of maximum wind, to the S of the jet core." This is the layered structure of the STJ.

❌ (b) N: horizontal shear is more to the N, but the two layers of max wind are to the S. ❌ (c) SW: not the stated direction.

Q14. The STJ strengthens:
(a) Northwards (b) Upstream (c) Downstream

✅ **Answer: (c) Downstream** — "The jet stream strengthens by 10–15 kt downstream, along the axis from Jodhpur to Guwahati."

❌ (a) Northwards: lateral direction, not along jet axis. ❌ (b) Upstream: opposite of the text — weakens upstream, strengthens downstream.

Q15. At and near the STJ the temperature gradient is very:

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(a) Small (b) Large (c) Moderate

✅ **Answer: (a) Small** — "Near the STJ the temperature gradient is very small or negligible." The strong gradient exists BELOW the jet (up to 300 hPa); it weakens at the jet level and reverses above.

❌ (b) Large: the large gradient is BELOW the jet (causes the jet), not AT the jet level. ❌ (c) Moderate: the gradient there is "very small or negligible."

Q16. Wind shear near a jet core

Wind shear means a change in wind speed or direction over a distance. Strong vertical shear can occur above and below a jet core. The typical cold-side turbulence pattern near or below a jet axis must not be treated as a universal statement that every subtropical jet has greater vertical shear below its core.

Q17. TJ prevails over Indian Peninsula from:
(a) May to Jun (b) Jun to Aug (c) Sep to Oct

✅ **Answer: (b) Jun to Aug** — The TJ (Easterly Jet Stream) is seen over India from June to August and disappears by September.

❌ (a) May–Jun: too early; TJ develops with monsoon onset in June. ❌ (c) Sep–Oct: TJ disappears by September; STJ returns in October.

Q18. TJ is located over India approximately at a height of:
(a) 11–12 km (b) 15–16 km (c) 8–10 km

✅ **Answer: (b) 15–16 km** — "It is at a mean height of about 15 to 16 km... The axis in July-August is near latitude 13°N (Chennai Lat.), at 150 hPa, over India."

❌ (a) 11–12 km: this is the height of the STJ (Sub-tropical Jet, at ~12 km). ❌ (c) 8–10 km: too low for a jet stream; PFJ is at 9–12 km.

Q19. TJ is located over India approximately at a height of 15–16 km. It is in:
(a) July–Aug (b) Sep–Oct (c) Jun

✅ **Answer: (a) July–August** — "The axis in July-August is near latitude 13°N (Chennai Lat.), at 150 hPa." The TJ is active June–August but at maximum strength in July–August.

❌ (b) Sep–Oct: TJ disappears by Sep; STJ returns Oct. ❌ (c) Jun only: TJ starts in June but the axis specification is for Jul–Aug.

Q20. In the TJ the wind shears are much less than the STJ:
(a) True (b) False

✅ **Answer: (a) True** — "In the TJ the wind shears are much less than the STJ." The STJ has much stronger associated wind shears (and therefore stronger CAT potential) than the TJ.

❌ False: TJ wind shears are less than STJ.

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Q21. Recognising a jet stream

Wind-blown cirrus wisps can suggest a jet stream when cloud conditions permit. Upper-wind and significant-weather charts provide clearer evidence. Lenticular clouds are a separate clue to mountain waves, so they should not be combined with cirrus streaks as though the two terms mean the same thing.

Q22. Aircraft track and jet winds

A westerly jet flows from west towards east. It can provide a tailwind for an eastbound aircraft and a headwind for a westbound aircraft. Crosswind and headwind or tailwind components depend on the aircraft track and actual wind. Nearness to the jet alone does not guarantee an increasing tailwind.

Q23. Strongest CAT occurs near the jetstream on the:
(a) Warm side (b) Cold side (low pressure side) (c) On the jet axis itself

✅ **Answer: (b) Cold side (low pressure side)** — "Strongest CAT occurs near or just below the axis on colder (low pressure) side with a secondary maxima above the Jetstream." In the NH, this is to the LEFT looking downstream.

❌ (a) Warm side: some CAT but less severe. ❌ (c) On axis: maximum wind but not necessarily maximum CAT; shear zones are above/below and to the cold side.

Q25. When and where tropical jet stream occurs: (a) All year along India (b) High level dart (c) In summers over SE Asia and Central Africa

✅ Answer: (c) In summers over SE Asia and Central Africa — The tropical (easterly) jet stream occurs in the summer monsoon months over Asia and Africa only, not over the Atlantic or the Pacific.

❌ (a) It is seasonal, not all year. (b) is not a place or season.

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Chapter 16: Clear-Air Turbulence

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1. Definition & Necessary Conditions

Definition: Clear Air Turbulence (CAT) is the bumpiness experienced by aircraft at high altitudes in cloud-free conditions, *excluding Ci clouds*. It is caused when bodies of air moving at widely different speeds meet. The Upper Troposphere (23,000–39,000 ft) near the Tropopause is the most susceptible region. CAT is most frequently found in regions of jet streams. Unstable atmosphere enhances CAT. At lower altitudes it may occur near mountain ranges.

Necessary Conditions

1. **Marked horizontal or vertical wind shear (wind speed or direction changing over distance)**

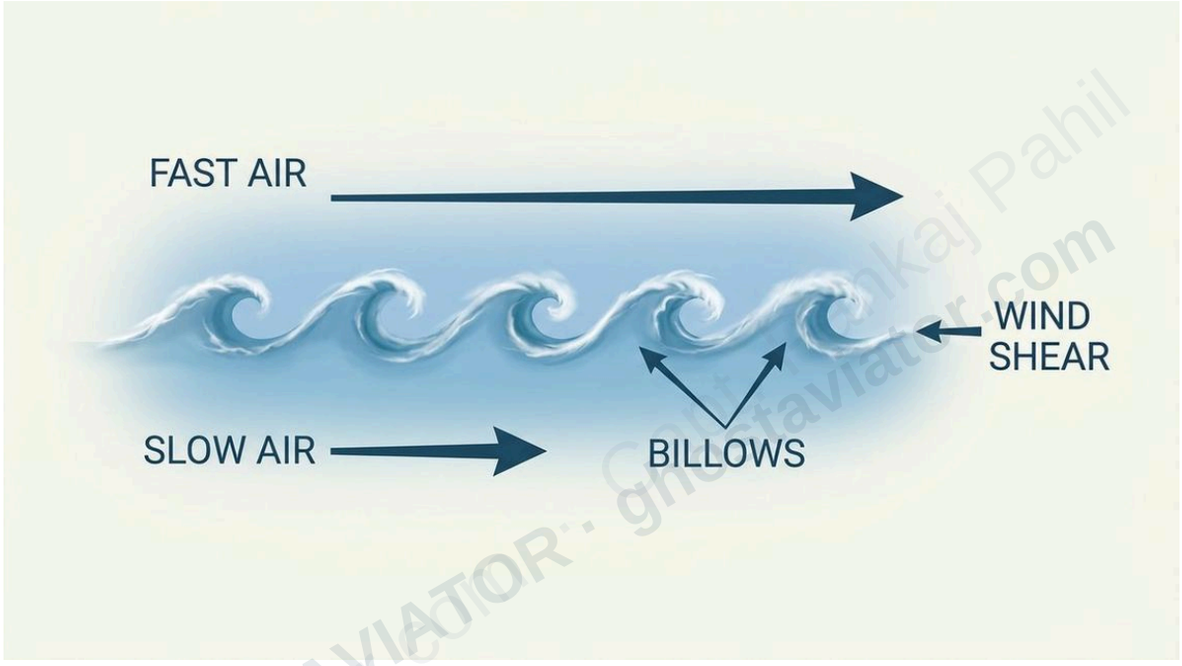
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2. **Steep lapse rate (temperature decrease with increasing height)** in the atmosphere

Instructor's Note: Remember "SHEAR + STEEP LAPSE = CAT". Both conditions are normally present just below the tropopause, especially near the jet stream core.

2. Causes & Sources of CAT

Lower Stratosphere: Even when the lapse rate is small or negative, a marked decrease of wind speed with height may give rise to temporary zones of CAT.



Kelvin-Helmholtz billows

Sources of CAT

Warning: CAT is often associated with strong wind shear near jet-stream boundaries and upper-air fronts. A jet core must not be assumed turbulence-free; assess the surrounding shear, forecasts and recent aircraft reports.

Mountain Wave + Jet Stream Combined Effect: Mountain waves amplify gravity waves and propagate them vertically towards the stratosphere. These waves may generate smooth up and downdrafts or may breakdown and cause CAT. Severe CAT is more frequent on the **leeward side of high ground** than over the plains.

Thunderstorm-induced CAT: CAT occurs when strong winds encounter the tops of thunderstorm clouds, resulting in waves that extend well downstream from the cloud. Sometimes a train of lenticular clouds can be noticed on the leeward side of the CB cloud.

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3. CAT Over India

Aspect	Details
Frequency & Regions	Highest over N, Central, NE India from October to May — due to Sub Tropical Jet stream (STJ). Over S India in July and August — due to peak activity of Tropical Jet stream (TJ).
Intensity	Most CAT in India is **moderate**. Severe CAT can occur from December to February when STJ is at its peak.
Horizontal Dimensions	~120 km N–S direction; ~250 km E–W direction (patchy phenomenon)
Vertical Extent	A CAT zone may be about 2000 ft, but may extend through a deep layer.

4. Aviation Hazards

Aviation Hazards of CAT:

1. CAT causes **discomfort** to crew and passengers.
2. In CAT, **difficulty may be experienced in controlling** an aircraft.
3. CAT is encountered **suddenly** and is amplified by the mountain waves.

5. Avoidance

Avoidance Procedures:

1. Avoid flying close to a **Jet stream**.
2. A change of altitude or route may help avoid CAT. Choose it from the forecast and recent aircraft reports, considering aircraft limits and the required ATC coordination; a compass direction alone cannot guarantee an exit.
3. No fixed height above the tropopause guarantees freedom from CAT. Use current forecasts and aircraft reports when assessing an alternative altitude.

7. Practice Q&A

Q1. CAT (Clear Air Turbulence) is most frequently found in which region of the atmosphere?
(a) Surface layer (b) Near the jet streams at upper troposphere (c) In the stratosphere above 50,000 ft (d) In cumulus cloud regions

✅ Correct Answer: (b) Near the jet streams at upper troposphere (23,000–39,000 ft near tropopause)

❌ Distractors: (a) Surface layer has friction turbulence, not CAT. (c) CAT is mainly found in the upper troposphere near the jet streams and the tropopause, not above 50,000 ft. (d) CAT specifically excludes cloud regions (Ci excluded).

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Q2. Most CAT in India occurs on the fringes of the jet stream. The maximum frequency of CAT over North India is during:
(a) July–August (b) October–May (c) June–September (d) March–June

✅ Correct Answer: (b) October–May — due to Sub Tropical Jet stream (STJ) activity over N, Central and NE India

❌ Distractors: (a) Jul–Aug is when Tropical Jet stream is active over S India. (c) Jun–Sep is SW Monsoon — STJ retreats. (d) Partially overlaps but Oct is when STJ starts strengthening.

Q3. CAT zones vary

Clear-air turbulence occurs in patches whose size and intensity vary. A teaching example of horizontal dimensions does not provide an escape distance. No fixed turn of 120 km north or south guarantees leaving turbulence. Use the latest forecasts and aircraft reports and follow aircraft-approved procedures.

Q4. Responding to clear-air turbulence

A route or altitude change may help avoid turbulence, but no fixed compass direction or height above the tropopause guarantees smooth air. Use current forecasts and aircraft reports, respect aircraft limits, and coordinate a required route or altitude change with ATC. Follow the aircraft-approved turbulence procedures.

Q5. Severe CAT in India is most likely during:
(a) June–July (b) December–February (c) March–May (d) August–October

✅ Correct Answer: (b) December–February — when the Sub Tropical Jet Stream (STJ) is at its peak intensity

❌ Distractors: (a) Jun–Jul = SW Monsoon, STJ weakens. (c) Mar–May = STJ retreating. (d) Aug–Oct = Monsoon withdrawal, STJ not yet at peak.

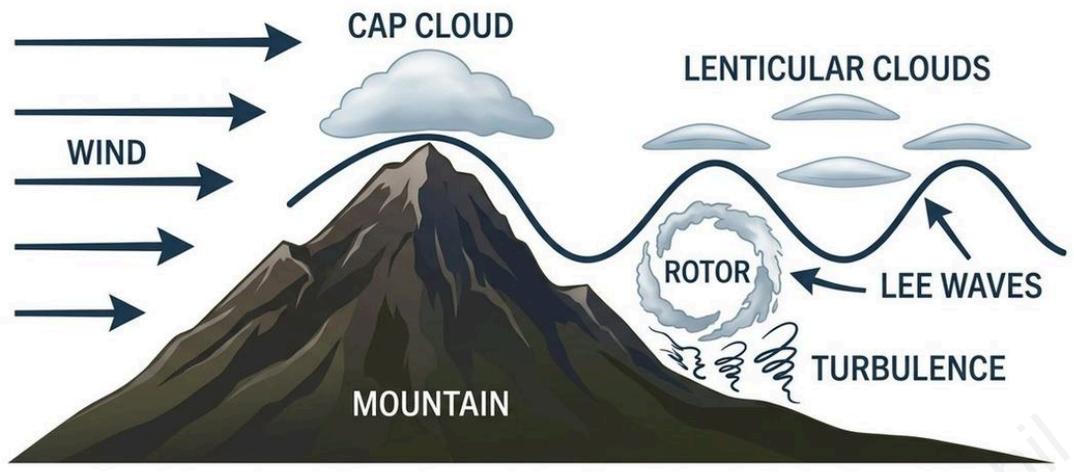
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Chapter 17: Mountain Waves

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1. Definition & Necessary Conditions

Definition: Mountain Wave is defined as oscillations of high scale and height due to disturbance in the horizontal air flow caused by high ground. When an airstream strikes a mountain range under suitable stability conditions, **a train of waves is generated on the leeward side of the range** — called *Mountain Wave*. Flying over such terrain is bumpy and unsafe.



A mountain wave and its features

Necessary Conditions for Mountain Waves

Wind:

- A definite air flow across the ridge within 30° perpendicular to the ridge.
- Wind speed of at least 7 m/s for smaller mountains and 15 m/s for larger mountains.
- A deep steady directional flow over a significant height band, strengthening with height.

Stability:

- A marked stable layer near summit height, such as an inversion or isothermal layer, with less stable air above and below.
- This is a classic favourable profile; stable air below the ridge alone does not describe the complete pattern.

2. Vertical Currents

In mountain waves there are alternate areas of rise and sink. **Areas of sink occur immediately downward of the ridge.** Vertical currents associated with mountain waves can be intense and extensive. Over large mountains, vertical currents of 10 to 25 m/s are not uncommon.

Critical Hazard: An aircraft flying *parallel* to a ridge, if caught in the **down current**, would suffer constant loss of height till the whole length of the ridge is traversed. An aircraft caught in up currents would constantly gain height. **Adequate terrain clearance is therefore essential while flying over mountains.**

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3. Associated Clouds

Cloud Type	Location	Description
Cap Cloud	Covers mountain top	Strong winds may sometimes sweep them down the lee slope — like Fohn Wall.
Rotor/Roll Cloud	Leeward side below crest, first wave	Most hazardous; associated with rotor streaming. Strongest rotor is normally in the first wave crest, level or slightly above the crest.
Lenticular Clouds	Leeward side, a few thousand feet above crests	Lens-shaped, form at successive crests of waves. Appear stationary (form on upwind side, dissolve on downwind side). Ragged edges = turbulence indicator. May look like "cloud fall" — Fohn Wall.
ST, SC, AS, CI, Nacreous	Various levels in the troposphere; nacreous cloud higher up, in the stratosphere (about 15–30 km)	Formed in humid atmosphere at various levels due to mountain waves extending to great heights.

Waterfalls: On large lee slopes, the clouds sometimes sweep down like a cloud fall — they resemble waterfalls and are called **Fohn Wall**. Mountains have a tendency to increase the cloud cover and also cloud depth.

Rotors: When winds are strong at lower levels but weaken or reverse at higher levels, violent rotors form on the lee slope and move downwind from the ridge. These are called **Rotor Streaming**.

4. Visual Recognition

Visual Recognition Guide:

1. **Lenticular clouds** on leeward side, few thousand feet above crests → **ragged edges indicate turbulence**
2. **Rotor Clouds** → Strongest rotor is in the **first wave crest**, level or slightly above the crest
3. **Cap Cloud** → Covers mountain top; strong winds may sweep it down the lee slope (Fohn Wall)

5. Turbulence

Flying over mountains is not always turbulent. It is smooth in certain regions and turbulent in others due to additional wind shears. However, **the transition from smooth to turbulent motion is extremely abrupt and without any warning.**

Severity:

- Wave cloud outlines that are not smooth but have ragged appearance = indication of turbulence.
- Severe turbulence occurs in cap cloud and rotor clouds.
- **Worst kind of turbulence** = in rotor clouds, similar to that of violent TS. Both vertical and horizontal gusts occur.

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- Acceleration of 2 to 4g is not uncommon; at times 7g has also been exceeded.
- Size, speed, and aerodynamic characteristics of aircraft determine bumpiness.
- In extensive mountain ranges, one wave system generated by one range is modified by further ridges downstream → may result in **violent turbulence**.

6. Altimeter Errors & Hysteresis Effect

Altimeter Errors in Mountain Waves: Aircraft in mountain waves are subjected to rapid height fluctuations due to up and down currents. The aneroid capsules of the altimeter are **unable to respond correctly** to these fluctuations. This is called the **Hysteresis Effect**.

- Altimeters also give erroneous readings if the temperature defers from ISA.
- If colder than ISA → readings are **too high**.
- If warmer than ISA → readings are **too low** (vice versa).
- **Hence altimeter readings are not reliable and deceptive when used for terrain clearance.**

7. Seasonal & Diurnal Variation

Diurnal: After sunset, atmosphere becomes stable at lower levels due to radiational cooling. Winds also weaken in lower layers. This results in formation of waves on the lee side of small and medium height mountains → called **Evening Waves**. These waves continue throughout, till warming up after sunrise destroys the stability structure. For large mountains, this effect is negligible.

Seasonal: In winters, winds are steady and strengthen with height, and the atmosphere has suitable stability → **Mountain waves are more frequent in winters.**

8. Extent

- Mountain waves may extend 100–200 km downstream and **vertically sometimes well into the Stratosphere**.
- Wavelength of mountain waves is about 10 km or more.
- Lee waves of long amplitude produce **rotors** in the crests of waves.
- If the mountain range is long, a succession or train of lee waves may be seen.

9. Aviation Hazards

Mountain Waves — Aviation Hazards:

1. **Strong up and down currents** may make it difficult to maintain altitude and loss of control.
2. **Turbulence**, often severe, up to great heights, may lead to **structural damage**.
3. **Unexpected winds** of large magnitude may cause loss or gain of altitude.
4. **Increase in frequency and intensity of icing.**

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5. **Errors in pressure altimeter** (Hysteresis effect).

11. Practice Q&A

Q1. For mountain waves to form there should be flow of air across the ridge, generally within …… of the perpendicular to the ridge.
(a) 30° (b) 45° (c) 60°

✅ Correct Answer: (a) 30°

❌ Distractors: 45° and 60° are too oblique — the flow must be nearly perpendicular (within 30°) to effectively create the wave pattern on the leeward side.

Q2. For mountain waves to form, the wind speed for small mountains should be at least:
(a) 15 m/s (b) 10 m/s (c) 7 m/s

✅ Correct Answer: (c) 7 m/s for small mountains

❌ Distractors: 15 m/s is the threshold for larger mountains. 10 m/s is not a specified value in the text.

Q3. For mountain waves to form, the wind speed for large mountains should be at least:
(a) 15 m/s (b) 10 m/s (c) 7 m/s

✅ Correct Answer: (a) 15 m/s for large mountains

❌ Distractors: 7 m/s is for small mountains. 10 m/s not specified.

Q4. Stability and mountain waves

A classic mountain-wave pattern has strong flow across the ridge and a marked stable layer near summit height, with less stable air above and below it. Stable means that displaced air tends to return towards its original level. “Stable air below the ridge” alone is an incomplete description of this pattern.

Q5. The worst kind of turbulence is encountered in the vicinity of rotor clouds.
(a) Rotor Clouds (b) Lenticular Clouds (c) Cap Clouds

✅ Correct Answer: (a) Rotor Clouds — worst turbulence, similar to violent TS

❌ Distractors: (b) Lenticular clouds indicate wave position and have ragged edges in turbulence, but not the worst zone. (c) Cap cloud has turbulence but rotor is worse.

Q6. In a mountain-wave system, rotor clouds form beneath:
(a) Wave troughs
(b) Wave crests
(c) Valleys

Answer: (b) Wave crests. A rotor cloud forms in the rising part of a low-level circulation beneath a wave crest. Lenticular clouds can form higher up at the crest.

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Locate the rotor below the wave crest. Its low-level circulation can cause severe turbulence.

Q7. Clear air turbulence is often encountered:
(a) At the boundary of a jet stream (b) In the wake of a passing airplane (c) In the wake of a larger airplane at take off and landing (d) All of the above

✅ Correct Answer: (a) At the boundary of a jet stream — CAT is found at the fringes/boundary of jet streams

❌ Distractors: (b) Wake turbulence from passing aircraft is wake turbulence, not CAT. (c) Wake turbulence at T/O & landing is mechanical, not CAT. (d) Not all — only (a) is correct for CAT specifically.

Q8. Wind shear near a jet

Strong wind shear near the edges of a jet and upper-level frontal zones can favour clear-air turbulence. The fastest wind at the core does not by itself prove turbulence, and the core is not guaranteed smooth. Axis and core should not be treated as interchangeable definitions.

Q9. CAT is the bumpiness experienced by aircraft at high altitudes. Over North India its maximum frequency is during:
(a) Oct–May (b) Jul–Aug (c) Jun–Sep

✅ Correct Answer: (a) Oct–May — due to STJ over North India

❌ Distractors: (b) Jul–Aug = TJ active over S India. (c) Jun–Sep = SW Monsoon; STJ displaced northward and weakened over India.

Q10. When approaching an area where mountain waves have been reported, a pilot should expect:
(a) Possible presence of roll clouds and lenticular clouds (b) Intense up draughts and down draughts on the lee side of the mountains (c) Moderate to severe turbulence as far as 20 to 30 nm from range on leeward side (d) All of the above

✅ Correct Answer: (d) All of the above

❌ Distractors: Each option alone is true — lenticular/rotor clouds, intense vertical currents, and turbulence extending far downstream are all characteristics of mountain wave areas.

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Chapter 18: Tropical Systems

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1. Tropical Discontinuities — ITCZ

Tropical Discontinuities / Convergence Zones: In the tropics, the temperature (or density (mass per unit volume)) contrast between two air masses is not as discernible as in higher latitudes. Hence fronts

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cannot be marked. However, a line of discontinuity can be drawn in the wind field in upper air charts and by inspection of the cloud pattern — called **Tropical Discontinuities** or **Convergence Zones**.

Inter Tropical Convergence Zone (ITCZ): A narrow zone along which Tm or Te air from the N meets the Tm or Em air from the S. Air masses within the tropics converge — hence the name ITCZ. ITCZ usually lies along the Equatorial Trough (ET). In summers it migrates up to 20° N and in winters up to Australia. ITCZ is active in certain portions, about 75–150 km length, where heavy showers occur and CB clouds predominate. On rare occasions it may be pushed up to S Bay of Bengal in winters and may initiate formation of cyclonic storms during the rainy season — ITCZ/ET coincides with Monsoon Trough.

Equatorial Front: Alternative term for ITCZ.

2. Easterly Waves

Easterly Waves: Troughs in tropical easterly winds forming on the ITCZ with fairly high frequency. They originate at sea at the boundary of Tm and Em air masses. A minor surge of Em air may distort the ITCZ and start the Easterly Wave. It moves in a more or less east-west direction.

- Weak at surface; better developed at 500 hPa level and above.
- Wavelength: about 1500 km.
- Easterly waves usually travel from east to west in the tropical easterly flow. Their speed and structure vary.
- Most weather occurs in **rear of the wave trough**.

Easterly Waves in India

In winters, first noticed in Andaman Sea as shallow lows with troughs in upper air. Move westwards across south peninsula and emerge into Arabian Sea at about 15 kt over sea, about 10 kt over peninsula. Further progression westwards beyond Arabian Sea cannot be easily traced, but also affect Africa. In winters ITCZ is south of equator — on rare occasions these waves move into extreme south Bay, possibly causing disturbed weather for 3–4 days over Bay Islands, Tamil Nadu, Kerala and Arabian Sea Islands. During pre-monsoon season, ITCZ advances further north — frequency of easterly waves over India increases, helping formation of premonsoon cyclonic storms over Indian seas. During monsoon, easterly waves from Pacific Ocean move across SE Asia and reach head of Bay of Bengal. Some move further WNW; others develop into depressions, move NW and weaken gradually. Very infrequently, an easterly wave may develop into a depression over north Arabian Sea.

3. Peninsular Discontinuity (PD) / Dry Line

Peninsular Discontinuity (PD): A discontinuity forming over peninsular India (south of Lat. 23° N) in April and May. Also called **Dry Line**. It separates hot and dry Tc air from the northern India from the comparatively cooler and moist air from the Bay of Bengal. This discontinuity gives rise to afternoon thunderstorms, at times with hail, strong northwesterly squalls and heavy showers. It remains more or less stationary for 3–4 days before dissipating.

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4. Surge

Surge / Pulse: Sudden strengthening of an air current in the same air mass (a broad body of air with similar temperature and moisture). At a surge the air arrives faster than it is removed → accumulation of air at the surge line → **velocity convergence**. This is a common and strong disturbance in the monsoon season over the Arabian Sea.

- To the **west** of the surge line, wind may be about 40 kt ahead of it; to the **east**, only about 10–15 kt.
- Vertical ascent at a surge line causes growth of CU clouds, showers and gusts or squalls.
- Surge line moves in the **direction of the wind**. It ultimately reaches over the head of the Bay of Bengal.
- When a surge arrives, sky becomes overcast with very frequent and sharp showers, and strong gusty or squally winds.
- The surge line **activates ITCZ** and favours formation of depressions or cyclones on the ITCZ.

5. Shear Lines

Shear Lines: Created when in the same air mass the wind speed decreases markedly from south to north or vice versa. Common over Arabian Sea during monsoon season. At a shear line, friction between air moving at different velocities causes a small-scale ascent. If shear extends vertically, ascent reaches to higher levels — stratiform layer clouds (ST, AS, NS) form, extending to about 500 km across. When a fresh surge arrives, large CU and CB clouds get embedded at the leading edge, which are at right angles to the shear line. **Surge and Shear Lines are a characteristic feature of vigorous monsoon.**

6. Low Pressure Systems Classification

IMD classification: These categories use maximum sustained surface wind, not the wind at 900 m. The wind averaging convention must be stated when comparing categories from different agencies.

System	Wind Speed Over Sea
Low Pressure	< 17 kt
Depression	17–27 kt
Deep Depression	28–33 kt
Cyclonic Storm	34–47 kt
Severe Cyclonic Storm	48–63 kt
Very Severe Cyclonic Storm	64–89 kt
Extremely Severe Cyclonic Storm	90–119 kt
Super Cyclonic Storm	120 kt or more

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7. Monsoon Depressions

Monsoon Depression: A low-pressure area with two or more closed isobars at 2 hPa interval. In India they form over NE India during monsoon and are termed **Monsoon Depressions**. Surface winds are 17–33 kt. Also form over Arabian Sea and land too. Land depressions mostly form over NE India. In June, July and August depressions form over head Bay of Bengal (N of 18 N). In September they form around 14 N. In Arabian Sea they originate 5° off coast, N of 12 N and move N/NW.

Average life of a monsoon depression: 3–5 days

Weather in Monsoon Depression:

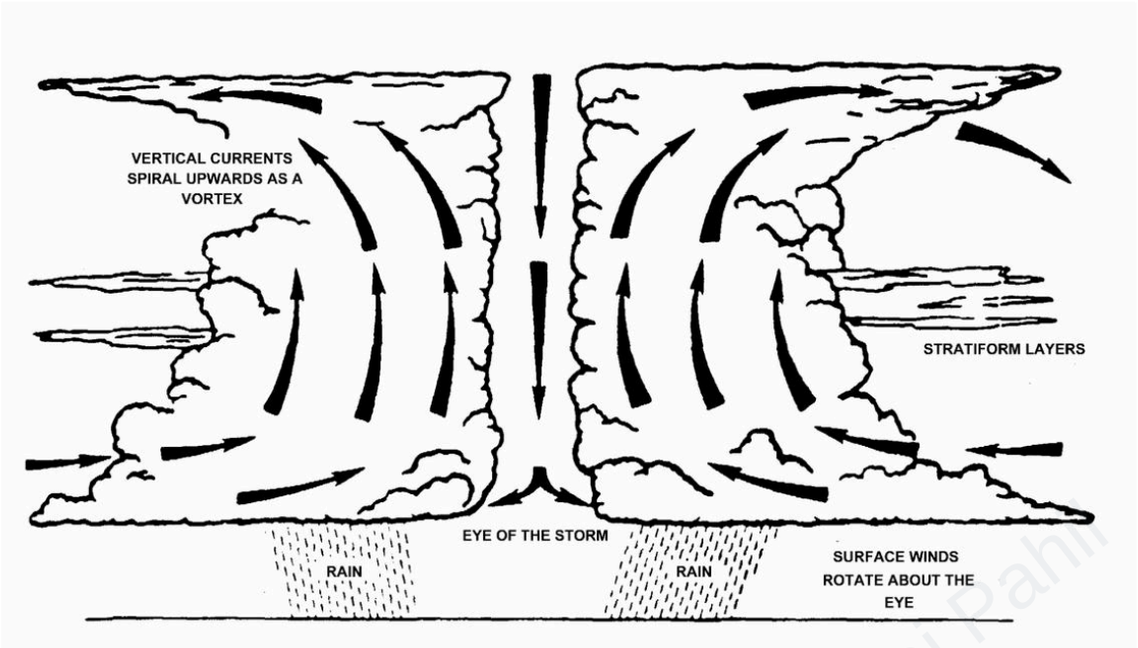
- Worst weather occurs in the **SW sector** — rainfall is steady and heavy in a belt of about 400 km width. Elsewhere it is scanty.
- At time of re-curvature, rain belt shifts to **N–NE sector**. Showery precipitation occurs in rear of depression when winds are strong S–SWly.
- Associated upper air cyclonic circulation extends up to about 300 hPa.
- Monsoon depression slope SWwards with height. E.g., if surface system is over Kolkata, it would be over Visakhapatnam at 300 hPa (9 km).
- **Fair weather over Assam during monsoon indicates intensification or formation of a depression over head Bay or vice versa.**

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8. Tropical Cyclones (TRS)

Tropical Cyclone (TC) / Tropical Revolving Storm (TRS): Non-frontal low-pressure system with numerous TS, strong winds and heavy rain. Form over warm tropical waters (temperature >26°C), between Lat 5–25°. Below 5° there is no Coriolis force → TC cannot form. Above 25° it is cold. **Maximum sustained wind** in a TC is 34 kt or more. As per IMD it is the **highest 3 min (some countries 1 and 10 min) averaged surface wind** within the circulation at 10 m height.

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Cross-section of a tropical revolving storm

Names by Region: Depending on location and strength — Hurricane (Atlantic), Typhoon (NW Pacific), Tropical Storm, Cyclonic Storm, Tropical Revolving Storm, simply Cyclone.

Necessary Conditions for TC Formation

1. **Warm sea** (>26°C) to a depth of 60 m to provide abundant evaporation (liquid water changes into vapour and absorbs latent heat).
2. **High RH** from the surface up to the middle troposphere (about 5 km) to facilitate condensation (water vapour changes into liquid water and releases latent heat) of water vapours into drops and release latent heat (energy absorbed or released during a change of state without a temperature change).
3. **Instability** for vertical development, and **Coriolis Force** to spiral wind inward.
4. **Very little Wind Shear (wind speed or direction changing over distance)**. WS preferably below 20 kt from 850 to 200 hPa. WS will distribute latent heat and intensification will suffer.

Characteristics — Mature Phase (4 Parts)

Zone	Size	Description
Eye	Mostly 30–65 km diameter	Light winds, often clear skies; warmest (warmer by 0–2°C at surface, up to 10°C at 12 km). Inversion from 1 to 3 km. Surrounded by eyewall — ring of towering CB clouds.
Inner Storm Area (Hurricane Core)	50–150 km	Winds >64 kt, violent squalls, torrential rains; thick dark CB clouds called **Eye Wall / Wall Clouds**. Most hazardous part. Pressure gradient very steep. TC exhibit concentric eye walls.
Outer Storm Area	About 400 km	Gale force winds (34 kt+) and occasional squalls. Winds decrease outwards. Spiral bands of clouds.
Edge of Storm	Beyond outer area	Winds are weak and clouding diminishes.

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Extent and Movement

- TC is a vast violent whirl of 150 to 800 km, spiraling around a centre.
- Moves over sea at 300 to 500 km/day at 15–20 kmh. Generally TC move westwards up to 25° N/S then recurve and move E.

Pressure

- Central pressure in TC: between 1000–900 hPa. The pressure at the centre is lower than the pressure around the storm: by about 6–10 hPa in a cyclonic storm, about 15 hPa in a severe cyclonic storm, 21–29 hPa in a very severe one, 40–66 hPa in an extremely severe one, and 80 hPa or more in a super cyclonic storm.
- Pressure gradient very steep up to a radius of 100 km.

Worst Weather Sector:

- In the **N hemisphere** → worst weather in the **right forward sector**.
- In the **S hemisphere** → worst weather in the **left forward sector**.

Upper Air Circulation

Cyclonic circulation in a TC may extend to 40,000 ft. Axis of circulation is almost vertical. In the inner storm area, winds of hurricane force persist up to about 7000 ft. Thereafter, winds weaken. Wind speeds at surface and upper air are stronger in the **right half** in N hemisphere.

Tropical Storms Over India

- Size varies from 50–2000 km radius, average 300–600 km.
- Form over Bay of Bengal and Arabian Sea → move inland.
- Damage mainly due to very strong winds, torrential rain and inundation — **storm wave and high tides**.
- Indian cyclones have a short life span (about 5–6 days, with hurricane intensity for 2–4 days; the longest on record lasted 14 days).
- Average life of Indian cyclones: 5–6 days.
- Notable recent storms: VSCS **Phailin** (2013, Odisha, max wind 220 kmh), **Hudhud** (2014, AP, max wind about 180 kmh, CP 950 hPa), **Vardah** (2016, Andaman/Chennai, peak 130 kmh), **Nilofar** (2014, Arabian Sea, max wind 215 kmh), **Biparjoy** (Jun 2023, Arabian Sea; became an extremely severe cyclonic storm on 11 June; crossed the Saurashtra–Kutch coast close to Jakhau Port, Gujarat, late on 15 June as a very severe cyclonic storm).

Flying Conditions in TC

Flying Conditions — Tropical Cyclone:

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- Clouds in spiraling bands — unbroken cloud mass or reach 18–20 km; severe turbulence in CB towers.
- Frictional turbulence in lowest layers due to strong winds.
- Strong up and downdraughts throughout storm.
- Rain in central area and lowest layers — torrential; winds very strong.

9. Frequency of Cyclones

- Frequency in Bay of Bengal and Arabian Sea is almost nil in **February and March**. Frequent in **May, June, October and November**. **Maximum frequency in October and November**.
- Tropical cyclones are rare in the South Atlantic, but they are not impossible there.
- In NW Pacific, TC occur throughout the year.
- Annual frequency of TC in Bay of Bengal is 4.5 and in Arabian Sea is 1.1.

Tropical cyclone position and avoidance: Use the analysed storm position, forecast movement and current aviation weather information. A wind from the port side alone, without hemisphere and aircraft track, does not establish a universal centre-location or avoidance manoeuvre.

10. Frontal Depressions

Formation of a Frontal Depression / Cyclone: A stationary or quasi-stationary polar front, in the N hemisphere, has warm tropical air to the south and colder polar air to the north. When warm air penetrates into a stationary polar front, a wave is generated. If this wave is stable, it moves eastwards along the front without intensification. If unstable, it intensifies. Pressure falls in the region where the bulge has occurred — cyclonic circulation develops into a depression. Ahead of the centre, warm air replaces cold air (**warm front**). Behind the centre, cold air replaces warm air (**cold front**). The depression moves with the warm front leading and the cold front following.

These depressions of middle latitudes are called **extra-tropical depressions**. Generally move towards E or ENE, being steered by the westerly currents in these latitudes. When a depression on the polar front occludes, the unoccluded portion trails at a lower latitude. Conditions then become favourable for secondary depressions — this may repeat at a still lower latitude → **family of 3 or 4 depressions** → **Cyclone Families**. They sometimes move eastwards together.

Non-Frontal Depressions: Thermal, Orographic and Secondary; Cold-Air Pools

Thermal depression. Surface air is heated and expands. Air flows out at height, so surface pressure falls and the air starts to turn cyclonically. These lows are most frequent over land in summer, and over warmer inland seas and lakes in winter. They are often shallow and weaken with height. If the air is already unstable, thunderstorms, widespread rain or squalls may result.

Orographic (lee) depression. When wind meets a mountain range at a large angle, much of the air flows round the ends and not over the top. This leaves a lack of air on the lee (downwind) side, so a low forms there.

- Dry, stable air: warm, clear and dry weather (Fohn effect).

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- Moist, unstable air: Cu and Cb, showers, possibly thunderstorms and hail.
- Cold front held back by the range, then breaking over it: heavy Cb, line squalls, very heavy showers and poor visibility.

Secondary depression. A small low inside the circulation of a larger (primary) low. Its isobars need not be closed. It forms in three places:

- On a trailing front from an occluded primary. This one may deepen and become the new primary.
- On a trailing cold front inside the primary circulation. It gives little weather of its own, but it may delay the cold front.
- At the tip of the warm sector of a partly occluded low.

Cold-air pool. A mass of cold air aloft, often behind a cold front. It **does not show on the surface chart**. Find it on an upper contour chart or thickness chart as an area of low height or low thickness. Weather is convective: showers, and thunderstorms over land in summer.

11. Flying Hazards in Tropical Cyclones

Never fly through the Eye Wall. The eye may appear calm but transitioning through the eye wall in either direction means traversing the most violent zone of the storm.

13. Practice Q&A

Q1. Wind speed in a tropical severe cyclone is:
(a) 27–33 kt (b) 48–63 kt (c) 17–27 kt

✅ Correct Answer: (b) 48–63 kt — Severe Cyclonic Storm category

❌ Distractors: (a) 28–33 kt = Deep Depression. (c) 17–27 kt = Depression. Candidates often confuse categories — use the table.

Q2. Fronts are characteristic of: (a) Tropical cyclone (b) Extra-tropical depressions (c) Monsoon depressions

✅ Answer: (b) Extra-tropical depressions — Fronts are a feature of extra-tropical (mid-latitude) depressions. Tropical cyclones and monsoon depressions form in a uniform warm, moist air mass and have no fronts.

❌ (a) and (c) are non-frontal.

Q3. In a mature tropical cyclone, the eye area is characterised by:
(a) Moderate weather (b) Stormy weather (c) Calm wind, little clouding and practically no rainfall

✅ Answer: (c) calm wind, little clouding and practically no rainfall — The eye of a mature tropical cyclone has calm or light winds, little cloud and almost no rain. The worst weather is in the eye wall surrounding it.

❌ (a) and (b) describe the eye wall and the rain bands, not the eye.

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Q4. Tamil Nadu cyclone landfalls

Tamil Nadu is exposed to Bay of Bengal cyclones during the post-monsoon season. The official landfall climatology identifies December as the month when these storms more often strike Tamil Nadu. This is a frequency pattern, not a rule excluding a landfall in another month.

Q5. Cyclone seasons over Indian seas

The main cyclone seasons over the Bay of Bengal and Arabian Sea are before and after the southwest monsoon. The regional operational plan identifies April–May and October–December as the periods containing most cyclonic storms. These are seasonal peaks, not a rule excluding storms in other months.

Q6. Where tropical cyclones form

Tropical cyclones usually form over warm tropical seas away from the equator, where the Coriolis effect can help organise rotation. A textbook latitude belt describes a common formation region, not an absolute boundary for every storm. Latitude alone is insufficient: moisture, a disturbance and favourable wind shear also matter.

Q7. During the months of July, August, Monsoon Depression form over:
(a) Over Head Bay (b) Central Bay (c) South Bay

✅ Correct Answer: (a) Over Head Bay — N of 18°N during June, July, August

❌ Distractors: (b) Central Bay = later in season. (c) South Bay = different systems.

Q8. In a monsoon depression maximum weather occurs in:
(a) SE sector (b) NW sector (c) SW sector

✅ Correct Answer: (c) SW sector — worst weather (steady heavy rainfall in ~400 km belt)

❌ Distractors: (a) SE sector = relatively fair. (b) NW sector = also relatively clear. SW sector = convergence zone with heaviest rainfall.

Q9. During re-curvature maximum weather in a monsoon depression occurs in:
(a) SE sector (b) NW sector (c) SW sector (d) NE sector

✅ Correct Answer: (d) NE sector — at time of re-curvature the rain belt shifts to N–NE sector

❌ Distractors: (c) SW sector = before re-curvature. At re-curvature the track changes and the rain belt migrates to N–NE.

Q10. Locating a cyclone centre

A tropical cyclone has an organised circulation around a low-pressure centre, with its most dangerous winds and convection near the eyewall. A single port-side wind observation, without the hemisphere and aircraft track, is insufficient for a universal centre-location rule. Use the analysed storm position and

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movement rather than an unqualified compass shortcut.

Chapter 19: Climatology of India

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1. Introduction — Indian Climate

India has variety of climate: Tropical in the South, Temperate and Alpine in the Himalayan North. The Himalayas receive sustained winter snowfall. Indian climate is strongly influenced by the Himalayas and the Thar Desert. The Himalayas act as a barrier to cold winds flowing down from Central Asia — this keeps most of the Indian subcontinent warmer compared to similar latitudes. The coastal regions are warmer and the rains are more frequent. India is influenced by **two seasons of rains** accompanied by seasonal reversal of winds from January to July. During winters, dry and cold NWly winds blow from the north. In summer the Indian land mass becomes hot → draws moist winds from the Arabian Sea and Bay of Bengal → causes reversal of winds → called the **SW Monsoon** — the most important feature controlling the Indian climate. About 75% of the annual rainfall is received during the short span of four months (**June to September**). There is large variation — W Rajasthan receives less than 13 cm, Mawsynram in Meghalaya gets as much as 1187.3 cm (467.4 in); Cherrapunji gets 1177.7 cm (463.7 in). The word **Monsoon** is derived from Arabic *"Mausim"* — it is the **seasonal reversal of winds and the rainfall**.

2. Six Seasons (Ritu) & IMD Four Climatological Seasons

#	Ritu (Sanskrit)	Months	Characteristics
1	Vasant (Spring)	Chaitra & Vaisakh (Mar–Apr)	Temps around 20–30°C; Vernal Equinox in mid-season
2	Grishm (Summer)	Jyeshtha & Asadha (May–Jun)	Very hot; around 40–50°C; Summer Solstice
3	Varsha (Monsoon)	Sravana & Bhadrapad (Jul–Aug)	Very hot, humid, heavy monsoon rains
4	Sharad (Autumn)	Ashwina, Asauj, Kartika (Sep–Oct–mid Nov)	19–25°C; autumnal equinox in mid-season
5	Hemanta (Fall)	Margashirsha, Agahan, Pausha (late Nov–Dec)	Very pleasant 19–25°C; ends with winter solstice
6	Shishira (Winter)	Magha & Phalguna (Jan–Feb)	Cold, below 10°C; pleasant with occasional sunshine

N India mostly conforms to the marked change of Ritus; Southern India (close to Equator) less so.

IMD Four Climatological Seasons

Season	Months
Winter Season / Cold Weather Season	January–February
Summer / Pre-Monsoon Season	March to May
SW Monsoon (Rainy Season)	June to September
Post-Monsoon / NE Monsoon Season	October–December

3. Winter Season / Cold Weather Season

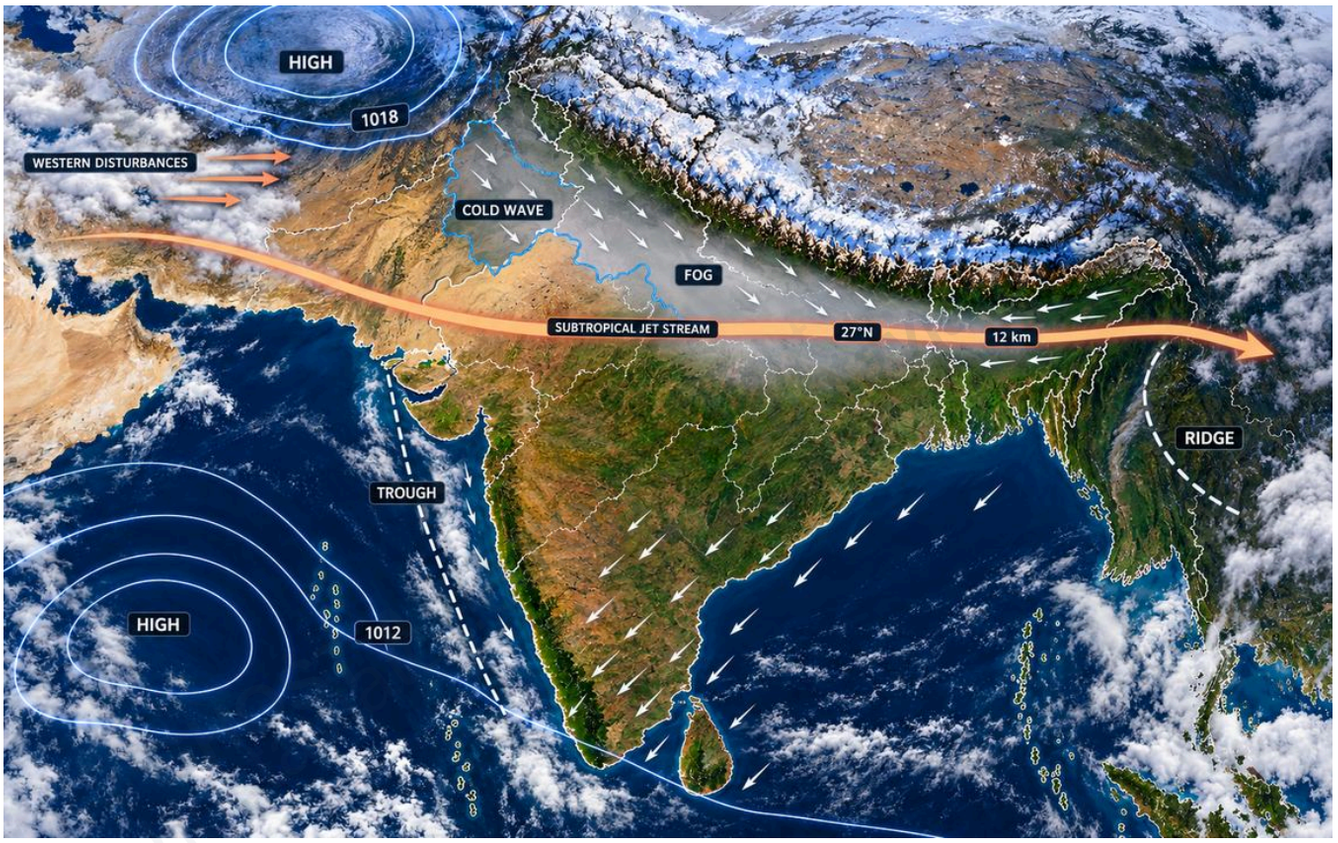


Figure 19.1. Winter weather over India. The picture combines common features for explanation; the pressure values and jet position are illustrative, not a simultaneous weather analysis.

Months: January–February (December–February for NW India)

Pressure Distribution

An extensive belt of high pressure from Sahara to Siberia. High is centered at 45° N and 105°E. India is at the southern periphery of this high. High pressure over the Indian ocean (on surface isobaric chart, about four isobars: 1012–1018 hPa, at 2 hPa interval). The 1012 hPa isobar (a line joining places with the same pressure at the same level and time) extends from Arabian Sea islands to Kerala to Gujarat along the West Coast. The 1018 hPa isobar over Pakistan and J&K. A shallow trough extends from Kerala to Gujarat along West Coast and a shallow ridge over adjoining Myanmar. NWly winds prevail over N India; NEly winds over S India. Under this pressure distribution, NWly winds prevail over N India and NEly over S India.

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Surface Temperatures

December and January are coldest months. Mean daily minimum temperatures range from 22°C in extreme south to 10°C in northern plains to 6°C in Punjab. Diurnal range of temperature is about 13°C; about 10°C over coastal areas; 5°C over coastal Tamil Nadu.

Surface & Upper Winds

Surface: Light, mainly NEly south of 25°N; NWly over N India; Ely over Assam. **Upper Winds:** Up to 10,000 ft NWly over N and Central India; NEly over S India. Thereafter Wly winds prevail aloft, strengthening with height (spreading S to tip of Peninsula). Sub Tropical Jet Stream (STJ) prevails over N India at height of about 12 km and 27° N, with strong Westerly winds of 80–100 kt, maximum being 200 kt.

Synoptic Systems

N India is affected by STJ and 5–7 WDs per month. A few CS and Easterly waves affect S India.

Weather



Figure 19.2. Radiation fog after overnight cooling. This generated illustration shows a typical shallow fog and inversion setting; it is not an observed airport image, a visibility measurement or an approach guide.

Western disturbances can bring cloud, rain or snow to northern India. Their arrival often raises the minimum temperature through cloud cover and moist air, while reducing the daytime maximum. After a disturbance moves away, clearing skies and colder northwesterly air can lower the minimum temperature and favour a cold wave. Fog may also develop where moisture and cooling are sufficient. The sequence and duration vary between events; use the cold-wave and cold-day criteria below.

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Cold wave and cold day in plains and hills: IMD considers these categories when the minimum temperature is 10°C or lower in the plains, or 0°C or lower in hills. For a cold wave, the minimum-temperature departure from normal is −4.5°C to −6.4°C; a departure below −6.4°C is severe. In the plains, an actual minimum of 4°C or lower also meets the cold-wave criterion, and 2°C or lower the severe criterion. A cold day instead uses the maximum-temperature departure: −4.5°C to −6.4°C, or below −6.4°C for a severe cold day. Coastal cold-wave criteria are separate.

Aviation Hazards — Winter:

- **Poor visibility** is the main aviation weather hazard during winter, specially over N India. Fog, mist and haze are all common all over the country in morning and evening.
- Smoke from industries and automobiles reduces visibility. Low level inversion favours reduction in visibility. Conditions most favourable for **Radiation Fog after passage of a WD** — over NW India, UP, Bihar, Central India, N Bengal and Assam (particularly S banks of Brahmaputra river). Sometimes fog prolongs, disrupting air traffic adversely, especially over polluted airfields like Palam (Delhi). On such occasions, there may be little improvement in visibility to 1000–2000 m during noon, but may again deteriorate in evening and night. Such conditions may last for a few days.
- **Advection (horizontal transport by moving air) Fog** occurs over coastal areas and where large water bodies are present. Generally lifts by 1000–1100 hr IST.
- **CAT** is experienced over NE India and in the vicinity of STJ. Mountain waves are also common over high terrain over N and E India.
- **Icing** — freezing level lowers in winter. Hence ice accretion may be experienced at lower levels.

4. Pre-Monsoon / Hot Weather / TS Season (Mar–Apr–May)

Pressure Distribution

Transition season — winter pattern changes over to monsoon pattern. Continuous rise in temperature and fall in pressure over the country. Weak pressure gradient — only two isobars (1006 and 1008 hPa) cover entire country. These run parallel almost along Indian coast line — trough along Peninsula with axis along 78°E longitude. In May the trough shifts to 79°E along Chennai coast. A weak low over NW Pakistan (upper Sindh) and a weak trough over E UP and Bihar. A low lies over Myanmar and a closed high over E Central Bay, W of Bay Islands.

Surface Temperatures

Due to N wards march of sun, land progressively gets heated after January. Temperatures are of the order of 30–35°C over central peninsula. Day temps highest in May in Andamans; in Rajasthan and Kashmir in June and July. Maximum temp highest in April in Saurashtra and in Rajasthan and Kashmir in July. **Heat wave (IMD):** considered when the maximum temperature reaches at least 40°C in the plains or 30°C in the hills. A departure from normal of 4.5°C to 6.4°C is a heat wave; more than 6.4°C is a severe heat wave. Whatever the normal, an actual maximum of 45°C or more is a heat wave and 47°C or more a severe heat wave. For coastal stations: maximum 37°C or more with a departure of 4.5°C or more.

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Surface Winds

In March N–NWly up to E India; April they are N–NWly, they shift to E of India. In May SWly prevail over Rajasthan and Ely from E UP to NE India. Over Orissa, WB, Bihar, Jharkhand and Chhattisgarh, they are Sly. In rest of country NWly to SWly winds prevail. In June winds are Sly and are W–SWly 10–15 kt over rest of country. On certain occasions winds become gusty and dust laden over Gangetic plains. **Dust Raising Winds (DRW)** mostly start at about 0900 hr and continue up to 1700 hr. Thereafter they weaken. Visibility on such occasions reduces considerably over a large area.

Upper Winds (Mar–May)

April: Wly strengthen with height over N India and Peninsula. At 300 hPa max speeds are 40 kt and at 200 hPa, 60 kt N of 22°N. Wind weakens slowly. In May Wly belt shifts to 30°N; Ely 15–20 kt aloft S of 25°N. In June winds are Wly up to 700 hPa and Ely aloft S of 25°N, which strengthen with height to 50 kt at 100 hPa.

Synoptic Systems & Weather



Figure 19.4. Nor'westers are pre-monsoon thunderstorms of eastern India. This generated scene illustrates possible squall, hail and lightning hazards; it is not an observed event or a storm-location forecast.



Figure 19.5. A thunderstorm outflow can lift dust over dry ground, including Rajasthan. This generated illustration depicts one possible severe reduction in visibility; the labelled distance is an example, not a universal value or a measurement. The aircraft is illustrative; the hazardous outflow and dust wall must be avoided.

Characterised by widespread dust haze and extremely high temperatures over N India. Dust storms over N India and Kalbaisakhi (Norwesters) over E and NE India. Triggered by WDs. A few tropical cyclones strike Bengal coast and move NW. STJ weakens and moves northwards. This is the most hazardous season for flying.

Aviation Weather Hazards — Pre-Monsoon (Most Hazardous Season for Flying):

1. **TS and DS Activity:** Violent TS are common over Rajasthan and NW India with all attendant hazards. Line squall type of development is most common over Rajasthan and NW India. **Norwesters** affect Eastern parts — line squall type of development is also reported by the aircrew. Poor visibility due to widespread dust haze is experienced over entire N India. Dust haze may extend to 7–8 km aloft.

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2. **Dust Raising Winds:** Strong surface winds of 30–50 kt between 0900 and 1700 hr. Visibility may reduce to a few hundred metres.

5. SW Monsoon / Rainy Season (June–September)

The most important feature of Indian climate. Spread over four months (**June to September**). Actual period varies from less than 75 days over SW Rajasthan regions to more than 120 days over the SW regions. **Onset of SW Monsoon:**

- Andaman and Nicobar Islands / South Andaman Sea: 22 May
- Kerala: 01 June
- Mumbai: normal onset 11 June.
- New Delhi: normal onset 27 June.
- Covers whole country by: 08 July

Withdrawal normals: The normal withdrawal date is 17 September at Jaisalmer and Bikaner, and 25 September at New Delhi. Dates differ between locations, and the actual withdrawal each year is diagnosed from the weather. A normal date is not a guaranteed date or the earliest permitted declaration.

Pressure Pattern — SW Monsoon

Low pressure area lies over Baluchistan (994 hPa). A trough — **Monsoon Trough (MT)** — lies over the Gangetic plains with its axis extending from Sriganganagar to the Head Bay, through Allahabad. A high pressure is over S Hemisphere. Isobars at 2 hPa interval run from 994 hPa over Baluchistan to 1008 hPa over Tenasserim Coast and S Hemisphere. Pressure gradient steep along west coast.

Surface Temperature — SW Monsoon

Temperatures over S parts are about 26°C and over Rajasthan 31°C. Spatial range is 9°C. Temperatures rise during break monsoon. Diurnal range is about 4 to 9°C.

Surface Winds — SW Monsoon

Ely to SEly prevail to the N of the MT and SW to Wly elsewhere. NWly winds prevail over Malabar coast. Wind speed is about 5–10 kt over land and over seas 15 kt. Strength of monsoon depends on strength of these winds.

Advance of Monsoon — Two Branches

The SW monsoon current advances in **two distinct branches**:

1. **Arabian Sea branch** — moves to Kerala
2. **Bay of Bengal branch** — moves to Bangla Desh and Bengal

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Upper Winds — SW Monsoon

Up to about 10,000 ft winds are W–SWly to the S of MT and to the N. At 200 hPa upwards Ely winds prevail over almost entire India. At 200 hPa and aloft an anticyclone lies over Tibet and another over Baluchistan. The winds are Ely from about 500 hPa upwards and grow stronger with height. The Ely jet stream (**Tropical Jet Stream**) prevails at 15–16 km with max speed of 70–100 kt (max. 150 kt) roughly along 13°N.

Monsoon Trough (MT)

Normal position on sea level chart passes through Ganganagar–Allahabad–Kolkata to Head Bay. The trough fluctuates N–S of its normal position and accordingly governs the rainfall. To the N of this trough lies the Bay branch. And to the S, the Arabian Sea branch. This trough more or less coincides with ITCZ. At 10,000 ft (700 hPa), it is almost parallel to 22°N; from Gujarat to Orissa and at 500 hPa it can be seen around 10–15°N across S Peninsula. Above that it is not traceable.

Break Monsoon

Monsoon Trough shifts northwards, running close to foot hills of Himalayas → drastic reduction in rainfall over the country outside the foot hills. During this period pressure rises over the country. Heavy rains occur over foot hills of Himalayas, particularly over Sub Himalayan West Bengal, Nepal and Bhutan → resulting in floods over NE India and Bihar. Also experienced over East UP, Bihar and adjoining states. During this phase the weather is of Pre-Monsoon type with widespread dust haze of moderate intensity.

Subtropical (Mid-Tropospheric) Cyclone

Sometimes during SW monsoon a cyclonic circulation establishes at Lat/Long 60–80°E and 18–28°N, between levels 500 and 700 hPa, and a trough at the surface. Heavy rainfall in W India and NE Arabian Sea are mostly attributed to this system. Particularly MP, Maharashtra, especially Mumbai received very heavy, unprecedented rainfall in year 2005. The heavy rainfall is due to thick stratiform clouds in which CB clouds are embedded. Such heavy build-up and rainfall occurs mainly to W and SW of the subtropical cyclone centre.

Weather — SW Monsoon

Rainfall oscillates between active spells of widespread rains and breaks. Heavy rainfall in mountains; uncomfortable weather due to high humidity and temperatures — the feature associated with the Breaks. Monsoon Depressions form in northern part of the Bay at frequency of about 2–3 per month moving Nly/Wly direction bringing well-distributed rainfall over central and northern parts.

Strength of Monsoon (Table 19.1)

Strength	Wind Speed Over Sea	Rainfall Over Land
Weak Monsoon	Up to 12 kt	Less than half the normal
Normal (Moderate) Monsoon	13–22 kt	Half to less than 1½ times the normal
Strong Monsoon	23–32 kt	1½ to 4 times the normal

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Strength	Wind Speed Over Sea	Rainfall Over Land
Vigorous Monsoon	>32 kt	>4 times the normal

El Niño and ENSO

El Niño = unusual warming of surface waters in eastern tropical Pacific Ocean = "warm phase" of El Niño-Southern Oscillation (ENSO). **La Niña** = "cool phase" = colder than normal sea surface in the central and eastern Pacific = High pressure in E Pacific and Low pressure in W Pacific. El Niño occurs irregularly at two to seven year intervals. ENSO cycle affects global climate. During El Niño: SE Trades become weak → weak SW Monsoon. During La Niña: SE trades are strong → active monsoon. Now evident that severe droughts occur in India in El Niño years, though opposite is not true. Trade winds push warm water to western Pacific; cooler waters rise up (upwelling) toward surface on coasts of Ecuador, Peru, Chile.

Aviation Hazards — SW Monsoon:

1. **Low Clouds:** Very low clouds (150–200 m agl) and strong surface winds are common during strong monsoon conditions and passage of Monsoon Depressions.
2. **Poor Visibility:** Generally good, but may deteriorate during precipitation. Dust haze in north; monsoon haze along west coast reduces visibility to a certain extent.
3. **CAT:** CAT is likely in the vicinity of TJ.
4. **TS Activity:** At onset of monsoon TS activity of longer duration is a common feature. During Break and then during revival of monsoon also TS are experienced.

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6. Post Monsoon / NE Monsoon Season (Oct–Dec)

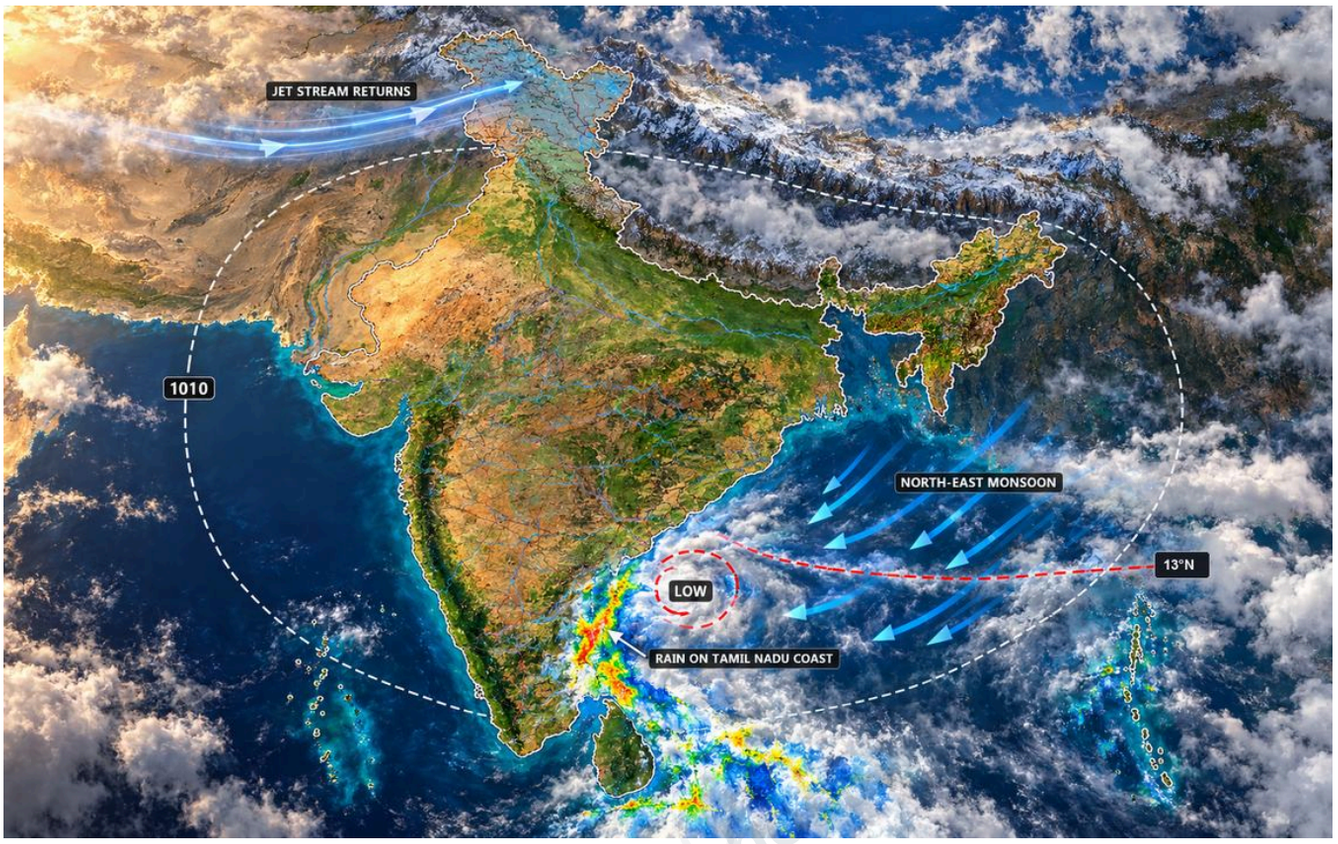


Figure 19.7. North-east monsoon and post-monsoon weather. Moist flow across the Bay can bring rain to Tamil Nadu and nearby areas. The drawn low is an example disturbance, not a permanent feature.

Also called **Retreating SW Monsoon Season**. Transition season — establishment of NEly wind regime over Indian subcontinent. **This is the major period of rainfall activity over coastal AP, Rayalaseema and TN-Pondicherry.** For TN this is the main rainy season, accounting to about 49% of annual rainfall. Rainfall mostly occurs between 2100 and 0300 hr in spells of 3 to 4 days duration.

Pressure Pattern

Summer pressure and wind patterns undergo slow but steady modification to finally change to winter pattern. One isobar (1010 hPa) surrounds the entire country including Bay Islands, Pakistan and parts of Myanmar. Monsoon trough shifts to Bay of Bengal with its axis along 13°N. Pressure field is flat over the country. A weak low (1008 hPa) lies off Chennai coast and another over Baluchistan.

Surface Temperature & Winds

Temperatures are nearly uniform, 18°C, over the country. Diurnal range is 6–7°C. Start falling sharply all over country. Mean temps over NW India show decline from about 38°C in Oct to 28°C in Nov. With withdrawal of monsoon from N and Central India and N Peninsula by first half of Oct, the lower level wind rapidly changes from SWly to NE to SW and get diagonally NE to SW, cross diagonally Bay of Bengal before reaching Cape Comorin and rest of TN. Due to this TN and some parts of Kerala get significant rainfall. Parts of West Bengal, Orissa, Andhra Pradesh, Karnataka and Mumbai also receive minor precipitation. Due to flat pressure distribution, surface winds are light and variable. Sea breeze pronounced along coasts up to 160 km inland.

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Upper Winds

Gradually Wly winds start establishing at all levels and pattern changes to that of winters. STJ reappears over J&K. Above 500 hPa level Ely remain confined to extreme S of country, which weaken to 20 kt.

Weather

A few (1–2 per month) WDs affect N parts. They cause clouding and precipitation over J&K and Punjab. **This is the best period for flying except over southern India.** Good visibility and clear skies are the main features. A few tropical storms form over the Bay of Bengal and the Arabian Sea and cause weather over coastal areas. Retreating monsoon winds cause occasional showers along the east coast of TN. In Tamil Nadu this is the main rainy season — accounts for nearly 60% of annual rainfall in coastal districts.

Maximum Cyclonic Storms

Maximum Cyclonic Storms occur in Oct over Bay of Bengal and in Nov over the Arabian sea. SCS are more in Nov than Oct. They usually form near Andaman Seas and move towards TN, AP and Orissa. Some move N and strike W Bengal or Bangla Desh. A few after crossing over peninsula, revive, recurve and strike Maharashtra or Gujarat coasts.

Aviation Hazards — Post Monsoon:

1. **Poor Visibility:** Generally good. May deteriorate near industrial areas during dawn and dusk. After passage of WD the S poor visibility is during precipitation. In S, CS over S India.
2. **Low Clouds:** Low clouds, very strong winds, heavy rain and TS activity is experienced during passage of a CS over S India.
3. **CAT:** CAT may be experienced due to STJ in the north.

Note: January and July Upper Wind Maps are in later pages.

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7. Route Winds



Figure 19.8. Seasonal upper winds, viewed towards the west: a dot means flow towards the viewer (westerly); a cross means flow away (easterly). Heights and latitudes are approximate typical positions, and the scene is not to scale.

The general pattern of winds over various routes in India is summarised below:

October to May (8 months)

Level	Wind
Above 3 km	Winds all over India are Wly. They become stronger with height. STJ (Wly/80–150 kt) over DLH–CAL Route at 12 km.
Below 3 km	NW–Wly over N India and NEly over S India

June to September (4 months)

Level	Wind
Above 6 km	Ely becoming stronger with height. TJ (Ely/60–100 kt) over MDS (Chennai) lat at 15 km.
Below 6 km	W–SWly over S India and SEly over N India

8. Present System of Forecasting SW Monsoon

IMD issues the forecast for SW Monsoon (Jun–Sep) rainfall over India in two stages:

- First stage forecast is issued in **April**

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- Second stage (update forecast): up to 2020 it was issued in June; since 2021 it is issued **by the end of May**
- Since 2021 the forecast uses a Multi-Model Ensemble (several coupled global climate models run together) along with the older statistical system
- Along with update, forecast for seasonal rainfall over four broad geographical regions (NW India, Central India, South Peninsula, NE India) and July rainfall over the country as a whole are also issued.

8 Predictors for Forecasting (Table 19.2)

S. No	Predictor (Period)	Used for F/C in
1	N Atlantic Sea Surface Temp (SST)(Dec+Jan)	April and June
2	Equatorial SE Indian Ocean SST (Feb+Mar)	April and June
3	East Asia MSL Pressure (Feb+Mar)	April and June
4	NW Europe Land Surface Air Temp. (Jan)	April
5	Equatorial Pacific warm Water Volume (Feb+Mar)	April
6	Central Pacific (Nino 3.4) SST Tendency (Mar Apr May–Dec Jan Feb)	June
7	N Atlantic MSLP (May)	June
8	North Central Pacific Wind at 1.5 Km ASL (May)	June

Model error of April forecast system is 5% and of June forecast system is 4%. **For forecasting July rainfall over the country,** a statistical model is used with 6 predictors (error of the model is 5%): (a) N Atlantic SST, (b) NNO 3.4 SST (May+Jun), (c) N Pacific MSL pressure (Mar), (d) E Asia MSL pressure (Feb+Mar), (e) N Atlantic MSL pressure (May), (f) Equatorial Indian Ocean MSL pressure (Nov of previous year). For forecasting seasonal rainfall over four broad geographical regions, multiple regression (MR) models based on separate set of predictors. All four multiple linear regression models have model errors of 8%. Since 2005, National Climate Centre also prepares an extended range forecast of onset of SW Monsoon over Kerala. Forecast is issued based on a statistical model with 6 predictors (accuracy is a few days, plus minus). IMD also prepares operational long range forecasts for Winter Precipitation (Jan–Mar over NW India) and NE Monsoon rainfall (Oct to Dec) over S Peninsula. Separate statistical models have been developed.

10. Practice Q&A

Q1. During winters the activity of radiation fog increases after the passage of a WD over N India. Which of the following is correct?
(a) Advection fog occurs over northern & central parts of India
(b) Radiation fog occurs over southern parts of India
(c) Activity of Radiation fog increases after the passage of a WD over N India

✅ Correct Answer: (c) Radiation fog increases after the passage of a WD over N India

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❌ Distractors: (a) Advection fog is coastal/water body area phenomena. (b) Radiation fog is predominantly N India phenomenon, not S India.

Q2. Low Temperatures and low humidity is the characteristic of:
(a) Pre Monsoon (b) Summer (c) Monsoon (d) Winter

✅ Correct Answer: (d) Winter — low temps and low humidity are winter characteristics in N India

❌ Distractors: (a) Pre-monsoon has high temps and low humidity. (b) Summer same as pre-monsoon. (c) Monsoon has high humidity and moderate temps.

Q3. The Summer Season period is:
(a) Jan–Feb (b) Mar–May (c) Jul–Sept (d) Oct–Dec

✅ Correct Answer: (b) Mar–May — Pre-Monsoon/Hot Weather/TS Season

❌ Distractors: (a) Jan–Feb = Winter. (c) Jul–Sep = SW Monsoon. (d) Oct–Dec = Post Monsoon + onset of winter.

Q4. Western disturbances in summer

Western disturbances can contribute to thunderstorms and dust storms over Punjab and Rajasthan during summer. It does not establish a universal northernmost track for every summer disturbance.

Q5. The monsoon current over the West Coast of India is:
(a) SWly (b) SEly (c) NEly (d) NWly

✅ Correct Answer: (a) SWly — the SW Monsoon current comes from SW direction and is called SW Monsoon

❌ Distractors: NWly winds prevail over Malabar coast at lower levels, but the monsoon current itself is SWly.

Q6. Monsoon period is:
(a) Jan to Feb (b) March to May (c) Jun to Sep (d) Oct to Dec

✅ Correct Answer: (c) June to September

❌ All other options are other seasons. Monsoon = Jun–Sep = 4 months delivering 75% of India's annual rainfall.

Q7. The monsoon advances with: (a) Bay of Bengal current only (b) Arabian Sea current only (c) Bay of Bengal and Arabian Sea currents

✅ Answer: (c) Bay of Bengal and Arabian Sea currents — The south-west monsoon advances in two branches: the Arabian Sea branch up the west coast and the Bay of Bengal branch towards Bengal and the north-east.

❌ (a) and (b) each name only one branch.

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Q8. What controls monsoon rainfall?

The monsoon trough influences where rain falls, but it is one part of the monsoon system. Bay of Bengal lows and depressions, the heat low and large-scale winds also affect rainfall. Do not treat the trough axis as the only cause.

Q9. Movement of the monsoon trough

A southward shift of the monsoon trough favours an active or vigorous monsoon over much of India. A northward shift towards the Himalayan foothills favours a break over much of India, with heavier rain near the foothills and the Brahmaputra valley. This does not imply uniform rainfall everywhere.

Q10. Break in monsoon occurs when the axis of the monsoon trough is along the:
(a) Gangetic plains (b) Foot hills of Himalayas (c) Bay of Bengal

✅ Correct Answer: (b) Foot hills of Himalayas — MT shifts northward → Break Monsoon

❌ Distractors: (a) MT along Gangetic plains = active/normal monsoon. (c) Bay of Bengal position is also active for Bay area.

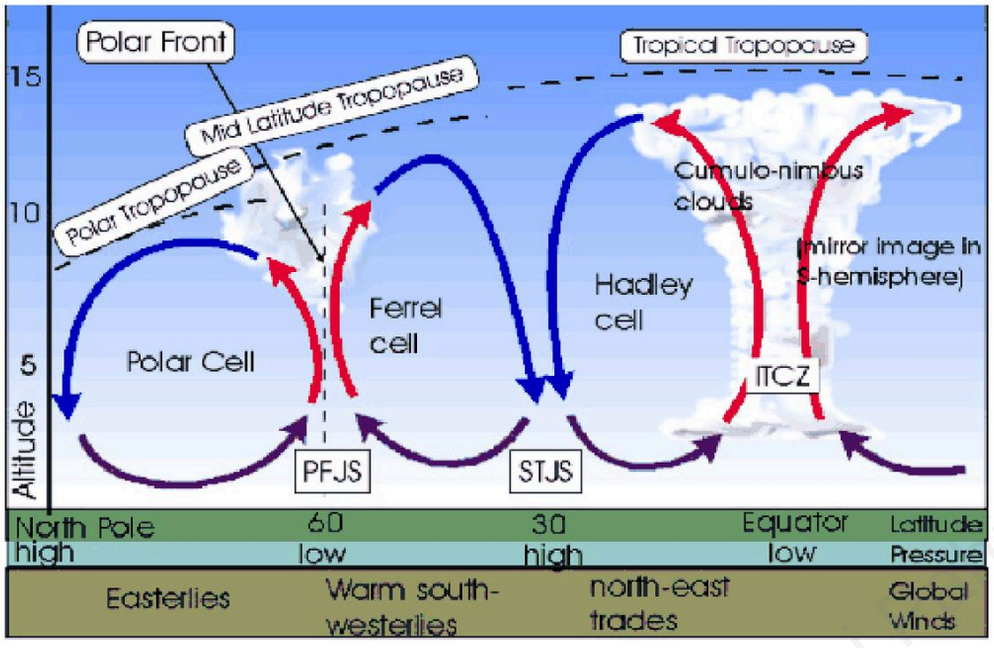
Chapter 20: General Circulation

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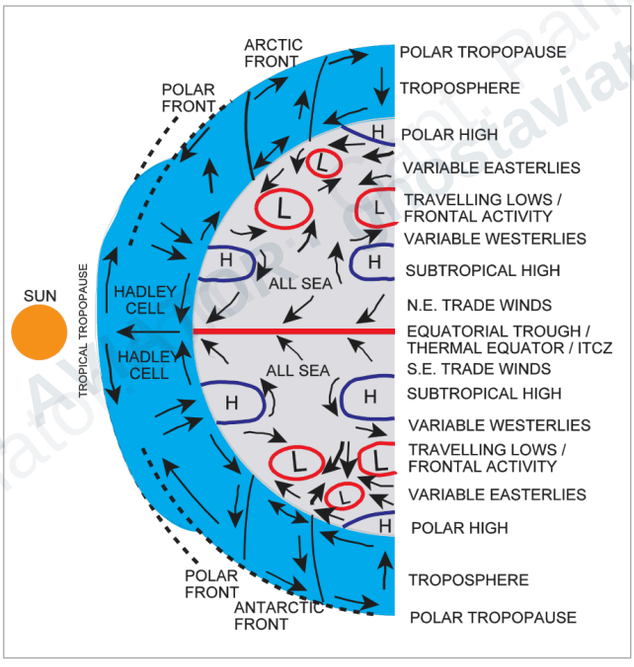
1. Three Cell Model of General Circulation

Definition: The Three Cell Model describes the average circulation of the atmosphere and the atmospheric energy transport. The energy balance over the globe is described in terms of three cells: **Hadley Cell, Ferrel Cell, and Polar Cell**.

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Vertical section of the cells and jet streams



The three cell model

2. Hadley Cell

Hadley Cell operates between the equator and approximately 30°N/S.

- Intense solar radiation at the equatorial region heats the surface and causes air to rise.
- Rising air forms convective clouds and heavy precipitation — this region is known as the **Inter-Tropical Convergence Zone (ITCZ)**.
- In this zone, NE Trades from the N hemisphere and SE Trades from the S hemisphere converge, hence the name ITCZ.
- ITCZ moves north and south following the sun during the year.

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- Due to solar heating, rising air reaches about 30°N and becomes a westerly wind due to the Coriolis force — these are **Zonal Westerlies**.
- The westerly winds keep strengthening with height and establish a **Subtropical Jet** just below the tropopause.
- Poleward-moving air piles up in the subtropical regions and forms a high pressure belt at the surface — **Subtropical High**.
- Subsidence inhibits cloud formation — this is why large deserts occur near 30°N and 30°S.

3. Ferrel Cell

Ferrel Cell operates between approximately 30°N/S and 60°N/S.

- Some of the diverging air at the surface near 30°N moves polewards and is deflected to the right by the Coriolis force.
- At about 60°N the Coriolis-deflected air rises, cools, and condenses — forms clouds.
- This rising air returns equatorwards at upper levels.
- This is the general region of the **polar front**.

4. Polar Cell

Polar Cell operates between 60°N/S and the poles (90°N/S).

- Sinking air at the poles warms and results in high pressure over the poles.
- At the surface, air flows away from the polar high towards lower latitudes. The Coriolis effect turns this equatorward flow to the right in the Northern Hemisphere and to the left in the Southern Hemisphere, giving the polar easterlies.
- The warm subtropical air meets the cold polar air and forms the boundary known as the **polar front**.
- The large temperature contrast at the polar front causes the **polar front jet stream** to occur in the vicinity of the polar cell.

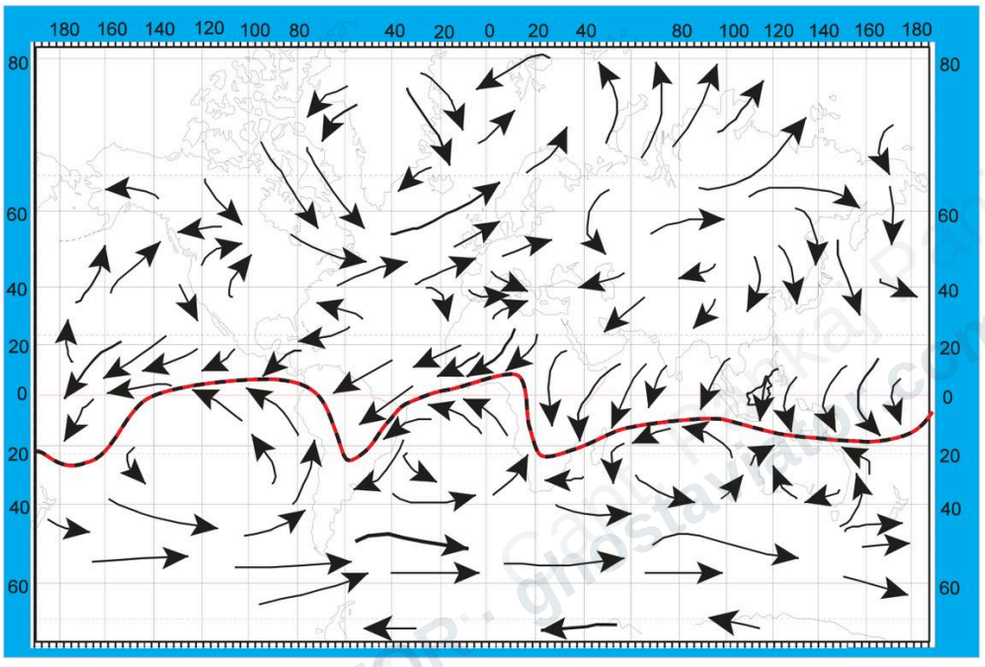
5. Surface Characteristics — Three Cell Model

Feature	Description	Location
Equatorial Doldrums	Rising air creates calms or doldrums	Equatorial region
ITCZ	Along the equator — rising air zone of convection (heat transfer through movement within a fluid)	Near equator (moves N/S seasonally)
Trade Winds	Steady NE winds in N hemisphere; SE winds in S hemisphere	0°–30° N/S
Horse Latitudes	Descending branch of Hadley cell — calm winds, high pressure	Lat 30–40°
Roaring Forties	Strong westerlies between 40° and 50°S. (The general belt of westerlies is about 35°–60° in both hemispheres.)	Southern Hemisphere oceans
Furious Fifties	Stronger westerlies between 50° and 60°S	Around 50°S

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Feature	Description	Location
Crying Sixties	Very strong, noisy westerlies between 60° and 70°S	Around 60°S
Polar Front	Boundary between cold polar air moving equatorwards and warm subtropical air moving polewards	Near 60° lat
Polar Easterly Winds	Cold polar air moving southwest meets prevailing westerlies to form polar front	Polar regions



Prevailing surface winds in January

Important: Over most of the globe the winds in the upper troposphere are westerly, known as **Zonal Westerlies**. Only in a narrow belt near the equator are the upper tropospheric winds **Easterlies**.

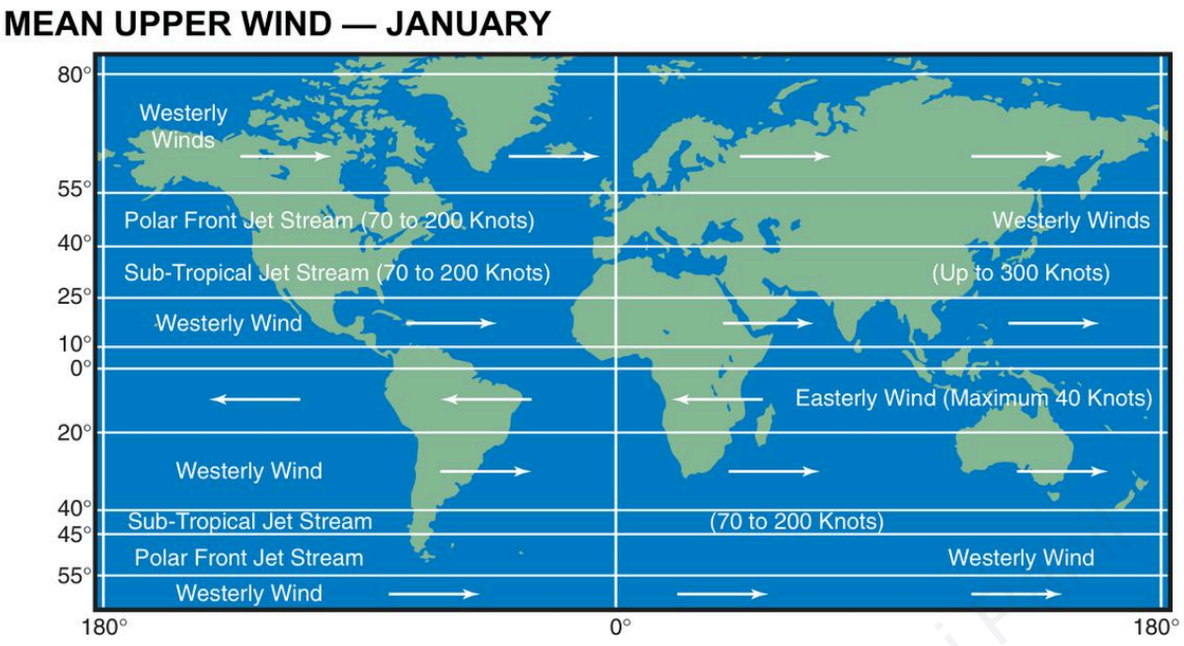
Upper Tropospheric Winds:

- The zonal flows in the upper troposphere govern the movement of disturbances in the pressure and wind patterns.
- Disturbances of **middle latitudes move from west to east** in the westerly current.
- Tropical disturbances form in the equatorial low pressure belt and move in a **westerly direction**, steered by the high level easterly flow.
- Tropical disturbances reaching the zone of transition in the upper level flow change course (or recurve) and begin to move in an opposite direction.

6. Standing Waves (Stationary Waves / Standing Eddies)

Standing Wave (also called Stationary Wave and Standing Eddies): A combination of two waves moving in opposite directions, each having the same amplitude and frequency. Due to interference, the waves are superimposed and their energies are either added or cancelled.

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Mean upper wind in January

- Atmospheric Standing Waves are **Zonally time averaged General Circulation**.
- Time period for averaging is over a month to a season.
- These waves arise due to mountains, continents, oceans, and different sea surface temperatures.
- N/S winds due to these waves transport cold and warm air which affect the overall climate.
- They advect moisture over the continents.
- **Stationary waves are capable of altering the storm tracks.**

7. Rossby Waves (Planetary Waves)

Rossby Waves (also known as Planetary Waves): A type of inertial waves that naturally occur in rotating atmosphere and oceans. They are giant longitudinal curves (troughs and ridges) in high-altitude winds.

- Associated with pressure systems and jet streams, and have a major influence on weather.
- The meandering jet streams are Rossby waves.
- Oceanic Rossby waves move along the boundary between the warm upper layer and the cold part of the ocean.
- Atmospheric Rossby waves are large-scale waves associated with rotation of the Earth. Their meanders help shape the jet stream and the tracks of weather systems. Gravity waves are a different wave mechanism.
- Named after the discoverer **Carl Rossby in 1930s**.

Exam Tip: Rossby waves = Planetary waves. The jet streams meander as Rossby waves. These influence storm tracks and weather systems in the mid-latitudes.

8. General Circulation of the Tropical Lower Stratosphere

The circulation in the lower stratosphere comprises large-scale vertical waves forced by the westward

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propagating Rossby waves — a phenomenon called **Rossby-wave pumping**.

- These waves have a period of a few weeks to a few months.
- The circulation affects surface climate through **radiative changes**.
- Circulation remains confined to the lower stratosphere only.

Practice Q&A

Q1. The polewards moving air piles up in the subtropical regions and forms high pressure belt at the surface, it is called:
(a) Subtropical high (b) Polar High (c) Equatorial high

✅ **Correct Answer: (a) Subtropical high**
At ~30°N/S, the poleward-moving Hadley cell air sinks, forming the Subtropical High pressure belt at the surface.

❌ (b) Polar High: This forms at the poles due to cold sinking air — not at 30°. ❌ (c) Equatorial high: The equator is a region of low pressure and rising air (ITCZ) — not high pressure.

Q2. _____ occurs over subtropical high.
(a) advection (b) convection (c) subsidence

✅ **Correct Answer: (c) subsidence**
Subtropical Highs are characterised by descending/sinking air (subsidence), which inhibits cloud formation and causes clear skies and deserts below.

❌ (a) Advection: Horizontal movement of air — not the defining characteristic here. ❌ (b) Convection: Rising air — this happens at the ITCZ/equator, opposite of what happens at subtropical highs.

Q3. The occurrence of large deserts near 30N and 30S are due to large scale:
(a) advection (b) convection (c) subsidence

✅ **Correct Answer: (c) subsidence**
Large-scale subsidence at the subtropical highs (~30°N/S) suppresses cloud formation and precipitation, creating desert regions (Sahara, Arabian Desert, etc.).

❌ (a)(b): Same reasoning as Q2 — only subsidence inhibits cloud formation.

Q4. A part of the sinking air over the subtropical highs flows towards the equator, turning west (in the northern hemisphere) due to the Coriolis force. This surface air is called:
(a) Trade winds (b) Westerlies (c) Doldrums

✅ **Correct Answer: (a) Trade winds**
The surface return flow from subtropical highs towards the ITCZ is deflected westward by the Coriolis force, creating the NE Trade Winds (NH) and SE Trade Winds (SH).

❌ (b) Westerlies: These blow in the mid-latitudes poleward of 30°. ❌ (c) Doldrums: These are the calm areas at the ITCZ, not a wind system from the subtropical high.

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Q5. The huge vertical circulations, one between the equator and 30N and another between equator and 30S, are called:
(a) Hadley Cells (b) Ferrel Cells (c) Polar cells

✅ **Correct Answer: (a) Hadley Cells**
Hadley Cells are the large convective cells between the equator and ~30°N/S, driven by solar heating at the ITCZ and sinking at the subtropical highs.

❌ (b) Ferrel Cells: These are in the mid-latitudes (30°–60°). ❌ (c) Polar Cells: These are between 60° and the poles.

Q6. Descending branch of the Hadley cell marked by calm winds and high pressure at the surface are called:
(a) Tropical Latitudes (b) Middle latitudes (c) Horse Latitudes

✅ **Correct Answer: (c) Horse Latitudes**
Horse Latitudes (approximately 30–40°) are characterised by calm winds, subsidence, and high pressure — the descending branch of the Hadley cell.

❌ (a) Tropical Latitudes: These are near the equator/ITCZ region, not at 30–40°. ❌ (b) Middle latitudes: These span 30°–60° and include the Ferrel cell — broader term, not specifically calm.

Q7. Upper winds

Upper winds: Mid-latitude upper winds are generally westerly. Tropical upper winds vary with region and season; over India they commonly become easterly during the southwest monsoon. A wind name tells us where the air comes from: a westerly comes from the west and an easterly from the east.

Q8. Movement of mid-latitude disturbances

Mid-latitude disturbances commonly move from west to east in the upper westerly flow. State the movement explicitly as “west to east”. Wind names describe where the air comes from; they should not be used as an ambiguous substitute for the direction in which a weather system travels.

Q9. The tropical disturbance which form in the equatorial low pressure belt move in a:
(a) Easterly direction (b) Westerly direction (c) Northerly direction

✅ **Correct Answer: (b) Westerly direction**
Tropical disturbances form in the equatorial low pressure belt and move westward (from east to west), steered by the high-level easterly flow. Here "westerly direction" means towards the west; it is not a wind name.

❌ (a) Easterly direction: This refers to the wind direction from east — but the movement is towards the west (westerly direction of movement). ❌ (c) Northerly: Not applicable for initial movement.

Q10. Latitudinally, on the average there is radiation …… in the tropics than in the polar regions.

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(a) surplus (b) deficit (c) balance

✅ **Correct Answer: (a) surplus**
The tropics receive more solar radiation than they emit (radiation surplus). The poles emit more than they receive (radiation deficit). This imbalance drives the general circulation.

❌ (b) Deficit: This applies to polar regions. ❌ (c) Balance: There is no balance latitudinally — this imbalance is the DRIVER of general circulation.

Q11. The systems like highs, lows, cyclonic and anticyclonic circulations etc. are associated with distinct types of weather. A study of the behaviour of these systems, known as:
(a) Synoptic Meteorology (b) Climatology (c) Physical Meteorology

✅ **Correct Answer: (a) Synoptic Meteorology**
Synoptic Meteorology is the study of weather systems (highs, lows, fronts, cyclones) and their associated weather, using synoptic-scale charts and observations.

❌ (b) Climatology: The study of long-term average weather patterns over a region. ❌ (c) Physical Meteorology: Deals with physical properties of the atmosphere — radiation, optics, thermodynamics.

Q13. Steady NE winds in the N hemisphere and SE winds in the S hemisphere are called: (a) Easterly winds (b) Trade Winds (c) Tropical Winds

✅ Answer: (b) Trade Winds

❌ (a) 'Easterly winds' is a general term. (c) 'Tropical Winds' is not a technical name.

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Chapter 21: Meteorological Services for Aviation

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1. WMO, WAFS and IMD

Objective of Meteorological Services: To contribute towards the safety and efficiency of air navigation by providing the required meteorological information to operators, ATC, airfield management and others.

World Meteorological Organisation (WMO)

- Specialised agency of UN; looks after weather aspects of aviation in collaboration with ICAO.
- Purpose: facilitate worldwide cooperation in establishing networks of observatories, promoting rapid exchange of weather information, research, application of meteorology.
- **World Weather Watch (WWW)**: coordinates world-wide weather activities using Global Observing System (GOS), Global Telecommunication System (GTS), and Global Data Processing System (GDPS).

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World Area Forecast System (WAFS)

WAFS purpose: To provide global aeronautical en-route weather forecasts to Met Offices.

- 2 World Area Forecast Centres (WAFC) — at **Washington** and **London**.
- 19 Regional Area Forecast Centres (RAFC) at various places; one at **New Delhi**.
- WAFC prepares digital and gridded global forecasts of Upper Winds and Temperatures, Humidity, FL of Tropopause and its Temperature, Max Wind, Icing, Turbulence and CB clouds.
- Also prepares SIGWX forecasts, accidental release of Radioactive material and Volcanic activity.
- Prepared 4 times a day. Upper wind and temperature forecasts are sent out within 5 hr of the observation time; SIGWX forecasts within 7 hr.

India Meteorological Department (IMD)

- National Met agency in India; web site: `https://mausam.imd.gov.in`
- Has 6 Regional Met offices (RMO) at Delhi, Mumbai, Kolkata, Chennai, Guwahati, and Nagpur.

2. Aviation Met Offices in India

Office	Total	Location/Details	Key Functions
AMO (Aerodrome Met Office)	18 (incl. 4 MWOs)	Major national and international airports	24h current weather; METAR/SPECI, RVR, weather forecasts, TAF, Landing Forecast, VOLMET, ATIS, Radar, Satellite pictures; provide forecasting cover to associated AMS
AMS (Aeronautical Met Station)	54	Smaller aerodromes	Current weather observations; aeronautical forecasts from AMOs during watch hours
MWO (Met Watch Office)	4	Delhi, Mumbai, Kolkata, Chennai	FIR meteorological needs; issue SIGMET
TCAC	—	ICAO designation	Monitor TC development; advisory on position, movement, central pressure and winds

OLBS (On-line Briefing System): AMOs provide web-based OLBS. Contents include:

- Wind and temperature forecast for FL50 to FL530
- 9 hr and 30 hr TAF
- National and WAFC SIGWX charts
- METARs of major airports
- Satellite imagery
- Synoptic features
- Route Forecast in T3 format for low level flights
- Local/Area forecasts for all airports in FIR
- Take-off data and SIGMET Warnings

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3. Notification — Advance Notice Requirements

Advance Notice Required (before ETD):

Flight Type	Office	Advance Notice
Non-scheduled flights, national or international	AMO (office with forecasting facilities)	3 hr before ETD
Non-scheduled flights, national or international	AMS (office without forecasting facilities)	At least 3 hr during watch hours, otherwise 18–24 hr before ETD

Notification given in respect of: time of departure, destination, route, level, type of met information required, time of briefing, and other necessary details.

4. Routine Met Observations

METAR

METAR (Aerodrome Met Report, in aeronautical meteorological code): Issued every half hour by Met offices. Appended with TREND type of landing forecast.

SPECI

SPECI (Aerodrome Special Met Report, in aeronautical meteorological code): Issued immediately for **deterioration** in weather to specified conditions. In case of improvement, issued after 10 minutes when conditions stabilise. Trend forecast is appended to each SPECI.

Aircraft Observations and Report (AIREP) / Pilot Report (PIREP)

- Routine aircraft observations during en-route and climb-out phase.
- When air-ground data link is used: automated observations made every 15 min during en-route phase, and every 30 sec during climb-out phase (first 10 min of flight).
- Special observations made whenever SIGMET criteria are encountered or observed.
- Non-routine aircraft observations (e.g. Wind Shear (wind speed or direction changing over distance), Icing, Turbulence) passed to ATS unit as soon as practicable.
- Complete report on Volcanic Activity submitted to AMO, transmitted to MWO of the region.

Aircraft is EXEMPT from making observations if:

- Aircraft is not equipped with RNAV equipment, OR
- Flight duration is 2 hr or less, OR
- Flight is 1 hr or less from the landing place, OR
- Altitude of aircraft is below 1500 m (5000 ft)

Contents of Air Reports (AIREP sections):

- **Section 1:** Information on Identification, Position (Lat/Long), Time, FL or Altitude
- **Section 2:** Operational Information — ETA and Endurance

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- **Section 3:** Met Information — Air Temp., Wind, TS, Turbulence, Icing and Humidity (if available)

Example: `AIREP IA120 BHP 1200 UTC FL350 MS54 270/40KT TS SCT CB TOP FL400=`

5. Types of Forecasts for Aviation

Forecast Type	Validity Period	Example
Nowcasting	Less than 24 hr	TREND, TAF, Aerodrome Warnings
Short Range Forecast	1 to 3 days	—
Medium Range Forecast	4 to 10 days	—
Long range / Extended Range Forecast	Beyond 10 days	IMD Forecast of Monsoon, onset date etc.

(a) TAF — Terminal Aerodrome Forecast

- Issued by **AMOs** for own and associated AMSs.
- TAF valid for at least 9 hr and a maximum of 30 hr.
- TAF valid for 9 hr: issued every 3 hr.
- TAF valid for 12–30 hr: issued every 6 hr, commencing 0000 UTC.

(b) TREND — Landing Forecast

- Issued by **all AMOs**, and by some AMSs where users need it.
- Appended to METAR or SPECI as trend forecast.
- Validity: 2 hr.
- **PROB not used** in TREND forecast.
- **NOSIG** = no significant change in weather.

(c) Forecast for Take Off

- Issued by AMOs, within 3 hr before ETD.
- Contents: **Expected surface wind direction and speed, with any variations**, Temperature, QNH (aerodrome pressure reduced to mean sea level using the standard atmosphere) and any other element as required.

(d) ARFOR — Area Forecast

Contains: Upper winds and temperatures at various levels, significant weather, clouds, freezing level, turbulence, icing, jet stream, tropopause, max wind, wind shear data, airfield warning and supplementary phenomena.

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(e) CODAR

Upper air report from an aircraft (other than weather reconnaissance aircraft), in Figure Code.

(f) WINTEM

Forecast upper wind and temperature for aviation.

(g) ROFOR — Route Forecast

Route forecast for aviation in coded form. Contains: upper winds/temps at various levels, significant weather, clouds, freezing level, turbulence, icing, jet stream, tropopause and maximum wind.

(h) Local/Area Forecast

- Issued 3 times a day for validity of 12 hr.
- IST issue times: 0700 (valid 0730–1930), 1300 (valid 1330–0130), 1800 (valid 1830–0930).
- Area Forecast covers 100 NM or more; Local Forecast covers 50 NM around.
- Contains: surface wind, visibility, weather, clouds, upper winds and temperature, sun rise/set, warnings, etc.

6. SIGWX Charts

Chart Type	Prepared By	Validity	Issue Frequency	Level Coverage
Global SIGWX (WAFS)	WAFCs	Read the forecast valid time	Since April 2025: T+6 to T+48 at three-hour intervals	Read the vertical coverage of the supplied product
National SIGWX (India)	IMD: the four MWOs (Chennai, Delhi, Kolkata and Mumbai)	24 hr	Based on 00, 06, 12 and 18 UTC; 4 times a day	Medium: FL100–FL250; High: FL250–FL630

Study Tip: Read the reference time, forecast lead time, valid time and vertical coverage separately. Since April 2025, multi-timestep WAFS SIGWX forecasts extend from T+6 to T+48 at three-hour intervals. A T+24 forecast is a forecast for a particular future time; it is not a blanket 24-hour validity period.

7. SIGMET

SIGMET: Information issued for aircraft in flight by MWOs (Mumbai, Kolkata, New Delhi, Chennai) for their FIRs, for occurrence or expected occurrence of en-route weather which may affect safety of aircraft operations.

- Issued when a listed phenomenon occurs or is expected, not more than 4 hr before its validity starts. Cancelled when the phenomenon ends.
- Validity: not more than 4 hr.

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- For Volcanic Ash and Tropical Cyclones: issued up to 12 hr before period of validity; updated every 6 hr.
- SIGMETs are passed to **FIC/ACC** for further communication to aircraft in flight.

SIGMET Criteria:

Phenomenon	Code	Description
Thunderstorm (Obscured)	OBSC TS	TS obscured in cloud
Thunderstorm (Embedded)	EMBD TS	TS embedded in cloud layers
Thunderstorm (Frequent)	FRQ TS	Little/no separation between adjacent TS (>75% area)
Squall line	SQL TS	Squall line with TS
Squall line with hail	SQL TSGR	Squall-line thunderstorms with hail
Tropical Cyclone	—	10-min mean wind ≥ 34 kt + cyclone name
Severe Turbulence	SEV TURB	Low level strong surface winds, rotor streaming, in cloud or not in cloud
Severe Icing	SEV ICE	In clouds other than convective clouds; severe icing from Freezing Rain (FZRA)
Severe Mountain Wave	SEV MTW	Down draught ≥ 3.0 m/sec, or SEV TURB observed or forecast
Duststorm	HVY DS	Heavy dust storm
Sandstorm	HVY SS	Heavy sandstorm
Volcanic Ash	VA	VA + Volcano name
Radio Active Cloud	—	Radioactive cloud

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TS Coverage Classification:

- **ISOL TS:** Individual TS — less than 50% area covered
- **OCNL TS:** Well separated TS — 50–75% area covered
- **FRQ TS:** Little or no separation between adjacent TS — greater than 75% area covered

Note: Separate SIGMETs issued to local ATC for Light or MOD Hail and MOD Icing for light aircraft — not disseminated outside the aerodrome of issue.

8. AIRMET and GAMET

AIRMET: SIGMET for low level flights; issued by **MWO**.

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GAMET: An area forecast in abbreviated plain language for a FIR or sub-area thereof; prepared by the **AMOs**; exchanged with Met offices in adjacent FIRs.

9. Warnings

Aerodrome Warning

- Issued by **AMOs** for expected weather which could adversely affect aircraft on ground and aerodrome facilities and services (safety, proper picketing and mooring of aircraft).
- Validity: not exceeding 4 hr.
- **Wind warning:** Speed 30 kt or more, OR Direction of 20 kt wind changes by 45 degrees.
- **Weather warning:** Squall (+ expected direction), TC, DS, SS, TS, GR, Frost, SN, Hoar Frost or Rime, FZ Precipitation, Raising Sand or Dust, VA, Tsunami.
- Warnings for **gliders and light aircraft and helicopters**: when wind expected to reach 17 kt or more; issued with prefix *"Warnings for Light Aircraft"*.
- Passed on to local ATS units; amendments and cancellations done as applicable; not disseminated beyond the aerodrome of origin.

Wind Shear Warning

- Issued by **AMOs** for observed or expected occurrence of wind shear adversely affecting aircraft on approach path, take off path, during circling, or on RW during landing/take off.
- Coverage: up to 500 m (1600 ft) above runway.
- WS above 500 m due to topography is also included.
- Also included as supplementary information in METAR, SPECI and SPECIAL reports (abbreviated plain language).
- Microburst reported by pilots is also included in the warning.
- Example: `WS WRNG MBST APCH RWY 30`
- Aircraft wind shear warning example: `WS WRNG B747 REPORTED MOD WS IN APCH RWY 28 AT 1000`

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Low-Level Wind Shear: Detection Systems, Warnings and Alerts

Low-level wind shear can affect an aircraft on the approach path, on the take-off path, during a circling approach and on the runway. Two products tell the pilot about it.

	Wind shear WARNING	Wind shear ALERT
Made by	The aerodrome meteorological office	Automated ground equipment
Covers	Observed *or expected* wind shear	Observed wind shear only
Ends	Should be cancelled when aircraft report that the shear has ended, or after two hours. The exact rule is agreed locally for each aerodrome.	No cancellation rule is laid down. The alert gives short, up-to-date information.

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Alerts support warnings. They do not replace them.

Where the evidence comes from:

- **Remote-sensing equipment** on the ground, for example Doppler radar.
- **Detection equipment** on the ground: an array (a set) of surface wind and/or pressure sensors watching a runway and its approach and departure paths. One system of this kind is the Low Level Wind Shear Alerting System (LLWAS), in which computer programs compare the wind readings from the sensors.
- **Aircraft reports** during climb-out or approach. If you meet wind shear, tell ATC as soon as practicable. Your report, with the aircraft type, is passed on to others.
- Other sensors on masts or towers near the aerodrome, or on nearby high ground.

If a microburst is observed, reported or detected, the warning and the alert should say so.

Suspect wind shear with: thunderstorms, microbursts, funnel clouds (tornado or waterspout) and gust fronts; strong surface wind with local topography; sea-breeze fronts; mountain waves and low-level rotors; low-level temperature inversions.

10. Exchange of Meteorological Messages

(a) VOLMET

VOLMET: A global network of radio stations, divided into regions, each having VOLMET stations that broadcast MET information for aircraft in flight on a published schedule with 5 minute intervals allocated to each station, on the same HF/VHF frequency. Two types:

- **D-VOLMET (Data Link VOLMET):** Provides current METAR, SPECI, 9hr TAF, SIGMET, Special AIREPs not covered by SIGMET, and where available AIRMET.
- **VOLMET Broadcast:** Provides METAR, SPECI, TAF, SIGMET by continuous and repetitive voice broadcasts. VOLMET Radio Telephony Broadcasts in India are made on HF Channel from Mumbai and Kolkata in turn every half an hour.

Mumbai broadcasts: METARs of Mumbai, Ahmadabad, Chennai, Colombo, Karachi and Male; TAFs of Mumbai, Male, Colombo.
Kolkata broadcasts: METARs of Kolkata, Delhi, Dhaka, Yangon and Katmandu; TAFs of Kolkata, Delhi and Ho-Chi-Minh.

(b) ROBEX (Routine OPMET Bulletin Exchange)

Routine METAR/SPECI of international aerodromes and their alternates are exchanged outside India under ROBEX scheme. Air Reports are also exchanged under ROBEX.

(c) ATIS (Automatic Terminal Information System)

Available at important aerodromes in India. Purpose: to continually broadcast semi-hourly current weather information for general use of the operators.

(d) Current Weather Display

Computer-based display systems at Kolkata, Delhi and Mumbai and Chennai for current weather information in ATC and Met Offices.

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Data Link Weather: D-VOLMET, D-ATIS and ACARS

Data link here means the air-ground data link between the aircraft and the ground. Weather information can be passed over it in place of voice.

Ground to aircraft:

- **D-VOLMET** (data link VOLMET) gives current METAR and SPECI, TAF, SIGMET, special air-reports not covered by a SIGMET and, where available, AIRMET.
- **D-ATIS** is ATIS passed by data link. In India the runway condition report is also passed on D-ATIS.

Aircraft to ground:

- Aircraft observations must be reported by air-ground data link. Where data link is not available or not suitable, special and other non-routine observations are reported by voice.
- A data-link air-report carries at least: wind direction, wind speed and air temperature. A special air-report also gives the condition that caused it.
- An aircraft with no air-ground data link is exempt from routine aircraft observations.

ACARS stands for Aircraft Communications Addressing and Reporting System. Airliners fitted with suitable sensors send automatic weather observations through it, mainly wind and temperature.

Remember: ATIS information can go out of date when the weather changes quickly. Always check the time of the report.

Practice Q&A

Study point for Q1 — Advance notice and supply time

Arrange advance notice with the responsible meteorological office for the particular flight and service. Advance notice is different from the deadline for supplying forecast information. Upper-wind, upper-temperature and significant en-route weather information should be supplied no later than three hours before departure; other information is supplied as soon as practicable.

Q2. For non-scheduled National Flights, advance notice to AMSs is required:
(a) 3 hr (b) 18–24 hr (c) 6 hr

✅ **Correct Answer: (b) 18–24 hr**
For non-scheduled flights, national or international, a station without forecasting facilities (AMS) needs at least 3 hours' notice during its hours of watch, otherwise 18 to 24 hours before ETD; an office with forecasting facilities (AMO) needs 3 hours before ETD.

Q3. The World Area Forecast System (WAFS) provides the Met Offices with: (a) High-quality en-route forecasts of winds and temperature (b) SIGMET (c) TREND

✅ Answer: (a) high-quality en-route forecasts of winds and temperature — WAFS supplies global forecasts of upper winds and temperature, and significant weather charts, to Met Offices.

❌ (b) SIGMET is issued by Met Watch Offices. (c) TREND is a landing forecast appended to a METAR.

Q4. IMD has …… Regional Met offices:

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(a) 4 (b) 5 (c) 6

✅ Answer: (c) 6 — IMD has six Regional Met Offices: Delhi, Mumbai, Kolkata, Chennai, Guwahati and Nagpur.

❌ (a) 4 and (b) 5 are too few.

Q5. There are …… Aerodrome Met Offices:
(a) 17 (b) 18 (c) 19

✅ Answer: (b) 18 — There are 18 Aerodrome Met Offices (AMOs).

❌ (a) 17 and (c) 19 are not the number.

Q6. There are …… Aeronautical Met Stations:
(a) 56 (b) 54 (c) 52

✅ Answer: (b) 54 — There are 54 Aeronautical Met Stations (AMS).

❌ (a) 56 and (c) 52 are not the number.

Q7. In AIREP, Met Information is contained in Section:
(a) 1 (b) 2 (c) 3

✅ **Correct Answer: (c) 3**
Section 3 of the AIREP contains Met Information: Air Temp., Wind, TS, Turbulence, Icing, and Humidity. Section 1 = Identification/Position/Time. Section 2 = Operational (ETA/Endurance).

Study point for Q8 — TAF validity

Short TAFs in India have nine-hour validity; long TAFs have thirty-hour validity. Always read the validity group of the actual forecast.

Q9. The validity of Landing Forecast (TREND) is:
(a) 1 hr (b) 2 hr (c) 3 hr

✅ **Correct Answer: (b) 2 hr**
TREND (landing forecast) appended to METAR/SPECI has a validity of 2 hr.

Q10. The Landing Forecast (TREND) is appended to: (a) METAR and SPECI (b) TAF (c) AIREP

✅ Answer: (a) METAR and SPECI — The landing forecast, called TREND, is appended to a METAR or SPECI and is valid for two hours after the time of the report.

❌ (b) A TAF is a separate aerodrome forecast. (c) An AIREP is an aircraft report.

Study point for Q11 — Local and area forecasts

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Where a continuous ATC watch is maintained, these forecasts are issued three times daily, each valid for the next eight hours.

Study point for Q12 — Coverage of local and area forecasts

Both cover the aerodrome and the area within 100 nautical miles. A nautical mile is a distance unit used in aviation.

Q16. CODAR is:
(a) Radar Report (b) Coded ARFOR (c) Upper report from an aircraft, other than weather reconnaissance aircraft

✅ **Correct Answer: (c) Upper report from an aircraft, other than weather reconnaissance aircraft**
CODAR is an upper air report in Figure Code from any aircraft other than weather reconnaissance aircraft.

Q18. SIGMET is a notice of severe weather for:
(a) actual (b) expected (c) both

✅ **Correct Answer: (c) both**
SIGMET is issued for occurrence OR expected occurrence of en-route weather which may affect safety of aircraft operations. So it covers both actual and expected phenomena.

Q19. SIGMET is issued for aircraft:
(a) in flight (b) on ground (c) both

✅ **Correct Answer: (a) in flight**
SIGMET is specifically issued for aircraft IN FLIGHT by MWOs for their FIRs. Aerodrome Warnings cover ground operations.

Q20. SIGMET is issued by:
(a) WAFC (b) AMO (c) MWO

✅ **Correct Answer: (c) MWO**
SIGMETs are issued by the 4 Met Watch Offices (MWO) at Mumbai, Kolkata, New Delhi, and Chennai for their respective FIRs.

Study point for Q21 — SIGMET validity and issue time

Ordinary SIGMET validity is at most four hours. Tropical-cyclone and volcanic-ash SIGMETs may be valid for up to six hours. Their issue time may be up to twelve hours before validity begins, and they are updated at least every six hours. Issue time and validity duration are different.

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Q22. SIGMET is not issued for:
(a) Rain (b) Volcanic Ash (c) SEV Icing

✅ **Correct Answer: (a) Rain**
Rain by itself is not a SIGMET criterion. SIGMET criteria include: TS (OBSC/EMBD/FRQ/SQL), TC, SEV TURB, SEV ICE, SEV MTW, HVY DS, HVY SS, VA, Radioactive Cloud. Plain rain is not included.

Q23. AIRMET is SIGMET issued by a MWO for the safety of:
(a) low-level aircraft (b) high-level flights (c) all

✅ **Correct Answer: (a) low-level aircraft**
AIRMET is SIGMET for low level flights, issued by MWO.

Q24. GAMET is an area forecast in abbreviated plain language for: (a) low level flights (b) high level flights (c) all level flights

✅ Answer: (a) low level flights — GAMET is an area forecast in abbreviated plain language for low-level flights in a FIR or sub-area, up to FL100.

❌ (b) and (c): high-level flights are served by SIGWX charts and SIGMET.

Q25. The validity of Airfield warnings is not exceeding:
(a) 3 hr (b) 4 hr (c) 6 hr

✅ **Correct Answer: (b) 4 hr**
Aerodrome (Airfield) Warning validity does not exceed 4 hr.

Q29. Wind Shear Warning is issued for observed or expected wind shear above runway (up to):
(a) up to 500 m (b) above 500 m (c) 300 m

✅ **Correct Answer: (a) up to 500 m**
Wind Shear Warning covers up to 500 m (1600 ft) above runway. WS above 500 m due to topography is also included as additional information.

Q30. VOLMET is a MET information for aircraft:
(a) on ground (b) in flight (c) both

✅ **Correct Answer: (b) in flight**
VOLMET is a global network of radio stations that broadcast MET information specifically for aircraft in flight.

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Q32. VOLMET consists of:
(a) SIGMET only (b) TAF only (c) TAF and METAR

✅ **Correct Answer: (c) TAF and METAR**
VOLMET broadcasts METAR, SPECI, TAF, and SIGMET. The basic components are METAR and TAF at minimum.

Study point for Q34 — SIGWX forecast times

A forecast cycle, its lead time and its valid time are different. Since April 2025, multi-timestep WAFS significant-weather forecasts extend from T+6 to T+48 at three-hour intervals. T+6 means six hours after the forecast reference time. Read the valid time of the product used for the flight.

Q35. Global SIGWX Charts for Medium-level are issued for Flight Levels:
(a) 150 & 200 (b) 100 & 250 (c) 100 & 150

✅ **Correct Answer: (b) 100 & 250**
Medium-level SIGWX covers FL100 to FL250. High-level covers FL250 to FL630.

Q40. Individual TS which cover less than 50% area is termed as:
(a) ISOL TS (b) FRQ TS (c) OCNL TS

✅ **Correct Answer: (a) ISOL TS**
ISOL TS = Individual TS, less than 50% area covered. OCNL TS = 50–75%. FRQ TS = >75%.

Q41. If TS cells have no separation between adjacent TS and cover an area more than 75%, it is termed as:
(a) FRQ (b) ISOL (c) OCNL

✅ **Correct Answer: (a) FRQ**
FRQ TS: little or no separation between adjacent TS, covering more than 75% of the area.

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Chapter 22: Weather Radar and Meteorological Satellites

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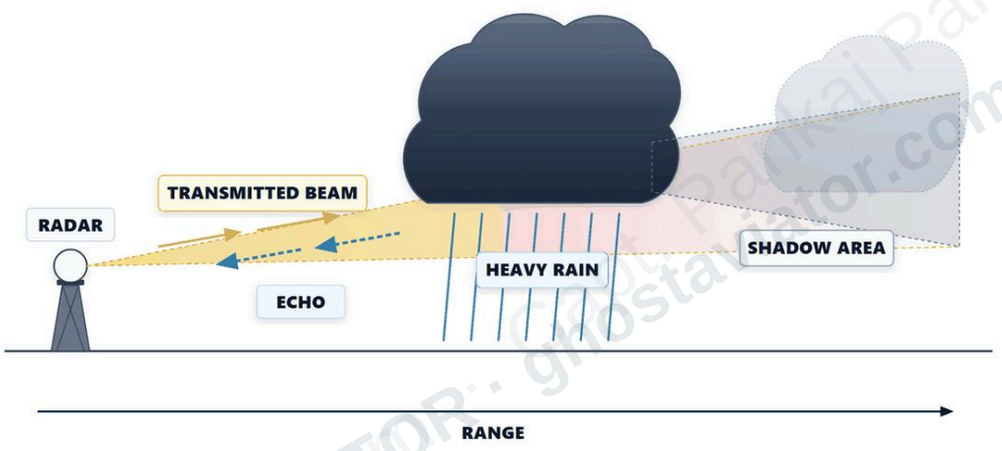
1. Weather Radar — Introduction

Weather Radar (Radio Detection and Ranging): An important tool for detection of rain-bearing clouds, especially CU and CB, their speed, direction of movement, and vertical extent. Used for issuing timely

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weather warnings so that precautionary measures are initiated for parked and moored aircraft. Radars along the coast line of India detect storms and enable issuing of storm warnings.

- IMD has used X-Band weather-cum-wind finding Multi Met Radars and Radiosonde (a balloon-carried instrument that transmits upper-air measurements) system since 1975.
- Good network of S-band cyclone detection radars covering the Indian coast line.
- Operational network of digital X-band radars, including Doppler Weather Radars (DWR) for wind finding, storm detection, and dual capability — planned to be replaced by state-of-the-art DWR in a phased manner.



Radar beam, echo and shadow area

Range: Limited by attenuation of radar waves by rain and earth curvature effect.

- Weather radars can detect up to a range of 500 km, but resolution is better up to a range of 250 km.
- Normally echoes from range within 25 NM are displayed.
- Echoes at long range up to 75 NM are also used.

2. Radar Scopes

Scope	Full Name	Function
PPI	Plan-Position Indicator	Presents a plan view of the echoes; provides the bearing and range of the echo
RHI	Range-Height Indicator	Enables direct reading of the altitudes of CB cells

3. Radar Wavelengths and Bands

Wavelength	0.75–2.5 cm	2.5–4 cm	4–8 cm	8–15 cm	15–30 cm
Band	K Band	X Band	C Band	S Band	L Band

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Band Uses:

- **K and X Band:** Used for detection of clouds.
- **X Band:** Used for storm detection.
- **S Band:** Used for rain, precipitation, cyclones, and depressions. Also called **Cyclone Detection (and tracking) Radar**.

4. Types of Weather Radars

Type	Wavelength	Purpose
Quantitative Radar	10.0 cm	Quantitative measurement of rate of precipitation, total precipitation, intensity of precipitation, and snowfall measurements
Ultra Sensitive Radar	—	Detecting convection (heat transfer through movement within a fluid) before and after cloud development; study CAT, gravity waves, and lee waves
Doppler Weather Radar (DWR)	—	State-of-art; Doppler Effect; wind finding, storm detection, dual capability

5. Doppler Weather Radar (DWR)

DWR is a state-of-art Radar that works on the principle of **Doppler Effect** and measures change of frequency of the return signal to determine if targets are moving towards or away from the radar station. It is a coherent radar: it compares the phase of the returned pulse with the phase of the pulse it sent out.

- A radar beam's reflectivity depends on the diameter of the target and its capacity to reflect.
- Snowflakes are large but weakly reflective; rain drops are small but highly reflective.
- When snow falls through a layer above freezing temperature, it melts; snowflakes become coated with water and radar sees them as very large droplets — can be mistaken for stronger precipitation. On PPI: shows up as an intense ring of precipitation — **Bright Band** at the altitude where the beam crosses the melting level.
- Drizzle does not show up on radar — radar returns are proportional to the **sixth power of droplet diameter**.

DWR is used for the following:

- **(a) Prediction of:** TS/DS/SS, Hailstorm and Tornadoes, HVY rainfall and floods
- **(b) Tracking of:** Cyclones and TS, and estimation of associated winds
- **(c) Study of structure of:** Clouds, Cyclones, TS, Hailstorm, Tornadoes
- **(d) Detection of:** Wind Shear (wind speed or direction changing over distance) and Turbulence, CAT, Gust Front, Microburst, Downburst (a strong descending current that spreads outward near the ground) and Tornadoes
- **(e) Genesis of:** Tornadoes
- **(f) Determination of:** Vertical Wind field and Horizontal Divergence (airflow spreading apart) field

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Additional Radar Facts:

- Most modern weather radars utilise a **pulse-Doppler technique** for estimating precipitation rate, droplet motion with respect to the radar, and radial wind speeds.
- More recently, **dual polarized weather radars** have been developed that enable more accurate determination of precipitation types and sizes.

6. Cloud Echoes and Echo Characteristics

Stratiform Cloud Echoes

- Echoes from Stratiform clouds and steady rainfall are **diffused and ill-defined**; generally of **uniform intensity**.
- **Bright band:** Enhanced radar reflectivity in the melting layer, where snow changes to rain. Wet ice particles reflect strongly. This effect does not determine storm severity.

Convective Cloud Echoes

- Clearly defined **sharp edges**.
- **Blurring of edges** = first sign of decay.
- In RHI: consists of one or more sharp-edged vertical levels.
- Developing shower areas distinctly visible in the echo.
- Squall lines depicted by **line of echoes**.
- If echo is exceptionally intense OR top extends to about −40°C temperature level = **CB cloud**.

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7. Radar Clues to Severe Storms

A hook-shaped reflectivity pattern can accompany a rotating thunderstorm. Doppler rotation and, where available, debris signatures add evidence. A hook or a shape resembling the number 6 does not by itself confirm a tornado.

Rapid growth and strong echoes deserve attention, but no single shape proves a particular hazard. Large drops, wet hail, attenuation and melting particles affect echo strength. A bright band is an enhanced echo from melting snow; it does not classify a storm as weak or severe.

8. Airborne Weather Radar

Airborne weather radar helps a crew locate precipitation and assess weather to avoid. Scan limits, beam width, tilt, colours and turbulence functions depend on the fitted equipment.

Read the radar legend and follow the aircraft equipment instructions. Reflectivity colours show echo strength; they do not directly measure turbulence strength. A weak or absent echo does not establish that the air is safe. Strong intervening precipitation can attenuate the beam (weaken the signal) and hide weather behind it.

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Storm Scope (Lightning Detector)

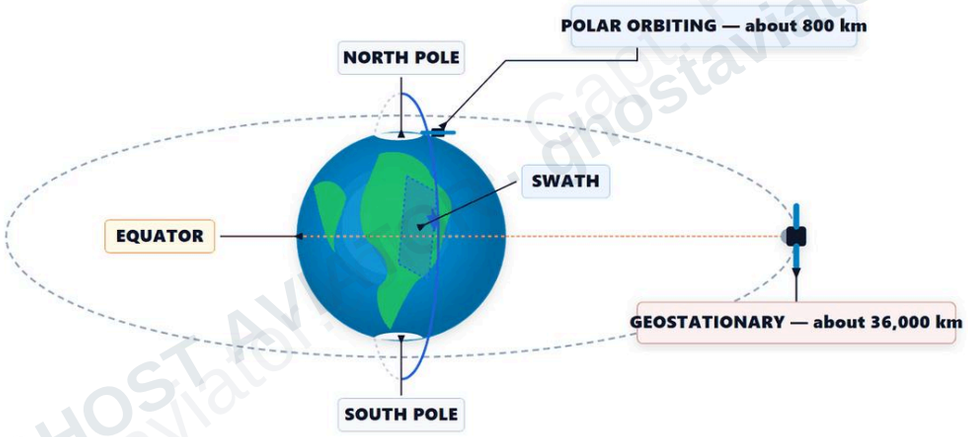
A **storm scope** is an airborne lightning detector. It detects, locates and maps areas of electrical discharge (lightning) inside thunderstorms. This helps the pilot to stay away from them.

It does a different job from weather radar:

Equipment	What it senses	What it tells you
Airborne weather radar	Precipitation (wet hail and rain reflect best)	Where the precipitation is, and how strong the echo is
Storm scope	Electrical discharge (lightning)	Where there is lightning activity inside a thunderstorm

Remember: heavy rain in front can attenuate (weaken) the radar beam and hide weather behind it. The storm scope senses lightning, not a rain echo. Follow the aircraft equipment instructions for both.

9. Meteorological Satellites



Geostationary and polar-orbiting satellites

Parameter	Polar Orbiting	Geostationary
Altitude	650–1500 km	~36,000 km
Earth coverage	Complete earth twice in 24 hr	Same spot always
Same point revisit	Every 12 hr	Continuous
Number of satellites	Multiple	A ring of several over the equator (5 in the classic ring). They cannot see the polar regions
Weather info	Periodic	Round the clock

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Parameter	Polar Orbiting	Geostationary
Examples	NOAA	METEOSAT, INSAT

Frequency of Satellite Imagery: Each INSAT imager takes a picture about every 30 min. With two satellites working in turn, a new picture reaches the forecaster about every 15 min.

10. Indian Meteorological Satellites

The INSAT meteorological series provides imagery and atmospheric sounding (measurements of the atmosphere through depth). Different channels observe different wavelengths, so their resolutions and uses differ.

For the INSAT-3D imager, nominal resolution at the point beneath the satellite is 1 km for visible and short-wave infrared, 4 km for mid-wave and thermal infrared, and 8 km for water vapour.

INSAT-3DS was launched on 17 February 2024 and first obtained meteorological images on 7 March 2024. Satellite position, operational status and image timing should be checked for the actual product rather than memorised as permanent facts.

11. Reading Satellite Imagery

Visible images show reflected sunlight. Clouds commonly appear bright, but brightness also depends on thickness, particle properties and the surface beneath. Visible imagery needs daylight; high clouds are not detected only by their shadows.

Infrared images show temperature inferred from emitted radiation and can be used by day and night. In an unenhanced greyscale image, colder tops commonly appear lighter and warmer surfaces darker. Enhanced colour images use their own legends. Cloud-top temperature can suggest height when temperature decreases with height, but it is not an exact height measurement.

Water-vapour imagery helps reveal moisture patterns and weather-system movement. Read its legend separately from the infrared cloud-top legend.

Practice Q&A

Q1. PPI scope provides:
(a) Plan view of echoes — bearing and range (b) Height of echoes (c) Intensity of echoes

✅ **Correct Answer: (a) Plan view of echoes — bearing and range**
The Plan-Position Indicator (PPI) presents a plan (top-down) view of echoes and provides the bearing and range of the echo.

❌ (b) Height of echoes: This is the function of the RHI (Range-Height Indicator) scope. ❌ (c) Intensity: While colour-coded intensity may be shown, the defining feature of PPI is plan view with bearing/range.

Study point for Q2 — Range–height display

An RHI (Range–Height Indicator) presents radar echoes by range and height along a selected direction.

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Q3. S band radar is used for:
(a) Detection of clouds (b) Storm detection (c) Cyclone detection and tracking (rain, precipitation, cyclones)

✅ **Correct Answer: (c) Cyclone detection and tracking**
S band (8–15 cm wavelength) is used for rain, precipitation, cyclones, and depressions. It is also called the Cyclone Detection (and tracking) Radar.

❌ (a) K and X Band = cloud detection. ❌ (b) X Band = storm detection. S Band is specifically the cyclone band.

Q4. Quantitative Radar measures rate of precipitation using wavelength:
(a) 5.0 cm (b) 10.0 cm (c) 3.2 cm

✅ **Correct Answer: (b) 10.0 cm**
Quantitative Radar (10.0 cm wavelength) is used for quantitative measurement of rate of precipitation, total precipitation, intensity, and snowfall.

Q5. DWR detects if targets are moving towards or away by measuring:
(a) Amplitude of return signal (b) Change of frequency of return signal (c) Time delay of return signal

✅ **Correct Answer: (b) Change of frequency of return signal**
DWR works on the principle of the Doppler Effect — measuring the change of frequency of the return signal to determine if targets (precipitation, wind, etc.) are moving towards or away from the radar station.

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Study point for Q6 — Bright band

A bright band is enhanced radar reflectivity in the melting layer, where snow is changing to rain. Wet coatings on ice particles increase the radar return. It identifies a melting-layer effect; it does not, by itself, classify a storm as weak or severe.

Study point for Q7 — Hook echoes

A hook-shaped echo can be associated with a rotating thunderstorm. It is a warning clue, but the shape alone does not confirm that a tornado is on the ground.

Q8. Spiral bands on radar indicate:
(a) Severe TS (b) Tropical Cyclone (c) Hailstorm

✅ **Correct Answer: (b) Tropical Cyclone**
Spiral bands on radar indicate tropical cyclones and last for a long time.

Study point for Q9 — Interpret airborne weather radar correctly

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Use the display legend and the instructions for the radar fitted to the aircraft. Strong precipitation echoes help identify weather to avoid, but echo strength alone does not measure turbulence strength. Some equipment has a separate Doppler-based indication of likely turbulence.

Q10. Geostationary satellites are placed at an altitude of approximately:
(a) 650–1500 km (b) 36,000 km (c) 500 km

✅ **Correct Answer: (b) 36,000 km**
Geostationary satellites are placed in near circular orbit at about 36,000 km in the plane of the earth's equator. They are earth synchronized with same period of rotation as earth.

❌ (a) 650–1500 km: This is the altitude range for Polar Orbiting satellites. ❌ (c) 500 km: Too low — this would be LEO (Low Earth Orbit).

Study point for Q11 — Satellite coverage

Geostationary satellites remain in a fixed position relative to Earth and repeatedly observe a large area. Their viewing angle becomes poor toward the poles, so no fixed number of geostationary satellites provides complete coverage of the entire globe. Polar-orbiting observations complement this coverage.

Q12. Kalpana-1 satellite is located at:
(a) 74°E (b) 93.5°E (c) 82°E

✅ **Correct Answer: (a) 74°E**
Kalpana-1 was launched on 12 September 2002 and is located at 74°E. INSAT-3A is at 93.5°E. INSAT-3D is at 82°E.

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Study point for Q13 — Infrared image legends

In a conventional unenhanced greyscale infrared image, colder features appear lighter and warmer features darker. Enhanced colour products use their own temperature legends; read the supplied scale.

Q14. Drizzle does not show up on radar because radar returns are proportional to:
(a) Square of droplet diameter (b) Fourth power of droplet diameter (c) Sixth power of droplet diameter

✅ **Correct Answer: (c) Sixth power of droplet diameter**
Radar returns are proportional to the sixth power of droplet diameter. Since drizzle droplets are very small, their sixth power is tiny — making them virtually invisible to radar.

Q15. INSAT-3D was launched from:
(a) Sriharikota (b) Baikonur (c) French Guiana Space Centre

✅ **Correct Answer: (c) French Guiana Space Centre**
INSAT-3D was launched on 26 July 2013 (Indian date) from Kourou, French Guiana (Ariane-5 launch site). It is an advanced weather satellite for operational, environmental and storm warning.

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Chapter 23: Meteorological Instruments

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Met Instruments — Master Table (Table 23.1)

This chapter is a concise but critical reference chapter for DGCA examination. The entire chapter is presented as Table 23.1 — a complete summary of meteorological elements, the instruments used to measure them, and their units of measurement. **Every row is examinable.**

S.No	Element	Instruments Used	Unit of Measure
1	**Pressure**	a. Mercury barometer (an instrument that measures air pressure) b. Aneroid barometer c. Barograph (Self-recording)	Hectopascal (hPa) or Millibar (mb) or Inches (in) of mercury
2	**Air Temperature (TT)**	Dry, Max (Mercury) thermometer Min (Alcohol) thermometer	Degree Celsius (°C)
3	**Dew Point (temperature to which air must cool to become saturated at unchanged pressure and water-vapour content) Temperature (TdTd)**	Dry & Wet bulb (Mercury) thermometers	Degree Celsius (°C)
4	**Relative Humidity (RH)**	a. Dry & Wet bulb thermometer b. Hygrometers c. Hygrograph (Self-recording) d. Psychrometer	Percentage (%)
5	**Humidity Mixing Ratio (HMR)**	No direct instrument; worked out from dry-bulb and wet-bulb (or dew point) readings and pressure	gm/kg
6	**Precipitation (Rain, Snow)**	a. Raingauge b. Self-recording rain-gauge c. Hyetograph d. Snowgauge	mm or cm (depth) Amount (by melting for snow)
7	**Surface Wind (Direction & Speed)**	a. Wind Vane (for Direction) b. Anemometer (an instrument that measures wind speed) c. Anemograph (Self-recording)	Direction: Degrees from True North in clockwise direction Speed: Knots (KT)

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S.No	Element	Instruments Used	Unit of Measure
8	**Cloud Base**	a. Search light / Laser beam b. Ceilometer / Ceilograph c. Ceiling Balloon d. Nephroscope (direction of movement) e. Alidade	Feet (ft) or m 8 points of compass
9	**Visibility / RVR**	AVRA Transmissometer or Scopograph Forward Scatterometer Manually — Visibility Landmarks	m or km
10	**Upper Winds**	a. Optical Theodolite b. Hydrogen filled Pilot Balloon c. Radar Wind (RAWIN)	Direction: in degrees Speed: Knots (KT) or km/hr (KMH) or m per sec (MPS)
11	**Upper Air Pressure, Temperature and Humidity**	Radio Sonde Dropsonde (used in cyclones)	Hectopascal (hPa) Degree Centigrade (°C) % (Relative Humidity (actual moisture compared with saturation at the current temperature, expressed as a percentage))
12	**Clouds**	Weather Satellites RADAR	—

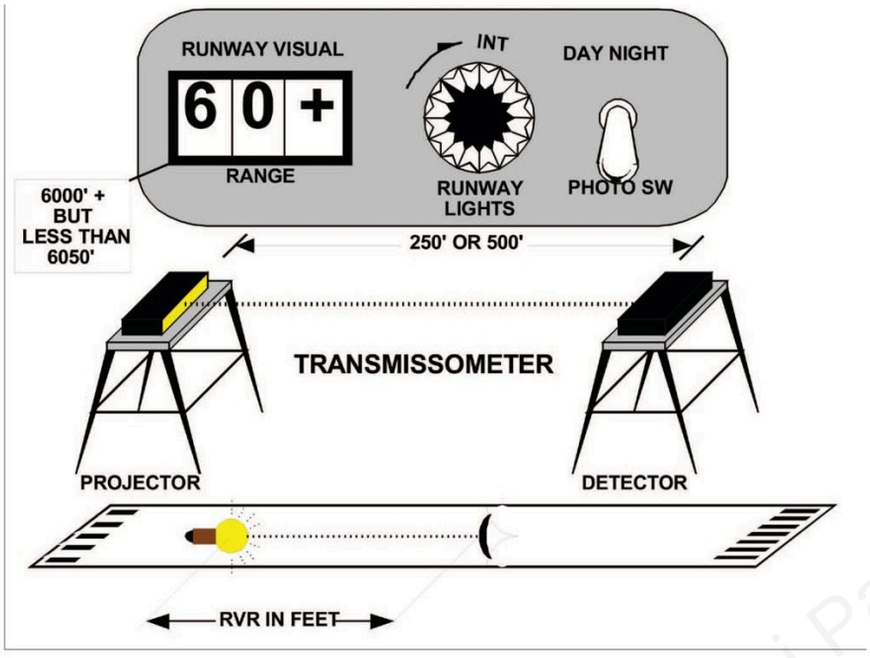
Instrument Classification by Category

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Laser ceilometer

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Transmissometer: a projector and a detector a fixed distance apart beside the runway

Self-Recording / Automatic Instruments

- **Barograph** — Pressure (self-recording)
- **Hygrograph:** a continuous record of relative humidity, normally in percent. Mixing ratio is commonly expressed in g/kg of dry air and is a different quantity.
- **Self-recording rain-gauge** — Precipitation
- **Anemograph** — Surface wind speed (self-recording)
- **Ceilograph** — Cloud base (automated)
- **AVRA** — Automated Visibility and RVR
- **Transmissometer / Scopograph** — Visibility / RVR

Upper Air Instruments

- **Radiosonde (a balloon-carried instrument that transmits upper-air measurements)** — Measures upper air pressure, temperature, humidity (balloon-borne)
- **Dropsonde** — Like Radiosonde but *dropped from aircraft into cyclones* to gather upper air data
- **Optical Theodolite** — Tracks pilot balloon to determine upper wind direction and speed
- **Hydrogen filled Pilot Balloon (PIBAL)** — Tracked optically for upper winds
- **RAWIN (Radar Wind)** — Radar tracking of balloon for upper winds

Cloud Base / Ceiling Instruments

- **Search light / Laser beam:** Projects beam upward to measure cloud base height
- **Ceilometer / Ceilograph:** Automated instrument for cloud base measurement
- **Ceiling Balloon:** Balloon released and timed until it disappears into cloud base
- **Nephroscope:** Determines direction of cloud movement — 8 points of compass

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- **Alidade:** Sighting instrument for angle/direction of cloud features

Humidity Instruments

- **Psychrometer:** uses dry- and wet-bulb readings to determine humidity.
- **Hygrometer:** measures humidity.
- **Hygrograph:** records relative humidity continuously, normally on a percentage scale.
- **Mixing ratio:** mass of water vapour divided by mass of dry air, commonly expressed in g/kg. It can be derived using water-vapour and pressure measurements; it is not the direct trace of a hygrograph.

Pressure Instruments

- **Mercury barometer:** Standard; measures atmospheric pressure in hPa or inches of mercury
- **Aneroid barometer:** Uses a sealed metal capsule (no liquid); portable
- **Barograph:** Self-recording aneroid barometer — continuous pressure trace

Temperature Instruments

- **Dry thermometer:** Measures ambient (dry bulb) temperature — Mercury type
- **Max thermometer:** Records maximum temperature — Mercury (has constriction to retain maximum)
- **Min thermometer:** Records minimum temperature — Alcohol (freezing point much lower)
- **Wet bulb thermometer:** Wet wick around bulb; combined with dry bulb to find dew point and RH

Practice Q&A

Q1. Pressure is measured in units of:
(a) Knots (b) Hectopascal (hPa) or Millibar (mb) (c) Percentage (%)

✅ **Correct Answer: (b) Hectopascal (hPa) or Millibar (mb)**
Atmospheric pressure is measured in Hectopascals (hPa), Millibars (mb), or inches of mercury. Note: 1 hPa = 1 mb.

Q2. Minimum temperature is measured by:
(a) Mercury thermometer (b) Alcohol thermometer (c) Aneroid thermometer

✅ **Correct Answer: (b) Alcohol thermometer**
The minimum thermometer uses alcohol (not mercury) because alcohol has a much lower freezing point (~−114°C) than mercury (−39°C), allowing measurement of very low temperatures.

❌ (a) Mercury thermometer: Used for Max temperature and Dry/Wet bulb (air temp and dew point). Mercury freezes at −39°C and cannot measure Arctic/polar minimum temperatures. ❌ (c) Aneroid thermometer: Does not exist — aneroid refers to pressure instruments (barometer).

Study point for Q3 — Humidity instruments

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A hygrometer measures humidity. A psychrometer uses dry- and wet-bulb readings to determine it. A hygrograph records relative humidity, commonly in percent; it does not directly record the humidity mixing ratio (mass of water vapour divided by the mass of dry air, commonly in g/kg).

Q4. Cloud base is measured by:
(a) Hygrograph (b) Ceilometer / Ceilograph / Ceiling Balloon (c) Anemograph

✅ **Correct Answer: (b) Ceilometer / Ceilograph / Ceiling Balloon**
Cloud base instruments include: Search light/Laser beam, Ceilometer/Ceilograph, Ceiling Balloon, Nephroscope (direction of movement), and Alidade. Units: feet or metres; 8 points of compass (for direction).

Study point for Q5 — Upper-air measurements

A radiosonde ascends beneath a balloon. A dropsonde descends after release from an aircraft. Both can measure pressure, temperature and humidity; tracking with GPS can also provide wind.

Study point for Q6 — Measuring upper winds

Upper winds can be obtained by tracking a radiosonde with GPS and from aircraft reports. A radiosonde is an instrument package carried by a balloon.

Q7. Visibility / RVR is measured by:
(a) Wind Vane (b) Transmissometer / Scopograph / Forward Scatterometer (c) Nephroscope

✅ **Correct Answer: (b) Transmissometer / Scopograph / Forward Scatterometer**
Visibility and RVR are measured by AVRA, Transmissometer or Scopograph, Forward Scatterometer, or manually using Visibility Landmarks. Units: metres (m) or km.

❌ (a) Wind Vane: Measures wind direction at the surface. ❌ (c) Nephroscope: Measures direction of cloud movement (8 compass points).

Q8. Surface wind direction is reported from True North in which direction?
(a) Anticlockwise (b) Clockwise (c) From South

✅ **Correct Answer: (b) Clockwise**
Wind direction is reported in degrees from True North in the clockwise direction. For example, 090° = East, 180° = South, 270° = West.

Study point for Q9 — Mixing-ratio units

Mixing ratio is the mass of water vapour divided by the mass of dry air. It is commonly expressed in g/kg. Relative humidity is a different quantity, expressed as a percentage. A hygrograph records relative humidity.

Study point for Q10 — Dropsonde uses

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Dropsondes are used in tropical-cyclone reconnaissance and in other atmospheric research. The original alternatives overlap, so they do not form a valid single-answer question.

Chapter 24: Station Model

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1. Introduction to Station Model

Definition: A *station model* is a standardised symbolic representation of surface weather observations plotted on a synoptic weather chart. Each symbol, number, and code occupies a fixed position relative to a central circle that represents the observation station.

Met stations worldwide make standardised surface observations and encode them using WMO codes. These are then plotted as station models on synoptic charts used by forecasters and pilots. Reading a station model is a core DGCA exam skill.

🎯 **Exam Tip:** The station model is always tested in DGCA exams — usually 3–5 questions per paper. Memorise parameter positions, ww codes, and cloud symbol tables.

2. Station Model Layout & Parameter Positions

Standard Station Model Layout

TT ↗
Temperature °C

↖ **PPP**
Sea Level Pressure QFF (last 3 digits, tenths)

Wind

↓

⊙

Station

T_dT_d
Dew Point (temperature to which air must cool to become saturated at unchanged pressure and water-vapour content) °C

ww (left) **W_1W_2** (right)

PP
Pressure Change in 3 hr / 10

VV
Visibility code 90–99

N circle
(Total Cloud)

a
Pressure Tendency Symbol

C_L
Low Cloud type

h
Height of base of C_L or C_M

RRR
Rainfall since 0300 UTC in code

C_M
Medium Cloud type

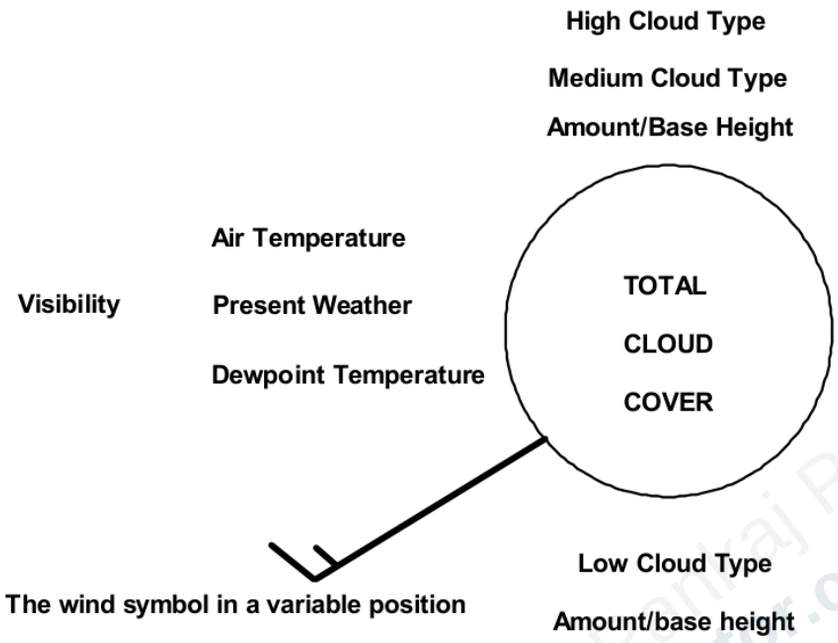
N_h
Amount of Low/Medium Clouds

P_1P_2
Pressure Change in 24 hr

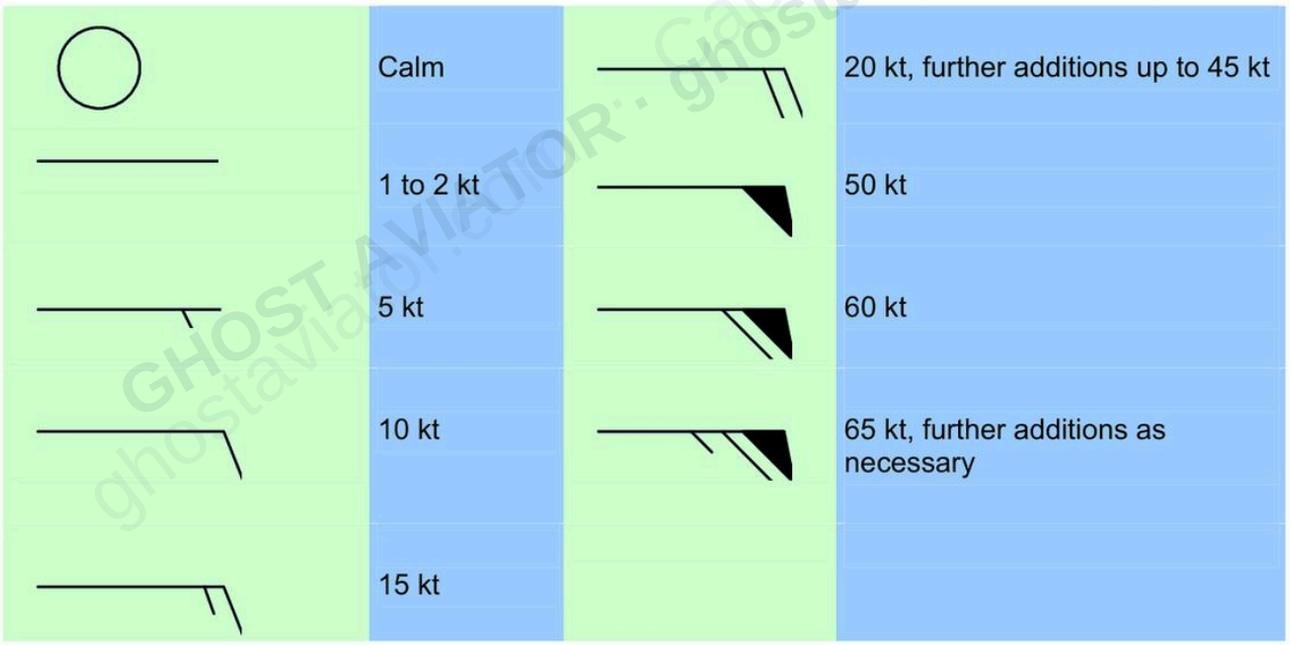
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C_H — High cloud type | **GG** — Time of observation (full hr UTC) | $t_s t_s$ — Sea Surface Temp | $H_w H_w$ — Wave height | $P_w P_w$ — Wave period



Layout of the station model



Wind speed symbols

Complete Symbol Key

Symbol	Parameter	Units / Description
TT	Temperature	°C
$T_d T_d$	Dew Point Temperature	°C
VV	Visibility	Code 90–99 (see Table 24.4)
ww	Present Weather	Code 00–99 (Table 24.5–24.7)

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Symbol	Parameter	Units / Description
W_1W_2	Past Weather	Symbols
PPP	Sea Level Pressure QFF (station pressure reduced to mean sea level using actual temperature)	Last 3 digits (tenths of hPa); prefix 9 or 10
RRR	Rainfall since 0300 UTC	Code (see Table 24.4)
PP	Pressure Change in 3 hr	÷ 10 (tenths of hPa)
a	Pressure Tendency Symbol	Rising/Falling/Steady etc.
t_st_s	Difference in Sea Surface Temp & TT	°C
H_wH_w	Height of First (swell) Wave	Repeated for subsequent waves
P_wP_w	Period of First Wave	Seconds
C_H	Type of High Clouds	Code figure 0–9
C_M	Type of Medium Clouds	Code figure 0–9
C_L	Type of Low Clouds	Code figure 0–9
N	Total Cloud Amount	Octas 0–9 (circle fill)
N_h	Amount of Low/Medium Clouds	Octas
h	Height of base of C_L or C_M	Code 0–9 (see Table 24.4)
GG	Time of Observation	Full hour UTC
P_1P_2	Average Speed of Ship in 3 hr	Knots

3. Main Symbols — ww Abbreviations

Key present weather symbols used in the simplified symbol table (Table 24.2):

Symbol	Meaning	Symbol	Meaning
RA	Rain	DS	Dust Storm
DZ	Drizzle	FG	Fog
SN	Snow	SH	Shower
TS	Thunderstorm	GR	Hail (Grains)
BR	Mist	HZ	Haze
FU	Smoke	SA	Sand
PO	Dust/Sand Whirl	SS	Sandstorm
IC	Ice Crystals	PE	Ice Pellets

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Symbol	Meaning	Symbol	Meaning
BLSN	Blowing Snow	DRSN	Drifting Snow
FZRA	Freezing Rain	FZDZ	Freezing Drizzle

Past Weather Codes W_1W_2 (simplified):

Code	W_1 or W_2
0	Cloud covers half the sky or less for the whole period
1	Cloud covers more than half the sky for part of the period
2	Cloud covers more than half the sky for the whole period
3	Sandstorm, duststorm or blowing snow
4	FG (Fog)
5	DZ (Drizzle)
6	RA (Rain)
7	SN (Snow)
8	SH (Shower)
9	TS (Thunderstorm)

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4. Present Weather (ww) Code Tables

ww Code Ranges:

- 00–19 — No precipitation at station at time of observation
- 20–29 — Weather during last hour but NOT at time of observation
- 30–39 — DS, SS or blowing SN at time of observation
- 40–49 — Fog at time of observation
- 50–59 — Drizzle at time of observation
- 60–69 — Rain at time of observation
- 70–79 — Solid precipitation (Snow etc.) at time of observation
- 80–89 — Shower at time of observation
- 90–99 — 90: hail shower without thunder. 91–94: rain, snow or hail now, with a thunderstorm in the past hour but not now. 95–99: thunderstorm at the time of observation

Key ww Codes to Memorise

ww	Present Weather	ww	Present Weather
00	No Cloud	50	Slight intermittent drizzle

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ww	Present Weather	ww	Present Weather
01	Clouds decreasing	51	Slight continuous drizzle
02	Clouds unchanged	52	Moderate intermittent drizzle
03	Clouds increasing	53	Moderate continuous drizzle
04	Smoke haze	54	Heavy intermittent drizzle
05	Moist haze	55	Heavy continuous drizzle
06	Dust haze	60	Slight intermittent rain
07	Dust raising winds	61	Slight continuous rain
08	Dust devil	63	Moderate continuous rain
09	Distant sandstorm	65	Heavy continuous rain
10	Mist	80	Slight shower
11	Shallow fog in patches	81	MOD/HVY shower
12	Shallow fog	82	Violent shower
13	Lightning	83	Slight shower with snow
14	Rain not reaching ground	85	Slight snow shower
17	Thunderstorm without rain	89	Slight SH of GR without TS
18	Squall	90	MOD/HVY SH of GR without TS
19	Funnel cloud	91	Slight RA after TS ceased
20	Drizzle (last hour)	95	Light TS with RA
21	Rain (last hour)	96	Light TS with GR
22	Snow (last hour)	97	HVY TS with RA/SN
28	Fog (last hour)	98	TS and DS
29	Thunderstorm (last hour)	99	HVY TS with GR

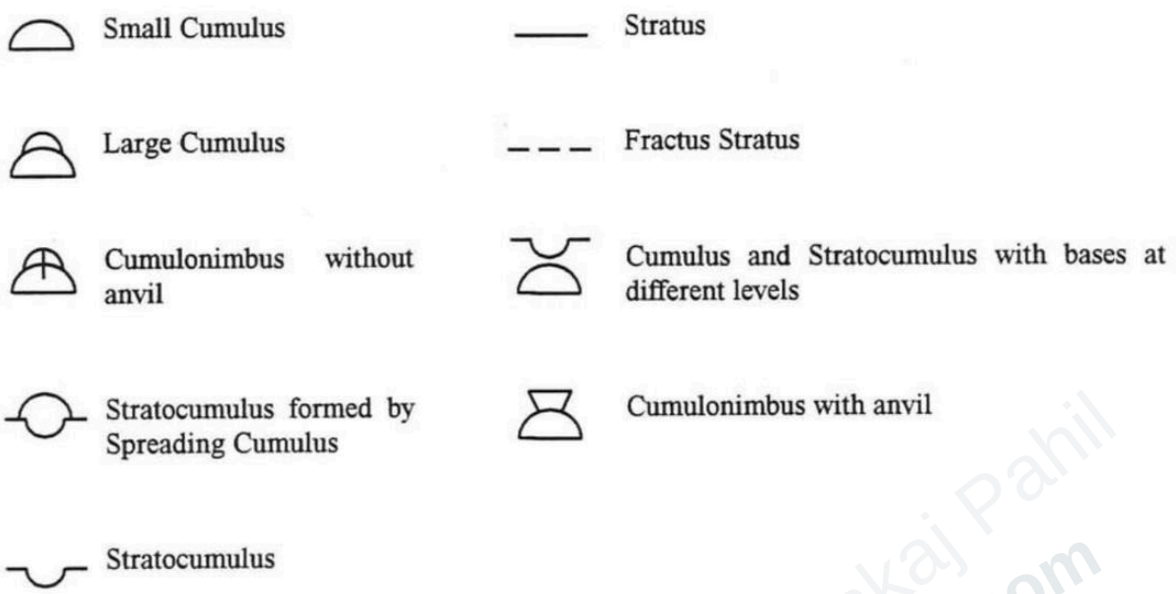
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⚠ **Critical Distinction:** Codes 20–29 mean the weather occurred during the last hour but is **NOT present at the time of observation**. Codes 50–99 mean the weather IS occurring at the time of observation.

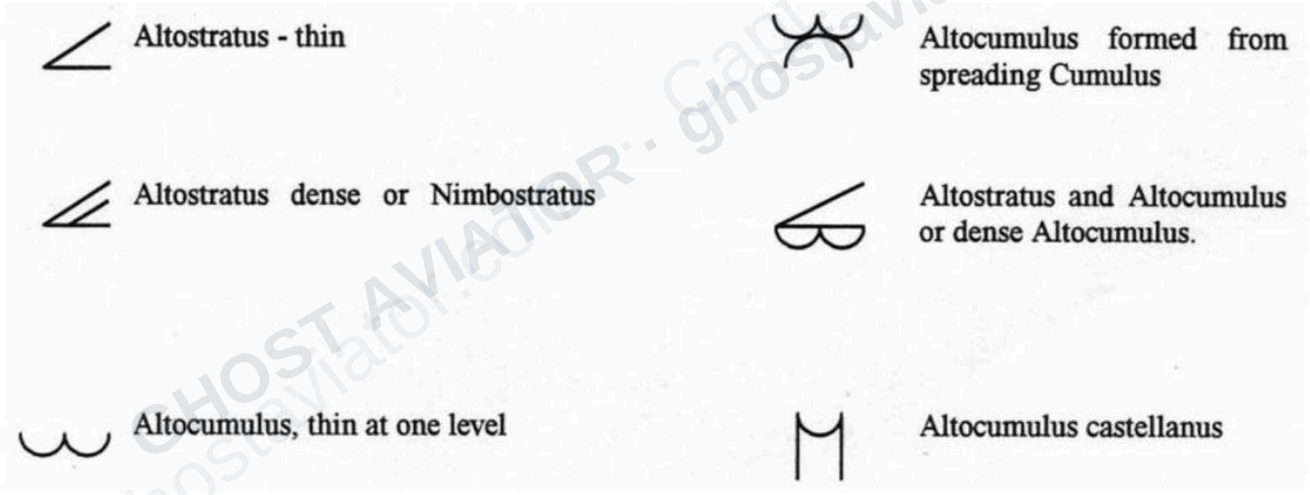
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5. Cloud Codes — C_L, C_M, C_H

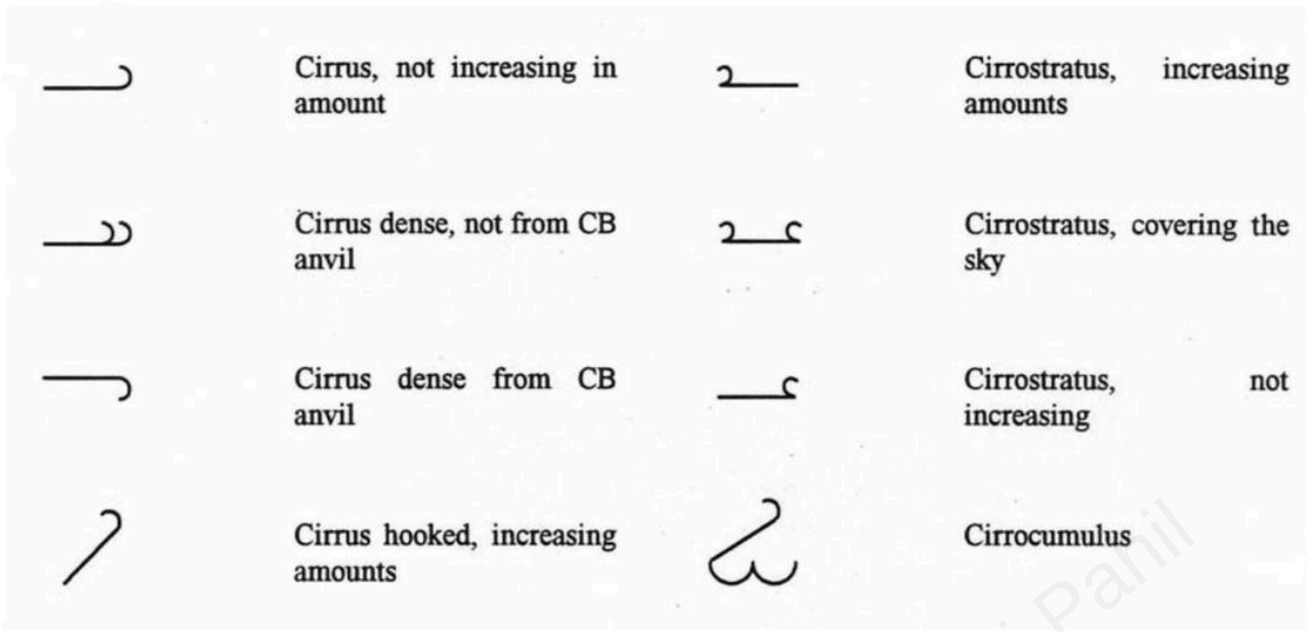


Low cloud symbols



Medium cloud symbols

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High cloud symbols

Low Clouds (C_L)

Code	Kind of Low Cloud
0	No low cloud
1	Fair weather Cumulus (CU)
2	CU of large vertical development
3	CB without anvil
4	Stratocumulus (SC) formed by the spreading out of CU
5	Stratocumulus (SC) not formed from CU
6	ST of fair weather
7	ST of bad weather
8	SC and CU at different levels
9	CB with anvil

Medium Clouds (C_M)

Code	Kind of Medium Cloud
0	No medium cloud
1	Thin AS
2	Thick AS or NS
3	AC at single level
4	AC at different level

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Code	Kind of Medium Cloud
5	AC in bands; Thickening and Spreading
6	AC resulting from spreading of CU
7	AC with AS or NS
8	AC with turreted structure
9	AC in chaotic sky

High Clouds (C_H)

Code	Kind of High Cloud
0	No high cloud
1	Cirrus in filaments
2	Dense Cirrus
3	Dense CI remnant of dissipated CB
4	CI becoming denser
5	CI and CS and progressively increasing
6	CI and CS progressively increasing covering most of the sky
7	CS covering entire sky
8	CS not covering entire sky
9	CC predominate

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6. Codes for h, RRR, and VV (Table 24.4)

h code	Cloud Base (m)	RRR code	Rain Amount (mm)	VV code	Visibility (m)
0	0–49	0	Nil	90	< 50 (Very Thick FOG)
1	50–99	…	Trace	91	50 to <200 (THICK FOG)
2	100–199	—	Trace	92	200 to <500 (MOD FOG)
3	200–299	1	1–1.5	93	500 to <1000 (FOG)
4	300–599	2	1.6–2.4	94	1000 to <2000 (THICK MIST)
5	600–999	3	2.5–3.5	95	2000 to <4000 (SLIGHT MIST)
6	1000–1499	4	3.6–4.4	96	4000 to <10000

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h code	Cloud Base (m)	RRR code	Rain Amount (mm)	VV code	Visibility (m)
7	1500–1999	5	4.5–5.5	97	10000 to <20000
8	2000–2500	6	5.6–6.4	98	20000 to <50000
9	No Cloud below 2500	7	6.5–7.5	99	50000 or more
/	Base not known / Or below hill Stn.	8	7.6–8.4	—	—
—	—	9	further extrapolate values	—	—

⚠ **Note on VV Codes:** VV code 90 = visibility <50 m (very thick fog). VV code 99 = 50 km or more. The scale is NOT linear — memorise the breakpoints.

⚠ **Note on h code:** h code / (slash) means the cloud base is not known, or the cloud base is below the station and the cloud top is above it (a hill station inside the cloud).

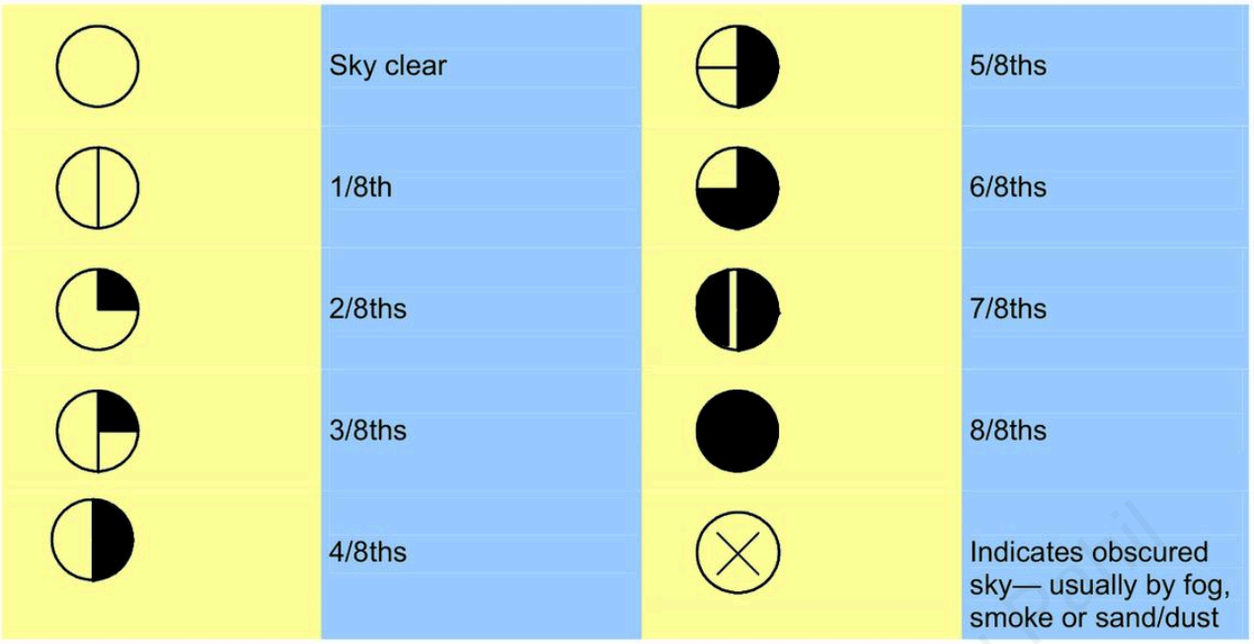
7. Total Cloud Amount (N) Coding

N is coded in Octas (eighths of sky covered) and plotted by filling the central station circle:

Octas (N)	Sky Cover	Symbol
0	Clear	Open circle ○
1	1/8	Circle with small wedge
2	2/8 (1/4)	Circle, quarter filled
3	3/8	Circle, 3/8 filled
4	4/8 (1/2)	Half-filled circle
5	5/8	5/8 filled
6	6/8 (3/4)	3/4 filled
7	7/8	Almost full
8	Overcast	Fully filled ●
9	Sky obscured	× symbol
/	Data not available	Diagonal slash

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Total cloud amount symbols

Exam Tip: N = 9 means sky is obscured (fog, smoke etc.) — not overcast! N = 8 is overcast. The / symbol means data not available.

8. Worked Example — Decoding a Station Model

Given Station Plot:

```
Wind: 330/15 kt (barb NNW, 15 kt)
     34    ↗    998
     93 ●      15
     27   3/2   …
```

Reading	Parameter	Decoded Value
Wind barb direction + speed	Wind Direction & Speed	330°/15 kt
34 (top left)	Temperature (TT)	34°C
93 (middle left)	Visibility (VV code)	Code 93 → 500 to <1000 m (FOG)
27 (bottom left)	Dew Point (T_dT_d)	27°C
998 (top right)	Sea Level Pressure QFF	999.8 hPa (prefix 9 as >950)
15 (middle right, small)	Pressure change in 3 hr	+1.5 hPa
● (filled circle)	Total Cloud Amount (N)	8/8 Overcast
3/2 (below circle)	N_h / h	N_h=3 → 3/8 of low cloud; h=2 → cloud base 100–199 m

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Reading	Parameter	Decoded Value
ww symbol (●) left	Present Weather	FOG at observation
Past Weather (right)	W_1W_2	Fog in past

Complete Decoding of Worked Example:
Wind Direction & Speed: 330°/15 kt | Temperature: 34°C | Type of Low Clouds: Fracto Stratus | Visibility: 500m but not 1000m | Amount of Low Clouds: 3/8 | Present Weather: Weather Fog | Height of base of Low Clouds: 100–199 m | Dew Point: 27°C | Sea Level Pressure QFF: 999.8 hPa | Pressure Change in 3 hr: +1.5 hPa | Past Weather: Fog | Total Cloud Amount: 8/8 | Type of Medium Clouds: AS/NS | Rainfall: Trace

Pressure Decoding Rule

PPP = Last 3 digits of pressure in tenths of hPa
Rule: If PPP > 500 → prefix 9 (e.g., 996 → 999.6 hPa)
If PPP ≤ 500 → prefix 10 (e.g., 024 → 1002.4 hPa)

Example: PPP = 962 → 996.2 hPa (prefix 9)
Example: PPP = 024 → 1002.4 hPa (prefix 10)

Quick Rule: PPP > 500 → add 9 in front; PPP ≤ 500 → add 10 in front. Then divide the full 4-digit number by 10 to get hPa.

9. Practice Q&A

Based on the example station model: Wind 090/ω; TT=34, VV=●, T_d=93, Nh=27, [circle=●], h/Nh=3/2, PPP=962, PP=15, ww=●, W_1W_2=…, 5×

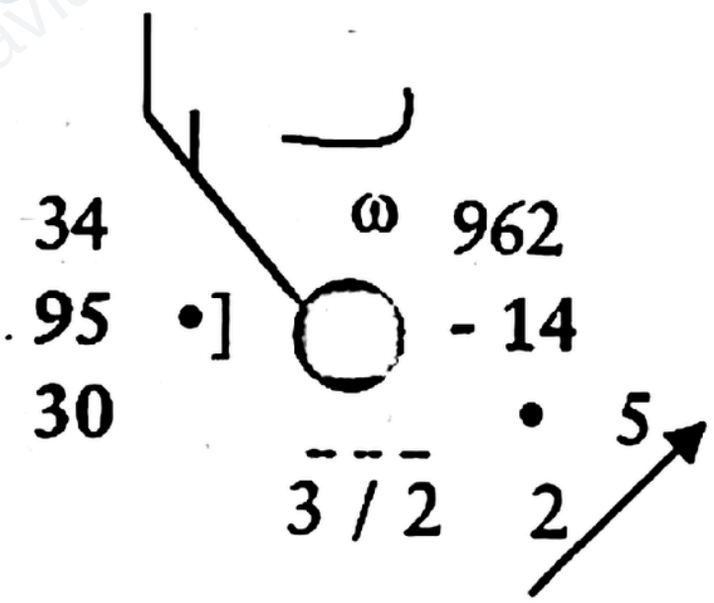


Figure 24.A Station model for Questions 1–20

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Q1. The direction of surface wind is: (a) NW (b) NE (c) SE

✅ Answer: (a) NW — The wind staff is drawn from the north-west of the circle: the wind blows FROM the north-west.

❌ (b) NE and (c) SE point to the wrong side of the circle. The staff always points to the direction the wind comes FROM.

Q2. The range of wind speed is: (a) 20–25 kt (b) 16–22 kt (c) 18–22 kt

✅ Answer: (c) 18–22 kt — The staff carries barbs that add up to 20 kt. A plotted wind speed is read as a band of about ±2 kt, so the speed lies in 18–22 kt.

❌ (a) 20–25 kt and (b) 16–22 kt are not centred on the plotted 20 kt.

Q3. The type of medium cloud is: (a) AC (b) AS (c) AC &AS

✅ Answer: (a) AC — The curved symbol above the circle is the altocumulus symbol for the medium cloud (CM) group.

❌ (b) AS and (c) AC & AS have different symbols in the CM table.

Q4. Low clouds are: (a) CU (b) SC (c) ST

✅ Answer: (c) ST — The dashed line under the circle is the stratus (ST) symbol for the low cloud (CL) group.

❌ (a) CU and (b) SC have their own symbols (a dome and a bar with scallops).

Q5. High clouds are: (a) CC (b) CI (c) CS

✅ Answer: (b) CI — The hooked curved line above the circle is the symbol for cirrus (CI) filaments, a high cloud (CH).

❌ (a) CC and (c) CS have different symbols in the high-cloud (CH) table.

Q6. Height of base of low clouds is: (a) 200 m (b) 250 m (c) 100 to 199 m

✅ Correct Answer: (c) 100 to 199 m — h code 2 = 100–199 m

❌ h code 2 gives a RANGE, not a single value. (a) and (b) are specific values, not ranges.

Q7. Amount of lowest cloud is: (a) 2/8 (b) 4/8 (c) 3/8

✅ Answer: (c) 3/8 — The group under the circle reads 3/2: 3 is the amount of the lowest cloud in eighths (3/8) and 2 is its height code.

❌ (a) 2/8 uses the height code as the amount. (b) 4/8 is not plotted.

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Q8. Dew Point temperature range is: (a) 29.5 to 30.4°C (b) 29.1 to 30.4°C (c) 30.6 to 30.4°C

✅ Answer: (a) 29.5 to 30.4°C — The dew point is plotted as 30. A plotted whole degree stands for the range from half a degree below to just under half a degree above: 29.5 to 30.4 °C.

❌ (b) 29.1 to 30.4 and (c) 30.6 to 30.4 are not valid rounding ranges.

Q9. Temperature 34°C is: (a) Dry (b) Wet (c) Dew Point

✅ Correct Answer: (a) Dry — TT (top left) is the dry bulb temperature

❌ (b) Wet bulb is not plotted on standard station model; (c) Dew Point is bottom left (T_dT_d).

Q10. 962 represents: (a) QFE (b) QFF (c) QNH

✅ Correct Answer: (b) QFF — PPP on station model is always Sea Level Pressure = QFF

❌ QFE = aerodrome pressure; QNH = altimeter setting. Station model plots QFF (Mean Sea Level Pressure).

Q11. Pressure value in hPa is: (a) 996 (b) 996.2 (c) 1096.2

✅ Correct Answer: (b) 996.2 hPa — PPP=962 → 9 prefix (since >500) → 9962 → ÷10 = 996.2

❌ (a) 996 misses the tenth; (c) 1096.2 uses wrong prefix.

Study point for Q12 — Pressure tendency in this station model

The pressure-tendency figure describes the preceding three hours. Its number is in tenths of a hectopascal: −14 represents a fall of 1.4 hPa. Keep the sign, magnitude and time period together when interpreting the plot.

Study point for Q13 — past weather

The past-weather symbol in this exercise indicates rain. It does not establish a rainfall intensity such as light, moderate or heavy. Read past weather separately from the present-weather symbol.

Q14. Rainfall amount is: (a) 2 mm (b) 1.5 mm (c) 1.6 to 2.4 mm

✅ Answer: (c) 1.6 to 2.4 mm — The rainfall is plotted as 2 mm, which stands for the band 1.6 to 2.4 mm.

❌ (a) a single value of 2 mm and (b) 1.5 mm are not the reported band.

Q15. Rainfall reported is from: (a) 0300Z yesterday to 0300Z today (b) 0100 to 2400Z today (c) 0530 to 0830 IST

✅ Answer: (a) 0300 UTC yesterday to 0300 UTC today — IMD measures daily rainfall for the 24 hours ending 0830 IST (0300 UTC), so the plotted rainfall runs from 0300 UTC of the previous day to

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0300 UTC of the day of the chart.

Q16. Speed of Ship is: (a) 5 KMH (b) 5 MPS (c) 5 KT

✅ Correct Answer: (c) 5 KT — Ship speed on station model is in knots

❌ (a) KMH and (b) MPS are not used for ship speed on synoptic charts.

Q17. Direction of movement of Ship: (a) NW (b) SW (c) NE

✅ Answer: (c) NE — The arrow shows the direction in which the ship is moving, drawn towards the north-east.

❌ (a) NW and (b) SW do not match the arrow.

Q18. Speed of Ship reported is averaged for: (a) 3 hr (b) 6 hr (c) 12 hr

✅ Answer: (a) 3 hr — The ship's speed is the average over the last 3 hours.

❌ (b) 6 hr and (c) 12 hr are not the averaging period.

Q19. Surface visibility is: (a) 500 to <1000 m (b) 1000 to 2000 m (c) 2000 to <4000 m

✅ Answer: (c) 2000 to <4000 m — The visibility code 95 stands for 2000 m to under 4000 m (code 94 is 1000–2000 m, 96 is 4000–10,000 m).

❌ (a) and (b) belong to codes 93 and 94.

Q20. Weather reported is: (a) Rain at time of observation (b) Rain during last one hour (c) Rain not at station but within 5 km

✅ Correct Answer: (b) Rain during last one hour — ww code 21 = Rain last hour but NOT at time of observation

❌ (a) Rain at time of obs = ww 60–69; (c) Rain within 5km = not a standard station model category.

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Chapter 25: Aerodrome Reports: METAR, SPECI and TREND

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1. METAR and SPECI — Introduction

METAR = Meteorological Aerodrome Report (routine)
SPECI = Aerodrome Special Met Report (issued when criteria are met)

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Both encoded using **WMO Manual on Codes (WMO 306)**. Used by aircrew, operators, and ATC. Both have the **same code format**.

- METAR is issued at half hourly or hourly intervals.
- SPECI is issued any time criteria are met — for deterioration **immediately**, for improvement after 10 minutes.

Full METAR/SPECI Code Form

METAR or SPECI CCCC YYGGggZ ddffGfmfmKT/MPS/KMH dndndnVdxdxdx VVVVDv VxVxVxVxDv RDRDRi/VRVRVRVRi w'w' or or CAVOK RDRDRi/VRVRVRVRi VRVRVRVRVRi NsNsNshs hshs T'T'/Td'Td' QPhPnPnPn REw'w' WSDRWYDrDr

2. METAR/SPECI Code Structure

Code Group	Meaning	Example
METAR or SPECI	Report type identifier	METAR / SPECI
CCCC	ICAO location indicator (4-letter)	VIDP (Delhi)
YYGGggZ	Date/Time UTC — Day(YY), Hour(GG), Min(gg), Z=UTC	241130Z = 24th, 11:30 UTC
dddff(Gfmfm)KT/MPS/KMH	Wind direction (ddd), speed (ff), gusts (Gfmfm), units	27010KT, 32020G45KT
dndndnVdxdxdx	Variable wind direction (min V max)	170V050
VVVV or CAVOK	Visibility (metres) or CAVOK	6000S, CAVOK
RDRDRi/VRVRVRVRi	RVR — Runway, instrument, RVR value, tendency	R27/1200D
w'w'	Present weather phenomena	+SHRA, TSRA, FZFG
NsNsNshs hs hs	Cloud amount (Ns) and base height (hs hs hs)	FEW020, SCT100, BKN250
VV///	Vertical Visibility (sky obscured)	VV///
SKC or NSC	Sky Clear / No Significant Cloud	SKC, NSC
T'T'/Td'Td'	Temperature / Dew Point (°C)	34/27
QPhPnPnPn	QNH (aerodrome pressure reduced to mean sea level using the standard atmosphere) pressure (hPa)	Q0999, A3005
REw'w'	Recent weather	RERA (recent rain), RETS
WSDRWYDr	Wind Shear (wind speed or direction changing over distance) on Runway	WSRWY09, WS ALLRWY
(WT/SS)	Sea Surface Temp / State	W19/S4
RMK	Remarks (appended at end)	RMK...

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3. Reporting Met Elements — Detailed Rules

(a) Surface Wind

Direction and Speed reported as: 27010KT (India uses KT; MPS is the only other unit now allowed)
Variable wind (speed < 3KT): VRB02KT
Stronger wind — direction changed ≥ 180°: VRB10KT
Calm (less than 1 kt): 00000KT
Gusty wind (30020 gusting 40KT): 30020G40KT
Extreme directions (variation 60° or more but less than 180°, and wind speed 3 KT or more): e.g., 270V050
Wind ≥ 100 KT: reported as 24 0P99KT

(b) Visibility

- Reported in steps of 50 m (when vis 0–800 m), e.g., 50, 100, 150…750, 800 m
- In steps of 100 m when vis is 800 m to 5000 m — e.g., 900, 1000…4900, 5000 m
- In steps of 1 km when vis is 5000 m to 9000 m — e.g., 6000, 7000, 8000, 9000 m
- Reported as 9999 when visibility is ≥ 10 km or more

(c) RVR — Runway Visual Range

RVR observations are made when visibility or RVR < 1500 m. Reported in steps of:
25 m — when RVR is 000–400 m
50 m — when RVR is 400–800 m
100 m — when RVR is 800–2000 m
RVR 1500–2000 m is reported under RMK and **not disseminated internationally**.

RVR Tendency Code (i): U = Increasing, D = Decreasing, N = No change

Code i	Meaning
R	R = RVR; DR = Runway; i = tendency in last 10 min
i as N	Nil — as D (decreased)
U	RVR increased
D	RVR decreased
N	No change

When RVR is more than 2000 m it is reported as P2000, for example R27/P2000 (prefix P = more than). Below 50 m it is reported as R27/M0050 (prefix M = less than). Older reports and many past questions show P1500, the earlier upper limit

(d) Cloud Base — Reported in Steps

- 30 m — up to 3000 m
- 300 m — above 3000 m

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(e) Cloud Reporting Terms

Code	Octas	Meaning
FEW	1–2	Few
SCT	3–4	Scattered
BKN	5–7	Broken
OVC	8	Overcast
SKC	0	Sky Clear (older code, still seen in some reports). When an automatic system detects no cloud the code is NCD
NSC	—	No Significant Cloud

Cloud Layering Rules:

1. Lowest layer reported **regardless of amount** (FEW, SCT, BKN or OVC)
2. Next layer reported when cloud amount is 3/8 or more (SCT, BKN or OVC)
3. Next layer reported when cloud amount is 5/8 or more (BKN or OVC)
4. In addition **CB is always reported** whenever observed

(f) Temperature

Involving 0.5°C or more are rounded up to next higher value. e.g., +2.5°C reported as 03; −2.5°C as M 02.

(g) Pressure (QNH)

Rounded down to nearest whole hectopascal lower value. e.g., QNH 996.6 hPa reported as QNH 996 hPa. Range of pressure over the world is 850 hPa to 1100 hPa.

(h) Fog (FG)

Reported when visibility is < 1000 m due to water drops or ice crystals or both, except when prefixed by MI, BC, PR or VC.

(i) Mist (BR)

Reported when visibility is 1000 m but not more than 5000 m.

(j) HZ, SA, DU, FU, VA

Reported when visibility is 5000 m or less, except SA when specified by DR (Low Drifting).

(k) Present Weather Phenomena (Full List)

Drizzle (DZ), Rain (RA), Snow (SN), Snow Grains (SG), Ice Pellets (PL), Ice Crystals (IC), Hail ≥5mm (GR), Hail <5mm (GS), Fog (FG), Mist (BR), Sand (SA), Dust (DU), Haze (HZ), Smoke (FU), Volcanic Ash (VA), Dust Devil (PO), Squall (SQ), Funnel Cloud (FC), Dust Storm (DS), Sandstorm (SS), Shower (SH), Thunderstorm (TS)

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(I) Supplementary Information

Recent significant weather also included in abbreviated plain language. e.g., *WIND 32010 KT, WIND AT 60 M 36025 KT IN APCH OR MOD TURB AND ICE IN CLIMOUT, WS RWY 12.*

4. CAVOK — Definition and Conditions

CAVOK (pronounced KAV-OH-KAY) = Ceiling And Visibility OK
Issued when ALL three conditions are met simultaneously:

1. Visibility ≥ 10 km
2. No clouds of operational significance
3. No weather of significance

CAVOK **replaces** Visibility, RVR, Present Weather and Clouds in METAR.

Minimum Sector Altitude (MSA): The lowest altitude which may be used which will provide a minimum clearance of 300 m (1000 ft) above all objects located in the area contained within a sector of a circle of 46 km (25 NM) radius centred on a significant point, the aerodrome reference point (ARP) or the heliport reference point (HRP).

5. General Criteria for Local Special Reports and SPECI

Report scope: Learn the criteria separately from dissemination. Indian service guidance records cessation of SPECI dissemination from 15 November 2015, alongside local special and ADDITIONAL reports. A trigger criterion does not by itself mean that an internationally disseminated SPECI is issued at an Indian aerodrome.

SPECI Issuance Timing:
For **deterioration** → issued immediately
For **improvement** → issued after 10 minutes

6. ADDITIONAL Reports in India

The procedure introduced in 2015 identifies these reports with the prefix **ADDITIONAL**. They are supplied to local ATC and are not disseminated beyond the originating aerodrome.

- **Visibility:** changes to or through 550, 1200, 2000, 2500 or 4000 m.
- **Cloud base:** a layer covering more than half the sky changes to or through 90 or 120 m (300 or 400 ft), at stations equipped with a ceilometer or ceilograph.

Runway Condition Report (RCR) and SNOWTAM

A wet or contaminated runway (a runway covered by water, ice or snow) gives poor braking. The pilot must know the runway state before landing or take-off. The aerodrome operator checks the runway and writes a **Runway Condition Report (RCR)**. In India this method has been required from **4 November 2021**.

- The runway is split into **three thirds**. Each third gets a **Runway Condition Code (RWYCC)**, written like 2/6/5.

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- In the Indian examples a higher code goes with a better surface: dry = 6, wet = 5, slippery wet = 3, standing water = 2.
- The report also gives the **coverage** (per cent), the **depth** (mm) and the type of contaminant for each third. NR means not reported.
- The report is written starting from the lower runway number. ATC always speaks of the first, second and third part *in your direction of landing or take-off*.

How it reaches you: by D-ATIS or ATC voice. If the runway has **standing water or wet ice**, the report is also sent out as a **SNOWTAM**. A runway that is only wet is passed by D-ATIS or ATC voice only.

Old books: an eight-figure runway-state group was once added to the end of a METAR. Runway-state information in the METAR was used only until 3 November 2021. Do not expect it in a METAR today.

Pilots report braking action as GOOD, GOOD TO MEDIUM, MEDIUM, MEDIUM TO POOR, POOR or LESS THAN POOR.

7. Explanation of METAR/SPECI Code Groups (Table 25.1 & 25.2)

Visibility Groups

Code	Meaning	Coded As
VVVV Dv	Min Visibility 6000m in South	6000S
VxVxVxVxDv	Visibility <1500m in one direction and >5000 in other then: N, NE, NW	1400SW 6000E / 3000NW
CAVOK	Visibility, Cloud Ceiling, Present Wx better than prescribed values	CAVOK
9999	Visibility 10 km or more	9999

RVR Group

Code	Meaning	Example
R	RVR identifier	R
DRDRDRi/VRVRVRVRi	Runway, tendency in last 10 min / RVR value / tendency	R27/1200D
i = N (nil)	As D — decreased	—
U	RVR increasing	R27P1500U
D	RVR decreased	R21P1500D
RVR >1500m	Has increased: R27P1500U	—
RVR below instrument	R27M0150	—

Pressure QNH

Code	Meaning	Example
QPhPnPnPn	QNH rounded down to whole hPa	Q0999
APhPnPnPn	QNH in inches (US format)	A3005 = 30.05 inches

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Recent Weather (RE)

Code	Meaning
RERA	Recent Rain
RETS	Recent Thunderstorm

Wind Shear

Code	Meaning
WSRWY09	Wind Shear Runway 09
WS ALLRWY	Wind Shear All Runways

8. TREND Forecast (Landing Forecast)

TREND = Statement of expected significant changes in any one or more of the weather elements: surface wind, visibility, weather and clouds.
When no significant change is expected: NOSIG
TREND is **appended to a METAR or SPECI** and is valid for TWO hours.

TREND Code Form

TTTTT or NOSIG TTGgg ddffGfmfmKT/MPS/KMH VVVV or CAVOK TTTTT w'w' or NSW NSNS NS hs hs or V/// or SKC or NSC (RMK)

Change Group (TTTTT)

Code	Meaning
BECMG	Becoming — gradual and permanent change (starts at…UTC, completes at…UTC)
TEMPO	Temporary change for less than one hour AND aggregate of such changes < half period indicated by GGgg. TEMPO lasts < 1 hour; total TEMPO time < half the TREND period.
NOSIG	No Significant Change

TEMPO Rule: TEMPO lasts for less than one hour and the aggregate of such repeated changes is **less than half** the period indicated by GGgg.
BECMG: Gradual permanent change — starts at …UTC and completes at …UTC.

Time of Expected Occurrence of Change (TTGGgg)

Code	Meaning	Example
FM (From)	From time TL	—
TL (Until)	Until time	—
AT (At)	At exact time	—

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Code	Meaning	Example
GGgg	Time in Hr and Min	eg. 0430, 0800

e.g., BECMG FM1230 TL1300 TEMPO FM0800 | BECMG TL1230 OVC005 | BECMG AT1030 OVC010

Visibility in TREND

Visibility 4000 m in Rain and Showers: 4000 RASH
CAVOK as in METAR

Weather in TREND (NSW)

NSW = No Significant Weather (used to indicate weather has ceased)

9. W' W' Significant Present & Forecast Weather (Table 25.3)

Qualifier	Description	Precipitation	Obscuration	Other
– Light	MI Shallow	DZ Drizzle	BR Mist	PO Dust/Sand Whirl (Dust Devil)
Moderate (no qualifier)	BC Patches (random, discontinuous coverage)	RA Rain	FG Fog (FU = Smoke)	SQ Squall
+ Heavy (well developed; substantial coverage)	PR Partial (covers part of aerodrome)	SN Snow	VA Volcanic Ash	FC Funnel Clouds (Tornado or Water Spout)
VC In the Vicinity (not at station, but within 8 km)	DR Low Drifting	SG Snow Grain	DU Widespread Dust	SA Sand Dunes
	BL Blowing	IC Ice Crystals, Diamond dust	SA Sand	SS Sandstorm
	SH Shower	PL Ice Pellets (Sleet)	HZ Haze	DS Duststorm
	TS Thunderstorms	GR Hail ≥5mm		
	FZ Freezing	GS Small Hail <5mm		

Key Notes on W' qualifiers:
FZ (super-cooled water droplets or precipitation) is used with DZ, RA and FG: FZDZ, FZRA, FZFG;
DR is used with DU, SA, or SN raised by wind to < 2 m agl;
BL is used with DU, SA or SN raised by wind to ≥ 2 m agl;
MI, BC and PR are used with FG only, e.g., BCFG, MIFG, PRFG

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10. hs hs hs Cloud Base Code (Table 25.4)

hs hs Code	000	001	002	003	004	etc.	011	099
Cloud Base m	<30	30	60	90	120	…	330	2970

Each code increment = 30 m. Code 001 = 30 m, Code 004 = 120 m, Code 011 = 330 m.

Exam Tip: hs hs hs in METAR = cloud base in hundreds of feet × 100 (e.g., FEW020 = FEW at 2000 ft). In TREND, hs hs hs code is given as 3-digit code where each unit = 30 m.

11. Worked METAR/SPECI/TREND Examples

Example SPECI Decoded

SPECI VIDP 230300Z 30015G35KT 270V050 1200NW 6000S R27/P1500 +SHRA FEW020 FEW025CB SCT100 BKN250 34/27 Q1004 RERA WS RWY27

Decoded:
SPECI VIDP → Aerodrome Special Met Report for Palam (New Delhi)
230300Z → 23rd day at 0300 UTC
30015G35KT → Surface Wind 300°/15 KT gusting to 35 KT
270V050 → Wind direction varying from 270° to 050°
1200NW → Visibility 1200 m in NW direction
6000S → Visibility 6000 m in South
R27/P1500 → Runway Visual Range R/W 27 more than 1500 m
+SHRA → Weather: Heavy Shower and Rain
FEW020 → Clouds 1–2/8 at 2000 ft
FEW025CB → 1–2/8 CB at 2500 ft
SCT100 → 3–4/8 at 10,000 ft
BKN250 → 5–7/8 at 25,000 ft
34/27 → Temperature 34°C and Dew Point (temperature to which air must cool to become saturated at unchanged pressure and water-vapour content) 27°C
Q1004 → QNH 1004 hPa
RERA → Recent Rain
WS RWY27 → Wind shear on Runway 27 (supplementary information, not a warning message)

Example TREND Decoded

TEMPO FM0830 TL0900 24025G45KT 2000 TSRA FEW020 FEW025CB SCT100 BKN250 BECMG FM0930 27005KT 6000 NSW.

TEMPO FM0830 TL0900 → Temporary from 0830 until 0900 UTC:
Surface Wind 240/25 KT Gusting to 45 KT, Visibility 2000 m in thunderstorm with rain, Clouds 1–2/8 at 2000 ft, 1–2/8 CB at 2500 ft, 3–4/8 at 10,000 ft, 5–7/8 at 25,000 ft.
BECMG FM0930 → Becoming from 0930 UTC:
Wind 270/5 KT, Visibility 6000 m and No Significant Weather.

METAR Examples with TREND

METAR VIAM 211500Z 33010KT 6000 FEW040 FEW050CB 29/25 Q1004 NOSIG = METAR VEGK 1600Z 23015KT 6000 LT FEW010 SCT020CB BKN100 22/Q1002 TEMPO FM 1630 TL 1725 30015G28KT 2500

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TSSH SCT020CB OVC150 = METAR VIJO 1230Z 23015KT 5000 SA SKC 37/18 Q1004 BECMG 1314 24015G28KT -DS 1500 FEW015 SCT020CB BKN100= METAR VIAR 1300Z 25015KT 2500 HZ SKC 39/19 Q1001 BECMG AT1400 25010KT CAVOK =

12. Practice Q&A

Worked METAR exercise

Use this METAR / TREND for Questions 1–22

```
METAR VIDP 160230Z 30005KT 290V050 1500S 5000N R15/P1500U BR FEW020 FEW025CB
SCT120 BKN300 32/29 Q1003 REFG
TEMPO FM0330 22015G25KT 3000 +TSRA FEW010 SCT025CB BKN150 BECMG AT0415 27008KT
CAVOK=
```

Q1. The METAR has been issued on day: (a) 15th (b) 16th (c) 17th

✅ Correct: (b) 16th — YY = 16 in YYGGggZ = 160230Z

❌ (a) 15th and (c) 17th are wrong day codes.

Q2. The METAR has been issued at: (a) 0630 IST (b) 0230 UTC (c) 0230 IST

✅ Correct: (b) 0230 UTC — GGgg = 0230, Z suffix = UTC

❌ IST = UTC + 5:30 hrs. 0230 UTC = 0800 IST, not 0630 or 0230 IST.

Q4. Wind direction is varying from: (a) 290 to 050° (b) 050 to 290° (c) 200 to 050°

✅ Correct: (a) 290 to 050° — Variable wind group: 290V050

❌ The format is always dndndnVdxdxdx = min direction V max direction.

Q5. Visibility 1500 m is towards: (a) N (b) S (c) All over the airport

✅ Correct: (b) S — 1500S = 1500 m in South direction

❌ Direction suffix gives compass bearing of minimum visibility: S = South.

Q6. Visibility towards N is: (a) 1500m (b) 5000 m (c) 3000 m

✅ Correct: (b) 5000 m — 5000N = 5000 m in North direction

Q7. Height of base of low clouds above station level is: (a) 2000 ft (b) 2500 ft (c) 2500 m

✅ Correct: (a) 2000 ft — FEW020 = FEW at 020 = 2000 ft (heights in METAR are in hundreds of feet)

❌ METAR cloud heights are in hundreds of FEET, not metres.

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Q8. Runway Visual Range is: (a) 1–2/8 (b) More than 1500m (c) 500–1000m

✅ Correct: (b) More than 1500m — R15/P1500U: P = more than, 1500 = value, U = increasing

❌ P prefix = Plus = more than; M prefix = Minus = less than.

Q9. RVR is: (a) Decreased (b) Increased (c) Remained same

✅ Correct: (b) Increased — tendency 'U' = Increasing

Q10. The poor visibility is due to Present Weather which is: (a) Fog (b) Brown Dust (c) Mist

✅ Correct: (c) Mist (BR) — BR is reported when vis is 1000–5000m

❌ FG = Fog (vis <1000m); BR = Mist (1000–5000m). Here vis = 1500m → BR is correct.

Q11. Amount of lowest cloud is: (a) 1–2/8 (b) 2–4/8 (c) 5–7/8

✅ Correct: (a) 1–2/8 — FEW = 1–2 octas

Q12. Amount of CB cloud is: (a) 1–2/8 (b) 2–4/8 (c) 5–7/8

✅ Answer: (a) 1–2/8 — FEW025CB is a FEW layer of cumulonimbus: FEW means 1–2 oktas.

❌ (b) 2–4/8 and (c) 5–7/8 would be SCT or BKN.

Q13. Height of base of CB is: (a) 2500 m (b) 3000 ft (c) 2500 ft

✅ Answer: (c) 2500 ft — FEW025CB: the cloud base is 025 × 100 = 2,500 ft above the aerodrome.

❌ (a) 2500 m confuses feet with metres. (b) 3000 ft does not match the group.

Q14. The height of topmost layer of cloud is: (a) 3000 m (b) 30000 m (c) 30000 ft

✅ Correct: (c) 30000 ft — BKN300 = 300 × 100 = 30,000 ft

❌ Cloud codes in METAR are always in hundreds of feet.

Q15. The landing forecast appended to any METAR is valid for: (a) 1 hr (b) 2 hr (c) 3 hr

✅ Correct: (b) 2 hr — TREND (landing forecast) is valid for 2 hours

❌ TAF is valid for 9–30 hrs. TREND is only 2 hours.

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Q16. A two-hour TREND begins with a report at 0230 UTC. In TEMPO FM0330, no TL end time is given. What is the implied end of this temporary-fluctuation period?
(a) 0400 (b) 0430 (c) 0415

Answer: (b) 0430 UTC — when TEMPO has FM but no TL, the fluctuations are forecast from the stated start until the end of the TREND period.

Q17. Expected visibility after 0415 UTC is: (a) 6000 m (b) 3000 m (c) 10 km or more

✅ Answer: (c) 10 km or more — After BECMG AT0415 the trend gives CAVOK, which means visibility of 10 km or more, no significant cloud and no significant weather.

❌ (a) 6000 m and (b) 3000 m are not forecast by the trend.

Q18. In the exercise, air temperature is 32°C and dew point is 29°C. The atmosphere is: (a) Very Dry (b) Moist (c) Saturated

Answer: (b) Moist — the dew point is high, but lower than the air temperature, so the air is not saturated.

Study point for Q20 — recent weather

FG means fog, but REFG is not a permitted recent-weather group in this code. Do not decode this malformed group as a valid report. Recent-weather groups are restricted to specified operationally significant phenomena; ordinary fog is not included.

Q21. Range of pressure as Q1003 is: (a) 1002.5 to 1003.5 hPa (b) 1003.0 to 1003.5 hPa (c) 1003.0 to 1003.9 hPa

✅ Correct: (c) 1003.0 to 1003.9 hPa — Q1003 means pressure was rounded DOWN to 1003, so actual value is 1003.0 to 1003.9

❌ Pressure is rounded DOWN (not rounded to nearest). So Q1003 = actual 1003.0 to 1003.9.

Q22. Temperature reported as 32°C is: (a) 31.4 to 32.3°C (b) 32.1 to 32.4°C (c) 31.5 to 32.4°C

✅ Correct: (c) 31.5 to 32.4°C — temperature is rounded UP to next higher for 0.5°C increments

Q23. Visibility is reported in steps of 50 m when visibility is: (a) 800 m to 5000 m (b) 0 to 800 m (c) 5000 m to 10 km

✅ Correct: (b) 0 to 800 m — steps of 50 m for 0–800 m range

❌ (a) 800–5000m uses 100 m steps; (c) 5–9 km uses 1 km steps.

Q24. Visibility reported in steps of 100 m when visibility is: (a) 800 m to 5000 m (b) 0 to 800 m (c)

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5000 m to 10 km

✅ Correct: (a) 800 m to 5000 m

Q25. Visibility reported in steps of 1000 m when visibility is: (a) 800 to 5000 m (b) 0 to 800 m (c) 5000 m to 9999

✅ Correct: (c) 5000 m to 9999 — 1 km steps for 5–9 km range

Q26. Visibility reported 9999 when visibility is: (a) 800 to 5000 m (b) 9000 to 9999 m (c) 10 km or more

✅ Correct: (c) 10 km or more

Q27. Temperature +2.5°C is reported as: (a) 02 (b) 03 (c) 2.5°C

✅ Correct: (b) 03 — +2.5°C rounded UP to 3°C = coded as 03

Q28. Temperature −12.5°C is reported as: (a) −12 (b) −13 (c) M 12

✅ Answer: (c) M12 — −12.5 °C is rounded to the next higher value, −12 °C, and a negative temperature carries the prefix M: M12.

❌ (a) −12 is the value without the M prefix. (b) −13 rounds the wrong way.

Q29. Pressure rounded to nearest whole number in hectopascal: (a) exact value (b) upper value (c) lower value

✅ Correct: (c) lower value — QNH always rounded DOWN

Q30. QNH 1002.6 hPa is reported as: (a) Q1002 (b) Q1003 (c) Q1002.6

✅ Correct: (a) Q1002 — rounded DOWN to lower whole hPa

Q31. QNH 29.92 inches is reported as: (a) Q2900 (b) A2992 (c) A 3000

✅ Correct: (b) A2992 — A prefix for inches; 29.92 → A2992

Q32. Fog is reported when visibility is: (a) <1000 m (b) 1000 m (c) >1000 m

✅ Correct: (a) <1000 m — FG reported when vis <1000 m due to water droplets/ice crystals

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Q33. Mist is reported when visibility is: (a) <1000 m to 2000 m (b) 1000 to 5000 m (c) >1000 m

✅ Correct: (b) 1000 to 5000 m — BR (Mist) range

Q34. CAVOK signifies Visibility, Cloud… and present weather better than the prescribed values or conditions: (a) ceiling (b) base (c) amount

✅ Correct: (a) ceiling — CAVOK = Ceiling And Visibility OK

Q35. CAVOK is issued when visibility is: (a) 9–10 km (b) 10 km or more (c) 10 km only

✅ Correct: (b) 10 km or more — CAVOK requires visibility ≥10 km

Study point for Q36 — All CAVOK conditions must be met

Visibility is at least 10 km; there is no cloud of operational significance; and there is no significant weather. Operationally significant cloud includes a base below 1500 m (5000 ft) or the highest minimum sector altitude, whichever is greater, and cumulonimbus (a deep cloud associated with thunderstorms) or towering cumulus at any height.

Study point for Q37 — general special-report criteria

Under the general wind criteria, any one trigger is sufficient: direction change of at least 60° with mean speed at least 10 kt before and/or after; mean speed change of at least 10 kt; or gust variation increase of at least 10 kt with mean speed at least 15 kt before and/or after.

Apply the Indian dissemination distinction explained above.

Study point for Q38 — general special-report criteria

Under the general cloud criteria, either a relevant BKN/OVC base crossing 30, 60, 150, 300 or 450 m, or an amount change below 450 m between SCT-or-less and BKN/OVC, can trigger a report. The two changes need not occur together.

Apply the Indian dissemination distinction explained above.

Study point for Q39 — general special-report criteria

When the sky is obscured, a measured vertical visibility crossing 30, 60, 150 or 300 m is a general special-report criterion. Do not treat obscuration alone as the threshold-crossing criterion.

Apply the Indian dissemination distinction explained above.

Q40. SPECI is issued when surface temperature has increased by: (a) 3°C (b) 4°C (c) 2°C

✅ Correct: (c) 2°C — SPECI when surface temperature increases by ≥2°C from last observation

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Q41. WSRWY28 indicates that the wind shear has been reported for: (a) take off path (b) approach path (c) both take off and approach paths

✅ Correct: (c) both take off and approach paths — WSRWY (runway) applies to both

Q42. R26/M0150 in a METAR indicates that: (a) maximum RVR is 150 m (b) minimum RVR is 150 m (c) 150 m is the lowest RVR instrument can measure and RVR < 150 m

✅ Correct: (c) 150 m is the lowest the instrument can measure and RVR is <150 m — M prefix = less than/below minimum instrument range

❌ (a) P prefix = maximum instrument range; (b) M = below minimum instrument range.

Chapter 26: Aviation Forecasts: TAF, ARFOR and ROFOR

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1. TAF — Aerodrome Forecast

TAF = Terminal Aerodrome Forecast
The 9-hour (short) TAF is used for domestic flights and VOLMET; the 30-hour (long) TAF is used for international flights.
Valid for at least 9 hr and a maximum of 30 hr.
TAFs valid for 9 hr are issued every 3 hr; 12–30 hr TAFs issued every 6 hr, commencing at 0000 UTC.
9 hr TAF is **not disseminated internationally** but used for domestic flights and VOLMET.

Key TAF validity rules:
9 hr TAF: issued every 3 hr, commencing 0000 UTC
12–30 hr TAF: issued every 6 hr, commencing 0000 UTC
Amended TAF identified by: TAF AMD

2. TAF Code Form & Groups

TAF CCCC YYGGggZ $Y_1Y_1G_1G_1/Y_2Y_2G_2G_2$ dddffGfmfmKT/MPS/KMH VVVV or CAVOK NsNsNshs hs hs or V/// or SKC or NSC PROB C_xC_x YYGGggZ Change likely to occur: (W_1W_2) (TXTrTr/G_1G_1Z TNT′rT′r/G_2G_2Z) Max & Min temperatures

Code Groups Explained

Code	Meaning	Example
TAF	Report type	TAF
CCCC	ICAO Location Indicator	VECC (Kolkata)

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Code	Meaning	Example
YYGGggZ	Date/Time of issue (Day/Hr/Min UTC)	102100Z = 10th, 2100 UTC
$Y_1Y_1G_1G_1/Y_2Y_2G_2G_2$	Validity: from Day Y_1 Hr G_1 to Day Y_2 Hr G_2 UTC	1100/1206 = 11th 00Z to 12th 06Z (30 hours)
dddff(Gfmfm)KT	Wind direction, speed, gusts, units	09012KT
VVVV or CAVOK	Visibility or CAVOK	2500 BR, CAVOK
NsNsNshs hs hs	Cloud amount and base	FEW040 SCT150
BECMG / TEMPO / FM / PROB30, PROB40	Change groups (NOSIG is used only in TREND)	BECMG 1102/1104
PROB C_xC_x	Probability of occurrence: 30% or 40%	PROB30 or PROB40
YYGGggZ (in change)	Change likely to occur between timings GG and G_4G_4	1809/1812 = change on 18th between 0900–1200 UTC
TXTrTr/G_1G_1Z	Maximum temperature / Time UTC	TX42/1110Z = Max 42°C on the 11th at 1000 UTC (the current form carries the day as well as the hour)
TNT'rT'r/G_2G_2Z	Minimum temperature / Time UTC	TN05/1201Z = Min 05°C on the 12th at 0100 UTC

PROB Rule: Only PROB30 or PROB40 are used — never PROB50 or higher (that would imply the change is likely, so it becomes BECMG or TEMPO). PROB30 or PROB40 can be combined with TEMPO to describe the probability of temporary conditions.

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3. TAF Decoded Examples

Example 1

TAF VECC 102100Z 1100/1109 09012KT 2500 BR FEW040 SCT150 BECMG 1102/1104 16015KT 6 000 TEMPO 1107/1109 FEW030CB =

VECC → Kolkata
102100Z → Issued on 10th at 2100 UTC
1100/1109 → Valid 11th day 0000 UTC to 11th day 0900 UTC
09012KT → Surface Wind 090°/12 KT
2500 BR → Visibility 2500 m due to Mist
FEW040 SCT150 → Clouds: 1–2/8 at 4000 ft, 3–4/8 at 15,000 ft
BECMG 1102/1104 → Becoming between 0200 and 0400 UTC on the 11th
16015KT 6000 → Surface Wind 160°/15 KT, Visibility 6000 m
TEMPO 1107/1109 FEW030CB → Temporarily from 0700–0900: 1–2/8 CB at 3000 ft

Example 2 — with AMD

In TAFs the date and time of issue of TAF is included after CCCC group. Letter **Z** is an indicator for time of issue in UTC. **KT** for wind speed.
Amended TAF identified by: TAF AMD

Sample TAFs for Study

TAF VOMM 031500Z 0318/0424 26005KT DZ/HZ SCT015 FEW025CB BKN080 TEMPO 0321/0403 1500 RA/BR SCT010 SCT015 FEW025CB OVC080 BECMG 0405/0406 07005KT 7000 TEMPO 0409/0415 2000 TSRA SCT008 FEW025CB BKN080 BECMG 0416/0417 00000KT 4000 HZ= TAF VOCI 141500Z 1418/1524 14005KT DZ/HZ SCT015 FEW025CB BKN080 TEMPO 1421/1503 1500 RA SCT010 FEW025CB OVC080 TEMPO 1509/1515 2000 TSRA SCT008 FEW025CB BKN080 BECMG 1516/1517 12010KT SCT120 7000=

4. TAF Validity & Issuance Rules

TAF Type	Validity	Issuance Interval	Commencing	Dissemination
Short TAF	9 hr	Every 3 hr	0000 UTC	Domestic + VOLMET only (not international)
Long TAF	12–30 hr	Every 6 hr	0000 UTC	International dissemination

TAF Validity — Example: 1200/1306 = Valid from day 12 time 0000 UTC to day 13 time 0600 UTC
1606/1615 = Valid on day 16 time 0600 to time 1500 UTC

5. ARFOR — Area Forecast for Aviation

ARFOR = Area Forecast for Aviation in figure code for a specific area.
The ARFOR covers meteorological data at **multiple altitude levels** including winds, temperatures, cloud, turbulence, icing, and jet stream data.
Example area description: *VECC AND 50 NM AROUND*

6. ARFOR Code Sections & Indicators

ARFOR Code Form

ARFOR (YYGGggZ) $Y_1Y_1G_1G_1/Y_2Y_2G_2G_2$[KMH or KT or MPS] AAAA (VVVV) ($W_1W_1W_i$) SECTION 1 (N.CC.lhslhs.hs or VV/// or SKC or NSC) (dddff)($_{eee}$) SECTION 2 (1111) $QL_iL_iL_iL_i$ h'h'(f ff) SECTION 3 (22222 h'ah'$_m f_af_af_a$ d_ad_aVV) SECTION 4 9innn

Section Explanations

Section	Content
Section 1	Code identification and time group of ARFOR
Section 2	Symbolic Figure: 11111, and Jet Stream Data (optional)

Section	Content
Section 3	Symbolic Figure: 22222, and Data of Max wind and Vertical wind shear (optional)
Section 4	Supplementary weather phenomena — lowest msl pressure, type of Front, change at Lat/Long, BECMG, FROM, TEMPO etc.

Indicator Figures Used in ARFOR

Indicator	Meaning
7hhhhhhh	7 Indicator for altitude hhhh = Altitude of cloud layer; hhhh = Altitude of Freezing Level
6Ihhhht	6 Indicator for Icing; L = Type of icing; hhhh = altitude; t = Thickness
5Bthsbh$_s$	5 Indicator for Turbulence
4hbhbT$_s$T$_s$	4 Indicator for Wind and Temperature
2h$_t$h$_r$T$_s$T$_r$	2 Indicator for Tropopause level pressure and temperature

AAAA = Description of area in plain language, e.g., VECC AND 50 NM AROUND
VVVV = Forecast visibility, e.g., 6000
Cloud Group: (N.CCIhslhs.hs or VV/// or SKC or NSC), e.g., 2/8 AC at 10000 ft reported as 2AC100
Cloud Group Reporting Rules:
1st Group: Lowest layer (N, 1/8 or more)
2nd Group: Next higher layer (N, 3/8 or more)
3rd Group: Next higher layer (N, 5/8 or more)
In addition CB if not already included

7. ARFOR Turbulence, Icing & W'W' Codes (Table 26.2)

B (Turb)	Type of Turbulence	L (Icing)	Type of Icing	W'W'	Reported as
0	Nil	0	Nil	111	Thunderstorm (TS)
1	Light	1	Light	222	Tropical Cyclone (TPS)
2	Moderate CAT OCNL	2	Light Icing in Cloud	333	Severe Line Squall (SQ)
3	Moderate CAT Frequent	3	Light Icing in Precipitation	444	Hail (HAIL)
4	Moderate in Cloud Occasional	4	Moderate Icing	555	Marked Mountain Waves (MTW)
5	Moderate in Cloud Frequent	5	Moderate Icing in Cloud	666	Widespread Sandstorm (SAND)
6	Severe CAT OCNL	6	Moderate Icing in Precipitation	777	Widespread Duststorm (DUST)
7	Severe CAT FREQ	7	Severe Icing	888	Freezing Rain (FZRA)

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B (Turb)	Type of Turbulence	L (Icing)	Type of Icing	W'W'	Reported as
8	Severe in Cloud Occasional	8	Severe Icing in Cloud	—	—
9	Severe in Cloud FREQ	9	Severe Icing in Precipitation	—	—

8. ARFOR Decoded Example

ARFOR VIDP 091400 AREA FCST UPTO FL460 VALID 1000/1100 FOR AREA 28N72E 30N80E 20N90E 15N75E 28N72E SYNOPSIS TC CENTRED 23N74E AT 0300UTC MOV NW 20KT INTNSF=

ARFOR VIDP → Area Forecast Issued by Delhi Palam
091400 → On 09th at 1400 UTC Area Forecast
UPTO FL460 → Valid 1000/1100 UTC Area Forecast Upto FL460
VALID 1000/1100 → Valid on 10th Day from 0000 UTC to Next Day 0000 UTC
FOR AREA: 28N72E → 30N80E → 20N90E → 15N75E → 28N72E (defining the area polygon)
SYNOPSIS TC CENTRED 23N74E AT 0300UTC → Tropical Cyclone centred at 23N 74E at 0300 UTC
MOV NW 20KT → Moving NW at 20 KT
INTNSF → Intensifying

9. ROFOR — Route Forecast for Aviation

ROFOR = Route Forecast for Aviation
Provides route-based weather data for en-route flight planning including winds, temperatures, cloud, turbulence, icing, jet stream, and maximum wind information at multiple altitude levels along a specific route.

10. ROFOR Code Form & Sections

ROFOR (YYGGggZ) $Y_1Y_1G_1G_1/Y_2Y_2G_2G_2$[KMH or KT or MPS] CCCC Oiizzz (VVVV) ($W_1W_1W_i$) N.CCl.hs.hs 7hhhhhhh.hs. 6lhhhht. SECTION 2 (1111) $QL_iL_iL_iL_i$ h'h'(fff) SECTION 3 (22222 h'ah'$_m f_a f_a f_a$ $d_a d_a$VV) SECTION 4 9innn

ROFOR Sections

Section	Content
Section 1	Code identification and time group of ROFOR
Section 2	Symbolic Figure 11111, Jet Stream Data (optional)
Section 3	Symbolic Figure 22222, Max Wind and Vertical Wind Shear (optional)
Section 4	Supplementary weather phenomena — lowest MSL pressure, type of Front, change at Lat/Long, BECMG, FROM, TEMPO etc.

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Note on ROFOR Encoding:
Height of winds and temperature are prefixed by **INDICATOR 4** followed by direction in 2 digits (e.g., 28 for 280°) and speed in 3 digits (e.g., 20 kt is coded 020).
Indicator 5 = MODERATE TURBULENCE IN CLEAR AIR at 33,000 ft, thickness of turbulence 6000 ft (1800 m)
Indicator 6 = LIGHT ICING IN PRECIPITATION at 12,000 ft; thickness 2700 m or 9000 ft
Indicator 11111 = JET STREAM location (quadrant, lat, lon), altitude, wind speed
Indicator 22222 = MAX WIND altitude, direction, speed and VERTICAL WIND SHEAR (wind speed or direction changing over distance) per 300 m

11. ROFOR Decoded Example

ROFOR 060300Z 060612 KT VIDP 12580 2AC120 4CI300 7///180 403028 27010 405020 28015 407015 30020 410010 33025 523306 631209 11111 12775 40120 22222 38150 27115=

Full Decoded ROFOR:
ROFOR → Route Forecast
060300Z → Issued on 06th Day at 0300 UTC
060612 → Valid on 06 Day from 0600 to 1200 UTC
KT → Winds Speed in KT
VIDP 12580 → Route: Palam (Delhi) to (Quadrant 1) 25N 80E
2AC120 → Clouds: 2/8 AC at 12,000 ft
4CI300 → 4/8 CI at 30,000 ft
7///180 → Indicator 7: Freezing Level 18,000 ft (/// = cloud altitude not reported)
403028 27010 → Indicator 4 for 3000 ft: Temp 28°C, Wind 270°/10 KT
405020 28015 → At 5000 ft: Temp 20°C, Wind 280°/15 KT
407015 30020 → At 7000 ft: Temp 15°C, Wind 300°/20 KT
410010 33025 → At 10,000 ft: Temp 10°C, Wind 330°/25 KT
523306 → Indicator 5 (TURB): Moderate Turbulence in Clear Air at 33,000 ft, Thickness of Turbulence 6000 ft (1800m)
631209 → Indicator 6 (ICING): Light Icing in Precipitation at 12,000 ft; Thickness 2700 m or 9000 ft
11111 12775 40120 → Jet Stream: Location (Quadrant 1) 27N 75E at 40,000 ft; Wind Speed 120 KT
22222 38150 27115 → Max Wind: At 38,000 ft, Speed 150 KT, Direction 270°, Vertical Wind Shear 15 KT per 300 m

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Interpret the ROFOR (Second Example)

ROFOR 070530Z 070915 KT VECC VANP 4000 444 3CB020 7///170 641809 5416009 405016 16012 410006 24015 415003 28025 4200M04 27045 4300M25 28080 11111 12881 40150 22222 42160 28 25=

Hints: 444 = stands for Hail; 64 (Moderate icing, medium), 54 (Mod Turbulence in Cloud Occasional)
(Note: See ARFOR for explanation of terms)

12. Weather Symbols Used in Forecasts

Symbol	Weather	Symbol	Weather
∞	Widespread Haze	↗	Widespread Blowing Snow

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Symbol	Weather	Symbol	Weather
∿	Widespread Smoke	◠∿	Freezing Precipitation
≡	Widespread Mist	⚡ (R)	Thunderstorm
≡≡	Widespread Fog	6	Tropical Cyclone
∇	Freezing Fog	—v—	Severe Line Squall
S	Severe Sand or Dust Haze	◠	Moderate Turbulence
S+	Widespread Duststorm or Sandstorm	∧	Severe Turbulence
,	Drizzle	∿∿∿	Marked Mountain Wave
/// ///	Rain	⌣	Light Ice Accretion
* *	Snow	⌣⌣	Moderate Ice Accretion
△	Hail	⌣⌣⌣	Severe Ice Accretion
▽	Shower	⌂	Volcanic Eruption

13. Practice Q&A — TAF

This invented TAF is a code-reading exercise, rather than an operational forecast.

Use this TAF for Questions 1–16

```
TAF VILK 241800Z 2500/2509 09008KT 0800 FG NSC BECMG 2504/2505 09015KT 6000 NSW
SCT008 BKN120
TEMPO 2506/2508 12015G30KT 3000 TSRA FEW012 FEW025CB BKN100 BECMG 2508/2509
09010KT 7000 NSW FEW030 SCT120 BKN280=
```

Q1. The TAF has been issued on: (a) 23rd (b) 18th (c) 24th

✅ Correct: (c) 24th — YY in YYGGggZ = 24

Q2. The TAF has been issued at: (a) 2330 IST (b) 1830 UTC (c) 24 UTC

✅ Answer: (a) 2330 IST — The issue time is 1800 UTC (241800Z). IST is UTC + 5 h 30 min, so 1800 UTC = 2330 IST.

Q3. The expected surface wind speed is: (a) 0 kt (b) 6 kt (c) 08 kt

✅ Answer: (c) 08 kt — The initial forecast wind is 09008KT: 090° at 8 kt.

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Q4. Initially expected wind direction is: (a) 120° (b) 090° (c) 050°

✅ Answer: (b) 090° — The initial forecast wind is 09008KT: from 090° at 8 kt.

Q5. TAF is Valid for Date: (a) 23rd (b) 24th (c) 25th

✅ Correct: (c) 25th — Validity 2500/2509 = 25th 00Z to 25th 09Z

Q6. Lowest forecast visibility in TAF is: (a) 1500m (b) 800m (c) 0200m

✅ Answer: (b) 800m — The lowest forecast visibility is the initial 0800 m in fog (FG). The later groups forecast 6000 m, 3000 m (in TSRA) and 7000 m.

Q7. Height of base of lowest clouds in TAF is: (a) 1000 m (b) 1000 ft (c) 0800 ft

✅ Answer: (c) 0800 ft — SCT008 is the lowest cloud: 008 × 100 = 800 ft.

Q8. Weather TSRA is expected after: (a) 0600 UTC (b) 0600 IST (c) 0800 UTC

✅ Answer: (a) 0600 UTC — The TEMPO group 2506/2508 forecasts TSRA from 0600 UTC on the 25th (to 0800 UTC).

Q9. Direction of gusty wind is: (a) 090° (b) 100° (c) 120°

✅ Answer: (c) 120° — The gusty wind is in the TEMPO group: 12015G30KT, from 120° at 15 kt gusting 30 kt.

Q10. Expected weather up to 2504 0400UTC is: (a) Fog (b) TSRA (c) Mist

✅ Answer: (a) Fog — The initial forecast is 0800 FG: fog, with visibility 800 m. It lasts until the BECMG 2504/2505 improves the visibility to 6000 m.

Q11. Amount of lowest cloud is: (a) 1–2/8 (b) 3–4/8 (c) 5–7/8

✅ Answer: (b) 3–4/8 — SCT008 is the lowest layer: SCT means 3–4 oktas.

Q12. Amount of CB cloud is: (a) 1–2/8 (b) 2–4/8 (c) 5–7/8

✅ Correct: (a) 1–2/8 — FEW025CB = FEW CB

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Q13. Height of base of CB is: (a) 2500 ft (b) 3000 m (c) 2500 m

✅ Answer: (a) 2500 ft — FEW025CB: the cloud base is 025 × 100 = 2,500 ft above the aerodrome.

Study point for Q14 — Cloud-base height

In the TAF group BKN280, the base of the broken layer is 28,000 ft above aerodrome elevation. This is a cloud base, not its top and not a pressure flight level.

Q15. Period of validity of TAF is: (a) 23 to 00 IST (b) 00 to 09 UTC (c) 20 to 12 UTC

✅ Correct: (b) 00 to 09 UTC — Validity: 2500/2509 = 25th 0000 UTC to 25th 0900 UTC

Q16. Group 1500/1509 in a TAF indicates: (a) TAF issued on 15th at 0000 UTC (b) TAF is valid from 15th 0000 to 0900 UTC (c) TAF is valid from 1500 to 0900 UTC

✅ Correct: (b) TAF is valid from 15th day 0000 UTC to 15th day 0900 UTC

Q17. What is true of a TAF? (a) 9 hr TAF for international dissemination (b) TAF valid for 9 hr is issued every 6 hr (c) TAF valid for 12–30 hr is issued every 3 hr (d) TAF for national use valid for 9 hr and issued every 3 hr

✅ Correct: (d) — 9 hr TAF for national (domestic) use, issued every 3 hr

❌ (a) 9 hr TAF is NOT for international dissemination; (b) 9 hr TAF issued every 3 hr, not 6; (c) 12–30 hr TAF issued every 6 hr, not 3.

14. Practice Q&A — ROFOR

Worked ROFOR exercise

Use this ROFOR for the ROFOR Questions

```
ROFOR 010000Z 010610 KT VECC VILK 2SC030 2CB030 3AC100 2CI300 7///170
621800 541501
405022 28015 407010 28020 410005 29030 420M05 27045 440M41 27105
11111 12870 380120
22222 36140 2825=
```

Q1. Time of issue of ROFOR is: (a) 0610 UTC (b) 0000 IST (c) 0000 UTC

✅ Correct: (c) 0000 UTC — YYGGggZ = 010000Z = 01st day, 0000 UTC

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Q2. Period of validity of ROFOR is: (a) 0000 to 0000 IST (b) 0600 to 0100 IST (c) 0600 to 1000 UTC

✅ Correct: (c) 0600 to 1000 UTC — Validity 010610 = Day 01, 0600 to 1000 UTC

Q3. Wind speed in ROFOR is in: (a) KT (b) MPS (c) KMH

✅ Correct: (a) KT — wind unit specified as KT in the code

Q4. Base of CB cloud is: (a) 300 m (b) 300 ft (c) 3000 ft

✅ Correct: (c) 3000 ft — 2CB030 = 2/8 CB at 030 = 3000 ft

Q5. Height of Freezing Level? (a) 1700 m (b) 17,000 ft (c) 1.7 km

✅ Answer: (b) 17,000 ft

Q6. Icing group in the ROFOR is: (a) 7///170 (b) 621800 (c) 54//609

✅ Answer: (b) 621800

Q7. Type of Icing? (a) Nil (b) Light (c) Light icing in cloud

✅ Correct: (c) Light icing in cloud — L=2 in 621800 = Light Icing in Cloud (from Table 26.2)

Q8. Indicator figure for turbulence group in the ROFOR is: (a) 7 (b) 6 (c) 5

✅ Correct: (c) 5 — Turbulence group starts with Indicator 5

Q9. Height at which Icing is expected: (a) 21,000 ft (b) 18,000 ft (c) 15,000 ft

✅ Answer: (b) 18,000 ft

Q10. Height of Icing expected to be: (a) 300 m (b) 600 m (c) Up to Cloud Top

✅ Correct: (c) Up to Cloud Top — Thickness code 0 in icing group = extends to cloud top

Q11. Height at which Turbulence is expected: (a) 15,000 ft (b) 5,000 ft (c) 12,000 ft

✅ Answer: (a) 15,000 ft — The turbulence group 541501: indicator 5, type 4 (moderate turbulence in cloud, occasional), height 15 = 15,000 ft, thickness code 1.

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Q12. Thickness of turbulence expected is? (a) 2000 m (b) 300 m (c) Up to Top of cloud

✅ Answer: (b) 300 m

Q13. Wind at 5000 ft expected is? (a) 280/10 kt (b) 280/15 kt (c) 280/20 kt

✅ Answer: (b) 280/15 kt

Q14. Wind at 20000 ft expected is? (a) 270/45 kt (b) 270/45 KMH (c) 270/40 kt

✅ Correct: (a) 270/45 kt — 420M05 27045: at 20,000 ft, Temp M05°C (−5°C), Wind 270°/45 KT

Q15. Temperature at 20000 ft expected is? (a) −05°C (b) 05°C (c) 265 K

✅ Correct: (a) −05°C — M05 in 420M05 = Minus 05°C

Q16. Temperature at 40,000 ft expected is? (a) −45°C (b) −41°C (c) 233 K

✅ Correct: (b) −41°C — 440M41 27105: at 40,000 ft, Temp M41°C = −41°C

Q17. Date of issue of forecast is? (a) 10 (b) 01 (c) 02

✅ Correct: (b) 01 — YY in 010000Z = 01 (1st day of month)

Q18. Indicator group for Jet stream in ROFOR is: (a) 22222 (b) 11111 (c) 111111

✅ Correct: (b) 11111 — Jet Stream data starts with indicator 11111

Q19. Indicator group for Maximum Wind in ROFOR is: (a) 22222 (b) 11111 (c) 111111

✅ Answer: (a) 22222

Q20. Maximum Wind speed expected is: (a) 120 kt (b) 130 kt (c) 140 kt

✅ Correct: (c) 140 kt — 22222 36140 2825: Max wind at 36,000 ft, speed 140 KT, direction 280°

Q21. Maximum Wind speed expected at height: (a) 40,000 ft (b) 36,000 ft (c) 38,000 ft

✅ Answer: (b) 36,000 ft

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Q22. Jet stream is expected at Lat/Long: (a) 27 N70 E (b) 28 N75 E (c) 28 N70 E

✅ Answer: (c) 28N70E — The jet stream group 11111 12870 380120 gives quadrant 1, latitude 28, longitude 70: 28°N 70°E, at 38,000 ft and 120 kt.

Q23. Vertical Wind Shear per 300 m expected is: (a) 25 kt (b) 30 kt (c) 38 kt

✅ Answer: (a) 25 kt — In 22222 36140 2825 the last two digits give the vertical wind shear per 300 m: 25 kt.

Q24. Jet stream core speed is expected to be: (a) 125 kt (b) 120 kt (c) 140 kt

✅ Answer: (b) 120 kt — In the jet stream group 11111 12870 380120 the speed of the core is 120 kt, at 38,000 ft.

Q25. Jet stream expected at height of: (a) 40,000 ft (b) 36,000 ft (c) 38,000 ft

✅ Answer: (c) 38,000 ft — The jet stream group 11111 12870 380120 places the jet core at 38,000 ft.

Q26. Wind at 40,000 ft expected is? (a) 270/105 kt (b) 270/115 KMH (c) 270/140 kt

✅ Correct: (a) 270/105 kt — 440M41 27105: at 40,000 ft, Wind 270°/105 KT

Chapter 27: Radar Reports, SIGMET and Satellite Bulletins

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1. Radar Report (RAREP) — Introduction & Code

RAREP = Radar Report. Reports the location, intensity, movement, and height of radar echoes (precipitation areas, thunderstorms) from a ground-based weather radar station.
RAREP is normally reported only at **synoptic hours**. In case of any break in observations or rapid development, additional messages are transmitted as necessary.

RAREP Code Form

RAREP FEBB Iiiiii YYGGg CHARACTER (b.b.b.b.b.f.t.t) b.b.b.h.h./f.t.t. Intensity Tendency: d.d.f.f. ALTD (bbb/H.H/rrr)

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Code Elements Explained

Code	Meaning
RAREP	Radar Report identifier
FEBB	Station Index number
YYGGg	Date and time of RAREP
Iiiiii	Station Index number
bbb	Azimuth in three digit degrees of points on the periphery of an echo area
rrr	Range: three digit in kilometres
ALTD	Indicator of Echo Height Information
H.H.	Height of Top of Echo amsl in km (may be repeated as required)
d.d.	Direction in tens of degrees in which echo is moving
f.f.	Speed in km per hour of the echoes

2. RAREP — Character of Echoes & Intensity Terms

Character of Echoes

Code	Character	Description
EYE	Eye	An echo of eye wall of a tropical cyclone
SPRL BAND	Spiral Band	A continuous or broken curved line of echoes of spiral band associated with a cyclone
SQL LN	Squall Line	Having length 60 km or more and length to width ratio about 10 to 1
BRKN LN	Broken Line	Broken line of echoes
SLD	Solid	An area fully covered with echoes
BRKN	Broken	An area 4/8 to 7/8 covered with echoes
SCT	Scattered	An area 1/8 to 4/8 covered with echoes
WDLY SCT	Widely Scattered	An area less than 1/8 covered with echoes
ISLTD	Isolated	Isolated solid mass of echo
ECHO ALOFT	Echo Aloft	Echoes seen only at elevations higher than the beamwidth

Intensity Tendency

Code	Meaning
INCG	Increasing

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Code	Meaning
DEC	Decreasing
NO CHG	No Change

Bright Band & Anomalous Propagation (AP) are reported in **plain language**.
Bright Band = A radar echo caused by enhanced reflectivity of partially melted snowflakes — occurs at the melting level.
AP (Anomalous Propagation) = False echoes caused by abnormal bending of radar beam near the ground (temperature inversions etc.)

3. RAREP — Important Notes

1. Groups within brackets () are repeated as many times as necessary.
2. To define the shape of **line echoes**, spiral bands and eye wall: as many bbb/rrr points as necessary are used. Points given along the line in **anticlockwise direction**.
3. To define the shape of **areas**: as many bbb/rrr points as necessary. Points given in **anticlockwise order** starting from the northernmost point. The first point is repeated as the last point to indicate it is a closed area.
4. If an echo system with a distinct characteristic is partly or wholly embedded in another, the two systems are reported in **separate groups**.
5. Reports of heights are restricted to a maximum range of 200 km from the station.

4. RAREP Decoded Examples

Example 1

RAREP 42375 15030 270/050 300/060 350/025 280/ 010 SLD INCG 1525 ALTD 270/09 050 340/10/060 280/12/ 010 =

Radar Report for 42375 (International Identification No.) (say) Palam Date 15th at 0300
UTC Area of Solid echo from and 270 deg 50, 300 deg 60 km, 350 deg 25 km, 280 deg 10 UTC
Area of Solid echo Moving in 150 direction with speed of 25 KMH
Altitude of echoes in 270 direction 9 km at 50 km
In 340 direction 10 km at 60 km; in 280 direction 12 km at 10 km = Intensity: **INCREASING**

Example 2

RAREP 43003 10143 270 / 030 / 330 / 040 050 / 020 MDT INCG RPD NW / 012 ALTD 260/ 15 / 030 BRIGHT BAND 280 / 30 AP 300/ 330 /150 Angles between the Azimuth 270° to 340° Range 15 to 20 nm (AP – Anomalous Propagation)

Station 43003, Date 10th at 14:30 UTC
Echo area defined by azimuths 270°/030 km, 330°/040 km, 050°/020 km (anticlockwise from northernmost)
MDT INCG RPD NW/012 → Moderate intensity, Increasing Rapidly, Moving NW at 012 km/h
ALTD 260/15/030 → Echo height: At azimuth 260°, top at 15 km, at 030 km range

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BRIGHT BAND 280/30 → Bright Band at 280° azimuth, 30 km range
AP 300/330/150 → Anomalous Propagation between azimuth 270° to 340°, Range 15–20 nm

5. SIGMET Message

SIGMET = Significant Meteorological Information
Prepared in **abbreviated plain language** using ICAO approved abbreviations. Contains information about significant en-route weather hazards.

SIGMET types: WS identifies phenomena other than tropical cyclones and volcanic ash; WC identifies tropical-cyclone SIGMETs; WV identifies volcanic-ash SIGMETs. These are bulletin data-type designators.

6. SIGMET Content & Structure

SIGMET Element	Abbreviation	Meaning
Sequence	SIGMET 2	2nd SIGMET = 2nd SIGMET in series
Validity	VALID 221215/221600	Valid from 22nd 1215 UTC to 22nd 1600 UTC
Severity	SEV	Severe
Frequency	FRQ	Frequent
Observed/Forecast	OBS / FCST	Observed at time / Forecast at time
FL	FL250, FL390	Flight Level (×100 ft)
Tops	TOPS FL390	Cloud/echo tops at FL390
Intensity change	INTSF / WKN / NC	Intensifying / Weakening / No Change
Movement	MOV E 25KT	Moving East at 25 knots
Outlook in the historical example	OTLK	Outlook; not a cloud-top code

7. SIGMET Examples

Historical tropical-cyclone SIGMET: reading practice

This archived example concerns 19 May 2020. It is a reading exercise, not current flight information.

VECF SIGMET 1 VALID 190600/191200 VECC – VECF KOLKATA FIR TC AMPHAN OBS AT 0000Z N1536 E08642 CB TOP FL 520 WI 200 NM OF CENTRE MOV NNE 12KT NC FCST 1200Z TC CENTRE N1724 E08706

Read the groups in order:
VECF / VECC: Kolkata FIR / originating meteorological watch office.
SIGMET 1: message sequence number 1.
190600/191200: valid on the 19th from 0600 to 1200 UTC, a six-hour interval permitted for a tropical-cyclone SIGMET.
OBS AT 0000Z: the cyclone position was observed at 0000 UTC.

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N1536 E08642: centre at 15°36′N, 86°42′E.
CB TOP FL520: cumulonimbus (a deep cloud associated with thunderstorms) tops at flight level 520.
WI 200 NM OF CENTRE: within 200 nautical miles of the centre.
MOV NNE 12KT NC: moving north-northeast at 12 knots; no intensity change expected.
FCST 1200Z TC CENTRE N1724 E08706: centre forecast for 1200 UTC at 17°24′N, 87°06′E.

For ordinary SIGMET phenomena, validity is at most four hours. The six-hour maximum applies to tropical-cyclone and volcanic-ash SIGMETs.

SIGMET Sub-types — Key Phenomena:

Phenomenon	Code in SIGMET
Thunderstorm	OBSC TS, EMBD TS, FRQ TS, SQL TS (add GR for hail)
Severe Turbulence	SEV TURB
Severe Icing	SEV ICE
Tropical Cyclone	TC
Volcanic Ash	VA
Severe Mountain Wave	SEV MTW
Heavy Duststorm / Heavy Sandstorm	HVY DS / HVY SS

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8. Satellite Bulletin

Satellite imageries are analysed and supplied as **Bulletins in abbreviated form**.
Format: SAT BLTN BASED ON METSAT PIC OF YYGGgg UTC

Satellite Bulletin Abbreviations

Abbreviation	Meaning
CONVTE	Convective
CLDS	Clouds
EMBDD	Embedded
MOD	Moderate
INT CONVTN	Intense Convection (heat transfer through movement within a fluid)
SKM	Sikkim
EC BAY	East Central Bay (of Bengal)
AFSN PAK	Afghanistan Pakistan
RPT	Repeat

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Abbreviation	Meaning
IDS	Islands
COT	Coast
ADJ	Adjoining
EXT	Extending
S AREA	Sea Area
BET LAT	Between Latitude
LONG	Longitude
W COT	West Coast
SB	Scattered to Broken (cloud coverage)

9. Satellite Bulletin Decoded Example

Original Satellite Bulletin (coded)

SAT BLTN BASED ON METSAT PIC OF 271200 UTC. THE CONVTE CLOUDS CURVED BANDS OF VORTEX OVER EC BAY HAS REDUCED NOW CENTERED OVER 16.5N/91.0E RPT 16.5N/91.0E(.) INTENSITY T REDUCED MOV 1.5 RPT T 1.5 (.) ASSTD BKN LOW MED CLOUDS EMBDD SCT MOD TO INT CONVTN (AT10.5N TO 14.0N LONG 85.5E TO 91.5E AND LAT 14.0N TO 18.5N EAST OF LONG 87.5E ARAKAN COT AND S MYANMAR(.) SCT M/LAYERED CLOUDS OVER NW JK AND AREA BET LAT 37.0N TO 16.0N LONG 60.0E TO 100.0E(.) SCT LOW /MED CLOUDS OVER SKM NE STATES BAY IDS KER(.) SCT MED/HIGHCLOUDS OVER AFSN PAK REST JK HP UTRCL PJB AND NW GUI(.) SCT TO BKN LOW/MED CLOUDS EMBDD, MOD TO INT CONVTN OVER S AREA BET LAT 5.0N TO 14.8N LONG 60.0E TO 68.0E SE ARSEA ADJ INDIAN OCEAN BET LAT 2.0N TO 8.0N LONG 69.0E TO 76.0E S SRILANKA GULF OF MARTABAN SUMATTRA ADJ W COT BORNEO JAVA IDS AND SEA EXT. N MADAGASCAR(.)=

Interpretation (Plain Language)

Satellite Bulletin based on METSAT picture of 27th day at 1200 UTC:

The convective clouds — curved bands of vortex over **East Central Bay** (of Bengal) — has reduced. It is now centred over 16.5N/91.0E. Repeat 16.5N/91.0E. Intensity 1.5. Associated broken low and medium clouds embedded, scattered to moderate to intense convection at 10.5N to 14.0N long 85.5E to 91.5E and lat 14.0N to 18.5N east of long 87.5E — Arakan coast and S Myanmar.

Scattered multi-layered clouds over **NW J&K** and area bet lat 37.0N to 16.0N long 60.0E to 100.0E. Scattered low/medium clouds over **Sikkim, NE states, Bay Islands, Kerala**. Scattered med/high clouds over **Afghanistan, Pakistan, rest J&K, HP, Uttaranchal, Punjab and NW Gujarat**.

Scattered to broken low/medium clouds embedded, moderate to intense convection over sea areas between lat 5.0N to 14.8N long 60.0E to 68.0E, **Arabian Sea**, adjoining Indian Ocean between lat 2.0N to 8.0N long 69.0E to 76.0E, **S Sri Lanka, Gulf of Martban, Sumatra**, adjoining west coast of **Borneo, Java, islands** and sea extending N of **Madagascar**.

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Chapter 28: Meteorological Documentation and Briefing

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1. Pre-Flight Briefing

Definition: Pre-flight briefing is the provision of meteorological information to the flight crew before departure, covering weather conditions relevant to the planned route and aerodromes.

Briefing for **Low Level flights** and those under **Visual Flight Rules (VFR)** is given up to:

- FL100 (or up to FL150 above in mountains)
- Poor visibility — 5000 m or less
- Any phenomena likely to lower widespread visibility to less than 5000 m
- Clouds which may affect flight
- SIGMET information

📌 **EXAM TIP — VFR Briefing Ceiling:** "FL100 on plains, FL150 in mountains" — remember: mountains are higher, so briefing ceiling is higher too. Both have the 5000 m visibility threshold.

2. Oral Briefing

Oral Briefing: Provided when adequate advance notice has *not* been given to the Met Office for preparation of formal documentation.

Notice Requirements

Recipient	Advance Notice Required
AMOs (Aerodrome Met Offices)	3 hours
AMS (Aeronautical Meteorological Station, no forecasting facilities)	At least 3 hours during watch hours, otherwise 18 to 24 hours before ETD

Where adequate notice has not been given, oral briefing covers all the Met information listed above. Details are recorded in the **Briefing Register** with remarks:

"Briefed Capt/Mr ..... documentation not provided for want of adequate notice" + signature of the crew obtained.

⚠ **Important:** Oral briefings may be **avoided as far as possible**. Formal documentation is always preferred.

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3. Flight Documentation — Forms

Flight Documentation is issued as Charts, Tabular, or Cross-section forms, and contains the following:

- (a) Upper Wind and Temperature, Tropopause height, Maximum upper Wind, Jetstream data, Expected en-route WX
- (b) SIGWX phenomena
- (c) METAR, SPECI and TAF of Aerodromes of departure, destination, en-route and alternates; Area/Local Forecast
- (d) Relevant SIGMET, AIREP, Cyclone advisory, VA report

4. Tabular Form (Met T-3) and Cross Section Form (Met T-4)

Met T-3: Tabular form for flights up to 500 nm
Met T-4: Cross-section form for flights beyond 500 nm

Information Provided in Both Forms

#	Content	T-3	T-4
(a)	Synoptic situation and Surface pressure data	✅	✅
(b)	Outlook of en-route weather	✅	✅
(c)	Amount and type of cloud (Pictorially depicted in T-4)	✅	✅ (pictorial)
(d)	Upper winds and temperatures, Height of Freezing Level	✅	✅
(e)	Tropopause data	✅	✅
(f)	Surface visibility (only for flights below FL100)	✅	✅

5. Significant En-Route Weather Phenomena

The following are included in Met T-4 and T-3:

Phenomenon	High Level Flights	Low Level Flights
TS (Thunderstorm)	✅	✅
TC (Tropical cyclone)	✅	✅
SQ Line (Squall Line)	✅	✅
Hail	✅	✅
MOD/SEV Turbulence	✅	✅
Icing	✅	✅
FZ Precipitation	✅	✅
Widespread DS/SS	✅	✅

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Phenomenon	High Level Flights	Low Level Flights
Marked Mountain Waves	✅	✅
Fog, Precipitation reducing visibility < 5000 m	❌	✅ *(additional)*

⚠ **Low Level EXTRA Items:** Fog and Precipitation reducing visibility below 5000 m are included *only* for Low Level Flights — these are additional to the 9 high-level phenomena.

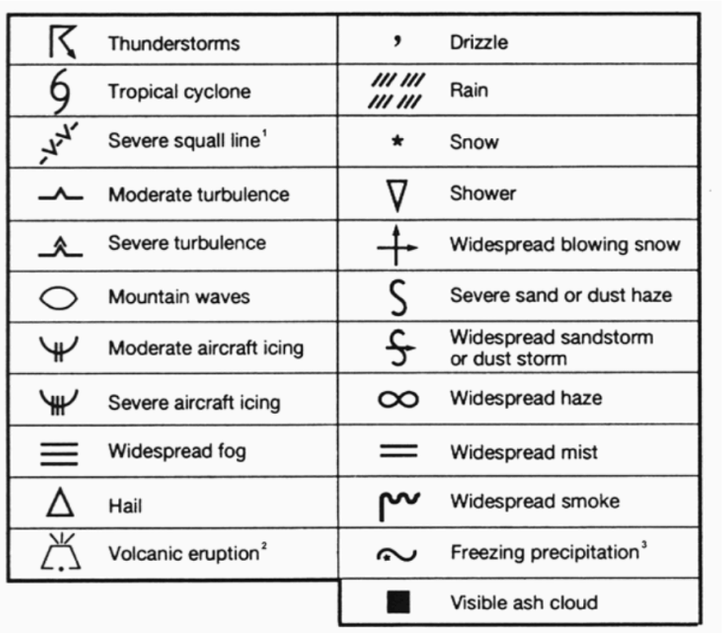
6. Set of Charts Provided

Flight Type	Charts Provided
International Flights	• SIGWX charts (WAFC forecast) between FL250 and FL630 • Forecast 250 hPa wind and temperature charts
National Flights	• National SIGWX charts • WAFC upper wind and temp. charts • For LL Flights: altitudes 600, 1500 and 3000 m (2000, 5000, 10,000 ft)
Fax/Internet NOT available	Met T-3 or Met T-4 form is provided

📌 **Key Values:**
International SIGWX: FL250–FL630 | International temp chart: 250 hPa
National LL altitudes: 600m / 1500m / 3000m = 2000 / 5000 / 10,000 ft

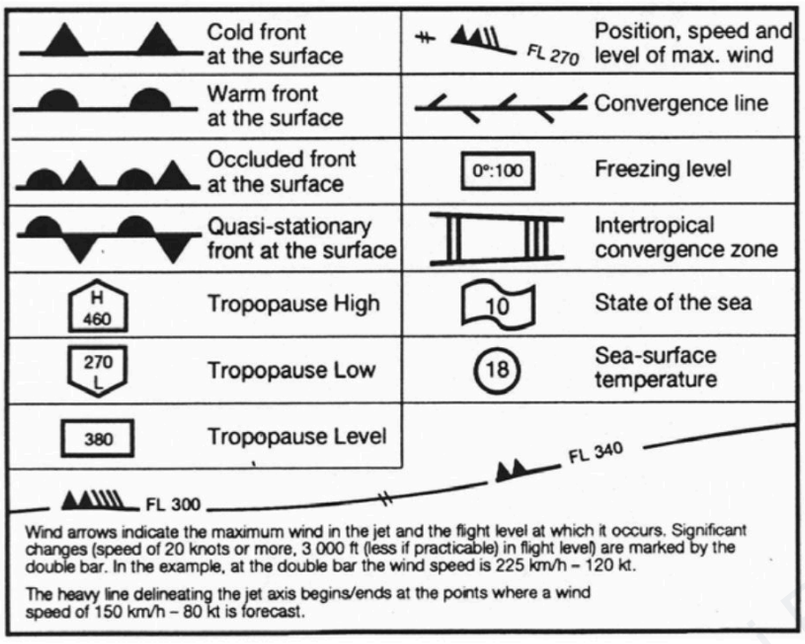
7. National Significant Weather (SIGWX) Charts

Preparation: The national numerical weather prediction division produces national significant-weather charts for medium and high levels. Check the currently issued product and its validity before use.

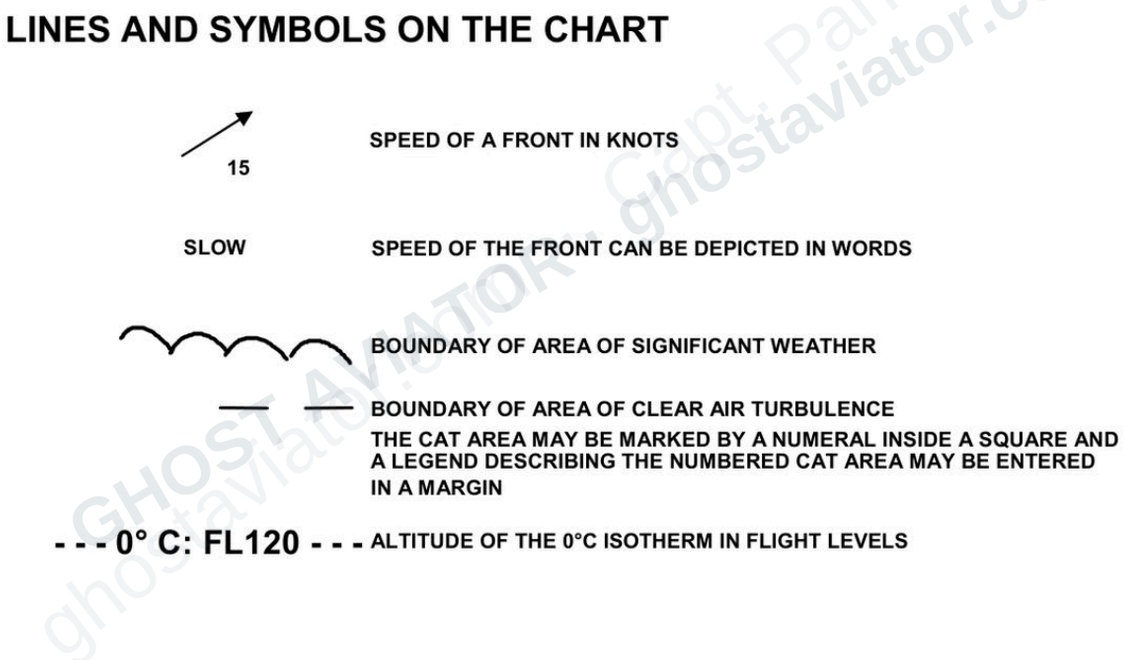


Symbols for significant weather

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Fronts, convergence zones and other chart symbols



Lines and symbols on a significant weather chart

Chart Levels

Level	Altitude Range	Special Feature
Medium Level	FL100 to FL250	Zero degree isotherm included
High Level	FL250 to FL630	—

Issuance Schedule

Based on Observation Times (UTC)	Issued	Valid for
00, 06, 12, 18 UTC	09 hr after observation time	24 hours

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→ SIGWX charts are made 4 times a day

Contents of National SIGWX Charts

- (a) Tropical Cyclone — wind 34 kt or more
- (b) CB associated with TS, TC, Line squall, Hail, MOD/SEV Icing and Turbulence, CAT, Marked Mountain Waves, Widespread DS/SS
- (c) Height of Freezing Level, FL of Tropopause
- (d) Location of Volcanic eruption and accidental release of radioactive material significant for air operations
- (e) Jetstream

8. Height Indications in Flight Documentation

Conditions	Height Expressed As
En-route Met conditions (e.g. upper winds, turbulence, cloud base/top)	Flight Levels
Aerodrome Met conditions (e.g. cloud base and top)	Above aerodrome elevation

9. In-Flight Met Information (VOLMET)

VOLMET / D-VOLMET: Continuous broadcast of meteorological information for aircraft in flight, transmitted by **Mumbai** and **Kolkata**.

In-Flight Information Provided to Operators / Crew

- METAR, SPECI, TAF, SIGMET, AIREP
- Upper winds and temperature
- VA (Volcanic Ash) and TC advisory

For in-flight planning: Information is supplied to Operators by the Met Office. For aircraft already in-flight, the information above is supplied directly.

Practice Q&A — Chapter 28

Answers: 1-c, 2-a, 3-c, 4-a, 5-b, 6-a, 7-b, 8-a, 9-c, 10-a

Study point for Q1 — National significant-weather forecasts

The checked service manual records a decision dated 23 June 2020 assigning production of national medium- and high-level significant-weather charts to the national numerical weather prediction division. This is a dated production description; confirm the current issuing arrangements from the actual product.

Study point for Q2 — Freezing level in the national medium-level chart

In the national significant-weather chart specification recorded with the decision of 23 June 2020, the medium-level chart covers FL100–FL250 and includes the 0°C isotherm (the freezing level). This dated specification does not mean that freezing-level information can never appear in other products.

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Question 3
Fog, Precipitation and weather which reduce visibility below 5000 m is considered Significant weather for:
(a) All level flights (b) High level flights (c) Low level flights

✅ Correct Answer
(c) Low level flights — Fog and poor visibility phenomena are significant only for low-level operations. High-level flights are above such phenomena.

❌ Distractor Analysis
(a) All level flights — Incorrect. These phenomena are not operationally relevant to high-level cruise flight.
(b) High level flights — High-altitude flight is above the cloud/fog layer where 5000m visibility thresholds apply.

Study point for Q4 — Supply of pre-flight weather information

Upper-wind, upper-temperature and significant en-route weather information should be supplied no later than three hours before scheduled departure. Other meteorological information is supplied as soon as practicable. Do not confuse this supply requirement with an operator's advance-notice requirement or the actual briefing time.

Question 5
SIGMET information is provided to:
(a) Low level flights (b) All level flights (c) High level flights

✅ Correct Answer
(b) All level flights — SIGMET is part of Flight Documentation and is included in briefing for all flights regardless of level.

❌ Distractor Analysis
(a) Low level flights only — Incorrect; SIGMETs cover significant meteorological hazards at all levels.
(c) High level flights only — Incorrect; even low-level flights need SIGMET awareness.

Question 6
For Low Level flights, and those under Visual Flight Rules, briefing is given up to:
(a) FL150 or above in mountains (b) FL110 above the sea (c) FL120 or above the plains

✅ Correct Answer
(a) FL150 or above in mountains — VFR/low-level briefing ceiling is FL100 on plains, FL150 in mountainous terrain.

❌ Distractor Analysis
(b) FL110 above sea — Not a defined level; trap option mixing up numbers.
(c) FL120 — Not the correct value; FL100 (plains) and FL150 (mountains) are the correct values.

Question 7
The height indications in Flight Documentation in all en-route Met conditions:
(a) Above aerodrome elevation (b) Flight Levels (c) As requested by the Pilot

✅ Correct Answer

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(b) Flight Levels — En-route conditions (upper winds, turbulence, cloud base/top) are expressed in Flight Levels.

❌ Distractor Analysis
(a) Above aerodrome elevation — This applies to aerodrome (local) conditions, not en-route.
(c) As requested by the Pilot — No such provision exists in standard documentation.

Question 8
The height indications in Flight Documentation in all aerodrome Met conditions:
(a) Above aerodrome elevation (b) Flight Levels (c) As requested by the Pilot

✅ Correct Answer
(a) Above aerodrome elevation — Cloud base and top at aerodromes are referenced to the local aerodrome elevation.

❌ Distractor Analysis
(b) Flight Levels — Used for en-route, not aerodrome conditions.
(c) As requested by the Pilot — Not a standard.

Question 9
For National Flights the Documents provided are:
(a) National SIGWX charts only (b) WAFC upper wind and temp. charts only (c) Both WAFC upper wind and temp. charts and National SIGWX charts

✅ Correct Answer
(c) Both WAFC upper wind and temp. charts AND National SIGWX charts — National flights receive both document types.

❌ Distractor Analysis
(a) National SIGWX only — Incomplete; WAFC temp/wind charts are also provided.
(b) WAFC charts only — Incomplete; National SIGWX charts are also provided.

Question 10 For National Flights the winds and temperatures are provided for LL Flights for altitudes: (a) 600, 1500 and 3000 m (b) 3000 ft, 5000 ft, 10,000 ft (c) 600, 1000 and 500 m

✅ Answer: (a) 600, 1500 and 3000 m — The levels are 600, 1500 and 3000 m, equivalent to 2000, 5000 and 10,000 ft.

❌ (b) 3000 ft is not one of the levels. (c) has the wrong middle and lower values.

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Chapter 29: Flight Forecasts: Tabular and Cross-Section Forms

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1. Overview — Met T-3 vs Met T-4

Met T-3 (Tabular Form): Used for flights up to 500 nm. Presents weather data in tabular columns for waypoints along the route.

Met T-4 (Cross Section Form): Used for flights beyond 500 nm. Presents data as a vertical cross-section of the atmosphere along the route, showing weather at different flight levels.

Feature	Met T-3 (Tabular)	Met T-4 (Cross Section)
Distance covered	≤ 500 nm	> 500 nm
Format	Table — columns per waypoint	Vertical cross-section diagram + table
Heights in	Feet amsl	Flight Levels
Cloud depiction	Text (amount, type, height)	Pictorial (cloud symbols)
Surface visibility	✅ Included	✅ Included (in Remarks)
Freezing level	✅ Height of 0° isotherm (feet)	✅ Included
QNH (aerodrome pressure reduced to mean sea level using the standard atmosphere)	✅ Forecast QNH (hPa)	✅ MSL pressure

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2. Met T-3: Flight Forecast Tabular Form — Annotated Sample

Below is an annotated breakdown of the sample Met T-3 (Table 29.1), dated **04 April 2023**:

📋 FLIGHT FORECAST (TABULAR FORM) — MET T-3 | Table 29.1

Route:VECC → VIDP via VILK (Kolkata → Delhi via Lucknow)
Valid for Departure:0300 UTC to 0500 UTC
Valid for Arrival:0600 UTC to 0800 UTC
Special Features:A WD LIES OVER W RAJASTHAN WITH UPPER AIR CYCIR UP TO 20,000 FT

Upper Winds and Temperatures — Sample Data

Height (ft amsl)	Temp (°C)	KOLKATA Wind	LUCKNOW Wind	DELHI Temp (°C)
30,000 ft	M33	300/60	270/70	M30
20,000 ft	M20	330/45	260/55	M18

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Height (ft amsl)	Temp (°C)	KOLKATA Wind	LUCKNOW Wind	DELHI Temp (°C)
10,000 ft	+12	050/15	120/25	14
5,000 ft	+22	050/10	160/15	24

Wind format: Direction(°)/Speed(kt) | M = Minus (negative temperature) | LOC = Locally

Cloud Forecast

Cloud Element	Details
Top/Base	SCT ST 1000 ft
	FEW SC 2500 ft
	SCT AC 12000 ft; ISOL EMBD CB 2000/35000 ft
	BKN CS 35000 ft
	3000 M IN RAIN AND [thunderstorm areas]

Other Parameters

Parameter	Value
Surface Visibility (general)	6 km
Surface Visibility (in TS)	1500 m in Thunderstorm
Significant Weather	ISOL THUNDERSTORM; ICING, TURB IN CB 25000/15000 ft
Height of 0° Isotherm	15000 ft (Kolkata) | 14000 ft (Lucknow)
Forecast QNH	1006 hPa (Kolkata) | 1008 hPa (Lucknow)

⚠ **Icing and Turbulence in CB:** Noted between 15,000 ft (base) and 25,000 ft (top). Aircraft must avoid CB areas.

3. How to Read a Met T-3

Wind Entry Format: `DDD/SSS` = Direction in degrees / Speed in knots
Example: `300/50` = Wind from 300° at 50 kt
`M33` = Minus 33°C (temperature)
`ISOL EMBD CB 2000/35000` = Isolated Embedded CB, base 2000 ft, top 35000 ft

✅ **Decision-Making from Met T-3:**

- Check Special Features first — they summarise the biggest hazard
- ISOL CB with EMBD = avoid — radar/pilot deviation required
- The freezing level is the height where temperature is 0°C. It helps assess possible airframe icing where liquid water is present; it is not an aircraft operating ceiling.

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- QNH tells you if pressure setting will differ from standard — plan altimeter corrections

4. Met T-4: Cross Section Route Forecast — Annotated Sample

The cross-section shows the atmosphere from **VIDP (Delhi)** → **VECC (Kolkata)** as a vertical slice, with cloud types depicted pictorially at their respective flight levels. (Table 29.2)

Cross Section Diagram — Cloud Symbols Identified

Symbol	Cloud Type	Typical Level
CI	Cirrus	High (FL300+)
CS	Cirrostratus	High (FL250+)
AC	Altocumulus	Medium (FL100–FL250)
AS	Altostratus	Medium (FL100–FL250)
SC	Stratocumulus	Low (below FL100)
ST	Stratus	Low (surface to FL050)
CU	Cumulus	Low-Medium
CB	Cumulonimbus (a deep cloud associated with thunderstorms)	All levels (vertical)

The diagram shows the route from VIDP to VECC with CB embedded near the eastern end, cirrus cloud at high levels throughout, and SC/ST near the surface.

Met T-4 Data Table — Sample

Remarks: VIS 8–10 km, 2000 m in Scattered TS

Lowest MSL Pressure: 1002 hPa

FL	Temp (VIDP side)	Wind (VIDP)	Wind Trend	Wind (VECC)	Temp (VECC side)
FL400	M38°C	270/80KT	WEAKENING	300/60KT	M39°C
FL300	M32°C	270/45KT	VEERING	300/35KT	M35°C
FL200	M15°C	260/30KT	VEERING	300/25KT	M16°C
FL150	05°C	210/20KT	VEERING	300/15KT	08°C
FL100	18°C	140/10KT	VEERING	300/20KT	20°C

📌 **Wind Trend Terms:**
VEERING = Wind direction changing clockwise (e.g. 260° → 300°) — describes a clockwise direction change
BCKING (Backing) = Wind direction changing anti-clockwise — describes an anticlockwise direction change
WEAKENING = Wind speed decreasing along route

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5. How to Read a Met T-4

Cross Section Format: Vertical slice of the atmosphere along the route. X-axis = route (departure to destination). Y-axis = Flight Levels (30 to 400+). Cloud symbols drawn at their actual position and level.

✅ **Key Operational Tips from T-4:**

- A clockwise change along the route is veering. This spatial change alone does not establish warm advection (horizontal transport by moving air) or an approaching front.
- Compare each direction carefully: 140° to 300° is a clockwise change of 160°, so the FL100 example is veering.
- CB shown near VECC end — route planning: plan diversion or delay if embedded CB
- When flying from higher to lower pressure without updating the altimeter setting, indicated altitude can be too high. Obtain the appropriate current setting; route minimum pressure is not automatically destination QNH.

6. Cloud Amount Abbreviations

Standard ICAO cloud amount code (Oktas = eighths of sky covered)

Code	Oktas	Meaning
SKC	0 oktas	Sky Clear
FEW	1–2 oktas	Few clouds
SCT	3–4 oktas	Scattered
BKN	5–7 oktas	Broken
OVC	8 oktas	Overcast

7. Frequency Abbreviations (ISOL / OCNL / FRQ)

Code	Area Coverage	Meaning
ISOL	< 50% of area	Isolated — individual features
OCNL	50–75% of area	Occasional — well-separated features
FRQ	> 75% of area	Frequent — little or no separation between features

⚠ **EMBD CB — Most Dangerous Scenario:** An Embedded CB (ISOL EMBD CB) is concealed within stratiform cloud layers. It cannot be visually avoided and may not always be clearly detected on weather radar. FRQ describes frequent embedded cumulonimbus with little or no separation. Plan to avoid hazardous convection (heat transfer through movement within a fluid) using current information and approved procedures; the code itself does not prescribe a particular deviation.

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8. Key Synoptic System Definitions

This section provides concise definitions of synoptic systems relevant to aviation in India. These are high-yield for DGCA written exams.

Term	Definition
Depressions and Cyclones	Intense LP systems with 2 or more closed isobars at 2 hPa interval and strong winds.
Trough in Westerlies	A wave in mid latitudes, moving from W to E over the globe. Affects northern parts of India.
Trough in Easterlies	A wave in the equatorial easterlies moving from East to West.
Western Disturbance (WD)	Low pressure area on surface or cyclonic circulation/trough in mid and lower tropospheric levels, originating over Mediterranean Sea, Caspian Sea, and Black Sea, moving across N India. Systems on the surface with 2 or more closed isobars are called **Western Depressions**.
Induced Low	Under the influence of WD, a low develops to the South of the system. Called an **Induced Low**. Cyclonic circulation in the upper air developed to the south = **Induced Cyclonic Circulation**.
Easterly Wave	A shallow trough in the equatorial region of tropics, more evident in upper level winds than surface pressure pattern. Moves from E to W. Causes clouds and showers. Affects S peninsular India.
High/High	Area in the atmosphere where pressures are higher than surroundings at the same level. Represented on synoptic chart by at least one closed isobar (a line joining places with the same pressure at the same level and time).
Shear Line	A line or narrow zone across which there is an abrupt change in the horizontal wind component. It is a line of maximum horizontal wind shear (wind speed or direction changing over distance).
Wind Discontinuity	A line across which there is an abrupt change in wind direction.

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Glossary

Plain-language definitions. Read the chapter for the full explanation.

Term	Simple meaning
Absolute humidity	Mass of water vapour per unit volume of air, commonly in g/m³.
Advection	Horizontal transport by moving air.
Air mass	A broad body of air with similar temperature and moisture.
Anabatic wind	Wind flowing uphill.
Anemometer	An instrument that measures wind speed.
Barometer	An instrument that measures air pressure.
Condensation	Water vapour changes into liquid water and releases latent heat.
Conduction	Heat transfer through direct contact.
Convection	Heat transfer through movement within a fluid.
Cumulonimbus	A deep cloud associated with thunderstorms.
Density	Mass per unit volume.
Dew point	Temperature to which air must cool to become saturated at unchanged pressure and water-vapour content.
Divergence	Airflow spreading apart.
Downburst	A strong descending current that spreads outward near the ground.
Evaporation	Liquid water changes into vapour and absorbs latent heat.
Front	A transition between different air masses.
Isobar	A line joining places with the same pressure at the same level and time.
Lapse rate	Temperature decrease with increasing height.
Latent heat	Energy absorbed or released during a change of state without a temperature change.
Mixing ratio	Mass of water vapour divided by the mass of dry air, commonly in g/kg.
QFE	Atmospheric pressure at the aerodrome reference point.
QFF	Station pressure reduced to mean sea level using actual temperature.
QNH	Aerodrome pressure reduced to mean sea level using the standard atmosphere.
Radiosonde	A balloon-carried instrument that transmits upper-air measurements.
Relative humidity	Actual moisture compared with saturation at the current temperature, expressed as a percentage.
Saturation	Water-vapour pressure has reached its saturation value at the current temperature.
Supercooled water	Liquid water below 0°C; droplets can freeze on an aircraft.
Temperature inversion	A layer where temperature increases with height.
Virga	Falling precipitation that evaporates before reaching the surface.
Wind shear	Wind speed or direction changing over distance.

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